

Andreas Gadatsch

Business Process Management

Analysis, Modelling, Optimisation and
Controlling of Processes

 Springer

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Preface to the First Edition

After the 9th edition of this textbook was published, process management has developed strongly through the trend of digitalization and as a result of the “pandemic”. Process management is unthinkable without digitalization in most cases. A new trend is the increased use of data science methods for process management, which is consequently referred to as “process science”.

Of particular importance are recent research results that have been published under the heading “exploratory process management”. They show that the first main phase of process management was primarily focused on the optimization of existing processes and business models. The exploratory business process management penetrates into the domains of information management, but also business field development and deals with innovative new business models for which processes have to be fundamentally redesigned.

New practice examples have been included in various places in the book, such as the migration strategies for the ERP system “SAP S/4 HANA”, which forms the basis for many industrial and service processes.

The chapter on process modeling has been updated and newer, more far-reaching methods, such as the business model canvas, have been included. The modeling examples have been supplemented on the basis of a small, consistent case study on the takeover of a general practitioner’s practice.

A quick test at the beginning of the book can be used by readers from professional practice to carry out a first analysis of their own situation. The corresponding file can be requested from the author. This also applies to the illustrations that can be used for teaching purposes. An email to andreas.gadatsch@h-brs.de is sufficient for this.

Despite the long running time of this textbook, it can be assumed that there will still be no error-free book after 22 years. The author is grateful for corrections and constructive suggestions for improvement, for example to the email address mentioned.

For the sake of better readability, we mainly use the generic masculine in this book. This always implies both forms, i.e. it also includes the female form.

Sankt Augustin
July 2022

Andreas Gadatsch

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Introduction to Business Process Management

1

Process management changes the world of work

Abstract

This introductory chapter first explains the concept and historical development of business process management. Subsequently, several basic concepts such as “function”, “business process”, “process”, “end-to-end process” and “workflow” are defined and distinguished from each other. The conclusion is formed by review questions and an exercise.

1.1 Concept Clarification

Why do we need business process management? This question is not only asked by students of business administration who go to a corresponding lecture with great expectations, but also by experienced practitioners. A look at history can help here a little bit. Since the beginning of the 19th century, the world of work has been characterized by a strong division of labor as a result of the previous industrial revolution. The *Taylorism* played an important role here, named after the US American Frederick W. Taylor (see the original work Taylor 1903).

The *business process management* (GPM) or simply *process management* was developed at the beginning of the 1990s in order to, among other things, eliminate the negative consequences of division of labor and poor coordination.

Business process management deals with the documentation, analysis and restructuring of workflows (processes). For a long time, the term “process organization” was common in German-language literature. The documentation of the processes is also referred

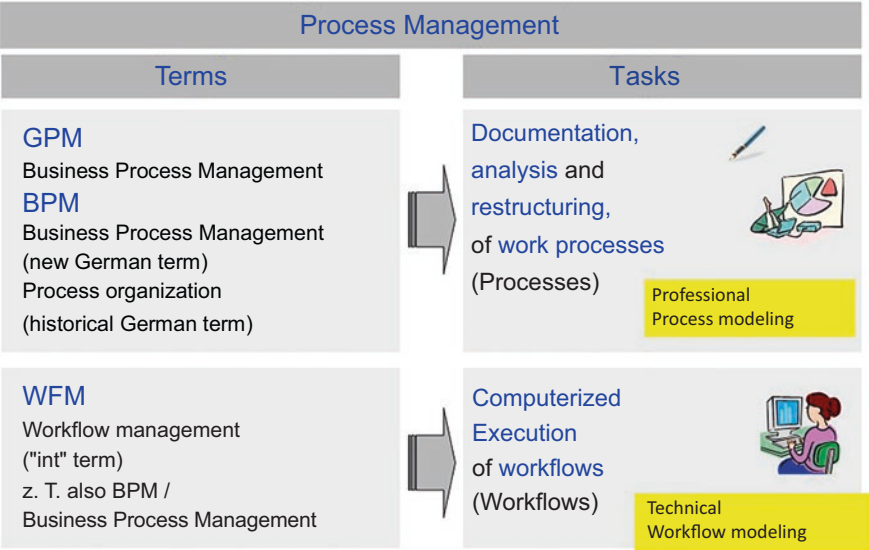


Fig. 1.1 Concept clarification process management

to as “technical process modeling”. In the international environment, the term “business process management (BPM)” is common.

To be distinguished from this is the term “workflow management” (WFM), which covers the computer-supported execution of workflows (so-called “workflows”). Here one also speaks of “technical workflow modeling”. In international usage, the terms “business process management and workflow management” are often not further differentiated, one usually speaks in both cases of “business process management (BPM)”. Figure 1.1 shows the terminology at a glance.

1.2 Historical Development

In the development of process management, four phases of development can be identified (see Table 1.1).

I. Phase: Breakdown of work into functions (Taylorism):

The early phase of process management begins with Taylorism, named after Frederic Winslow Taylor (1856–1915), who consistently separated planning and executive activities. This classical bureaucratic organizational structure prevailed in almost all companies of the 19th century and played a central role for departments (cf. Sua-Ngam-Iam and Kühl 2021, p. 46). According to the then prevailing business paradigm, the construction and operational organization were considered separately in this phase. This was first

Table 1.1 Development phases of process management

Phase 1	From 1900	Breakdown of Work into functions (Taylorism)
Phase 2	Approx. 1970–1980	Sequencing of functions (action-oriented data processing)
Phase 3	Approx. 1990 bis 2015	Process orientation: Formation of processes as an overarching structural element (Exploitative Process Management)
Phase 4	From 2015	Digitalization: processes become digital, development of innovations and new business models (Explorative process management)

made clear in the works of Nordsiek and Hennig around 1930 (Gaitanides 2007, p. 7). Nordsiek’s thesis published in 1931 is one of the first works in Germany to explicitly deal with business process modeling (Mendling 2021, p. 1).

The structural organization regulated the disciplinary structure (Who reports to whom ?) And defined the task assignment (who has which part task to fulfill?). This clarified the responsibility for partial sections of the service creation (functions). The operational organization served the breakdown of the work into small individual steps and ultimately the assignment to elements of the structural organization, i.e., the areas, departments, groups and persons. The advantage of this organizational concept, which was sensible for the time being, lay in the support of industrial mass production through the efficient use of resources (machines, employees, etc.).

However, the disadvantage was the fragmentation of the process into individual fragments. For the individual there was no overview of the entire process, but only of his own area of responsibility. This led to a restricted view and ultimately little interest in what happened before or after his own activity. In the extreme case, the results of the departments were simply “thrown over the fence” to the next department without checking whether the results were needed (cf. Sua-Ngam-Iam and Kühl 2021, p. 47). This laid the foundation for a long phase of promoting departmental thinking and egoism, which still hampers cooperation in companies today.

II. Phase: Sequencing of functions (action-oriented data processing)

Only with the further development of “Electronic Data Processing” (EDP) came movement back into the traditional, organizational separation of workflows and the structural organization.

In the 1980s, the concept of action-oriented data processing (AODV) as a predecessor of process management was developed to better use the possibilities of IT for controlling workflows (cf. Berthold 1983; Hofmann 1988). The basic idea was to control processes at the level of elementary work steps (cf. Berthold 1983, p. 20).

This was done using databases jointly used by the individual components of IT. So-called “action databases” contained information from application programs (e.g. minimum stock level for article number 4711 is 10 pieces below) and passed this on to the respective processor in the form of action messages (e.g. message to dispatcher: “Order

article number 4711”). The transmission of the messages to the employees took place via rudimentary electronic mail systems. The action database fulfilled the function of a mailbox for the employee, who could view and process the work stock and its priorities contained therein. Trigger databases also received structured information from programs (events) and in turn forwarded this to programs, thus triggering the execution of program runs. A trigger describes an action to be carried out and the event that triggers the action (cf. Scheer 1994, p. 72).

The goals of AODV were to shorten the processing times of processing objects, reduce the amount of paper and improve the use of resources. AODV was successfully implemented in large companies for procurement, customer order, master data and bill of material management in the years (cf. Berthold 1983, p. 25). Both the acceptance of the concept by the employees and the degree of target achievement were positive. Nevertheless, the concept could not prevail because the performance of information technology was not sufficient for larger data volumes at that time. The underlying idea was only successfully implemented later as “workflow management” (cf. Mertens 2006, p. 28).

III. Phase: Process orientation (Exploitative Process Management):

In the early 1990s, a “process management wave” was triggered in corporate practice by numerous publications by renowned researchers and practitioners. Well-known names were in the USA the authors Hammer and Champy (see Hammer 1990; as well as Hammer and Champy 1994), in Germany Scheer (see Scheer 1990) as well as in the Switzerland Österle (see Österle 1995). They triggered heated debates because the basic idea of the concepts was contrary to the then common practices in companies. They demanded in particular the reassembly of unrelated functions into an overarching overall process as well as the separation of process responsibility and organizational structure. In addition, the intensive use of the now significantly more powerful information technology as an “integration instrument” came into play. Many CEOs and managing directors use the new technologies to break down entrenched structures in companies and “force” an organizational change in their companies of using process-oriented application software. This was particularly beneficial for the business standard software system “SAP R/2” and later the successor product “SAP R/3” of the company SAP AG, Walldorf. The previously hardly methodologically supported modeling (structured description) of processes was supported by holistic scientifically founded concepts, such as the “Architecture of Integrated Information Systems (ARIS” by Scheer (see Scheer 1991) as well as first generations of powerful modeling tools for personal computers (e.g. “ARIS Toolset” by IDS Scheer, Saarbrücken or “Bonapart” by UBIS GmbH, Berlin), which were previously developed as prototypes at German chairs of business informatics (ARIS Toolset at Prof. Scheer, Saarbrücken, Bonapart at Prof. Krallmann, Berlin). Since the focus of this phase is mainly on the improvement of existing business models and their processes, in recent research literature also of the “exploitative business process management” is spoken (see for example Gross et al. 2019; Grisold et al. 2019).

IV. Phase: Digitalization (Explorative Process Management)

Around 2015, one can speak of the beginning of the “digitalization wave”. “Information technology” is upgraded and seen as an enabler of “digitalization” (cf. Winkelhake 2021). New concepts of information management such as *Cloud Computing*, *Big Data* and *Industry 4.0* influence process management in different ways. This has led to the formation of the term “Explorative Process Management”, which is characterized by the future-oriented development of new business models and the search and implementation of innovations (cf. Griesold et al. 2021).

In addition to organizational coordination (who does what?), technical coordination is now added (which processes are supported by which “apps”?). Cloud computing applications, for example, include modeling and execution of processes that have so far been carried out mainly on internally operated software. Typical applications for Cloud Computing are, for example, the real-time analysis of machine data with interventions in the maintenance process in case of irregularities or the “Active Customer Steering” through real-time analysis of sales and prediction of the current customer behavior (“We know what the customer buys tomorrow”). New business models increasingly lead to digitally implemented business processes that have not been possible so far (cf. Gadatsch 2014). In science, the connection between digitalization and process management was only intensively researched in 2019 and 2020 (cf. the detailed study by Allweyer 2020).

1.3 Classification of Selected Topics and Methods

Process management has become a multi-faceted term that is interpreted differently. Due to numerous publications and experiences in practice, different views and variants have emerged. Before going into detail on individual aspects, an attempt will be made to roughly classify important topics and associated methods of process management. The explanation of the terms will follow in the next chapters.

In Fig. 1.2 three planes are shown, the strategic, the professional and the technical process management.

Strategic process management

The strategic process management can also be referred to as business model management. It includes the analysis and design of business models and their structures. Business models are the basis of a company or an organization, they describe the purpose of the company, the pricing and marketing policy, the marketing model and the way of creating value (cf. Reinhart 2017, p. 7–8). Business models can be described using the Business Canvas Method (cf. Hansen et al. 2018).

In addition, decisions about the basic structure of the processes are necessary, which are laid down in the process strategy. The process strategy is increasingly formulated as a digital strategy, since manual processes are hardly relevant anymore. Typical methods of strategic process management are the process map (or value chain diagrams), IT

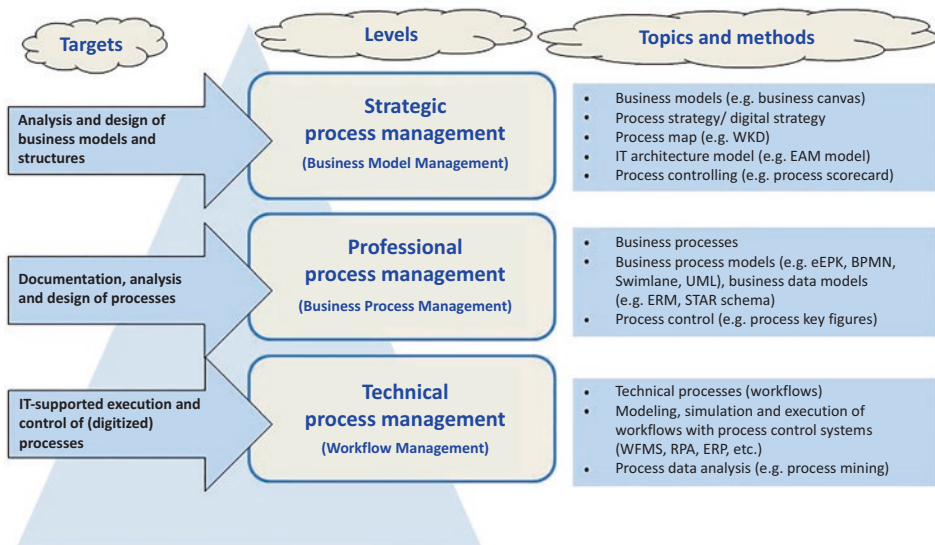


Fig. 1.2 Overview of selected topics and methods

architecture model (e.g. Enterprise Architecture Management/EAM) but also the Process Scorecard for process controlling.

Technical process management

The technical process management, also known as business process management or “operational process management”, is concerned purely from a technical point of view with the documentation, analysis and design of processes. For this purpose, technical process models are created using special modeling languages (e.g. eEPK, BPMN, Swimlane or UML). Depending on the level of detail, data models also must be created, e.g.: using the Entity-Relationship-Model (ERM) method for ERP systems or the STAR schema for Data Warehouse systems. For operational process control, key figures are usually used.

Technical process management

The first two levels in Fig. 1.2 relate to the business view of processes. The task of technical process management is to realize the IT support for the execution and control of processes. It is also referred to as “workflow management” because workflows represent executable digital processes. On this level, detailed technical modeling, simulation and execution of workflows is carried out using process control systems (workflow management systems/WFMS, enterprise resource planning systems/ERP and robotic process automation systems/RPA). The analysis of the processes carried out using process mining tools, which identify the actual process course from the log data of the systems.

1.4 Processes

1.4.1 Characteristics

Basic characteristics

Nowadays, many definitions and synonyms for the “business process” or simply “process” have become known, such as corporate process, performance creation process, core process, key process, main process, process chain, organizational process, etc. (see Schmelzer and Sesselmann 2013, p. 55). For the beginning, the following important characteristics of a process can be noted: A process supports a company-related goal that is aligned with the company’s or organization’s strategy, consists of several individual steps, takes place regularly, is often carried out by several people, departments, areas or companies in a division of labor, usually requires support by one or even several software systems and possibly other resources (e.g. telephone, copier, transport vehicle, machines, facilities), processes information (input) and leads to a desired result of the company (output). The overall context of processes and their division of labor results from Fig. 1.2.

Typical processes

The variety of processes in practice is unmanageable. Typical processes are:

- Processing customer inquiries and creating offers in an industrial company,
- Creating a production plan in an engine factory,
- Investigation and treatment of patients in a doctor’s practice,
- Conducting courses and examinations at universities,
- Production of baked goods in a bakery,
- Creating the annual profit and loss statement and balance sheet (annual financial statements),
- Purchasing, storing and selling goods in a supermarket.

A process differs from a once off project in that it is repeated several times. So, the introduction of a logistics system for a company or the celebration of a company the its 50th anniversary is a project and not a process.

- **Youtube video (2 min) for representing business processes: “Stephie’s Bakery”** On the Internet video **Youtube** under the address <http://www.youtube.com/watch?v=ehcYYLYHOul> there is an interesting and well-made video that compactly and impressively explains the essential elements of business processes using the example of “Stephie’s Bakery”.

1.4.2 Process Definitions

Hilmer has presented an extensive scientific study on the systematization of the process concept and identified 75 characteristics of processes (cf. Hilmer 2016, p. 267). He processed 101 sources (cf. Hilmer 2016, pp. 268 ff.), which indicates an active scientific discussion. The definitions selected here give an insight into the multi-year discussion about the business process concept without claiming to be complete.

Enterprise process according to Hammer and Champy

Hammer and Champy define the *enterprise process* as a set of activities for which one or more different inputs are required and which generate a result of value for the customer (Hammer and Champy 1994). As an example, they mention the development of a new product. An enterprise process is controlled by a process responsible person, who should come from the circle of the upper management level.

Business process according to Scheer and Jost

Scheer and Jost understand a *business process* as the model-based description of the functions to be carried out in a company in terms of content and time (cf. Scheer and Jost 1996). Functions are understood to mean individual tasks and activities that are linked to each other via triggering or generated events. Scheer equates the concept of the business process with the concepts of the process chain and the process chain (cf. Scheer 1990) and thus emphasizes the cross-functional character of the business process, which extends over several functional steps.

Business process according to Österle

According to Österle the *business process* is a sequence of tasks which can be distributed over several organizational units and whose execution is supported by information technology applications (cf. Österle 1995). A process is at the same time producer and consumer of services and pursues goals set by process management. As a special form of process organization the business process concretizes the business strategy and links it to the information system. Therefore the business process can be seen as the link between the corporate strategy and system development or the supporting information systems.

Berkau

The engineering sciences started formalizing processes and their systematic documentation much earlier in order to ensure consistent quality for repetitive activities carried out by different people. Processes can therefore be divided into technical processes and business processes (cf. Fig. 1.3) (cf. Berkau 1998, p. 27). Technical processes (e.g. milling a cylinder head, assembling an engine) are formally described by bill of materials and work plans (job production) or recipes (process production). Business processes relate to commercial activities, such as processing customer orders or hiring an employee.

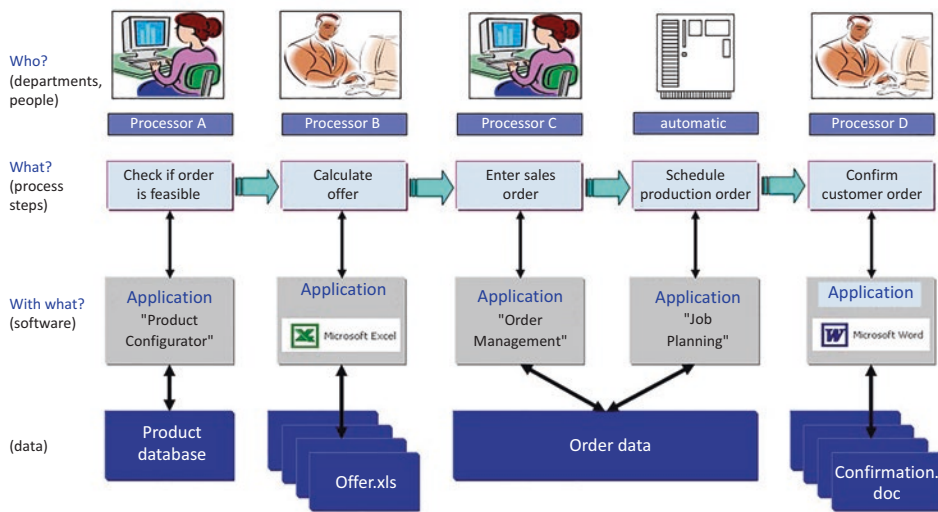


Fig. 1.3 Division of labor of processes—Schematic representation

They are documented using flowcharts or business process models and colloquially also referred to as “office processes”.

For the following explanations the definition of the business process according to Gehring (1998) is used:

Gehring

A business process is a purposeful, time-logical sequence of tasks that can be carried out by several organizations or organizational units using information and communication technologies. It serves to create services in accordance with the predetermined, process-oriented goals derived from the corporate strategy. A business process can be formally described at different levels of detail and from different perspectives. The maximum level of detail of the description is then achieved when the assigned tasks can be carried out by one employee without changing the workplace (cf. Gehring 1998).

1.4.3 Hierarchization of Processes

Processes can be considered at different levels of abstraction. Especially in very large companies it is important to identify these levels and use them for further work. The hierarchization of business processes is carried out in stages according to the “top-down principle”. Figure 1.4 shows the hierarchization principle, starting from the business process via business process steps to elementary business process steps that no longer require a change of operator for task fulfillment (cf. Fig. 1.4).

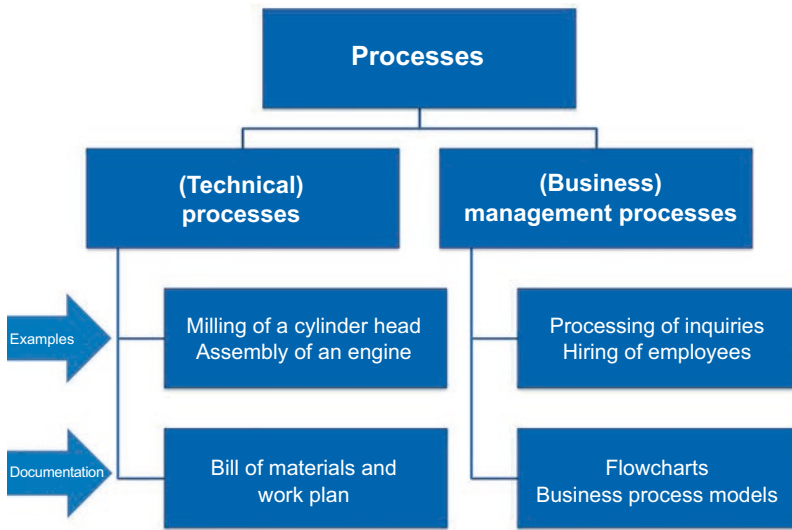


Fig. 1.4 Process variants according to Berkau (1998)

1.4.4 Categories of Processes

An important categorization of business processes is the subdivision according to the proximity to the core business of a company (cf. e.g. Seidlmeier 2002, p. 2 ff.). Accordingly, processes can be differentiated into “control processes” (alternatively “management processes”), “core processes” (also “primary processes”) and “support processes” (alternatively “cross-sectional processes”) (cf. Fig. 1.5).

Control process

Control processes actively intervene in the interaction of all business processes (e.g. strategy development, corporate planning). They are the entrepreneurial bracket over value-added and supporting processes and ensure a goal-oriented structure.

Core process

Core processes are business processes with a high value-added share. They characterize the essence of the company, are usually competition-critical and form value-added processes starting from the customer’s wish to the customer’s perceived delivery or service. Typical examples are order processing, product development, production, distribution and service.

Support process

Support processes have no or only a very small value-added share. They offer cross-sectional services for other processes, without which the value creation of the company is

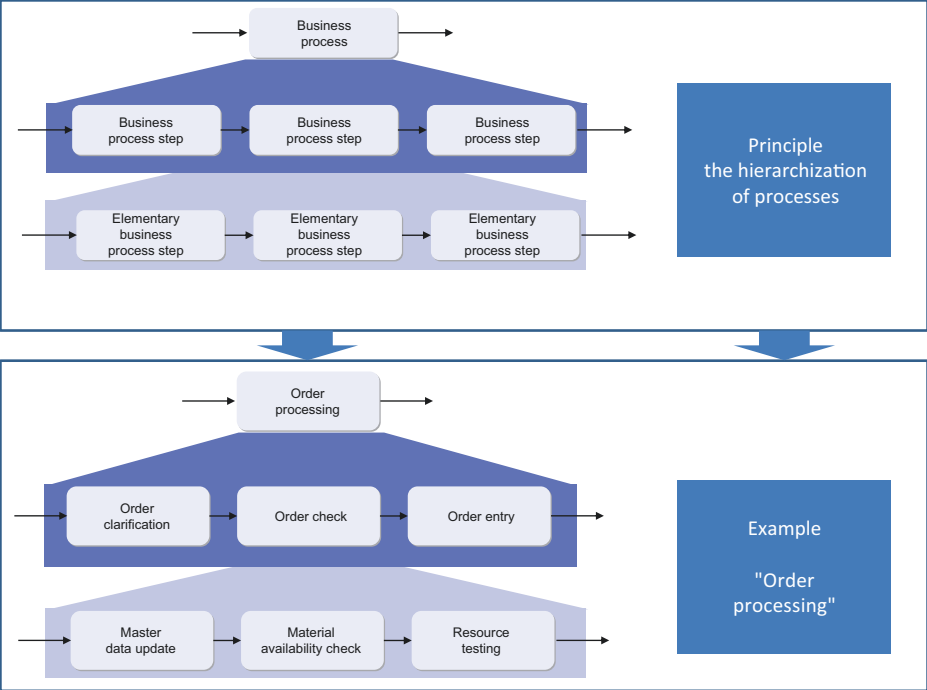


Fig. 1.5 Process Map

not possible. They are usually not competitively critical and do not appear directly in the customer's field of perception. Examples are accounting, cost accounting, reporting or human resources.

The Fig. 1.6 shows an example of the categorization of processes for a fictitious car dealership. In the upper part of the representation, four control processes are shown: strategy development, controlling, product planning and personnel control. Below, the two core processes ("car purchase" and "service") are depicted in detail, with the most

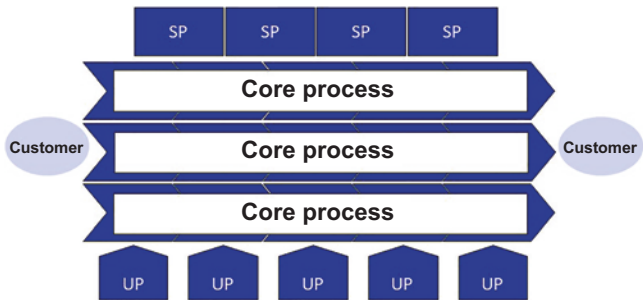


Fig. 1.6 Process categories

important business process steps being shown in a sequential form. At the bottom are the support processes marketing, accounting, customer service, information technology and administration. All support processes are not directly assigned to a process, but either generally effective for the whole company (e.g. information technology, administration) or for several processes (e.g. customer service for “car purchase customers” and for “service customers”).

1.5 Workflows

1.5.1 Central Terms of Information Processing

The increasing digitalization leads to the fact that more and more processes are carried out with computer support. The term “workflow” plays a central role here. Workflows are processes that are controlled on the basis of models and algorithms. They therefore require the possibility of at least partial automation of the process and its execution using software systems.

Before the term Workflows is dealt with in more detail, some selected term elements should be clarified in the context of hardware, software and business processes (cf. Fig. 1.7). The lower level of IT support consists of hardware (e.g. computers, printers) and other technical equipment (e.g. reading devices, scanners), which is referred to as a Hardwaresystem. Together with the software system consisting of Anwendungssoftware (e.g.: mail sorting, accounting) and Basissoftware (e.g. operating system), the Hardwaresystem

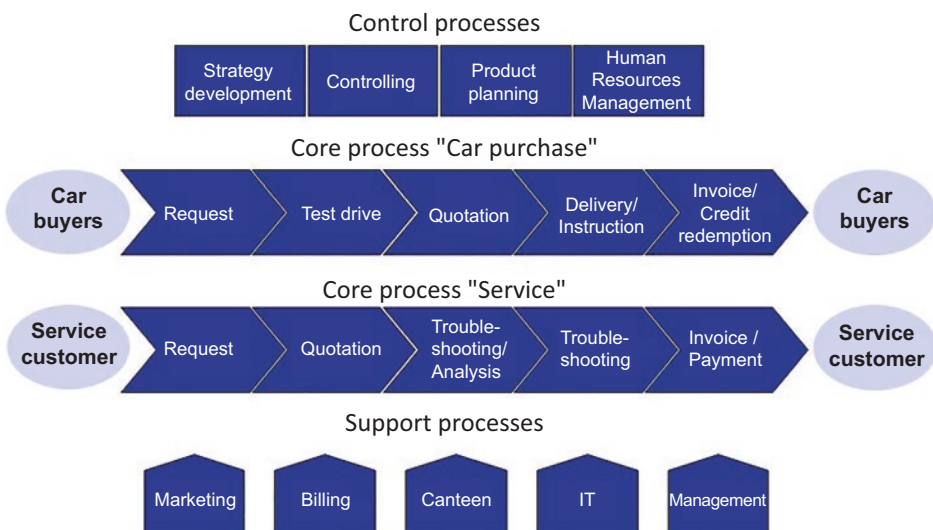


Fig. 1.7 Process Map of a car shop

forms the “Anwendungssystem”. If the Anwendungssystem is supplemented with organizational components (people and business processes), it is an “Informationssystem”.

1.5.2 Workflow Definitions

The digitalization of processes is currently being intensively discussed. Workflows are digitally executed and controlled by a software system using rules. The first workflow definitions date back far into the past:

- Galler and Scheer see in the workflow a technical refinement of the business process (cf. Galler and Scheer 1995). The degree of refinement is the automatability. The workflow must be usable as input and rule set for the control by a process control specialized software system (workflow management system).
- Similarly, Österle (1995) describes the workflow as a refined business process. Starting from a process design at the macro level and its successive decomposition into sub-processes, the micro level is then reached when the tasks are so detailed that they can be implemented by the process participants as work instructions. Based on the task chain, a manager can control the workflow. The workflow thus represents the detailed form of the micro-process. Instead of a manager, the computer now takes over the process control.

1.5.3 Delimitation Business Process and Workflow

Business processes and workflows describe workflows, but with different levels of detail. Business processes describe from a business point of view who does what. The workflow is a refinement of the business process in terms of information technology (cf. Fig. 1.8).

A clear distinction is not always possible because of the common subject of investigation and often leads to the terms being equated, although they pursue different goals. There are some essential differences, which are summarized in Fig. 1.9:

Business process: The business process describes “what” needs to be done to implement the given business strategy. The workflow describes “how” this should be implemented. The business process is the business-conceptual level, the workflow is the operational level. The required level of detail of a business process is reached when it describes the work steps that can be carried out by an employee in one go at a workstation. The business process is therefore a business term.

Workflow: The workflow level is reached when the level of detail can be understood by the executing employee as a concrete work instruction and the description for software-controlled work is given so specifically that it can be executed by an application system. A clear distinguishing feature is the executability by a human task performer

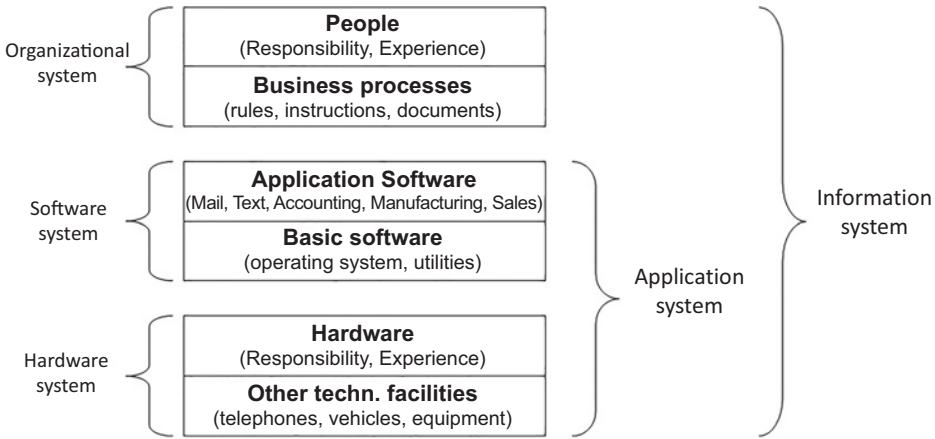


Fig. 1.8 Important IT terms (cf. Herzwurm and Pietsch 2009, modified)

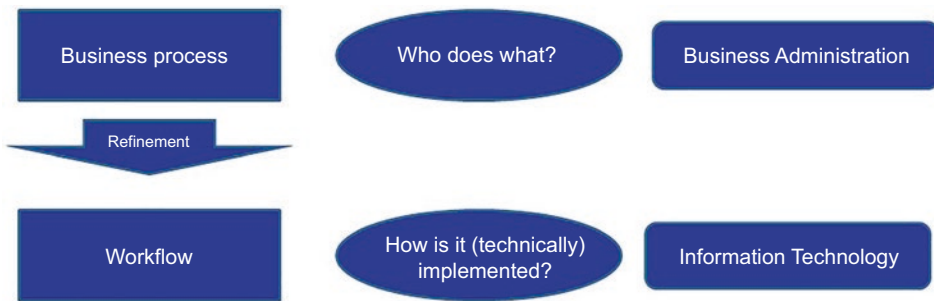


Fig. 1.9 Business process and workflow

(employee) or a computer program. The workflow is therefore a rather technical term with a strong connection to computer science.

► **Workflow definition** A workflow is a formally described, fully or partially automated business process. It includes the temporal, professional and resource-related specifications that are required for automatic control of the work process at the operational level. The work steps triggered by this are intended to be carried out by employees or by application programs. To be distinguished from the workflow as a type or template of a partially or automated work process is a workflow instance, which denotes a concrete execution of the workflow (cf. Gehring 1998).

In the workflow, the computer controls the process

The current digitalization trends lead to the fact that the difference between business processes and workflows is becoming increasingly less visible, since hardly any processes are conceivable without software support and the active process control by computer is increasing rapidly. In the workflow, the computer controls the sequence of activities, in the business process the human controls.

1.5.4 Workflow Types

Workflows can be distinguished according to the degree of structuring of the underlying processes and the degree of computer support for the processes.

Workflows according to the structurability of the process

The **general workflow**, which is also referred to as a production or transaction workflow, concerns well-structured workflows in organizations, such as travel expense accounting. General workflows are characterized by repetitive character and work steps that can be defined in detail in advance. They can be automated to a high degree or supported by information processing systems.

The **case-related workflow**, which is also referred to as a flexible workflow, characterizes workflows that are not fully standardized. An example of this is the processing of loan applications at banks. The transition from case-related to general workflow is fluid. In comparison to the general workflow, case-related workflows have higher degrees of freedom for the processors. Individual processes can be skipped or modified (e.g. waiver of individual test procedures as part of a credit processing or an assessment center when hiring an employee).

Ad hoc Workflows are unstructured process steps, the sequence of which cannot be determined in advance. In ad hoc workflows, the processor of a workflow instance determines the subsequent processor in his own responsibility (Scheer 1998, p. 90). Ad hoc workflows cannot be modeled (e.g. working group for the development of an advertising campaign). Another example is the processing of investment proposals in large companies. It is often only roughly structured, for example regarding the signature rules, and offers high degrees of freedom with regard to the participants and the course. Depending on the type of investment, different contacts and preliminary work may be necessary before the application is approved.

Workflow according to the degree of computer support

Workflows can be divided according to the degree of computer support. The **free workflow** is carried out completely manually by a personnel processor (e.g. checking the competence of an incoming request). In this case, only a process control of the process is possible, i.e. a check whether all sub-steps have been carried out in the correct order. The **partially automated workflow** is, supported by an information processing program,

carried out by a personnel processor (e.g. entering the master data of a new customer). The **automated workflow** is executed by a program without intervention by a personnel processor (e.g. printing an invoice after delivery). In the case of partially automated or automated workflows, in addition to process control, an execution control is also possible, i.e. it is ensured that, for example, a certain transaction has been executed.

1.6 End-to-End Processes

The process management developed in the 1990s focused on improving customer wishes. Processes serve directly (core process) or indirectly (support process, management process) to meet the needs, expectations or requirements of customers. The control of processes is carried out by a business process responsible, who specifies the objectives and key figures for controlling the process within the framework of the company's strategy.

An **end-to-end process** is a customer-focused process. The term "customer" can be extended here. "Customer" stands either for the external company customer, who, for example, orders goods, or for an "internal process customer", who uses the services of another process. So, the process "Leading employees" can request the service within the framework of the new filling of a position that a selected employee should receive an employment contract from the process "New employees".

The End-to-End Process begins with the trigger by the (process) customer and ends with the fulfillment of the customer's needs. Core processes of a company should be organized as end-to-end processes (cf. the schema in Fig. 1.10). In the case of the external customer, one speaks of "customer-to-customer processes".

► **Note:** In the end-to-end process, a customer need is at the beginning and the performance the customer receives is at the end.

An example of the end-to-end process "offer processing" described in tabular form in Table 1.2 can be seen in Fig. 1.11.

1.7 Function Versus Process

The organizational structure of a company serves the vertical mapping of the hierarchy. Each position (employee, manager) performs individual functions as part of the division of labor of the company. The function "check stock" is carried out in the disposition, the function "enter customer order" is carried out in sales. The entire process ("processing of customer orders") is not visible and there is no overall responsibility for all departments involved.

	Business process	Workflow
Destination	Analysis and design of work processes in terms of given (strategic) objectives	Specification of the technical execution of work processes
Design level	Conceptual level with link to business strategy	Operational level with connection to supporting technology
Level of detail	Work steps that can be performed in one go by one employee at one workstation	Specification of work steps with regard to work procedures as well as human and technological resources

Fig. 1.10 Business process versus workflow

Table 1.2 End-to-End Process Offer Processing

Customer's central requirements	Offer created promptly with realistic and matching delivery terms to the inquiry as well as favorable prices
Essential activities of service creation	Receipt of the inquiry, clarification of details, clarification of offer contents (products, dates, prices, additional services, resource check, if necessary, involvement of suppliers), creation and submission of offer, monitoring of the offer
Possible process goals	High-quality offer with realistic delivery dates and binding prices
Possible control indicators	Processing time Order rate (number of orders/offers)

Example of a description of an end-to-end process

The process is created by sensibly coupling individual functions, e.g. “enter order”, “check stock” under one unified leadership by the process responsible (e.g. “processing of customer orders” in Fig. 1.12) to a sensible whole (Fig. 1.13).

1.8 Quick Test Process Management—Self-evaluation

The following described “Quick Test” is intended for readers from practice. It can be used for a self-evaluation of a company or an authority. This gives the readers a simple help to roughly assess which areas of the organization still have development potential and where active intervention may be required.

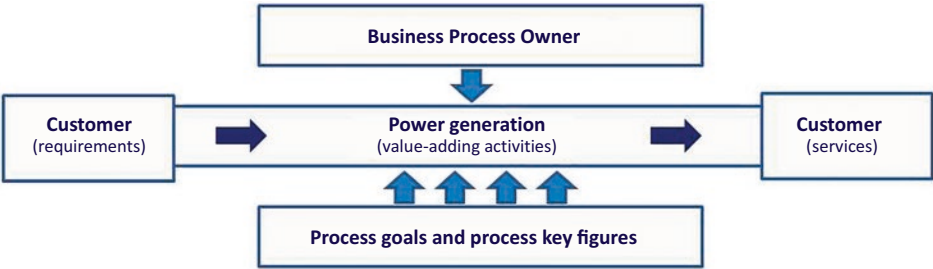


Fig. 1.11 Schema for an end-to-end process (Schmelzer and Sesselmann 2013, p. 53)

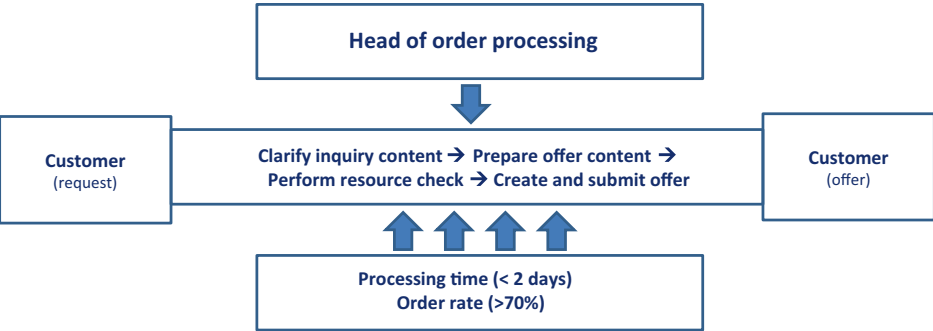


Fig. 1.12 End-to-end process (example according to Schmelzer and Sesselmann 2013, p. 53)

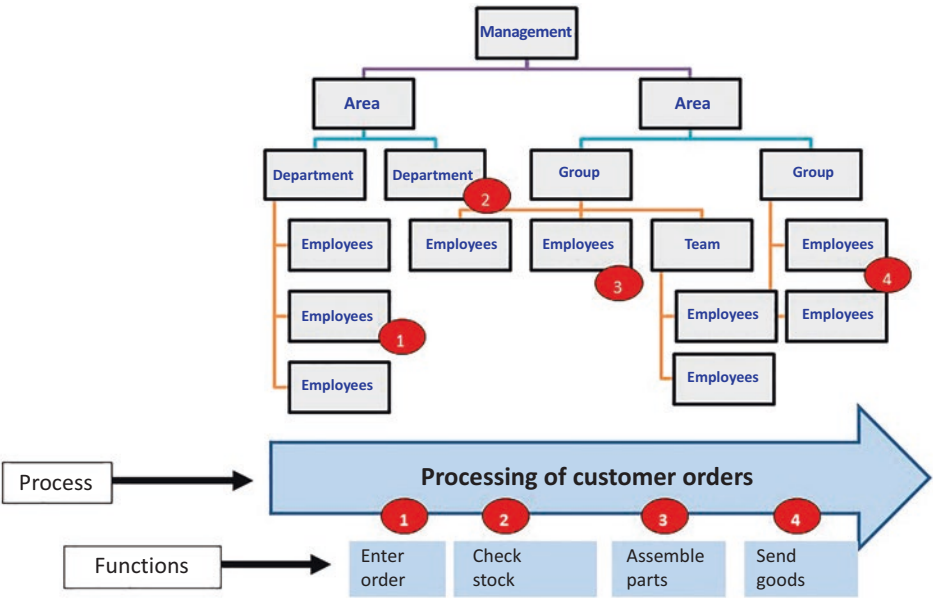


Fig. 1.13 Process versus function

A total of five questions are to be answered on a scale of 1 to 5, intermediate values are allowed. The result can be displayed as a network diagram and compared with other organizations.

Questionnaire Process Management Quick Test

Question 1: Experiences with process management

- 1 = Process management is largely unknown and no experience
- 2 = Process management is known (for example, from training or study), but not yet introduced in the company
- 3 = First activities have been started to introduce process management, for example, pre-project, pilot project, but no regular activities
- 4 = Processes are partially formally documented, for example, process map, some detail processes (Swimlane, eEPK, BPMN, UML, etc.), responsibility for processes is not yet or only partially defined and communicated
- 5 = Responsibility for processes (for example, definition of process responsible) is anchored

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- 5 = Responsibility for processes (for example, definition of process responsible) is anchored

Question 2: Strategic process management

- 1 = No process strategy available or processes are not part of the corporate strategy
- 2 = Process map is in work, process strategy is planned
- 3 = Process map is known and communicated, process strategy is formulated
- 4 = Process strategy is backed by process scorecard with corresponding goals and measures
- 5 = Measures for implementing the process strategy have been initiated or already started and are controlled by process controlling

Question 3: Professional process management

- 1 = Processes are not or only in exceptional cases or fragmentarily documented
- 2 = Individual processes are documented, first approaches to process optimization have been developed
- 3 = Important processes (e.g. core processes) have been analyzed and optimization concepts have been developed
- 4 = Processes are stored in an IT tool and described with other elements (e.g. profile). The contents are communicated, selected optimization processes run regularly
- 5 = The process management feedback loop is established (strategy, professional modeling, redesign / restructuring of processes, implementation, use and monitoring)

Question 4 Technical process management

- 1 = not known, not available
- 2 = Workflow tool or ERP/RPA system with workflow functions in selection
- 3 = Workflow tool is selected and first processes for digitization are selected
- 4 = Selected processes have been implemented with the workflow tool
- 5 = All important processes (e.g. core processes) are controlled by workflow tool

Question 5 Processor Organization

- 1 = functionally structured line organization, i.e. for example departments such as purchasing, warehouse, production, sales, shipping for all products or processes
- 2 = staff unit within a functional line organization
- 3 = staff unit with project organization
- 4 = matrix organization with functional responsibility and additional process responsibility (process manager)
- 5 = pure process-oriented structure of organizational units, possibly also with shared service centers for cross-task

Question 6 Process Modeling

- 1 = No documentation of processes available
- 2 = Processes are represented with simple flowcharts, for example with MS Visio or similar graphic programs
- 3 = Processes are documented with modeling tools, for example with BIC Design, ARIS, Signavio
- 4 = Processes are analyzed with modeling tools
- 5 = Processes are dynamically evaluated with simulation functions of the modeling tools in terms of times, costs, quantities, etc.

Question 7 Process Automation

- 1 = Processes are controlled purely manually (e.g. job tickets, by calling out)
- 2 = Processes are digitally supported, but control is via static aids (e.g. job tickets) or people (e.g. transfer of task to next person)
- 3 = Processes are partially rule-based controlled by systems (e.g. order requests are forwarded to different recipients depending on value)
- 4 = Processes are partially rule-based executed and controlled by control tools (e.g. workflow management systems, ERP systems, RPA tools)
- 5 = Processes are mainly executed and monitored by control tools

Question 8 Process Controlling

- 1 = no process-related key figures available
- 2 = key figures defined for individual processes
- 3 = key figures available for important processes, no key figure system in place (e.g. process scorecard)
- 4 = a key figure system (e.g. process scorecard) provides key figures for important processes centrally (e.g. data warehouse)
- 5 = the key figure system is used for operational and strategic process control

In Fig. 1.14 you will find an anonymized example from participating institutions. If you are interested in a self-evaluation according to this procedure, you can request a blank table (Microsoft Excel) at the email address andreas.gadatsch@h-brs.de.

1.9 Review Questions and Exercises

1.9.1 Questions

1. Distinguish between business process management and workflow management.
2. Describe the essential characteristics of processes.
3. Distinguish between a business process and a workflow.
4. Distinguish between a project and a business process.
5. Explain different categories of business processes and give an example from your chosen industry for each category.
6. Explain the difference between an operational function and a business process using an example of your choice.

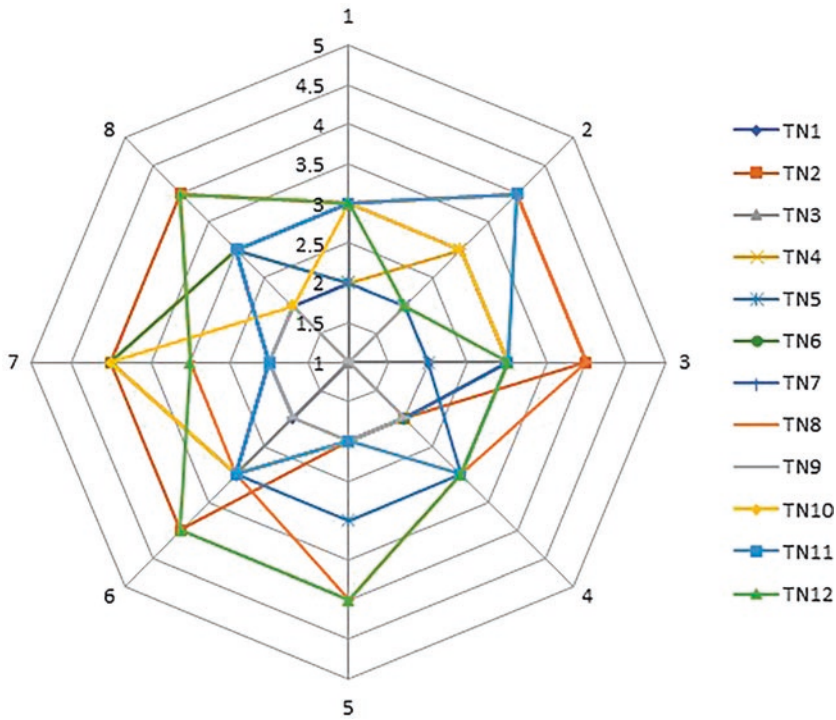


Fig. 1.14 Quick test process management results

1.9.2 Exercise “End-to-End Process”

Identify an “End-to-End Process” of your choice and create a clear process diagram that contains the following information: process name, central customer requirements, essential activities of service creation, possible process goals and possible performance indicators for control.

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Process management requires the use of methods

Abstract

This chapter first presents an integrated concept for business process and workflow management. It is described in terms of its elements (levels, phases and views). Standardized optimization concepts for business processes are then discussed and some well-known management concepts are introduced that pursue similar goals as process management. The conclusion is a section on reference models for process management as well as questions and an exercise.

2.1 Integrated Business Process and Workflow Management

Business processes and workflows are closely linked and cannot be developed independently of each other. Therefore, process management must be mapped out in a thorough, integrated concept. The design framework of the concept of integrated business process and workflow management shown in Fig. 2.1 comprises several levels: the development and control of corporate strategy (strategic level), process management in the narrower sense (functional-conceptual level), technically oriented workflow management (operative level) as well as the task areas of application system and organization design that are linked to process management (cf. Gehring and Gadatsch 1999, p. 70). The concept is used to reconcile with corporate strategy, the organizational design of processes and their technical implementation with suitable communication and information systems, and strategic and operational process control.

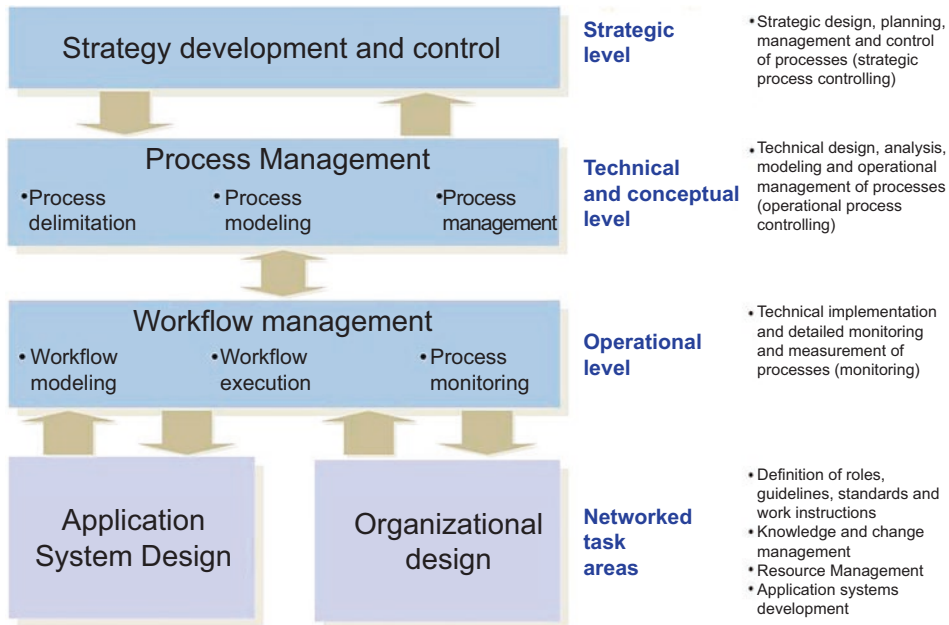


Fig. 2.1 Integrated Business Process and Workflow Management—Concept

Strategic Level (Strategy Development and Control)

On the strategic level, the business areas of a company, including the critical success factors effective here, are considered. The central processes of the company are identified, planned and implemented by means of suitable measures. Strategic process control monitors and controls the implementation and achievement of targets by means of strategic key figures and, if necessary, initiates corrective measures in the event of too great deviations from the target values.

On the lower functional-conceptual level, the processes are derived within the framework of process management. Process management thus represents the connection to corporate planning at the strategic level, while workflow management from the perspective of the lower level, operational implementation, involves application system and organization design.

Technical-conceptual level (process management *sensu stricto*.)

The process management includes the phases of process delimitation, process modeling and process management in the life cycle of processes:

- Process delimitation describes process formation. Based on business fields and strategically oriented specifications, such as product range, critical success factors, etc., process candidates for each business field are to be derived and evaluated in a stepwise manner. Finally, the processes to be modeled and implemented are to be selected.

- Process modeling is about mapping sections of reality from a business field into a business process from a technical-conceptual perspective. Depending on the strategic goals of a company, for example, complete redesign of processes or further automation of existing processes can be pursued. For example, the BMW Group develops special business strategies in tool and plant construction that explicitly take into account the increased environmental requirements with regard to CO₂ emission limit values and the associated reduction in consumption and safety requirements. These are then reflected in revised and adapted business processes (cf. Brunner et al. 2002, pp. 312 ff.).
- Process management refers to the phase of process implementation. Its goal is to align processes with predetermined process success measures, the so-called process management indicators. The management indicators of the processes are, if necessary in several steps, to be derived from the critical success factors of the respective business fields. Depending on the extent of identified success deficits, occurred weaknesses in the project course, etc., re-modeling or re-running of process modeling may be required.

Operative level (workflow management)

Workflow management is divided into the phases of workflow modeling, workflow execution and workflow monitoring. Workflow modeling follows business process modeling. In this case, the modeled business process is extended by specifications that are necessary for an automated process execution under the control of a workflow management system. This is followed by the phase of workflow execution; it includes the creation of process objects and the passage of process objects along the planned processing stations under the control of a workflow management system. The subsequent workflow monitoring serves the ongoing monitoring of the process behavior. The juxtaposition of process control variables and corresponding process actual variables at the level of workflows provides information on whether a process is already set correctly or whether corrective measures need to be taken.

Because of the support function for business process management, workflow management systems are also increasingly referred to as BPM systems (business process management systems) or process management systems (PMS) (Dadam et al. 2011, p. 364).

Networked areas of responsibility (application system and organizational design)

Organizational design complements process management as a general support function by specifying roles, guidelines, standards and specific work instructions for employees. In addition, it provides methods for knowledge and change management and controls the management of personnel and other resources.

Application system design provides process-oriented information systems. These can be developed individually for the company or used in the company in the form of standard software that has been adapted.

2.2 Structural Elements

2.2.1 Perspectives of the Process Cube

The structure of the process management can be divided into three perspectives (levels, phases and views) (see the “process cube” in Fig. 2.2). The abstraction levels of strategy, processes and workflow are considered (see Sect. 2.2.2). The phases of business modeling, technical modeling and deployment and monitoring of ongoing activities are considered (see Sect. 2.2.3). The modeling can be structured by the views of organization, function, data, software and process (see Sect. 2.2.4).

Application of the process cube

The structural elements of the process cube can be used to describe a process-oriented concept for corporate startup. The cube serves as a structure. Below are some examples of aspects to consider when starting an “online car dealership”.

- Levels of Abstraction:
 - **Strategy:** An online car dealership needs to differentiate itself from its competitors. It competes with factory-owned dealerships of car manufacturers, independent local dealers, and specialized EU importers. The advantage of the online car

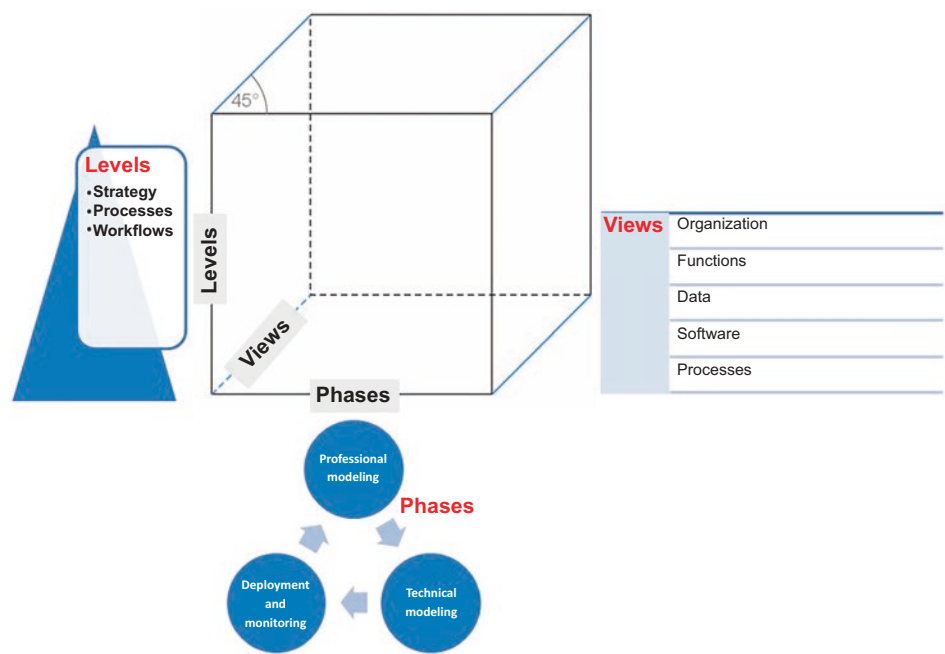


Fig. 2.2 BPM cube

dealership is the complete handling of the car purchase from consultation, selection to delivery and handover via digital media and therefore appeals especially to younger and media-interested buyers.

- **Processes:** All processes visible to the customer are to be digitized and accessible via any internet-enabled device (PC, tablet, smartphone). Paper support is to be avoided except for legally unavoidable exceptions.
- **Workflow:** For the online car dealership, numerous processes are to be realized as a workflow, i.e. the processes are to be controlled by a software system. Examples are: capturing customer data, capturing insurance data, searching the vehicle fleet, requesting vehicle information, selecting a vehicle, reserving a vehicle, registering a vehicle for the customer and insuring it, making a test drive, online consultation by employees, ordering, making an appointment for handover, withdrawal from purchase or cancellation.
- Phases:
 - **Expert modeling:** First, leadership, core, and support processes must be described expertly. The models should be visible to the respective employee (e.g., online sales consultant) in the intranet and later also serve as a reference work.
 - **Technical modeling:** The implementation of the processes at the workflow level is handed over to an external software house and programmed there. It is important that general IT standards and peculiarities of the automotive industry are mapped in order to obtain the connection to the data networks of the automobile manufacturers.
 - **Operation and monitoring:** After successful programming and testing, the applications are to be used in operation. For the monitoring of the processes, customer reactions (e.g. model comparisons, termination of online sessions) are to be reacted to in real time.
- Views:
 - **Organisation:** Internal departments (accounting, administration, marketing, IT), external departments (sales/consulting; service)
 - **Function:** Master data entry, vehicle data entry, price data entry, etc.
 - **Data:** Vehicle data, price data (own prices, competition prices), customer data, customer behavior, statistics (sales, storage time, average prices), etc.
 - **Software:** Internal software for accounting, administration, payroll, online portal with all customer-oriented functions, etc., customer app for the operational processing of consulting and purchase as well as customer loyalty
 - **Process:** Control processes (corporate planning, marketing strategy, controlling; core processes (providing vehicle information, processing purchase, processing complaints, delivering vehicle, etc.)), support processes (accounting, warehousing, personnel, IT provision, etc.)

2.2.2 Levels

The distinction between business processes and workflows leads to a differentiation according to abstraction and modeling levels (cf. Gehring 1998). Because of the different goals of the terms and their complexity, it makes sense to form modeling views and to divide them into phases in order to focus on the respective research questions in practical work. The professional analysis and modeling of processes has the goal of specifying which tasks are to be carried out by which organizational units. The technical analysis and modeling specifies how the process is to be carried out in detail using software systems. For modeling, therefore, two levels are to be formed: the level of professional-conceptual process modeling and the operational level of workflow modeling derived from it (cf. Fig. 2.3).

- In practice, the terms “of‘professional modeling’” and “technical modeling” are also used to delimit the abstraction levels. Under a “Round-Trip-Modeling”, the holistic approach is to be understood, in which is an executable model is successively refined from the professional model. This is, for example, possible with the modeling language BPMN (cf. Sect. 5.7).

Repository (model database)

In addition to the introduced context of the framework concept, the results of the design and modeling activities are added. These are stored in the form of models (process model, workflow model and additional information) on a permanent basis. The

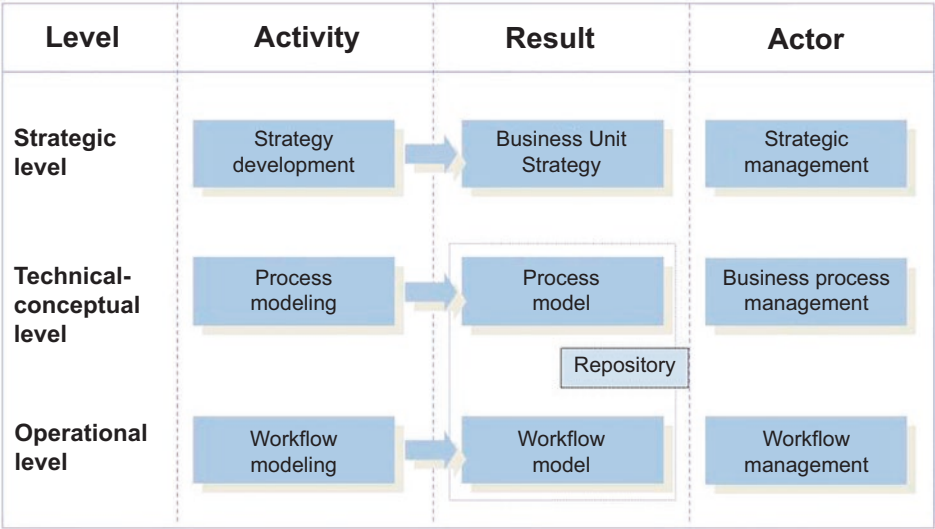


Fig. 2.3 Level concept (Gehring 1998)

repository represents a dictionary that serves to describe the model components and the relationships between the components. It captures business processes and connections between business processes and workflows. In addition, the interfaces to the model environment are described. The latter consists mainly of the respective business field strategy, the supporting information systems and the involved organizational units.

- Processes and workflows must be permanently documented in order to be used in the organization. Ideally, the workflows are derived from the processes, i.e. refined. When using modeling tools, it makes sense to use a single repository for processes and workflows. Within the scope of the use of modern software tools, which for example generate executable workflow models by means of the modeling language BPMN 2.0 (see Sect. 5.7), the repository is a prerequisite for the execution of the processes.

2.2.3 Phases

Process management is organized in projects in a division of labor (see Sect. 3.3). For this purpose, one uses phase or life cycle models to structure the complex activities, in particular within the scope of modeling tasks. The modeling can be carried out in one or two steps.

In single-stage modeling, the workflow model is created directly without requiring a business process model. In the two-stage approach, a workflow model is derived from a previously created business process model. The two-stage workflow modeling takes into account the fact that business processes and workflows serve different purposes, although a distinction is not possible in every individual case. In practice, the two-stage approach is often preferred because, in addition to the different purposes of the models, there are only a few software tools that support the single-stage approach so that the requirements of all participating groups of people are fully supported. In Fig. 2.4 a two-stage workflow life cycle is shown, which includes three partly meshed sub-cycles.

Sub-cycle (1)

Sub-cycle (1) includes business process modeling, analysis and restructuring, as well as business strategy development and can be classified into the strategic or functional-conceptual level of the integrated overall concept. The starting point for sub-cycle (1) is the collection and modeling of the actual business process models. These are then subjected to a business process analysis regarding their contribution to the fulfillment of the business process goals derived from the business strategy. In this context, unproductive or superfluous business processes and organizational structures are identified. The business process analysis can also have repercussions on the initially given business strategy of the company, which in turn affects the subsequent design and restructuring of the business processes. The newly designed and restructured business processes regarding the

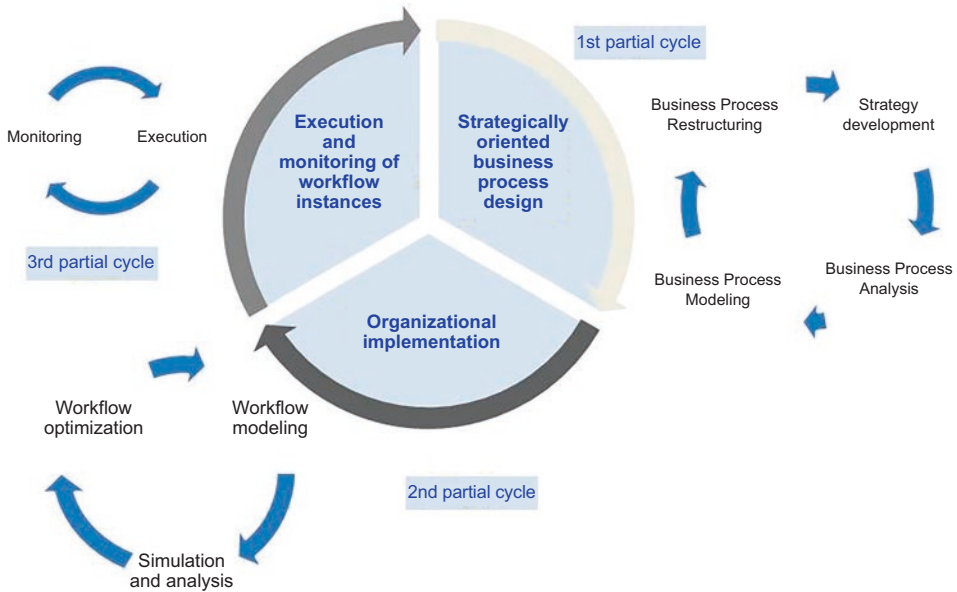


Fig. 2.4 Business process and workflow life cycle model

given or possibly adapted business goals are formally described as target business process models. A subsequent analysis of the target business process models can lead to further restructuring cycles until the design of the business processes is in conformity with the given or possibly adapted business goals.

Part cycle (2)

With the completion of partial cycle (1), the professional and conceptual design of business processes is completed. In the subsequent partial cycle (2), the business process models are refined down to the operational workflow level. The desired level of detail should allow for both automatic execution and simulation-based analysis of workflows. The workflow optimization that follows the analysis completes the second, possibly iterative, part cycle.

Part cycle (3)

The execution of workflows and their ongoing monitoring form the beginning of 3rd partial cycle, which is also attributable to the operational level. Depending on the degree of deviation of the process results from the expected results, feedback is given to partial cycle (1) or (2). Smaller deviations lead to incremental changes in the form of a re-run of the 2nd partial cycle, i.e. optimizations of the workflow models. Larger deviations from reference values indicate modeling deficiencies and may require re-modeling or a return to 1st partial cycle. Activity-triggering threshold values for monitoring workflow instances are to be specified as tolerance ranges for process control variables in the context of business process modeling.

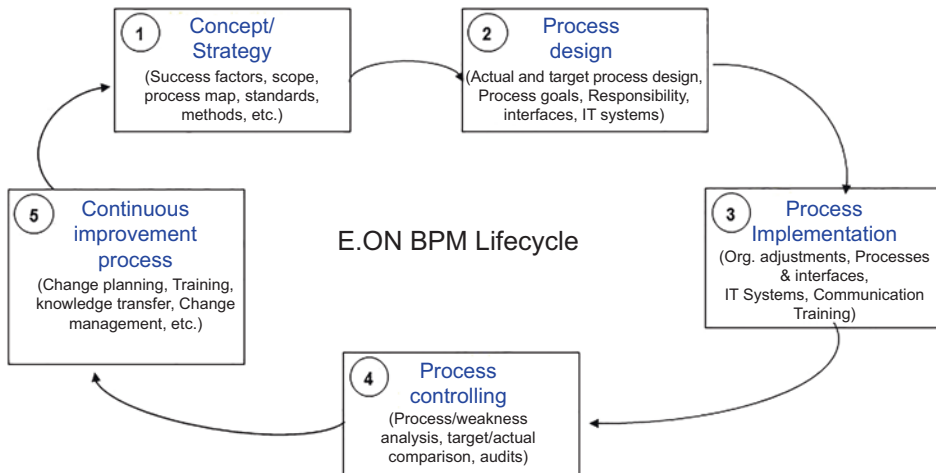


Fig. 2.5 Life cycle model of the company EON for business process and workflow management

Phase models are important for practice. Therefore, companies develop their own tailored process models. Fig. 2.5 shows, by way of example, the life cycle model of the company EON, which contains all relevant process steps for an integrated business process and workflow management (cf. von Büdingen and Schlaf 2011, p. 83).

A special feature of the EON model in Fig. 2.5 is the integration of a process step “process control” to emphasize the importance of these tasks. However, process control is to be understood as a control loop, as shown in Fig. 2.6. Based on the corporate strategy, a process strategy is derived. This is concretized with target values and measures. This is followed by the actual values from the measures taken, which are evaluated as part of a deviation analysis. The control life cycle therefore overlaps the process management life cycle in the sense of an overarching meta-control loop.

Summary: Role of phase models for process management

Phase models represent the temporal sequence of process management. Since process management is a recurring life cycle, loop diagrams are common, which describe the essential steps in the implementation of process management activities. In practice, the models are partly adapted to specific company requirements.

2.2.4 Views

In process modeling, it is not useful to map all modeling-relevant facts into a single representation, as the overview would be lost. To reduce the complexity of the representations and to improve transparency, the use of a view concept (cf. Sinz 1996) is recommended, which logically subdivides the aspects to be considered. Fig. 2.7 shows an

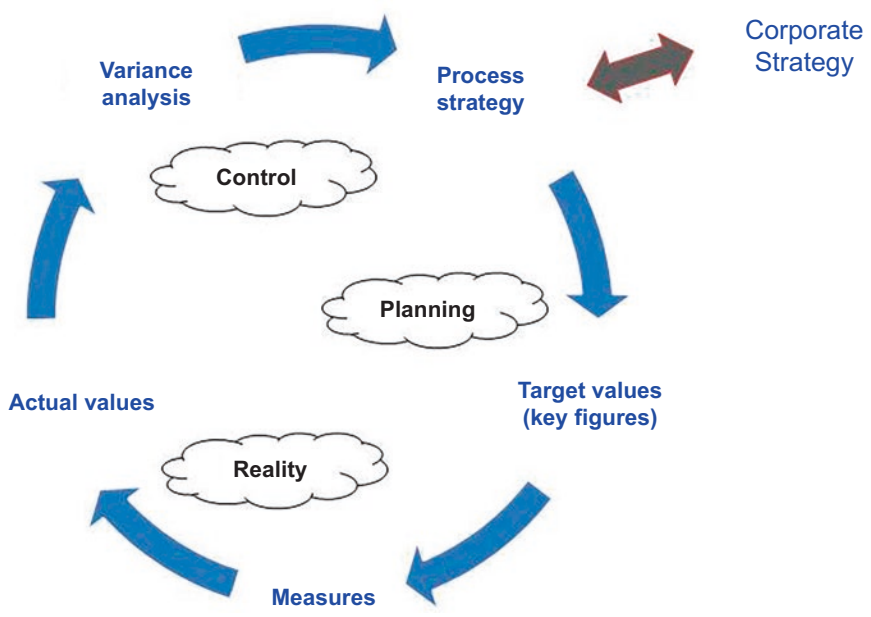


Fig. 2.6 Control loop

View concepts of business process modeling						
Becker	Ferstl/Sinz	Gadatsch	Gehring	Österle	Scheer	Weske
Organization	Performance View	Process View	Organization view	Organization	Organization view	Function Modeling
Business object	Steering View	Organizational Structure View	Functional view	Functions	Functional view	Information Modeling
Process	Sequence view	Activity - structure view	Data view	Data	Data view	Organization Modeling
Resource		Application - structure view		[Staff]	Control view	IT Landscape Modeling
		Information - structure view		[...]	Performance View	

Fig. 2.7 View concepts

overview of the views used in some selected process management concepts (extended representation based on Gehring 1998).

Becker et al.

The approach of Becker (cf. Becker et al. 2007, 2008) has the goal of providing an over-view of the process landscape and supporting measures for the reorganization of the

business processes under investigation. To support these goals, four views are proposed: organizational view (who does something?), business object view (what is processed/produced?), process view (what is done how?) and resource view (what is something done with?). These views are realized in four model types: business object model, organization model, process block and resource model.

Gadatsch

Gadatsch subdivides regarding the consideration of the workflow modeling into a central process view and four additional structural views (cf. Gadatsch 2000, pp. 179 ff.). The process view describes the modeling objects involved in a process from a process-oriented perspective. The structural views describe the structure of the modeling objects that are brought together in the process view.

Gehring

Gehring orients himself in the formation of views on the classical basic elements of process modeling, the process, the organizational structures and the data (cf. Gehring 1998).

Österle

Österle does not speak of views in his concept, but of design dimensions (cf. Österle 1995). He calls organization, data, functions and personnel as dimensions of business engineering, but does not include the personnel dimension in the concept. A separately shown “dynamic” dimension does not exist. However, dynamic aspects are considered in the representation of processes with task chain diagrams.

Scheer

Scheer makes a decomposition into five views. The primarily static description objects of business processes are mapped in the organizational, data, function and performance view. The dynamic aspects are summarized in the control view (cf. Scheer 1998a, b).

Weske

Weske divides into the modeling domains Function Modeling, Information Modeling, Organization Modeling and in IT-Landscape Modeling (cf. Weske 2007, p. 77). He thus considers the special importance of information technology.

The concepts of the authors Gadatsch, Gehring, Scheer, Österle and Weske, which were briefly sketched before, consider the process or the function as the central element, which transfers input data to output data using organizational units.

The object-oriented concept of Ferstl and Sinz (cf. Sinz 1996) considers the process as a whole and does not provide detailed views for the representation of data and organizational units.

2.3 From Function to Process Thinking

The traditional functional organization of companies (see Fig. 2.8) is strictly hierarchical. At the top of the organization is the leadership, e.g. the board or managing director of the company. Below that are differentiated areas according to professional criteria. They are divided into operational functions (e.g. purchasing, sales, warehousing, production, personnel, finance).

Processes are not the subject of a functional organization. They run across the entire organization, without having to be defined, described or even known. Often the employees also have no idea what a process is, they identify themselves with “their” task. However, in general, several organizational units are involved in the execution of processes, even if they are not defined. In fact, the processes already exist, but they are not formally manifested.

At the transitions between the units of the organization involved in a process, interfaces are created at which the process is constantly interrupted by the transfer of object information (e.g. order data). In addition, there is the danger of media breaks when existing data is re-entered or transformed. Ultimately, the process is not controlled across organizational units, as each unit is only responsible for its “process section”. The formally non-existent, but real processes “therefore find their way” through the functional organization of the company.

Summary: Functional thinking versus process thinking

Many companies are organized by functions (e.g. purchasing, production, sales, accounting, cost accounting), i.e. there are numerous organizational units (divisions, departments, etc.) which are divided by activity groups (functions). However, the processes in

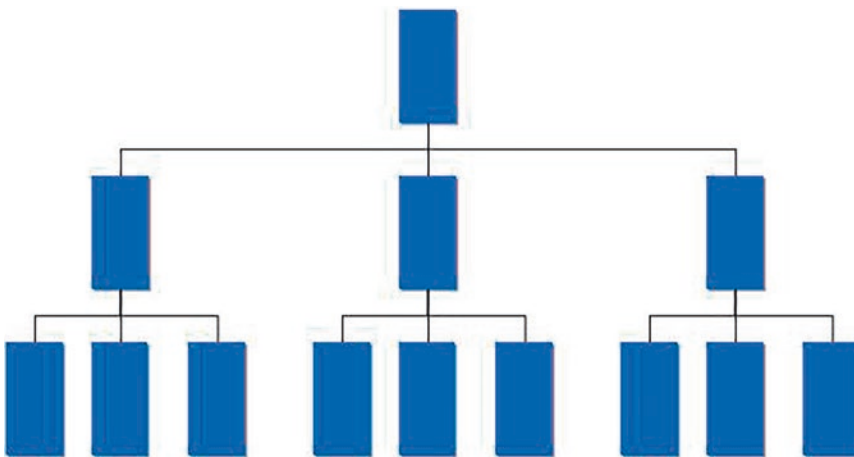


Fig. 2.8 Functional organization (schema)

such organizations usually pass through several of these organizational units, i.e. they are “department-spanning”. Functional thinking deals with the tasks of one’s own area, one’s own department, etc., process-oriented thinking includes the entire process chain, possibly also across several departments or areas.

Chimney effect

The functional organization does not pose a problem in small organizations because the employees know each other, are familiar with the interaction in the processes and can exchange and resolve conflicts directly. In growing organizations, however, many areas often only see their own area of responsibility. There is no overview of the whole. The areas become silos: large, thick and windowless (Osterloh and Frost 2003, pp. 28–29). They only see what is happening inside their areas. The functional thinking of the traditional organization leads to internal blockades and to “information silos” in which the internal communication between the departments takes place only via reporting, file notes and memos. The “chimney effect” occurs because cross-departmental problems are “pulled up” due to lack of horizontal communication to corporate management (cf. Fig. 2.10 based on Osterloh and Frost 2003, p. 29).

In a function-oriented organization, objectives are agreed for the heads of the functional areas and partly also implemented in terms of salary effects. For example, the head of logistics is given the target of keeping the stock level as low as possible in order to reduce capital costs. The head of sales, on the other hand, has the target of selling as many units as possible, which would be easier with high stock levels than with low stock levels.

In the process-oriented organization of a company, the attempt is made to bring process objectives and the resulting results into the foreground. As a rule, these are not

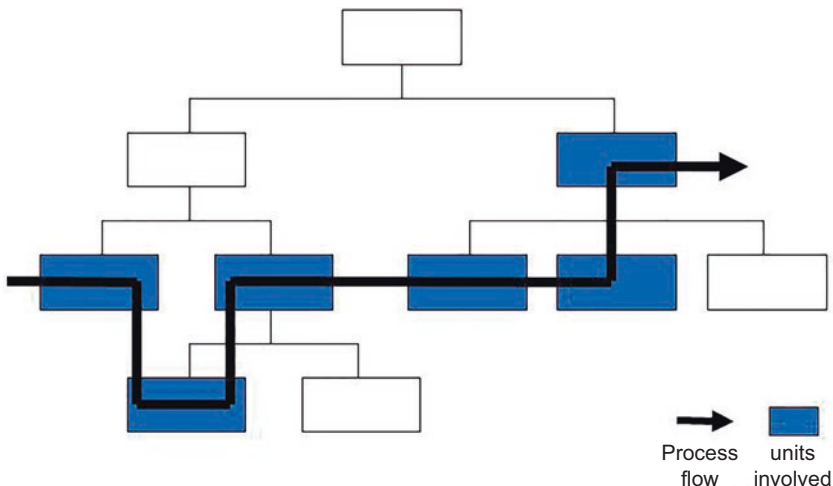


Fig. 2.9 Process course in functionally structured organizations (Dillerup and Stoi 2012)

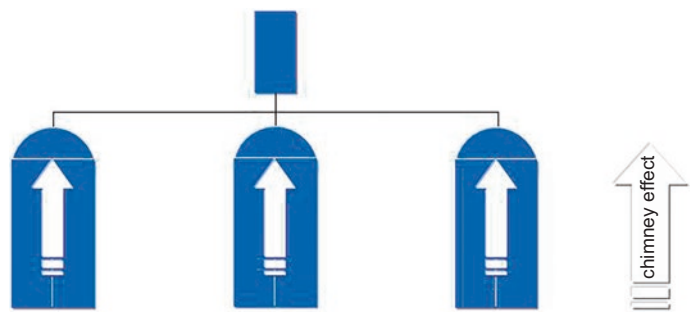


Fig. 2.10 Chimney effect (Osterloh and Frost 2003, p. 29)

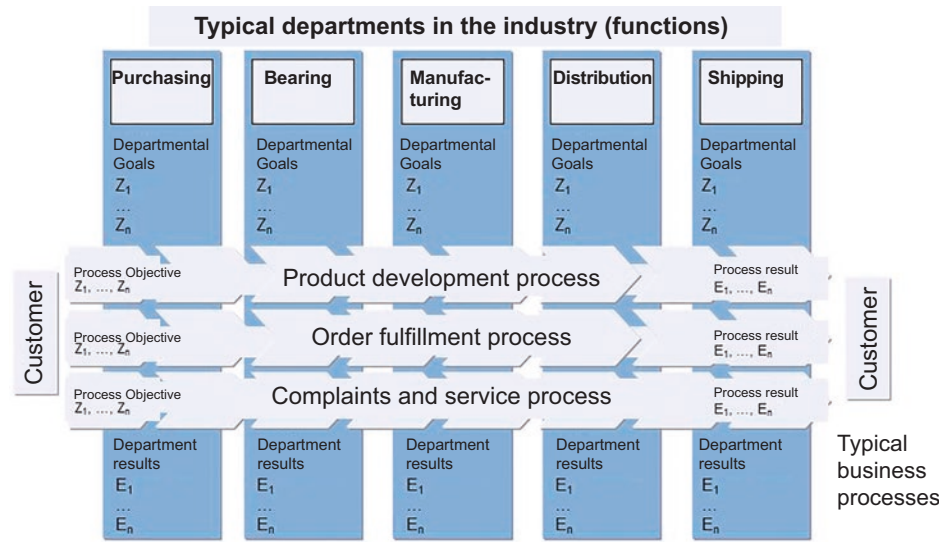


Fig. 2.11 Targets and target conflicts in functional organization

identical if they are compared with the departmental or divisional objectives and results of the classical functional organization (cf. Fig. 2.11).

Example: Classification of invoice verification in the procurement process

A typical example of the different views of process and functional thinking is the procurement of goods and services. In the context of designing procurement processes, the question regularly arises as to which area the sub-task of “invoice verification” should be assigned: logistics or accounting.

The fact that invoice verification carries out the qualitative and quantitative control speaks in favor of logistics. Logistics pursues, inter alia, the goal of transporting the

right goods in the right quantity and quality to the right recipient at the right time. Accounting often claims responsibility for checking postings and payment conditions. Accounting, inter alia, has the goal of preparing a balance sheet and profit and loss account in accordance with the law.

If the process is split, for example in the way that first the qualitative and quantitative control is carried out in logistics and only later, after the documents have been forwarded (e.g. delivery note), the commercial or financial check is carried out in accounting, delays are almost inevitable due to the change of processor. ◀

2.4 Optimization Concepts

2.4.1 Business Reengineering

The concept of Business Reengineering stands for a management approach to radical corporate restructuring that gained popularity in the early 1990s through the work of Hammer and Champy (see Hammer 1990; and Hammer and Champy 1994). The discussion took place primarily in corporate practice and there mainly in the field of management consulting. A scientific investigation of business reengineering took place only later. This development led to several developments of the original concept of Hammer and Champy (Hess and Österle 1995, p. 128). In this context, the terms “Business Process Reengineering”, “Business Engineering”, “Business Redesign”, etc. are sometimes used synonymously. The mentioned concepts mainly deal with the analysis and restructuring of primary processes with market and customer orientation, such as sales processes. However, there are also isolated examples of the use of such approaches in supporting cross-sectional processes, such as accounting.

Hammer and Champy define business reengineering as a “radical cure” for the company. They understand this to mean a fundamental rethink of the company and its business processes in order to essentially realize improvements in costs, quality, services, time and above all customer benefits. (Hammer and Champy 1994, p. 48). Business reengineering is, in their opinion, not an optimization of existing processes, but a new beginning, i.e. a complete rethink of the structures (Hammer and Champy 1994, p. 12). They outline their concept with the key words “fundamental”, “radical” and “dramatic”.

The key word “*fundamental*” stands for answering the question of the purpose and purpose of every activity in the company and also the way in which it is carried out.

The term “*radical*” stands for the willingness to implement even fundamental changes in the company, i.e. it is not about the optimization of existing processes (see also Hammer and Champy 1994, p. 12), but about a new beginning, i.e. a complete rethink of the structures.

The key word “*dramatic*” describes the demand for changes to the company and the efficiency of its workflows in quantum leaps. Hammer and Champy attribute a key role in task fulfillment to information technology (see Hammer and Champy 1994, pp. 113 ff.).

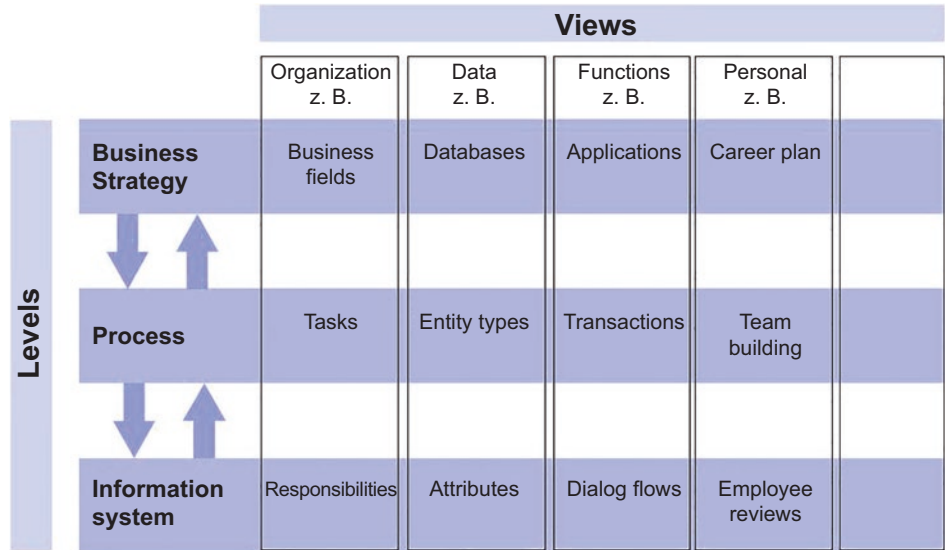


Fig. 2.12 Business Engineering according to Österle (1995, p. 30)

They are mainly concerned that the innovative possibilities of information processing are exploited.

In short, Business Reengineering means answering the question: “*How would we proceed if we started all over again from scratch?*”. It is the task of management to rethink how the work is carried out and how the organization is structured if they were to start all over again from scratch (Robbins 2001, p. 33).

The approaches of Business Reengineering have been taken up and intensively further developed by other authors. Terms used synonymously are Business Process Reengineering, Business Engineering, Business Process Redesign, etc. In German-speaking countries, the approaches of Scheer and Österle became known. Österle comprehensively defines Business Reengineering in the form of a top-down approach starting with the development of business strategy down to the level of information systems (Österle 1995, p. 24). He uses the term Business Engineering and understands this to mean the redesign of the informatory economy (Österle 1995, p. 14). In Fig. 2.12, Österle’s subdivision into the levels of business strategy, process and information system is shown (Österle 1995, p. 30).

The business strategy determines the global framework data for the company, such as the company structure and the business fields. The process level defines the organizational units and determines the company processes and their services. It also defines the rough entity types of information processing, such as customer or account. The information system level is specified in detail. The level view is supplemented by a view concept. Österle distinguishes between the perceptions, or rather views that different departments

in the organization may have such as, data and function (cf. Österle 1995, p. 30) and leaves room for the inclusion of other views, such as personnel, marketing or law.

Summary Business Reengineering

Business Reengineering is a radical concept of process management and is to be assigned to the strategic level. It is about the fundamental rethink of the organization and its processes in order to realize rapid improvements in costs, quality and customer benefits.

2.4.2 Business Process Process - Optimization

Business Reengineering and business process optimization are, although the terms are often used synonymously, different approaches to restructuring the business processes of a company.

The goal of business process optimization is the sustainable improvement of the competitiveness of a company by aligning all essential workflows with customer requirements. This means above all a focus on those business processes that are directly triggered by customer actions (e.g.: ordering, paying an invoice, complaint).

Practice examples of causes are:

- Media breaks in the workflow: Data entry into a PC database, which was taken from a printed report from the SAP ERP system.
- Operators change during the workflow: The invoice is received in the mailroom, then the invoice is forwarded by internal mail to accounting, after processing a copy is forwarded for review to purchasing.
- Double work: Data is captured twice because the responsibilities are not clearly defined.
- Waiting or waiting times: For the booking of a payment document, data from the finance department is required, the inquiry remains without success due to the absence of the employee.

Basic forms of process restructuring

The main objectives of business process optimization are the reduction of the processing time and the improvement of the process quality. Figure 2.13 shows, based on Bleicher (1991, p. 196), basic design options. Explanations can be found in Tab. 2.1.

Processes can be restructured in different ways. Purely organizational approaches such as “eliminating unnecessary activities” or technical measures such as “using a web portal” or mixed forms can be distinguished (cf. Table 2.1).

In addition to these basic methods, the concept of segmentation is often used. In this case, process variants are determined for which different processes are developed. The approach comes from military and disaster medicine: If many injured people have to be

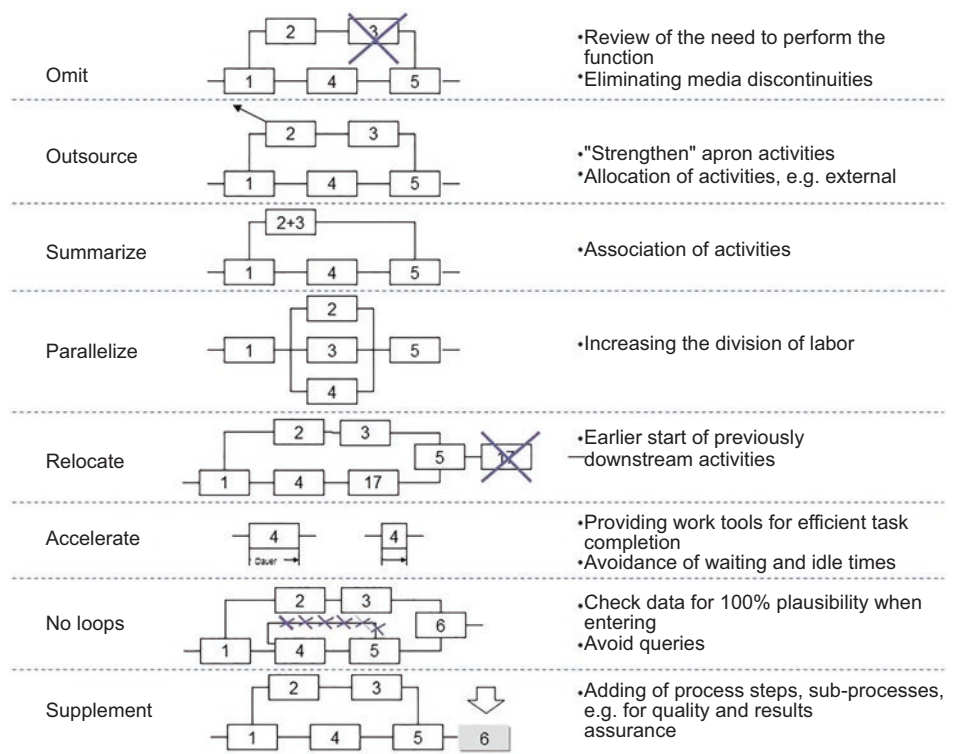


Fig. 2.13 Restructuring approaches according to (Bleicher 1991, modified)

treated at the same time, sequential processing can be fatal. Therefore, priorities have to be set. But even in the normal case, the method is used: After a clinical examination, it is decided whether it is a “standard course” or whether further examinations or treatments are necessary. After that, different process variants run (cf. Hellmann and Eble 2010).

2.4.3 Example Case: Restructuring Spare Parts Procurement

The procedure for business process optimization is to be shown using a deliberately exaggerated, but realistic, fictional example. The organizational structure and the business process before optimization are shown in Fig. 2.14.

The subject of the business process under consideration is the procurement of spare parts for a fictitious machine manufacturer.

Table 2.1 Basic forms of process restructuring according to Bleicher (1991), modified

No.	Concept	Explanation
1	Omit	Checking the necessity of processes or sub-processes for functional fulfilment, abolition of media discontinuities, abolition of pointless approval steps
2	Outsource	Assignment of sub-processes or complete process chains by external specialized service providers (e.g. bookkeeping and accounting by a tax consultant)
3	Summarize	Division of labor tasks are summarized so that an operator performs complete, related sub-processes without operator change (for example, customer consultation and order entry to the creation of the order confirmation)
4	Parallelize	Increasing the division of labor at parallelizable sub-steps (for example, examination correction by several examiners per subject area)
5	Transfer	Transfer of process steps so that tasks are performed early without later becoming a bottleneck (for example, complete capture of customer information when ordering)
6	Accelerate	Use of time-saving work aids (document management system replaces paper documentation), reduction of waiting and down times by increasing capacities
7	Avoid loops	Loop-free design of processes, i.e. avoiding repetition of partial steps of a process (e.g. online capture of all customer and order data as part of order capture and release of the order only after complete plausibility of the data)
8	Complement	Avoidance of downstream processes for “damage control” (e.g. supplementing a quality control after partial assembly with a possible “post-processing process” or a “recall of defective goods” to avoid)

Source: Own representation based on Bleicher (1991)

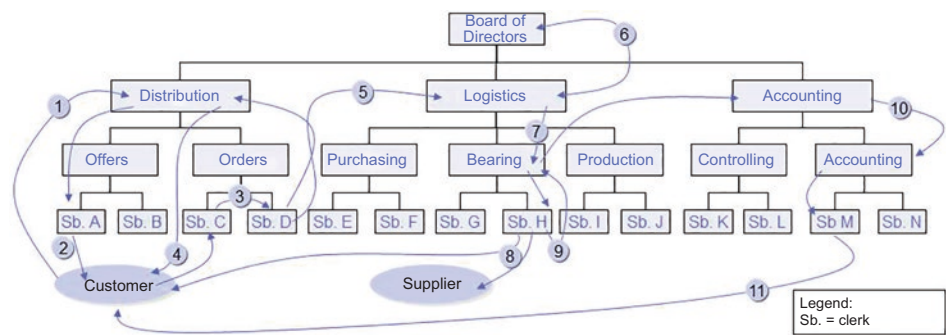


Fig. 2.14 Spare parts procurement before process optimization—before optimization

1. The process begins with the sales manager, who personally takes care of incoming customer inquiries.
2. After that, the offer from clerk A is sent to the customer. Before the offer is sent, it is checked by the sales manager. Since the sales manager is not always present, it can happen that an offer created by clerk A lies around for a few days.
3. When the customer makes an order, it is checked manually by clerk C and then entered into the order processing system by clerk D.
4. The customer receives an order confirmation after the sales manager has seen and released the order.
5. After the order has been entered, it goes to the head of the logistics department. This person decides personally whether a part can be taken from the warehouse, must be procured or even has to be produced.
6. If he is unsure, he asks the board.
7. The warehouse manager then receives the order to deliver the material. Since he is not present at the plant on that day, he does not give the order to one of his clerks until the following workday, e.g. H.
8. This clerk (here H) takes the part, sends it to the customer and initiates a reorder of the spare part from the responsible supplier.
9. After shipping, clerk H submits the departure booking to his superior in the warehouse. This checks the voucher and sends it to the head of accounting.
10. The head of accounting passes the voucher to the head of the accounting department and this in turn to one of his clerks. Since the head of accounting is often used by the board for planning tasks, the vouchers often remain unprocessed for several days.
11. In this case, clerk M creates the invoice and sends it to the customer.

The main weaknesses of the process are relatively easy to identify:

- Managers make decisions on operational matters up to and including the management,
- there is involvement of too many people with the resulting change of employees,
- there are few contacts at the level of the clerks, since the transfer of processes is often carried out by managers,
- in case of absence, obviously no substitute regulation.

This results in several possibilities for improvement in terms of process optimization, i.e. a change in small steps:

- the board of directors usually does not decide on operational questions of business processes,
- managers only intervene in the process in exceptional cases, the process is controlled throughout by the clerk level,

- the customer communicates directly with the (responsible) clerks,
- clerks pass information on to each other directly,
- employees complete a processing step completely.

If these principles are applied to the business process, an optimized version of the process could follow the course shown in Fig. 2.15.

The sequence of the revised business process is now as follows:

1. The process begins with the clerk in sales, who creates the offers independently on the basis of customer inquiries.
2. Then the offer is created by clerk A and sent to the customer.
3. If the customer places an order, it is checked by clerk C and then recorded directly in the order processing system.
4. Then clerk C informs the responsible buyer, warehouseman or production clerk, depending on how the business case is to be assessed (alternatives are sales from stock, self-manufacturing or outsourcing). The customer also receives an order confirmation with the delivery date.
5. In the case considered here, employee G in the warehouse receives the order to deliver the material to the customer. Since he is not present on this day, his deputy H takes over his task. He removes the part from the warehouse, sends it to the customer and initiates a reorder of the spare part from the responsible supplier.
6. Now employee H informs employee M in accounting.
7. Now clerk M creates the invoice based on the information received and sends it to the customer.

For the operational implementation of reengineering or optimization projects, the individual development of an analysis checklist with approaches for process optimization is recommended (Riekhof 1997, p. 15):

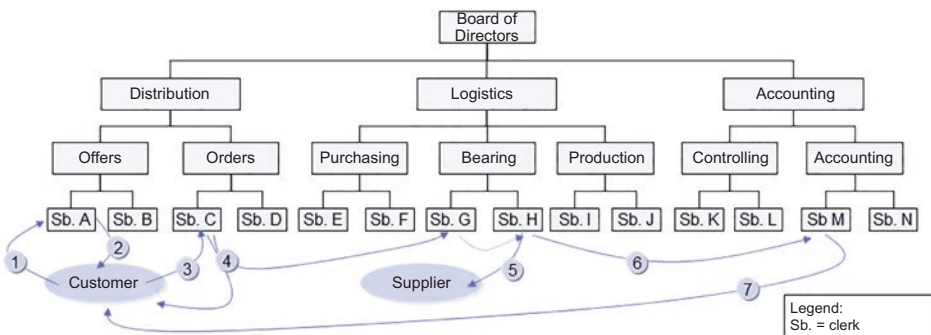


Fig. 2.15 Spare part procurement after process optimization

- Can double work or unnecessary administration be dispensed with?
- Can process elements be simplified and standardized?
- Can process elements be automated?
- Can the sequence of activities be optimized?
- Can process elements be designed to be fail-safe?
- Can non-value-added elements be eliminated?
- Can the division of labor between process customers and suppliers be optimized?

Summary of Business Process Optimization

The goal of business process optimization is a sustainable improvement of the processes by the successive introduction of coordinated measures. Typical is a thorough analysis of the situation with subsequent optimization and IT-supported implementation.

2.4.4 Case Study: Process Optimization Accounts Receivable Processing

2.4.4.1 Initial Situation

In a medium-sized engineering office, the processing of incoming supplier invoices is largely paper-based. The volume of invoices is about 250 per month. The process shown in Fig. 2.16 consists of two task areas: incoming processing and archiving, which are outlined below.

- Incoming: Manual processing of incoming paper invoices, printing of invoices for email incoming, partly picking up invoices in online portals of large suppliers
- Archiving: Printing of several copies (chronologically + alphabetically according to business areas and suppliers)

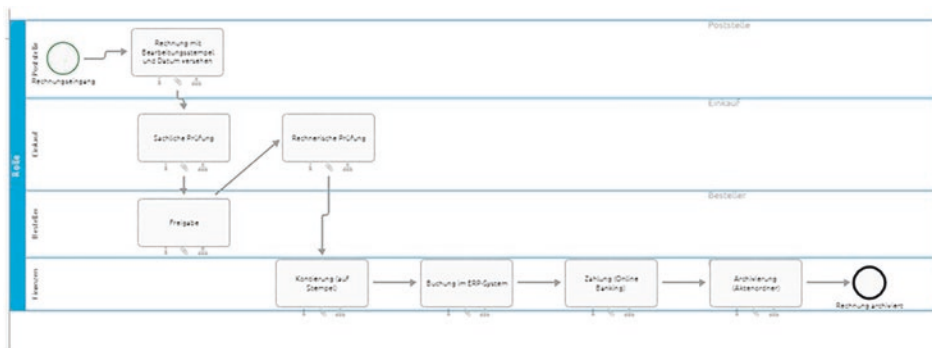


Fig. 2.16 Actual process for processing incoming invoices

A short analysis revealed numerous weaknesses, such as many interfaces between the process steps. The ERP system used in the company is only integrated late in the process (from the booking of the invoice). The paper-dominated work of the employees does not allow mobile work and causes many processing errors.

2.4.4.2 Problem Solving

The company has checked the process in a workshop using the “checklist” presented in Fig. 2.13 and came up with the following suggestions:

- Omit: Input and processing stamp (done automatically by new scan software), no manual “photocopies” for other departments
- Outsource: Consider using a cloud provider for the ERP system and the DMS
- Summarize: Factual examination, computational examination and release as well as accounting and payment
- Parallelize: Factual examination, computational examination (for complex invoices)
- Move: Archiving directly after scanning, update archiving continuously
- No loops: Only transfer data plausibly into the system
- Supplement: Evaluations and analyses of existing invoices for quality control

The derived target process is shown in Fig. 2.16, 2.17.

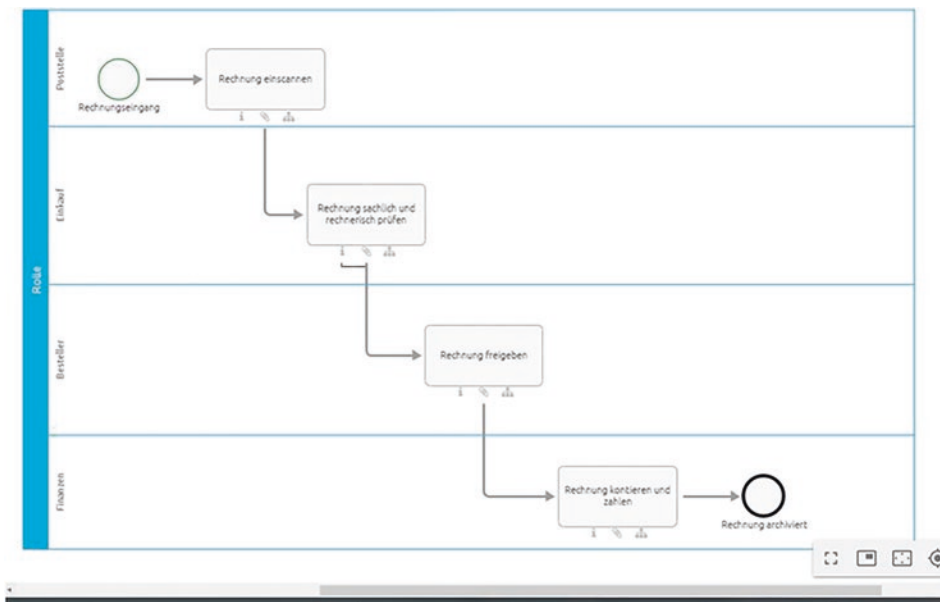


Fig. 2.17 Target process for processing incoming invoices

2.4.5 Example Case: Process Optimization of Order Processing IT Service

2.4.5.1 Initial Situation

The process relates to the processing of order requests in the IT service of a machine engineering company. It is shown as a simplified process model with the modeling tool ARIS-Express and the simplified eEPK notation (cf. Sect. 5.6) in Fig. 2.16 as an actual process and a target process.

It has traditionally been created over the years and has some weaknesses:

- The internal orders of the IT users (order requests, abbreviated “BANF”) are made via “free text” without a material number, so there are regularly time-consuming consultations between the IT purchasing department and the IT users
- If the goods are missing, a manual entry is made in the so-called “delivery database”, an “Excel table”
- The IT users do not receive an order confirmation from the IT service about the order received.
- There is no current order status, neither for the IT user nor for the IT service.
- Due to the partly manual processing, only consolidated invoices are issued.

Overall, the process has several media breaks, i.e. information that is available electronically in the SAP system is copied out and sent by e-mail, as well as the already mentioned “Excel table”.

2.4.5.2 Problem Solving

The process was optimized. The integration of the entire process into SAP as an automated workflow was chosen as the solution approach. IT users have been using the material number from the beginning of the process. For this purpose, a search function of the SAP system was provided. All data belonging to the process are recorded in the SAP system. IT users receive an automated order confirmation. The status of the order is always available (for IT users and IT purchasing). The process no longer contains any media breaks, all data is centrally available. Due to the integrated data storage, invoices are possible for each individual order (Fig. 2.18).

2.4.6 Case Study: Optimizing Applicant Management

2.4.6.1 Initial Situation

The subject of the case study is a large company operating worldwide. It strives to fill vacant positions as quickly as possible with suitable applicants. In the past, there have been cases again and again where the filling of vacant positions took several months and no optimal staffing decisions were made. For this reason, the head of human resources

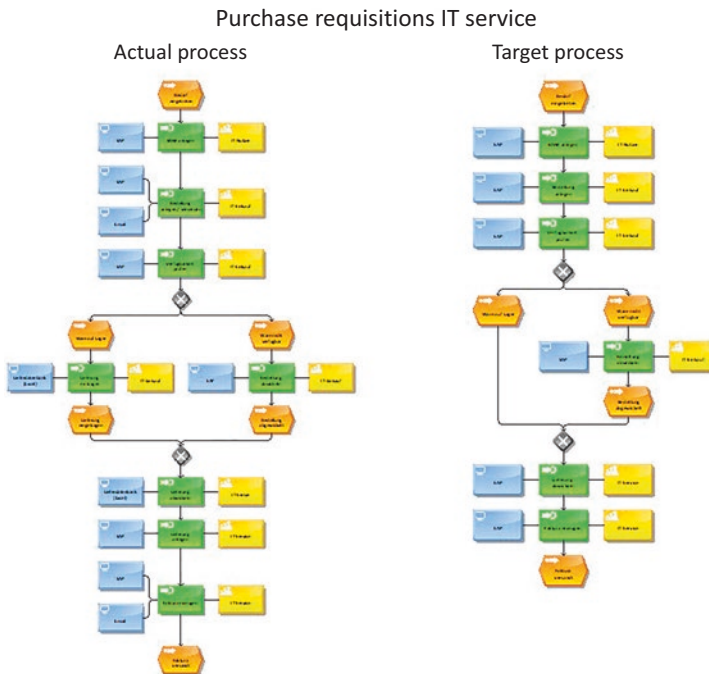


Fig. 2.18 Current and target process: order requests in IT service

has commissioned a process improvement team to examine the process of acquiring personnel and to make suggestions for optimization.

2.4.6.2 Problem Solving

First, the team looks at the duration of the job posting, measured from the decision “position can be filled” to the event “employment contract signed” (see Fig. 2.16, 2.19).

The duration of the job posting varies greatly depending on the hiring departments. The reasons are different. The Organization and Information Technology (Org/IT) department has a high demand for skilled workers, which obviously leads to accelerated hiring behavior. Other departments (e.g. purchasing, manufacturing or sales) take more time. The hierarchy level obviously has little influence on the hiring time.

The situation is different, however, depending on the contact medium with the applicant. Traditional media such as newspapers or university fairs lead to average or good values. However, these solutions are relatively expensive (especially newspaper ads) or labor-intensive (university fairs require staff and are only worthwhile if there are enough positions to be filled). Modern and supposedly “fast” media such as job boards or the company’s own website lead to a high number of applications, which are often of very poor quality (missing/contradictory information, often unsuitable candidates, etc.). This ties up a lot of capacity in the HR department and in the departments that are looking

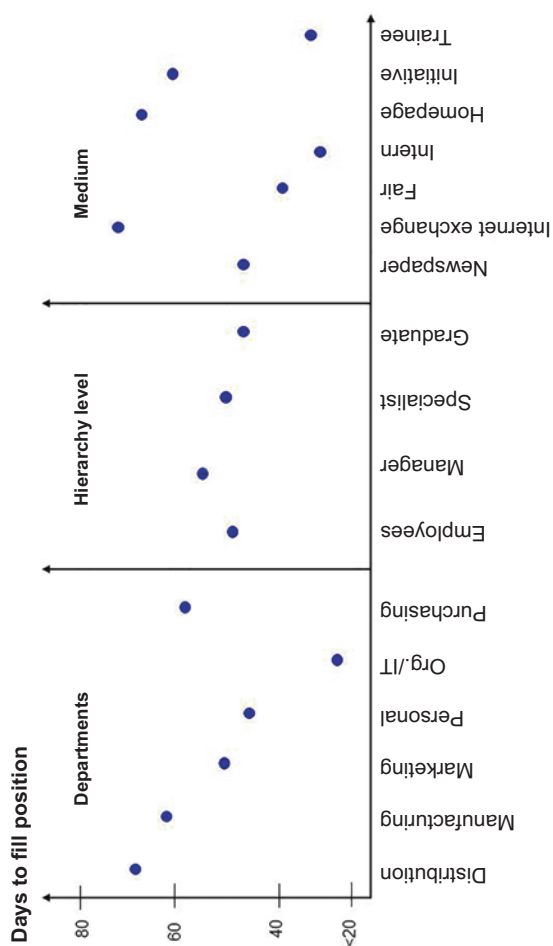


Fig. 2.19 Analysis of the duration of the job posting

for candidates. Appointments based on unsolicited applications are also very time-consuming because often there are no suitable positions for good candidates and these positions may have to be created. Relatively short periods of time for job postings are to be expected for university interns and participants in trainee programs. Since the people are already known in the company, the departments can assess their quality and usability very well and are interested in an accelerated appointment.

An investigation of the relationship between the duration of the job posting and the quality of the first personnel assessment by the direct supervisor produces a worrying picture. The shorter the time frame for the job posting, the higher the probability for a good assessment. Obviously, there is a danger that after a longer search for applicants, compromises will be made in the quality of the applicants in order to fill the position at all.

Proposal for process improvement

The analysis leads to the decision to significantly expand the pool of interns and intensify the trainee program for university graduates. In future, no advertisements will be placed on job boards on the Internet. Job advertisements in traditional print media will be limited to executives and specialists, as they cannot be recruited from the pool of interns and trainees. Only positions that are published simultaneously in print media will appear on the company website.

2.5 Related Management Concepts

In recent years, numerous management concepts have been developed and implemented in practice, which at least partially pursue comparable goals to process management. In particular, goals such as customer orientation, efficiency, reduction of interfaces and simplification of work organization are taken into account. These concepts include process performance management, lean management, kaizen/KVP and, more recently, agile methods of software development.

2.5.1 Process Performance Management

Process Performance Management (also “Business Performance Management, Scheer and Hess 2009, p. 145) has developed from process management (Oehler 2006, p. 50). The concept includes process management and emphasizes in particular the identification and analysis of key figures that are derived from the ongoing process. To implement this, powerful process performance management systems are required, which are used for data collection and analysis. The technical basis for process performance management systems are data warehouse systems. They take over the formal cleansing, content-related checking and condensation of the data. The process performance management

system accesses this and creates key figures and reports (cf. Schmelzer and Sesselmann 2013). Like process management, process performance management pursues the goal of increasing the performance of business processes across all areas, while business performance management is more global in scope and, in addition to processes, also focuses on the financial sector (Scheer and Hess 2009, p. 145).

2.5.2 Lean Management

A study by the Massachusetts Institute of Technology (MIT) at the beginning of the 1990s, in which Japanese, US and European car manufacturers were compared, led to the development of the Lean Management concept. Originally, the focus was on production, which was made clear by the term “Lean Production”. Later, an extension to the entire company followed. Lean management is to be understood as a lean management, the goal of which is high efficiency, speed and superior quality. (cf. e.g. Schmelzer and Sesselmann 2013).

2.5.3 Kaizen/Continuous Improvement Process (CIP)

A Japanese management philosophy is Kaizen (literally “improvement”) or CIP (Continuous Improvement Process), under which a continuous improvement process is to be understood with the involvement of employees. The strong process orientation is emphasized, i.e. the focus is not on the result, but on the process of creating the result and the involvement of employees and the use of their skills to solve existing problems in the processes. The aim is to achieve a permanent increase in process performance through an improvement in small steps (cf. e.g. Schmelzer and Sesselmann 2013).

2.6 Reference Models

A reference model is a ready-made model of an “ideal” process that can be used in one’s own company with or without few changes. Typical applications are the analysis and restructuring of business processes, in-house software development, selection of standard software and documentation of software or standard software by software developers or manufacturers. Sources for reference models are other comparable companies in their own industries, literature sources, management consultants and software providers.

The benefits of using reference models include the greater experience base that can be included in one’s own projects and the reduction of one’s own modeling effort. Risks include the loss of competitive advantages through the standardization of processes. The

company is deprived of the opportunity to positively differentiate itself from the competition. In addition, a high level of training is to be considered. Ultimately, there is the risk of lack of acceptance among employees, because they are confronted with a model, the development of which they were not involved in. Research work has already been published under the term “process acceptance theory” (cf. e.g. Drewes and Nissen 2022).

Characteristics of reference models

There is a distinction to be made in particular between global business reference models, software reference models and enterprise process models (cf. Schmelzer and Sesselmann 2013): Business reference model Reference models are general information models tailored to specific application areas (e.g. functions, industries). They serve as orientation and as a recommendation for the construction of company-specific models. The “SCOR model” (SCOR = Supply Chain Operations Reference for Supply Chains) can be cited as an example (cf. SCOR 2001).

Software Reference models are offered by standard software manufacturers and describe those processes that are supported by the standard software. They provide customers with templates that reduce the implementation effort of the standard software. The company SAP AG from Walldorf provides extensive model-based descriptions of the functionality of its software with the “SAP Solution Map” and the “SAP Business Scenario Map”.

Enterprise process models are tailored to the specific situation of companies. They contain guidelines and rules for how business units are to be structured and described. They are often centrally prescribed by the management as internal rules. Elements of the models include: hints for compliance with statutory regulations, rules for organizational documentation, guidelines for the selection of ERP systems and for customizing standard systems. The “V-Model” can be cited as an example, a development process model for software development in the public sector (IABG o. J., retrieval 28.07.2016).

2.7 Exploratory Process Management

The “exploratory process management” is a newer research approach to process management with a future-oriented view of innovations and new business models (cf. Grisold et al. 2019; Gross et al. 2019; as well as Recker and Mendling 2016).

The concept is based on the realization that the focus of the previous methods in process management was rather past-oriented and concerned mainly with the improvement (optimization) of existing processes. Therefore, the term “exploitative process management” is also used (cf. Grisold et al. 2019). Typical questions of the previous “exploitative process management” are:

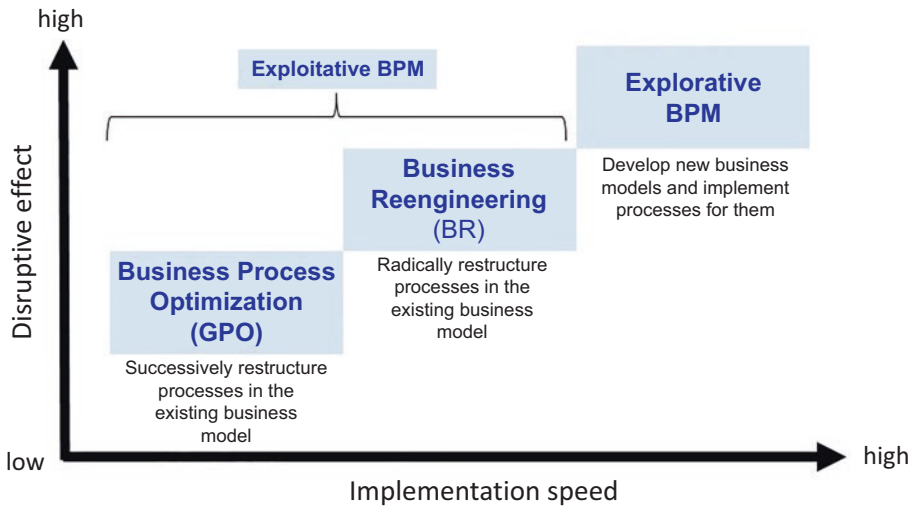


Fig. 2.20 Classification of different BPM concepts

- How can we reduce the number of process variants?
- How can we reduce the process runtime?
- How can we avoid downtime in the process?
- How can we reduce the cost of the process?
- How can we ensure that the “process rules” or “compliance rules” are adhered to?

The approach is therefore “from the inside out” according to the principle: “How can I fulfil customer wishes in an improved form with my current processes”. With exploratory process management, innovation is in the foreground in order to secure the “company’s profit of tomorrow”. You work your way “from the outside in” (cf. Grisold et al. 2019), in order to develop ideas for new, mostly digital, business models driven by opportunities such as new market trends.

In Fig. 2.20 the two exploitative BPM concepts business reengineering and business process optimization are compared with the exploratory approach. While both business reengineering and business process optimization deal with the optimization of the existing business model and the further development of the processes, the exploratory approach develops the business model and provides processes for it. The disruptive effect is therefore particularly great with the exploratory approach, while business process optimization only makes small changes.

Ideally, both concepts are combined, the “explorative process management” secures the strategic environment and thus the long-term competitiveness through the introduction of innovations, the “exploitative” process management secures the efficient economic implementation of the concepts (cf. vom Brocke et al. 2019).

2.8 Review Questions and Exercises

2.8.1 Questions

1. Explain the concept of holistic business process and workflow management.
2. Why is the two-stage process modeling (1. Business process, 2. Workflow) often preferred in practice?
3. Explain the essential steps of a process management life cycle model.
4. Why are views formed in the context of process management?
5. Explain under which conditions the “chimney effect” occurs in organizations.
6. Compare the optimization concepts of business reengineering and business process optimization.
7. Differentiate “explorative process management” from “exploitative process management”

2.8.2 Exercise “Process Cube”

You want to start a medical practice and proceed in a methodologically sound way. Use the process cube to create a concept for setting up a medical practice that is structured in terms of the ideas of process management. For each element of the levels, phases and views, write down typical examples from the environment of a medical practice that are to be described in more detail later.

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Organization and Introduction of Business Process Management

3

Process management is permanent project work

Abstract

The introduction of process management is a classic change project that affects the entire organization at all levels in every area. Many stakeholders, starting with the executive level (“Chief Process Officer”) through the middle management levels (“Process Manager”) down to the individual employee, the “Process Expert”, are to be involved. Changing and optimizing processes means “changing people” in the first place and motivating them to work in a process-oriented way. The chapter describes possibilities for process-oriented organization of companies and the optimization of processes. In addition, the different roles and actors are presented, which must be differently pronounced depending on the size of the company. A special focus is placed on the project organization, which is particularly important in the initial introduction of project management concepts. The chapter ends with repeating questions and an exercise.

3.1 Process-Oriented Organizational Forms

3.1.1 Design Forms

The introduction of process-oriented organizational forms (short: Process organization) has not yet arrived in practice, although it belongs to the traditional methods of business administration under the term “process organization” (cf. e.g. Olfert and Rahn 2021, p. 149). A scientific study on business process management yielded a sobering result: “For more than 80% of the participants, the improvement of quality and the increase of

transparency are the most important goals for process management in the companies. However, these goals are only achieved by less than 50% of these companies.” (Gadatsch et al. 2016).

The “optimal” organizational form for process management is just as difficult to realize as the design of “digitalization organizations” in public management, whether as a “super digital ministry” with extensive competencies or as a networked system of decentralized “digitalization officers” in “federal ministries or state governments” (cf. Mertens 2021 in detail).

Classical line organization

In many companies, classical functional divisions of the organizational structure still prevail. A simplified example can be seen in Fig. 3.1. (It shows a company that produces engines, brakes and gearboxes). The company is functionally divided into various departments (sales, purchasing, production, etc.), each of which deals with individual tasks (functions). However, the process flow for the core processes (production and sale of engines, gearboxes and brakes) runs horizontally, but without there being a responsible body for these processes. In other words: There is no process control. The processes are “divided” into individual, independent sections by the various departments.

Because of the lack of process control within the framework of the functional line organization, concepts for process-oriented organization have been developed. The organizational design of process management has a major impact on success in the company. Process management can be set up as a classical process organization, as a staff unit within a functional organization, or as a matrix organization (cf. Fig. 3.2).

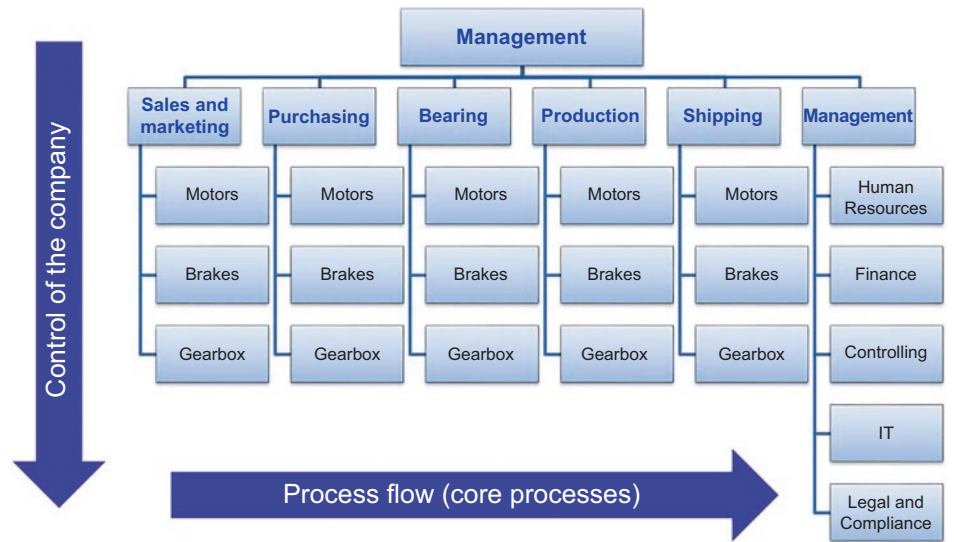


Fig. 3.1 Forms of process organization—classical line organization

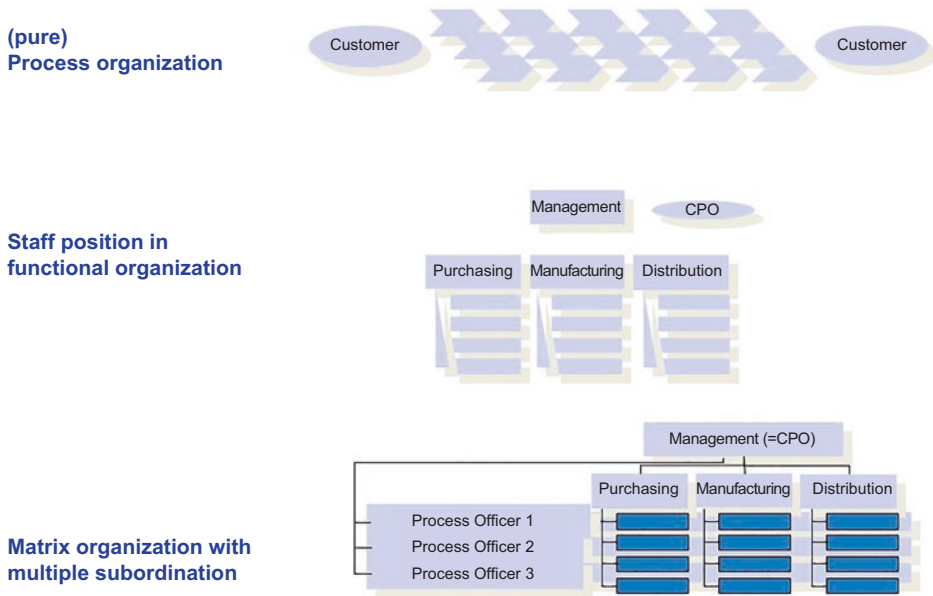


Fig. 3.2 Basic design forms of the process organization

In the **pure process organization** the activities are arranged so that they are aligned with the customer's requirements in terms of an end-to-end process as far as possible. The aim is to arrange the sequence of steps in such a way that the process can be carried out smoothly. Here, processes must be organizationally separated without overlap (e.g. private customer business, business customer business, mail order business).

Figure 3.3 shows the example from Fig. 3.1 as a pure process organization. For each core process (e.g. engine production and sales), there is a responsible party, the "Chief Process Officer" for the respective process. In this, all necessary resources (budget, personnel, machines, raw materials, buildings, etc.) are organized that he needs for the process.

Ongoing activities (e.g. joint purchasing, marketing) must be coordinated because there is no functional responsibility. They are also referred to as "share services" in practice. The process responsible persons assume the entrepreneurial responsibility for their process. The processes are the "departments" of the process organization.

For this purpose, the example of the pure process organization from Fig. 3.3 was slightly modified, the process steps for personnel, finance, controlling, IT and law / compliance were outsourced to a shared service center, i.e. a cross-process responsibility (cf. Fig. 3.4). The core processes are still structured according to the principle of the pure process organization.

The **staff organization** within a functional organization coordinates the processes within the company. However, the functional organization (divisions, departments,

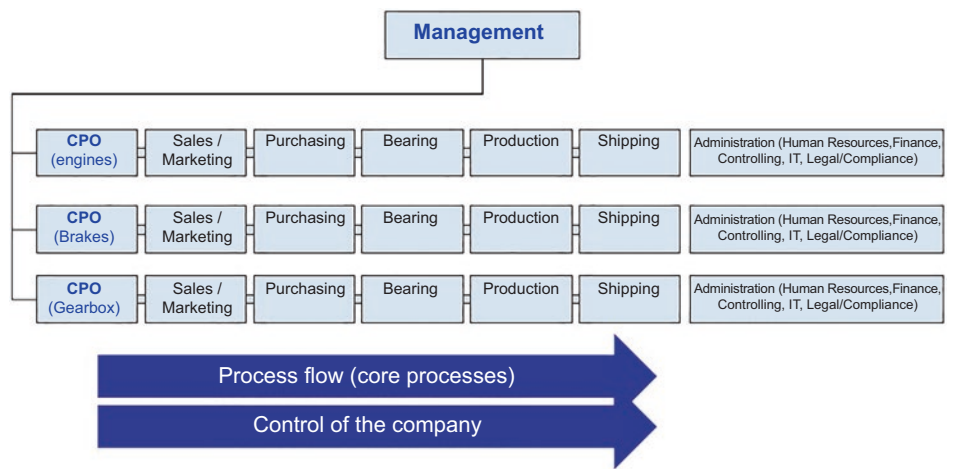


Fig. 3.3 Forms of process organization—pure process organization

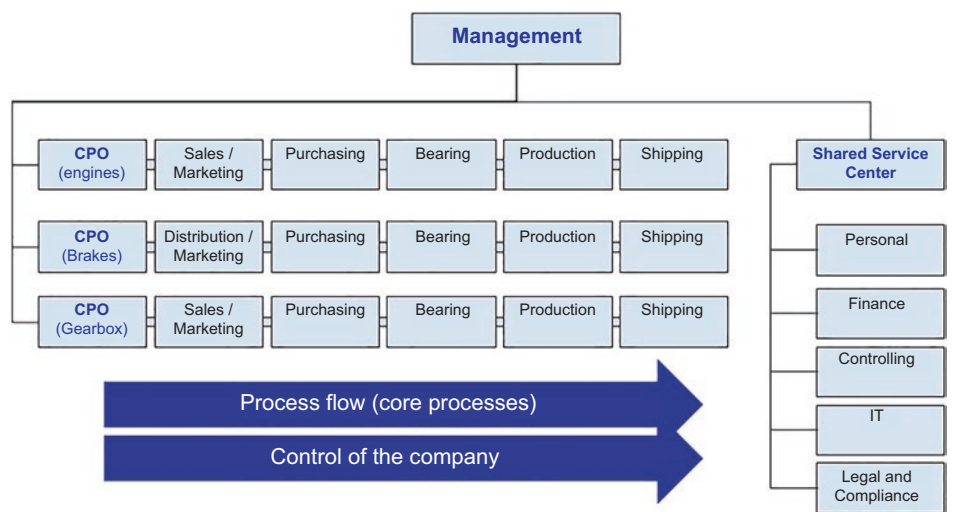


Fig. 3.4 Forms of process organization—process organization with shared service centers

groups, etc.) remains in place, i.e. the organization is basically oriented towards functions. The efficiency of this model is therefore not considered to be particularly high regarding process management, but can be a suitable alternative to process organization in the event of suitable leadership qualities of the Chief Process Officer (head of the staff department), as part of the initial introduction of process management.

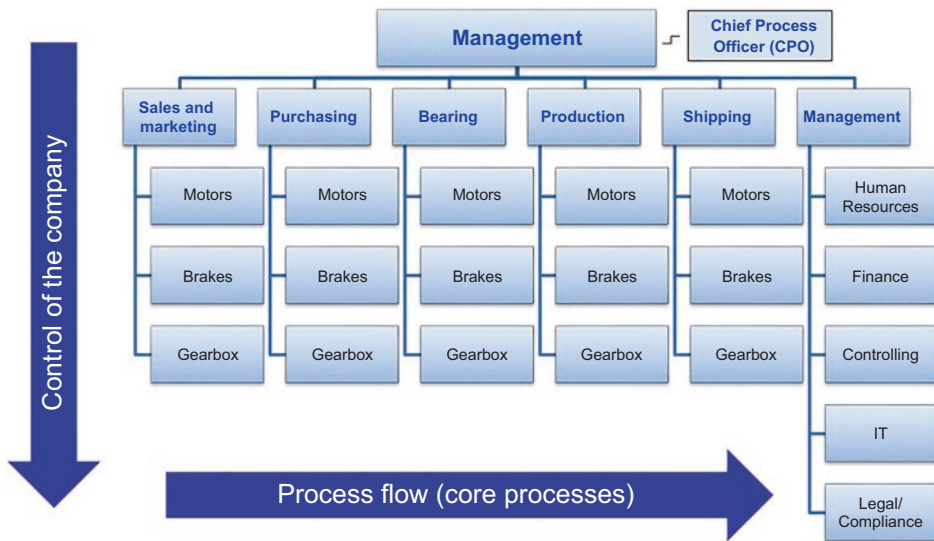


Fig. 3.5 Processor organization as a staff organization

Practice example DAK

The tasks of the CPO set up as a staff unit in corporate development at DAK (German Health Insurance) include the “moderation, documentation and derivation of concrete projects from the strategy”. The IT manager is still responsible for implementation and thus also significantly involved in process management (Vogel 2004, p. 22). ◀

The example from Fig. 3.1 is shown in Fig. 3.5 as a processor organization with a staff organization. The difference to the pure line organization is marginal.

The **matrix organization** knows two principles of structure: activity/function and object/process, according to which the activities are aligned. In this case, process managers (process officers) take on the task of aligning processes along the functional organization in such a way that the processes function smoothly. They compete with the heads of the functional departments for resources, which is intended to lead to permanent coordination conflicts.

The well-known example of the engine manufacturer from Fig. 3.1 is shown in Fig. 3.6, in a variant with shared service centers.

Another example of the use of matrix organization in the context of a hospital is shown in Fig. 3.7.

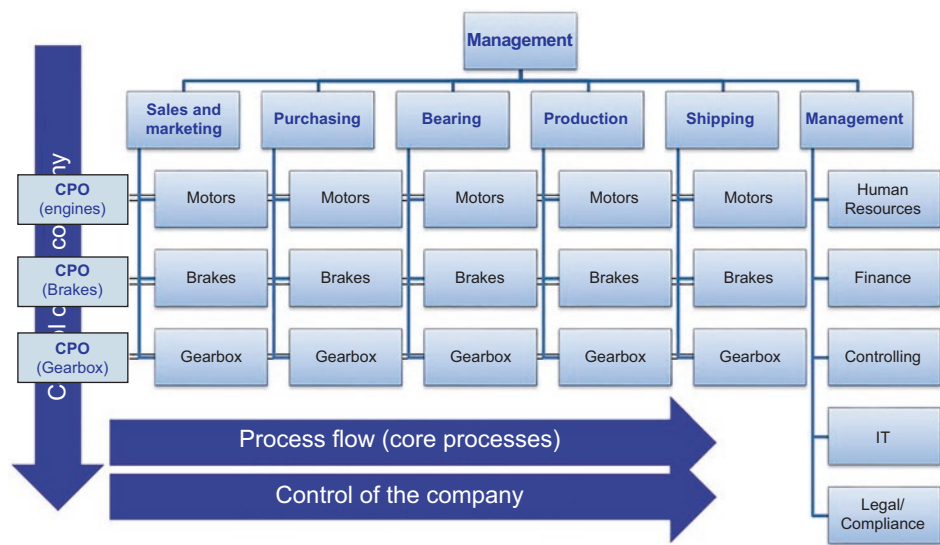


Fig. 3.6 Processor organization as a matrix organization

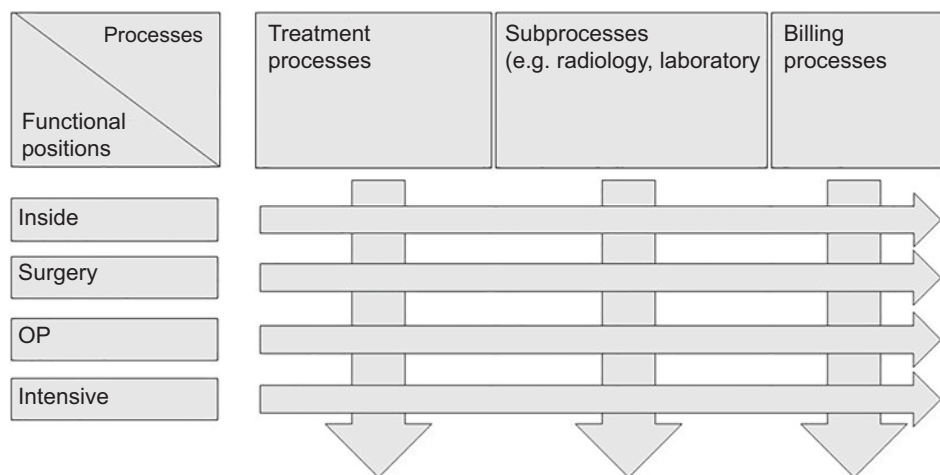


Fig. 3.7 Matrix organization in the hospital

3.1.2 Assessment

The **pure processor organization** is considered the ideal form, which is practically not realizable in practice, as it is very expensive. It is difficult to implement, especially in an existing functional organization. This is due to resistance and understanding problems among employees and managers. Its features are:

- Processes are customer-oriented (customer-to-customer processes),
- Processes are the primary organizational structure of the functional organization,
- Processes have their own organizational autonomy,
- Processes have their own resources (budget, personnel, machines, equipment, IT systems),
- CPO has full professional and disciplinary authority.

The effect can be summarized as follows: strong process orientation, very high process efficiency and low resource efficiency (cf. Schmelzer and Sesselmann 2013).

The **staff organization**, a process management office in a functional organization, is the entry-level solution for process management. Because of the limited influence of staff offices on decisions, this variant is also referred to as the “influence-process organization” by Schmelzer and Sesselmann (2013). It is easy to implement and widespread. Its features are:

- Processes are not customer-oriented (no customer-to-customer processes!),
- Functions are the primary organizational structure of the functional organization,
- Processes have only very limited organizational autonomy,
- Processes have only limited own resources (budget, personnel, machines, equipment, IT systems),
- Chief Process Officer (CPO) has only coordinating powers.

The effect of this variant is therefore limited: strong functional orientation, low process efficiency and high resource efficiency. In most cases, a more pragmatic solution is chosen in practice, which **matrix organization**. Its features are balanced:

- Processes are customer-oriented (customer-to-customer processes),
- Processes and functions are equal members of the organizational structure,
- Processes have limited organizational independence,
- Processes have comprehensive own resources (budget, personnel, machines, equipment, IT systems),
- CPO has full professional, but only limited disciplinary powers, because this lies with the functional responsibility.

The effect of the matrix organization is a balanced process and functional orientation, process and resource efficiency (cf. Schmelzer and Sesselmann 2013).

The main pro and contra arguments of the organizational design options are summarized in Fig. 3.8

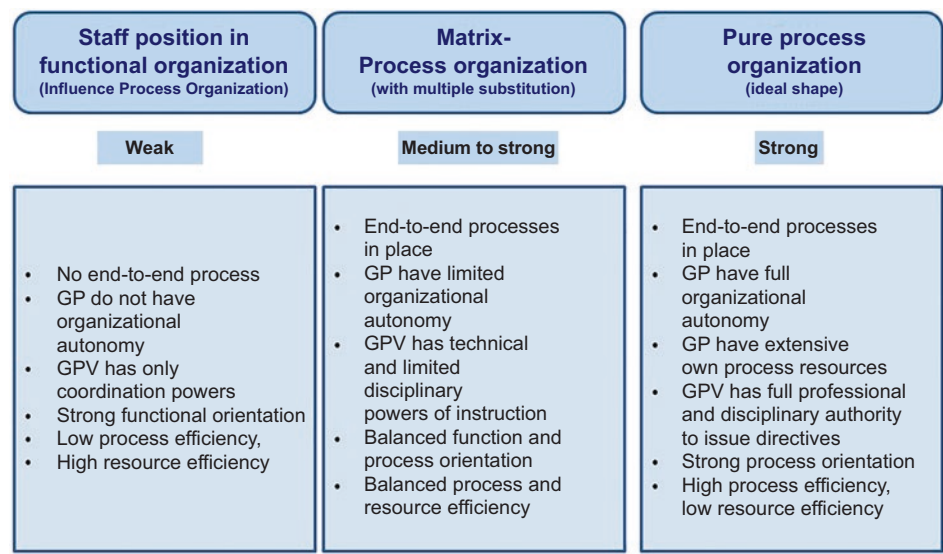


Fig. 3.8 Summary of the organizational forms of process management according to Schmelzer and Sesselmann (2013)

3.2 Roles and Actors

Process management is characterized by the interaction of many stakeholders in different roles on different levels of an organization, such as an industrial company or a university. The overview in Fig. 3.9 first lists the essential stakeholders in the previously introduced concept of business process and workflow management in an abstract form. Since there are a large number of German and English synonyms, the German and English terms are given where common. In the following, the terms most used in practice will be used more frequently.

The representation in Fig. 3.9 differentiates between roles in day-to-day business (run the business) and change projects (change the business) to improve efficiency and quality. In addition, there are cross-cutting roles such as the *Proccess Steering Board* or the *Process Auditors*.

The roles are described below based on Schmelzer (2005). They must be expressed in specific form in each company and transferred to specific positions. Depending on the situation in which the facility is located, this may also require the creation of new positions.

Head of Process Management or Chief Process Officer (CPO)

The person responsible for the processes of a facility is the Chief Process Officer (CPO). He/she ensures the enterprise-wide documentation, restructuring and monitoring of the

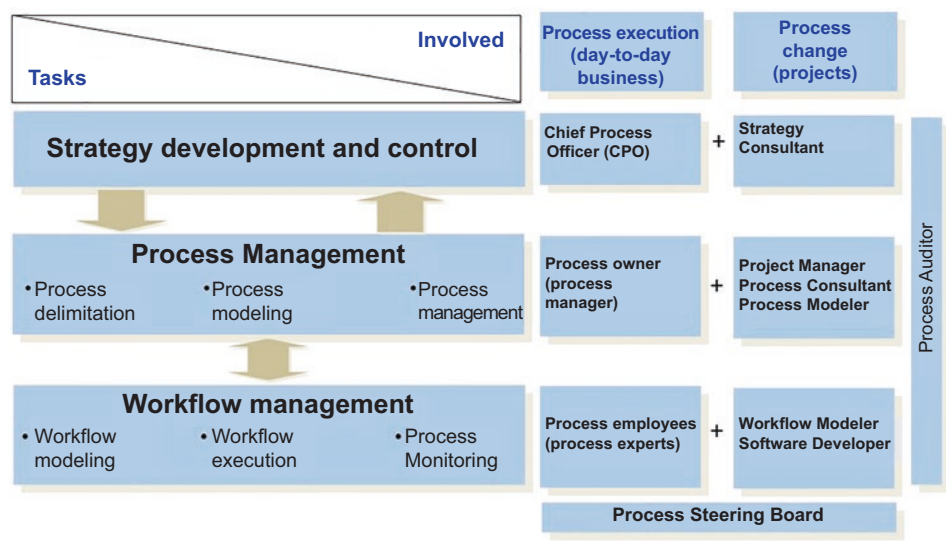


Fig. 3.9 Roles in process management

processes, consultation of the organizational units and the process-oriented design of the organization. He/she is not responsible for individual processes, but for the effective interaction with the customer or patient in mind. His/her tasks are:

- Process documentation: identification and description of relevant processes,
- Process analysis: business-oriented simulation and weak point analysis of business processes,
- Process optimization: identification, definition, initiation and monitoring of process improvements,
- Process monitoring: ongoing analysis of process indicators with a view to achieving process objectives,
- Design and implementation of a process-oriented corporate organization including the transfer of process responsibility to so-called process owners (Process Owner),
- Ensuring process-oriented IT systems through cooperation with the CIO (Chief Information Officer).

The actual occupation of the CPO -role falls in industry also differently out. Not all companies have corresponding positions with a dedicated job title “CPO” within their organizational structure. So the proportion of companies that have a role that has overall responsibility for all processes is only just under 28% (cf. Gadatsch et al. 2016). Often, the role of the CPO therefore remains in practice with the management. In smaller organizations this can still be considered a pragmatic and sensible solution. In larger

companies, however, this results in the management not being able to carry out this task to a sufficient extent.

Due to the increasing digitalization of processes, the role of the CPO is also taken over by the Chief Information Officer (CIO) or Chief Digital Officer (CDO), as the tasks in practice cannot always be separated (cf. the interviews with several CIOs in Brenner and Brenner 2022). In many cases, digitalization and thus also process design are considered tasks of the entire management as a leadership team and not only of dedicated roles such as those of a CIO (cf. Brenner and Brenner 2022).

Process owner or process manager:

Another central role is played by the process owner, also called process responsible or process manager. They are responsible for the strategic and operational control and restructuring of the processes. They set process goals and ensure their achievement by target-oriented leadership of the process-supporting employees. If both roles are named, the process owner is responsible for the strategic part of the tasks and the process manager (PM) for the daily business and its operational control.

Process employees or process staff/process experts or process expert/process participants or process participants:

The process employees or process experts (e.g. a clerk in sales, an employee in administration) support the initial implementation of business process management and further development in major restructuring of the process organization. They are generally involved as specialists in one or more processes and have more executive tasks in this role in relation to the process.

Process consultant or process consultant:

The execution of conceptual and executive project work packages, e.g. knowledge transfer of best practices for processes, use of special methods and tools, conduct of workshops and training, is the focus of the activities of internal or mostly external process consultants.

Process/workflow modeler:

The IT-based collection, modeling and specification of processes, detailed analysis and optimization as well as the implementation in workflow management systems (WFMS) is the task of the process or workflow modeler.

Project manager:

Project managers are recruited from internal or external experts or managers and take over the management of the business process management project, the coordination of project objectives, the achievement of target achievement, the leadership of project employees and the information of management. Ideally, the project manager has

interdisciplinary knowledge matching the respective industry (business administration, computer science, engineering, etc.).

Process Auditor:

The process auditor is responsible for the independent examination of workflows and process change projects. He should be involved as an external or independent internal role.

Process Steering Board:

The Process Steering Board is a group of senior managers from line management (functional view) and process management (process view) to clarify cross-cutting issues. It is usually formed in larger companies and corporations.

Process Controller:

The role of the process controller is dedicated to its own chapter (see Chap. 4). It is usually carried out by the process owner or process manager. For this reason, it is not explicitly included in the representation of Fig. 3.9. The tasks include, in particular, the key figure-based control and control of processes at strategic and operational level.

Assignment of roles in the life cycle:

Process management is teamwork. The previously defined roles work together in different constellations. The main assignment of roles in the life cycle of process management already introduced in Fig. 2.4 is shown in Fig. 3.10. The strategically oriented design of

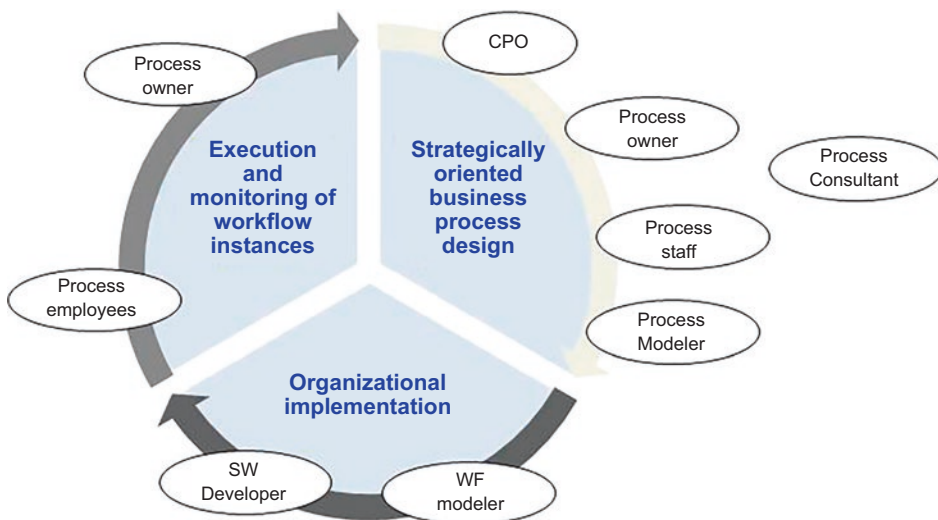


Fig. 3.10 Roles in the life cycle of process management

business processes is based on the process strategy of the Chief Process Officer. Process Owner, Process Employee and Process Modeler (possibly supported by external consultants) analyze the processes, identify weaknesses as well as change the process. The subsequent technical implementation of the processes is the responsibility of the software developers or workflow modelers. The execution of the processes is carried out by the process employees. The monitoring of compliance with the process objectives is the task of the Process Owner.

3.3 Project Organization for Process Management

3.3.1 Classical Forms of Project Organization

Indicators of the need for restructuring measures are falling profits, declining sales, rising inventories of finished products and business performance indicators (see Maurer and Versteegen 2001, p. 27). The measures are carried out in project form. For the implementation of projects with a focus on process management, reference is made to the method collection by Leyendecker and Pötters (2022). A practical example of the organization of a restructuring project is shown in Fig. 3.11.

The members of the steering committee are managing directors, board members, process responsible persons, project managers or external experts (consultants). Their tasks include providing the necessary resources, checking and releasing the project planning, solving cross-project problems and making necessary decisions. The position of the project manager is often occupied by the process responsible person or someone from his reference area. His tasks include planning, controlling and monitoring the project,

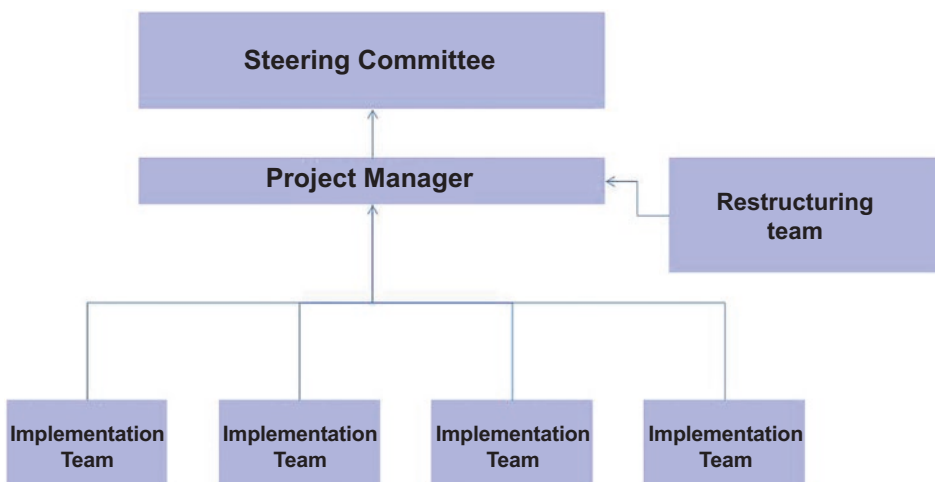


Fig. 3.11 Project organization for restructuring projects (cf. Schmelzer and Sesselmann 2013)

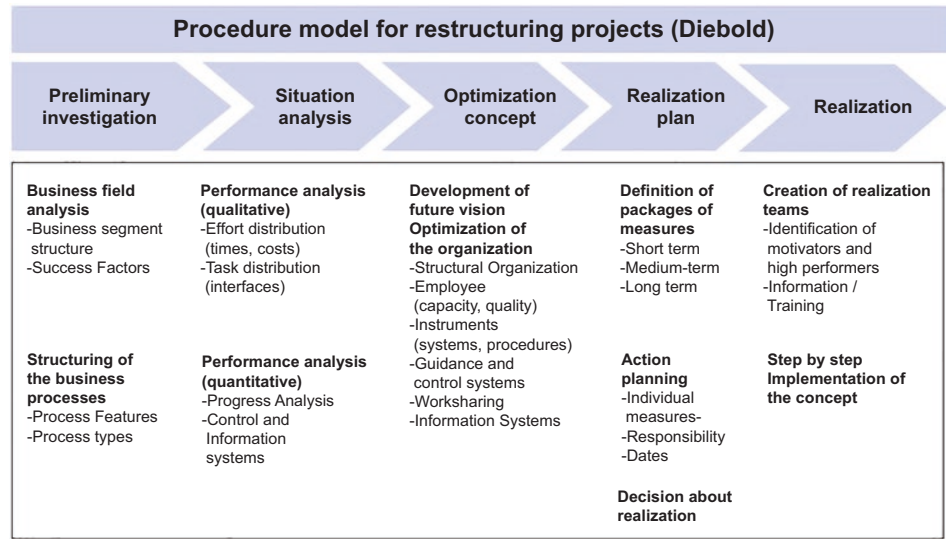


Fig. 3.12 Process model for restructuring projects (Diebold o. J.)

managing the use of resources and reporting to the steering committee. In addition, there are the communication and representation of interests of the project to the outside as well as the motivation of the implementation teams. The restructuring team is the full-time core of the project. It consists of the part-process responsible persons, the leaders of the implementation teams and possibly also external experts (consultants). The tasks of the team are mainly the analysis of the actual processes and the design of the target processes. It is common to divide the overall project into sub-projects for the divisional implementation of the overall concept. The members of the necessary implementation teams are employees from the sub-processes as representatives of the part-process responsible persons, external experts (consultants) and possibly also IT experts. Their tasks include the fine-tuning of the target process design, the implementation of the sub-projects, i.e. the introduction of the target processes (real operation) and the reporting to the restructuring team. The course of the projects takes place in several phases (cf. the proposal of the consulting firm Diebold o. J., p. 19 in Fig. 3.12).

Preliminary investigation

In the first phase, a “preliminary investigation” is carried out, which firstly develops the goals and fixes them together with the decision-makers.

Situation analysis

In the second phase “Situation analysis”, a performance analysis of the company is carried out, taking into account times and costs. In this phase, the involved information systems and information flows are also analyzed.

Optimization concept

The next phase “Optimization concept” serves to develop a vision of the future and the “optimization” or more precisely an “improvement” of the organization with regard to meaningful criteria such as process costs, cycle times, process quality, resource utilization, etc. In particular, a new organizational structure including the required capacity requirements for employees as well as the necessary information, leadership and control systems is designed.

Implementation plan

In the fourth phase “Implementation plan”, the concrete planning of short-, medium- and long-term scheduled individual measures is carried out to form a package of measures and submitted to the decision-makers for approval.

Implementation

The fifth phase “Implementation” completes the project and has the task of implementing the action plan concretely. This phase brings about the critical changes in the company and requires the full concentration of management. The key to the success of the implementation is to identify the relevant performers in the company, to motivate them to support and to prepare all the employees affected sufficiently for the changes.

3.3.2 Agile Methods of Project Organization**3.3.2.1 Software Development as an Initiator of Agile Methods**

Classical well-known concepts of software development, such as the waterfall model (see, for example, Pomberger and Blaschek 1993), assume that software customers know at the beginning what they need and software developers know how to implement these wishes and that no major changes to the planning take place during the course of the project. Unfortunately, reality is often different. Often, however, the software customer only knows during the course of the project what they need and the software developers work out solutions much later and with other methods and tools than planned. In addition, project plans are usually only up to date for a short time, they are quickly overtaken by reality.

Solution approach of agile methods

The basic consideration of agile methods is: If the classical methods do not work, they should not be applied. Instead, a mixed team with experienced members should be given the freedom to solve the problem. Transparency, freedom, daily or timely team coordination and decentralized responsibility replace detailed central planning.

Paradigm shift

The basic idea of all agile methods is a paradigm shift. One moves away from the idea of a “complete planning of a product, process or software” and its implementation (plan-built-try) to an “experimental iteration” in which continuously small, usable artifacts are

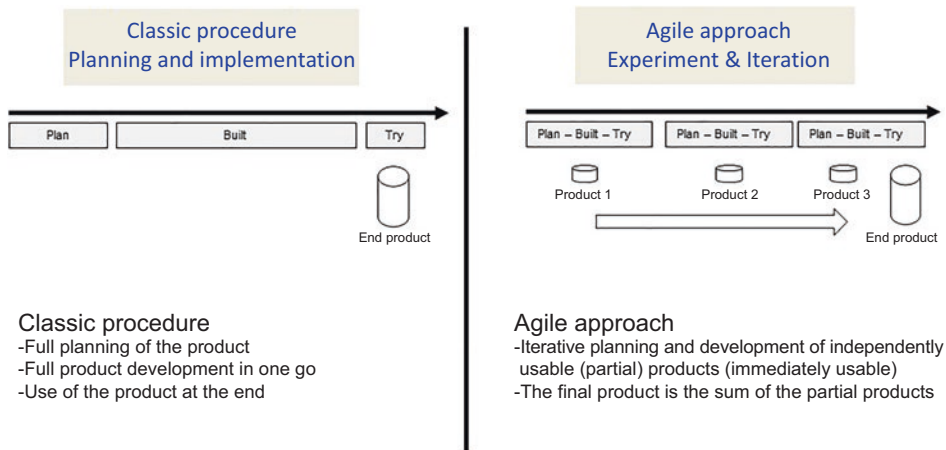


Fig. 3.13 Classical and agile methods in comparison

produced, which in sum then result in the finished end product, the complete process or the complete software (multiple, iterative plan-built-try), (cf. Fig. 3.13).

Agile Manifesto

In an agile manifesto known software development experts have written down revolutionary principles according to which, in their opinion, the weaknesses of classical approaches can be overcome (cf. Agile Manifesto [o. J.](#)). In the concept, a completely different approach to project implementation is proposed.

Occasionally, “agile” is equated with “aimless” in general discussion, but only the approach to planning has been turned around in order to get a handle on the problem of time overruns (Hanschke [2016](#), p. 70). This leads to criticism that “time and schedule compliance” is a higher goal than the quality of the software produced (Mertens and Bissantz [2021](#), p. 4).

In classical waterfall-oriented planning, the contents are specified by the client. The costs of the project are then derived from this by estimation. This results in the time budget of the project, which is often exceeded. In agile planning, the planning is done differently, the budget is given. Given resources (staff deployment), this results in the time budget and ultimately the deliverables (cf. Hanschke [2016](#)). Simply put, the client knows in agile planning what the project will cost and how long it will take. However, the delivered content is open and depends on the team’s performance.

The agile methods became known throughout the SCRUM methodology. SCRUM is a term from rugby. It describes the close togetherness of the team that tries to get the ball after the throw-in.

Transferred to process management, this means that the entire project team must try to achieve the goal of process restructuring. Here, emerging obstacles must be removed and not considered a problem.

SCRUM knows three central roles: the Product Owner, the team and the Scrum Master (cf. Sutherland o. J.). The Product Owner is comparable to the process responsible person, who is the “advocate” of the software product to be developed and responsible for the definition of work packages and their prioritization. The team works on the results of the work packages independently. The Scrum Master is responsible for the correct use of the methods and supports the team in methodological questions.

3.3.2.2 Agile Project Organization in Process Management

The more than 25 years of experience with classical methods of process management have shown that the strong orientation towards waterfall models from software development also has a negative effect on process management. Therefore, it is obvious to transfer the idea of agility to process management.

Classical process management approaches try to completely document, analyze, revise, adapt processes and, if necessary, provide new information systems for this purpose, in turn, to completely plan, design, develop and then test and deliver them. Since insufficient information is available at the beginning of the projects, feedback and delays are inevitable in practice and in the worst case the conditions at “delivery of the new process” are so changed that the revised process is outdated again.

Agile process management

The scope of agile methods such as SCRUM is shown in Fig. 3.14.

The use of SCRUM in process management mainly includes the phases of process documentation, analysis and restructuring. Fig. 3.15 shows an “agile” view of the classical BPM-Life-Cycle already presented in Fig. 2.4 and the associated potential applications.

The roles in the classical SCRUM concept only has to be adapted slightly for use in process management (cf. Fig. 3.16) while the SCRUM process itself can be adopted unchanged. The Product-Owner in standard SCRUM corresponds to the *Process Owner*.

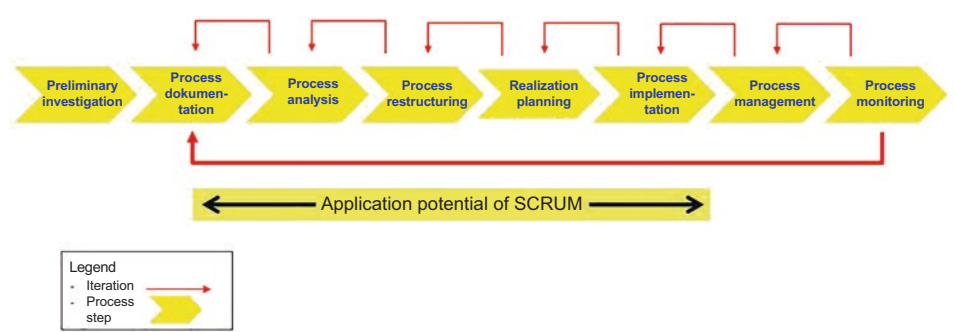


Fig. 3.14 Scope of agile methods in process management

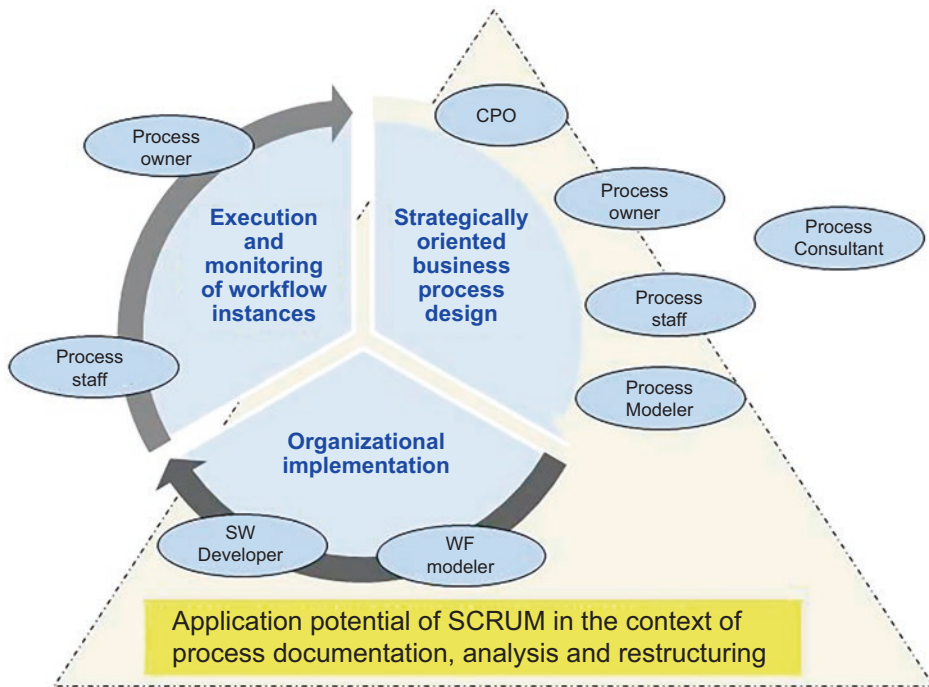


Fig. 3.15 Agile view of the BPM Life-Cycle

The role of the *SCRUM-Masters* remains unchanged. The SCRUM-Team (employees of the department, database experts, software developers and possibly consultants) is adapted slightly to the *BPM-Team* and consists of process employees, process and workflow modelers, software developers and possibly external process consultants.

Hybrid model for agile process management

The life cycle of agile process management is shown in the hybrid model in Fig. 3.17. The process is methodically accompanied by the SCRUM master. The *Process-Owner* creates a summary of prioritized requirements as a user story at the beginning of a cycle and stores it in the *Prozess-Backlog*. The process requirements are recorded in technical language (user story), e.g. flowcharts, EPK models/BPMN models or rough process sketches, depending on the level of detail.

This is followed by a discussion of the next steps by the BPM team and the process owner as part of the *Sprint Planning*. The results achieved by the team are recorded in the sprint backlog. This is followed by the work of the team in the *Sprint*. 2–8 weeks are planned per sprint. In a daily team meeting, the *Daily-Scrum* (Maximum 15 minutes), the team exchanges information briefly. Important topics are:

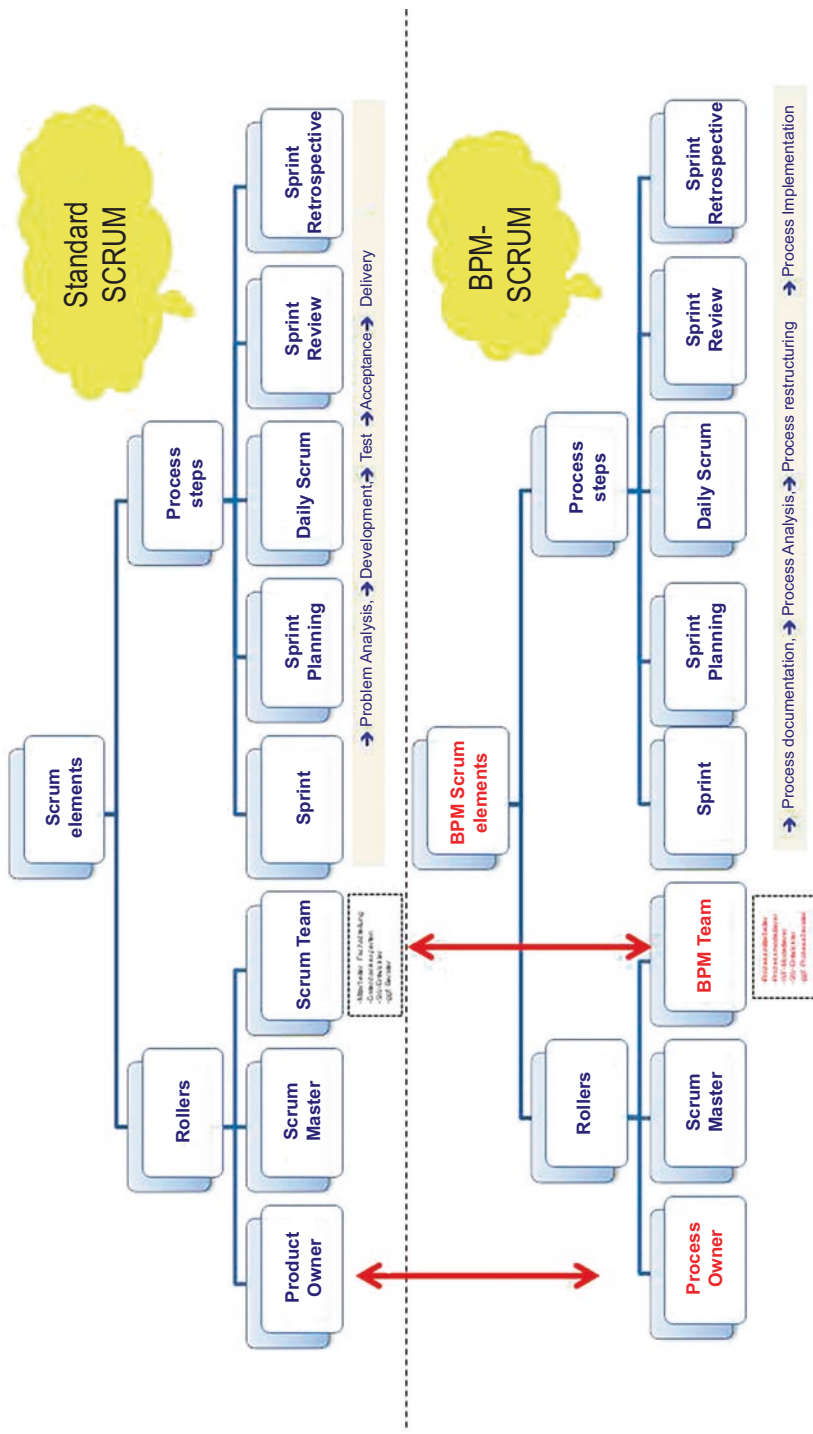


Fig. 3.16 BPM view of SCRUM elements

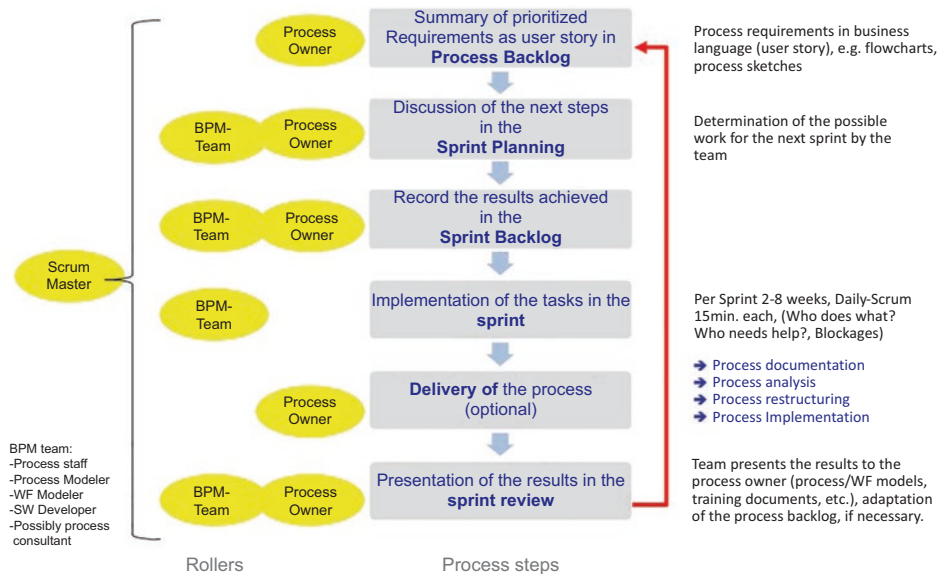


Fig. 3.17 Hybrid model of an agile process management life cycle

- Who is doing what in the team?
- Who needs help?
- Where are there bottlenecks?
- What is the status of process documentation, process analysis, process restructuring and process implementation?

The process owner can decide on a “delivery” of a new or revised process version after a process cycle. The results of the sprint are presented to the process owner in the sprint review. The team presents the process owner with the detailed results of the sprint (e.g. process/WF models, training materials, etc.). Together, they may adjust to the process backlog.

3.4 Review Questions and Exercises

3.4.1 Questions

- Describe the central roles of process management and distinguish them from each other.
- Explain why an influence process organization is not sufficient in the long run, but acceptable in the start phase.

- Why is the “pure” process organization an ideal that often fails in practice?
- Why is the matrix process organization a good compromise that is chosen by many companies for the implementation of process management?
- How do agile methods of process management differ from traditional approaches?
- Describe a procedure for introducing process management in the company.

3.4.2 Exercise Process Organization

Design a process-oriented organization for any company with multiple different products and various departments that operate across products (e.g. sales, marketing, human resources, accounting, production, shipping, information technology, accounting). Consider which disadvantages a pure process organization has in your example, e.g. in terms of cost, resource conflicts, personnel disposition.

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Process control supports the achievement of process management's strategic goals

Abstract

Process management is unthinkable without process control. Process control sets goals, regulates cooperation between project partners via process agreements and monitors compliance with target concepts using key figures. If targets are not met, IT controllers ensure that appropriate countermeasures are proposed. The conclusion is repeated questions and exercises.

4.1 Development of a Process Strategy

Term

Controlling is to be understood as a management concept for future-oriented corporate and profit control, but also as a strategy for existence and job security (Gadatsch and Mayer 2013, p. 1). It is thus a management task which supports the management of the company in its tasks.

Process controlling is a sub-task of controlling with a special focus on business processes. It ensures the achievement of process targets and the quality of process execution. The process controlling regularly analyses planned/actual deviations for this purpose and thus supports continuous business process management (Scheer and Hess 2009, p. 149).

A key starting point for the implementation of this task is a strategic specification by the management. Strategic process controlling is therefore based on a process strategy. It supports the achievement of strategic company objectives, i.e. the company strategy, by planning, implementing and monitoring suitable process-related measures.

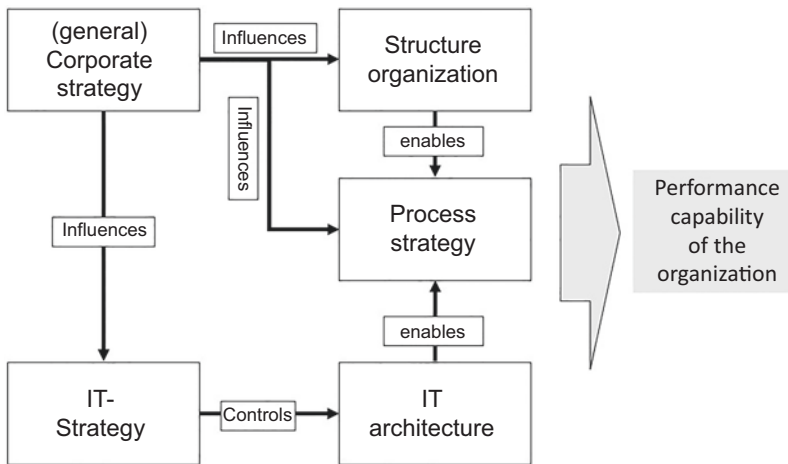


Fig. 4.1 Relationship between process strategy and performance (Krcmar 2005)

Procedure

Various strategies are systematically derived from the general corporate strategy, including the IT strategy and the process strategy, which are both important for the organization's performance (see Fig. 4.1).

A strategy consists of several elements, such as formulating a target state (where do we want to go?), identifying the need for action (what do we need to do? where are the weaknesses?), identifying options for action (what alternatives do we have?), setting goals and defining measures (what should be done concretely? when do the measures need to be completed? who will carry out the measures?), and specifying target values for goal monitoring and the question of goal achievement (when have we achieved the goals?).

Strategic process control

Strategic process control requires a holistic management cycle that starts with the previously defined business strategy and leads to a strategic process planning in which the essential processes are coordinated and described in detail with their goals in the form of performance indicators (cf. Fig. 4.2). The target achievement is monitored as part of the process execution. For this purpose, the process reporting is used as a data source and employed in the strategic process control (analysis of deviations) for process control. Process control acts directly on the executed processes by means of corrective measures, which also have an effect on process control in the form of actual values.

Strategic process control requires a strategy process in a four-stage hierarchy: First, the management formulates an abstract vision for the long-term orientation of the company (cf. Fig. 4.3).

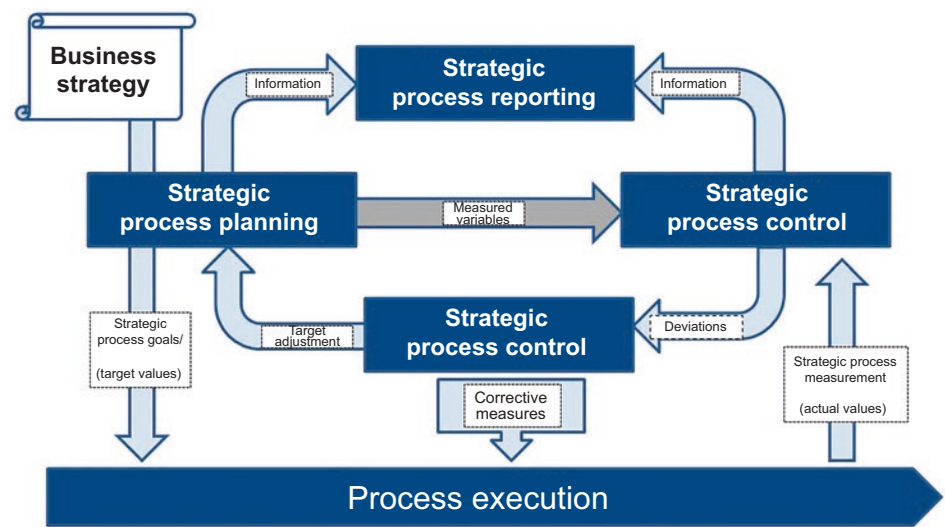


Fig. 4.2 Strategic process control as a management cycle according to Schmelzer and Sesselmann (2013)

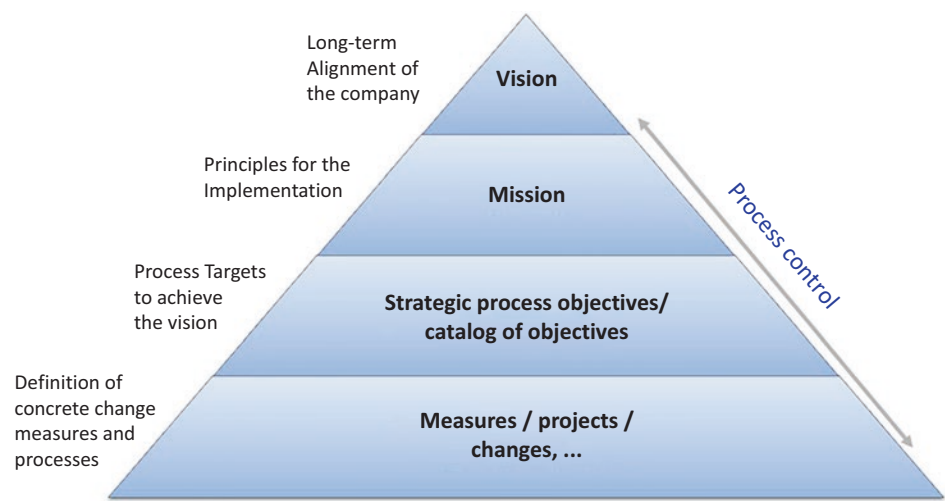


Fig. 4.3 From mission to measure

Example of a vision

For a university, the vision could be: “We will be the leading provider of industry-related digital university products in the region”. This statement is still very general. It needs to be further specified. This is done in the form of a mission, i.e. more specific intentions are formulated. These could be statements such as: “Establishment of a digital

university process management”, “Analysis and digital restructuring of leadership, core and selected support processes”, “Use of modern ICT technologies for student-oriented research and teaching”.

From this, process goals are set, which in turn are described by specific processes or measures to change the situation. A process goal would be that the enrollment of students should take place digitally, barrier-free and within competitively short processing times within the legal framework conditions. A concrete measure to improve barrier-free accessibility would be a project to revise the social media appearances of the university (website, Facebook appearance, etc.).

4.2 Process Scorecard

For the successful implementation of a process strategy, the link to the corporate strategy must be considered. The process strategy must be networked with the company-related elements on several levels, from strategy, via goals and budgets to the individual measures (cf. Fig. 4.4). For this purpose, the instrument of the Balanced-Scorecard was designed.

A process strategy must not only be formulated, but also monitored continuously regarding implementation. For this purpose, traditional key figures are used. The use of

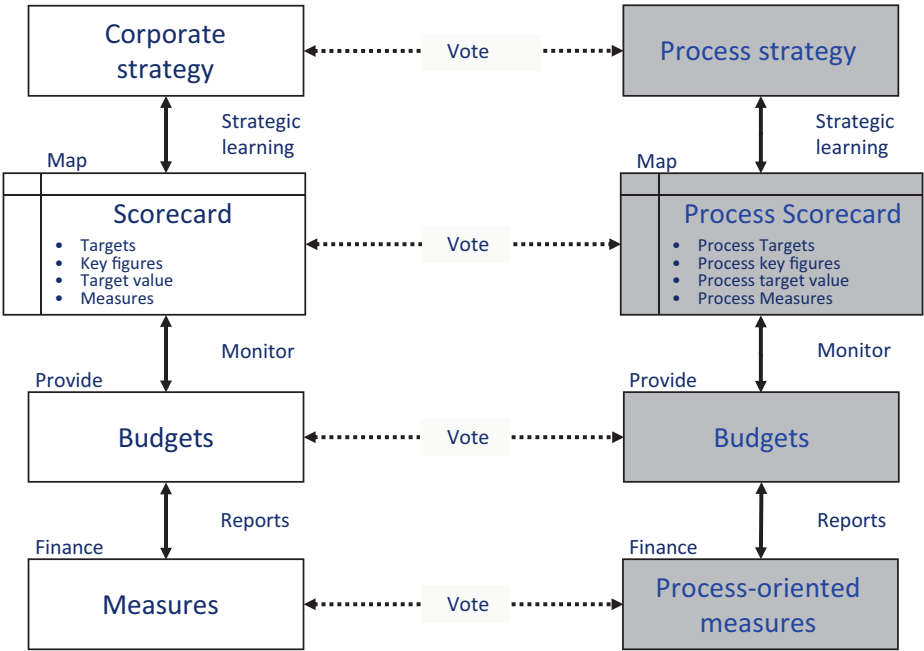


Fig. 4.4 Anchoring process strategy with corporate strategy

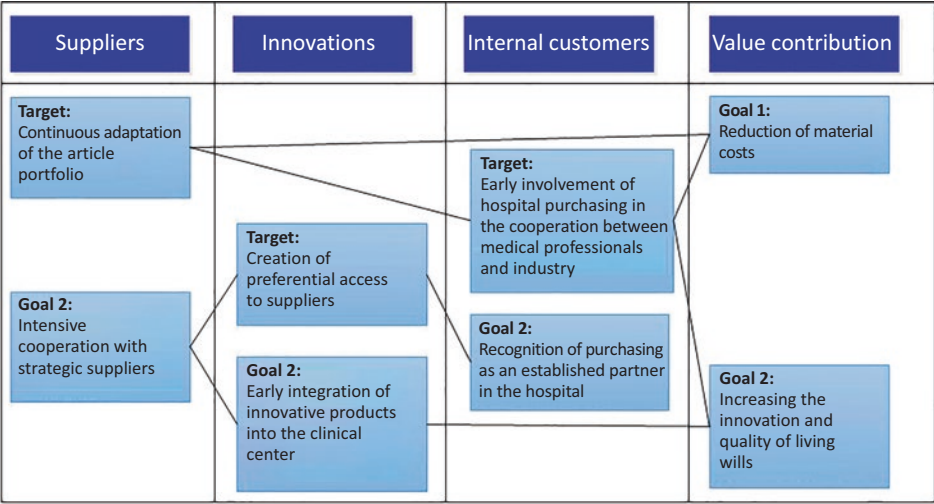


Fig. 4.5 Cause-effect chains in the hospital (cf. Stachel and Eltzholtz 2018, p. 92)

individual key figures has not proven itself because of the risk of misinterpretation and has led to the development of key figure systems that initially mainly covered financial and technical questions. Later, this approach was extended by the concept of the Balanced Scorecard (BSC). At the beginning of the 1990s, R. S. Kaplan and D. P. Norton developed the BSC as a new instrument for corporate controlling. The basis for the BSC were long-term research projects in cooperation with American companies. The Balanced Scorecard links the company strategy and operational planning measures via cause-effect chains in order to create and maintain financial balance (see the example of a hospital in Fig. 4.5).

The process scorecard is a variant of the general balanced scorecard, which was developed as a key figure-based management and control system for general corporate controlling (cf. Schmelzer and Sesselmann 2013). It consists of a mutually coordinated, interdependent target system, which is described by key figures, target values and measures in different views (cf. Schmelzer and Sesselmann 2013). The views, also called perspectives, describe areas of impact of the business processes, which should support the company’s goals in a balanced (“balanced”) manner. Examples of goals of the “process finance” view are process value contribution, process sales and process costs. The goals of the “process customer” perspective are, for example, customer satisfaction, customer loyalty. The “process performance” can be controlled, for example, by the goals process times, process quality, process deadline (cf. Schmelzer and Sesselmann 2013). In Fig. 4.7 a simplified process scorecard for the process “sales of products” is presented (Fig. 4.6).

Customer				Process performance			
Destination	Key figures	Target values	Measures	Destination	Key figures	Target values	Measures
Achieve high customer satisfaction	Sales of the customer to the previous year	+2%	Interview customers Analyze requirements	Performance better than competition	Lead time	< 1 day	Perform process analysis and benchmarking with competitors
	Quantity Customer-complain-the	Share < 1%	Analyze and track customer event complaints		Capacity	> 1000 operations per day	Automate processes
Resources/staff				Finance			
Destination	Key figures	Target values	Measures	Destination	Key figures	Target values	Measures
Personal an for-the-fair out-forms and insert-ready	Number of training days / employees	10 Days Per Year	Update job descriptions Match requirements with training level	Positive Dec-k-contribution	DB % / Sales per Customer	DBk > 30%	Perform product analysis Customer ranking
	Compliance with scheduling agreements	Share > 95%	Create training plan		DB % / each Product	DBp > 30%	Revise calculation

Fig. 4.6 Process scorecard example (sales of products)

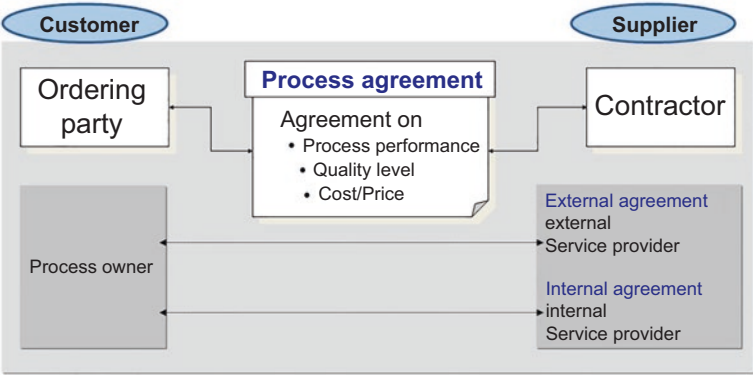


Fig. 4.7 Process agreement—schematic representation

4.3 Process Agreements

For operational control of the processes, business process agreements have established themselves. They mainly document the internal customer-supplier relationships of process management and describe the process performance, quality level and costs in detail, or, in the case of agreements with external partners, the price.

The term process agreement is also mainly known in the IT environment under the English name “Service Level Agreement” (short SLA). The principle of agreement of customer-supplier relationships on the basis of business process agreements is shown in Fig. 4.8.

Each process responsible person regulates with his internal “customers” and “suppliers” the services to be provided (e.g. number of examinations, number of operations, number of transportations) and the quantities required for this. The planning of the service relationships can be done annually as part of the planning and, if necessary, adjusted on a quarterly basis. If an internal cost and performance accounting is used, prices for the internal services have to be added in order to determine process costs based on this. An example of a business process agreement from the healthcare sector is shown in Fig. 4.9. It also demonstrates the possible level of detail of business process agreements. The business process agreement contains information about the process, the service to be provided including the required requirements, the participants and the contact persons. The service is to be documented in such a way that the participants are clear about the contents and the quality level.

The use of process agreements improves clarity about process inputs to be provided and the results of processes. It also makes it easier to coordinate between the parties involved.

4.4 Process KPIs

KPIs (or English Key Performance Indicators) are of high importance for controlling. They serve in the feedback loop of process controlling (cf. Fig. 4.10) as a basis for steering the implementation of strategy. Without KPIs, no controlling is possible.

Procedure

Starting from the corporate strategy, a process strategy is developed. For steering the implementation, KPIs are formed which are to be implemented by measures. The actual values from reality are compared with the target values of the KPIs and result in a deviation analysis. This can lead to various activities. For example, measures one could actively steer the measures that one wants to implement (change of resources, deadlines, goals, etc.), or changes to the process strategy can be made.

Structure of KPIs

There are different types of indicators. They can be structured in absolute and relative indicators (see Fig. 4.11). Absolute indicators refer to countable facts, such as the number of employees used in a process. They are only partially informative because they lack a benchmark. Relative indicators relate multiple indicators to each other and can thus describe relationships between different aspects. They differ again in structure, relationship and index indicators. Structure indicators represent shares of sizes of the same

Klinik-Logo					
Abt.					
Geschäftsprozess-Vereinbarung					
Prozess:	Operative Behandlung des Zervixkarzinoms			Prozess-ID:	OBZ
				Schnittstellen-ID:	OBZ 3-4
Schnittstelle	Teilprozess: Transport Teilprozess:		Lieferant Stationsleitung Pflege Kunde Stationsleitung OP		
Version vom:	10.06.2014	Gültig ab:	20.06.2014	Versions-ID:	OBZ 3-4 20120610
Modifiziert am:	25.09.2014			Modifiziert durch:	Prozessstamm
Review von:	Qualitätszweig	Review am:	15.07.2014	Genehmigt durch:	Direktor
Verteiler:	Prozessstamm, Mitarbeiter der Teilprozesse, Intranet				
Beteiligte Personen des Projektteams:					
	Name:	Funktion:	Abteilung:	Telefon-Nr.:	E-Mail:
Prozess-Owner:	Schmidt	Prozessbegleiter	Management	7234521	schmidt@klinik.net
Teilprozess-Owner, Lieferant:	Müller	Stationsleitung	Onkologie-Gynäkologie	7234560	müller@klinik.net
Teilprozess-Owner, Kunde:	Meier	Stationsleitung	OP-Pflege	7234567	meier@klinik.net
Weitere Permanente Mitglieder:					
1.	Schulz	Assistenzarzt	Onkologie-Gynäkologie	7234568	schulz@klinik.net
2.	Ralder	Krankenpfleger	Onkologie-Gynäkologie	7234569	ralder@klinik.net
Weitere fakultative Mitglieder:					
1.	Bucher	QMB	Qualitätssicherung	7234512	bucher@klinik.net
Leistungsumfang:					
Der Leistungsumfang erstreckt sich auf die Qualität, die Zuverlässigkeit, die geforderte Menge, die Kooperation, Beratungsmöglichkeiten und Flexibilität des Lieferanten					
Anforderungen an den Lieferanten:					
Patient:					
Körperpflege: Nagelreinigung, Nagelpflege/Nagelack entfernen					
Entsorgung von Bauch-, Intimbereich und Oberschenkeln					
OP-Hemd, Einmalstip					
Blasenentleerung, abführende Maßnahmen					
Schmuck- und Prothesenentfernung					
Patientenarmband mit Namen					
...					
Labor:					
Blutbild: Hb/Hk, Thrombozyten, Blutgruppe, BSG					
Leberstatus: SGOT, SGPT, Gamma-GT, alkalische Phosphatase, Transaminasen					
Gerinnungsstatus: Quick, PTT					
Elektrolytstatus: Kalium					
...					
Untersuchungen:					
Röntgen-Thorax in zwei Ebenen					
EKG					
Sonographie und intervenziöse Pyelographie					
Zystoskopie, Rektoskopie					
Anforderungen an den Kunden:					
Einhaltung des Übergabetermins an definierter OP-Schleuse					
Falls der Patient telefonisch abgerufen wird, ist der zeitliche Vorlauf von 45 Minuten zu berücksichtigen					
Abnahme der geleisteten Leistung durch autorisiertes Fachpersonal					
Nichterfüllung der Anforderungen:					
Ausrichtung der nächsten Sitzung					
Nächste Sitzung am:	25.11.2016				
Muster, den 25.09.2016					
Ort, Datum	Prozess-Owner				
	Teilprozess-Owner, Kunde		Teilprozess-Owner, Lieferant		

Fig. 4.8 Example of a business process agreement from the healthcare sector (Kölking 2007)

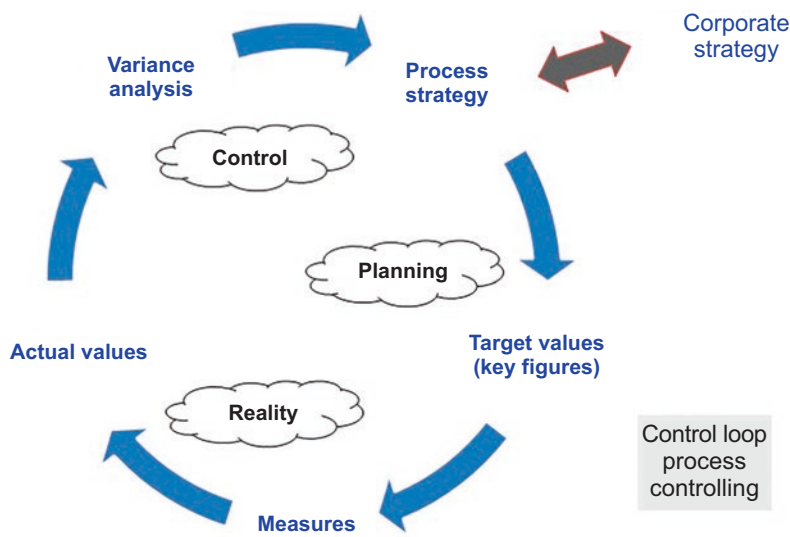


Fig. 4.9 Regulation cycle process controlling

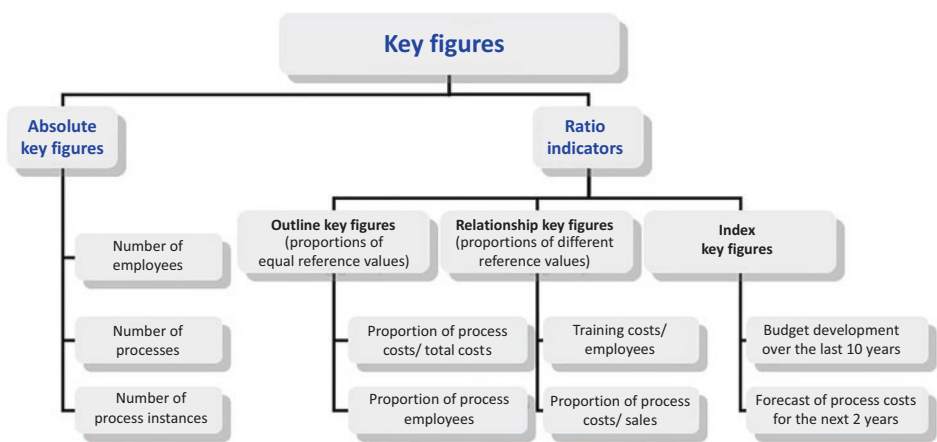


Fig. 4.10 KPI structure

dimension, e.g. the proportion of process costs to the total costs of the company. Relationship indicators relate sizes of different dimensions, e.g. process costs per employee. Indexes are normalized developments of indicators over longer periods of time, e.g. development of process costs for procurement of office supplies.

In the practice of process controlling it is important to assess indicators according to their quality, calculability and analyzability, economic efficiency and the possibility of

Quality	Predictability and analyzability	Economic efficiency	Organization
•What should be controlled with the key figure? •Does the metric measure the right effect? •What can be actively controlled with the key figure? •Are the key figures understandable for the recipient? •How is the quality of the basic data to be assessed (are preparations / corrections necessary)? •Does the metric measure objectives relevant to the IT strategy?a	• Can target and desired values or expected values be defined? • Can corresponding actual values be determined? Can tolerance values be defined? - What must happen in the event of an overrun/ underrun? - Who needs to take action and how? • Are the key figures "benchmarkable"? • How sensitive are the key figures to changes? • Can the necessary basic data be determined? • Are the metrics drill-down capable?	•Is the effort required to determine basic data for target/actual determination economically justified? •Is the effort required to determine the key figure offset by an appropriate benefit from the possibility of active taxation? •Can pragmatic surrogates be identified?	•Can responsible parties be named for data provision, calculation, reporting and for the content of the indicator itself? •Are the key figures tamper-proof? (are there control variables?) •How do metrics respond to organizational or technological change?

Fig. 4.11 Criteria for indicators. (After Kütz 2011, modified)

implementing them organizationally. For this purpose, possible candidates for indicators must be critically questioned. The checklist for indicators in Fig. 4.12 (created according to Kütz 2011) provides assistance. The results of the examination should be considered in the decision-making process for the selection of indicators.

Documentation

Indicators must be described precisely to allow for a meaningful use in the company. Otherwise, there is a risk of misinterpretation. For example, the indicator “order processing time” can be measured according to the interpretation by the participants from different areas as follows:

- Time span from the receipt of the order to the recording in the SAP system (IT view),
- Time span from the receipt of the order to the dispatch of the order confirmation to the customer (sales view),
- Time span from the receipt of the order to the dispatch of the goods to the customer (overall process view).

It has to be clarified how the indicator is to be calculated and interpreted. The description includes numerous aspects. These include, among other things, a meaningful professional description, the indication of the validity of the indicator, the persons responsible for the content, the target groups for the reporting, the reporters and the data suppliers. Further information relates to the indicator category (e.g. finance, production, IT), the desired target values (e.g. 90% order entry on the day of the order receipt), possible tolerance values for deviations and the associated escalation rules in case of outliers (e.g.

Process Name:	Enter sales order		
Process owner	Process Manager Sales		
Process Indicator:	Lead time		
Person responsible for the key figure:	N. N.		
Recipient of the key figure (reports):	Director Sales, Process Manager Sales, Sales Manager		
Objective of the key figure			
What should be controlled with the key figure?	Time from receipt of the order to handover to the shipping department for delivery		
Which target values are to be achieved?	Minimum processing times depending on the type of goods (10 min. to 60 min.)		
What happens if the tolerances are violated?	Information to Director Sales		
Determination of the key figure			
Calculation / Formula	Timestamp "Order entered" minus "Order received" (events BPMN model)		
Tolerance values	+/-20%		
Data sources	SAP document database, order type "O"		
Publication / Reporting			
Deployment frequency	Director Sales: monthly, by customer group	Process Manager, Sales Manager	
Level of detail / aggregation	Country, region, customer group	Customer group, customer	
Information Technology			
Source System(s): SAP Document Database (ERP)	Receiver system(s): SAP BI (Data Warehouse)	Reporting tool(s): Power BI	

Fig. 4.12 Process indicator profile—example

up to 90% target achievement information to the process owner, below 80% information to the management).

Of particular importance is the exact specification of the data collection to obtain uniform consistent indicators. Documentation of the data sources includes

- the data-providing IT system,
- the measuring method (manually if necessary, automatically at regular intervals, indirectly by derivation of complementary values), as well as
- the exact measuring points.

The following must be observed for the use of the indicators:

- Type of representation (text, graphics, numbers, ...),
- Periodicity of data provision (if applicable, hourly, daily, weekly, ...),
- The aggregation levels of the values (individual data, sum data per department, sum data per day/month/year) and
- The type and duration of archiving (location, medium, duration).

Optional information to be documented are complaints (e.g. wrong or contradictory values), feedback on the use of key figures and changes in the calculation logic in order to be able to trace the values retrospectively.

Process performance indicator profile

Process performance indicators should be documented consistently and centrally in the form of a process performance indicator profile in order to provide general information and ensure consistency and assignment of responsibilities. The profile serves the purpose of exact definition and description of a performance indicator for process management.

The following information can be recorded in the profile:

- Objectives and scope of the performance indicator
- Responsibilities and organizational questions
- Target values and threshold values
- Documentation of data sources and information distribution (reporting) including access rights
- Notes on the use of IT tools

Storage should take place in a central database. The profiles are to be published visibly in the intranet for process participants (CPO, process owner, process experts). An example of a process performance indicator profile for the performance indicator “cycle time” of the process “enter customer order” can be taken from Fig. 4.12.

Significance of individual performance indicators

The significance of isolated performance indicators must be critically questioned. In a cartoon with reference to controlling, two managers refer to a water glass as half full or half empty. The third opinion of the controller is: “For me it is twice as big as necessary”, i.e. it depends on the perspective when using performance indicators (Verlag für Controlling-Wissen 2006, p. 162). The performance indicator “IT costs in relation to sales” (cf. e.g. Gadatsch et al. 2013), which is frequently used in practice, can be used to represent the lack of significance of individual performance indicators.

Example

An example of misinterpretations based on the key figure “IT costs/sales” is documented by Kütz (2003, pp. 20–21). A comparison of two companies led to the interim result that the cost shares at company A amounted to 0.8% and at company B to 1.2%. From this followed a proposal for decision: Company B should take over the IT systems of A in order to reduce its IT costs. The further detailed analysis showed that company A operates an outdated IT architecture that has not been maintained for years. The IT costs consisted essentially of maintenance costs for the old systems. Company B, on the other hand, has a modern and much more powerful IT architecture. The takeover of the IT systems was then abandoned. ◀

Key figure systems

Individual key figures are, as described, only limitedly informative. Therefore, key figure systems have been developed which establish interdependencies between individual key figures and should provide the analyst with an overall view. A key figure system puts individual key figures into a logical context. General requirements for such key figure systems are:

- Objectivity and consistency, i.e. a suitable structure of the key figures supports consistent statements.
- Simplicity and clarity, i.e. the simple structure supports the dissemination and use in the company.
- Information compression, i.e. the key figures should be structured in management levels and allow top-down or bottom-up analysis. The individual values of the subordinate key figure values add up to the sum value of the next level.
- Multi-causal analysis, i.e. higher-level key figures can be split into different views at lower levels. The IT costs of the company are explained by different cost categories and quantities of the subordinate levels (projects, measures) (Gladen 2008, pp. 92–93).

Assignment of indicators to main and sub-processes

It makes sense for process controlling to align indicators with the objectives and requirements of the process and to define them. The indicators allow the process responsible

person to specifically control the process. This is made possible by the assignment of indicators to main and sub-processes or process steps.

An assignment of indicators to sub- and main processes is shown in Fig. 4.14 using Daxböck's "Order-to-Cash Process" as an example (Daxböck 2014, p. 60). The indicator "On Time In Full (OTIF)" controls the process steps "Availability Check", "Order Confirmation", "Create Delivery" and "Disposition and Transport" while the indicator "Order-to-Cash-Total Cost" affects the entire process (Fig. 4.13).

Example calculation of process indicators

Below is a calculation example that picks up some process indicators by Schmelzer and Sesselmann (2013). The basic data are summarized in Table 4.1, the formulas for the indicators can be found in Fig. 4.15.

The average process time for the three completed orders is 33 days. It is calculated from the total number of days difference (99 days) divided by the number of completed three orders (see Fig. 4.16).

The average punctuality of the orders is 66% ($2/3 * 100$), because two orders are completed without deviation and already completed. Three orders are completed in total (see Fig. 4.17).

The average process quality of the orders is 33% ($1/3 * 100$), because one order is without rework and three orders are finished (cf. Fig. 4.18).

4.5 Process Costing

The business analysis of processes requires the inclusion of cost information. As an instrument for this purpose, process costing is proposed (cf. Hirschmann and Scheer 1994, p. 189). Process costing was developed in the early 1990s as an addition to the traditional costing methods of job order costing, process costing, and activity-based costing to be able to assess processes. The traditional methods of costing allocate costs of indirect business areas that cannot be directly assigned to services. These are, for example, general and administrative costs incurred in the commercial and administrative area, which are charged to the company's services using flat-rate surcharges. In contrast, process costing tries to find allocation bases even for the costs of indirect areas, so that a differentiated allocation of the costs can take place.

In contrast to the traditional approaches to costing, process costing offers the possibility of assessing processes (cf., for example, Hirschmann and Scheer 1994, p. 190). It provides allocation rates for the assessment of the services provided by the processes. Process models are a good basis for process cost calculation (cf. Berkau and Flotow 1995, p. 203). Process costing receives allocation rates on the basis of process models, which contain the time- and quantity-related information for the assessment of the processes.

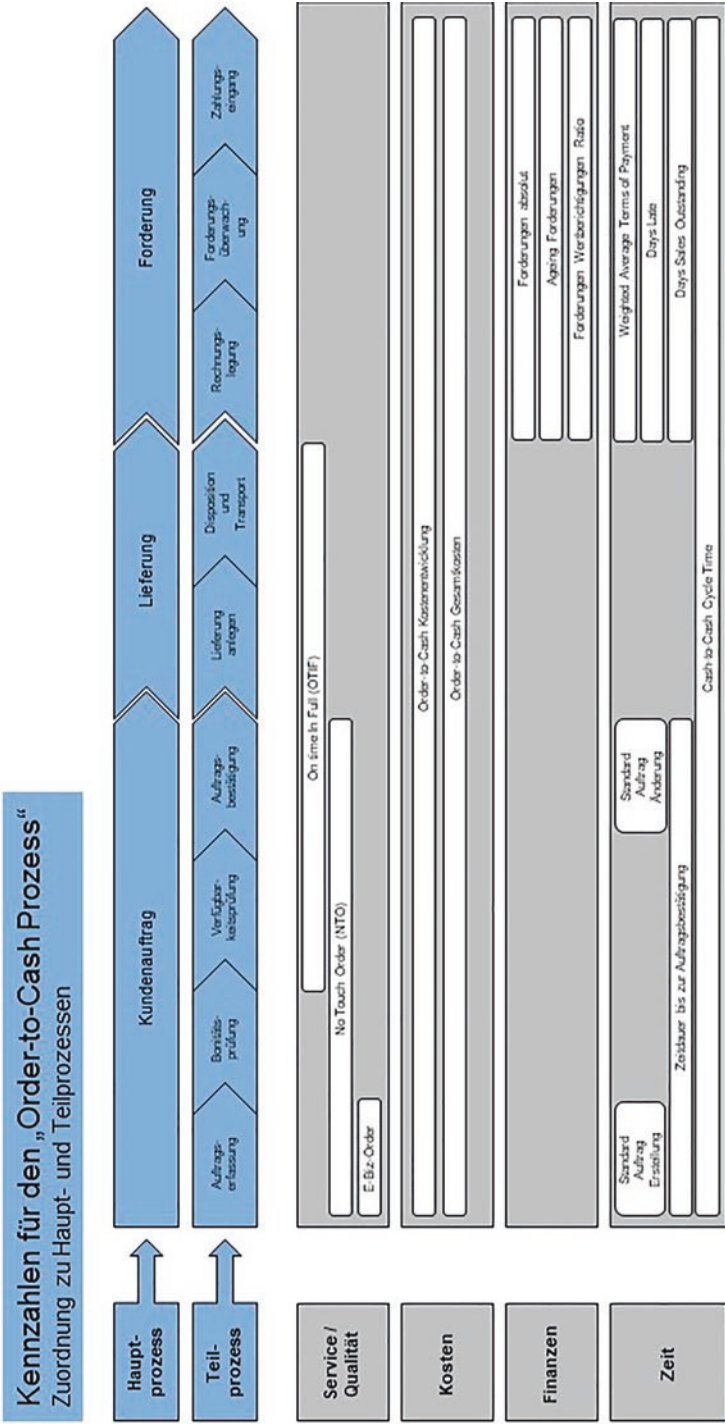


Fig. 4.13 Process indicators for an Order-to-Cash process (Daxböck 2014, p. 60, modified)

Average process time or cycle time (DLZ): (time required to process orders) in days

$$DLZ = \frac{\sum \text{End date} - \text{start date per completed order}}{\sum \text{Number of orders completed}}$$

On-time delivery (TT): Percentage of completed orders without schedule deviation

$$TT (\%) = \frac{\sum \text{Number of completed orders without deadline deviation}}{\sum \text{Number of completed orders}} * 100$$

Process quality = Percentage of completed orders without rework

$$\text{Process quality (\%)} = \frac{\sum \text{Number of completed orders without rework}}{\sum \text{Number of completed orders} * 100}$$

Fig. 4.14 Process indicators (formulas)

Order data					Analysis date 15.06.2020		
Order no.	Customer	Start date	Plandau	Planned endter	Actual end date	Rework Y/N	Deadline deviation Y/N
100	Berger	01.04.2020	40	11.05.2020	20.05.2020	J	J
200	Müller	01.05.2020	30	31.05.2020		N	J
300	Schmitz	01.02.2020	10	11.02.2020	11.02.2020	J	N
400	Zeppelin	01.03.2020	40	10.04.2020	10.04.2020	N	N
500	Meier	01.06.2020	30	01.07.2020		N	J
5					3	3	2
(total)					(completed)	(without NA)	(without dev.)

Fig. 4.15 Determination of process time. (After Schmelzer and Sesselmann 2013)

Table 4.1 Order Data (Date of analysis: 15.06.2016)

Auftrags-Nr	Kunde	Start-Termin	Plandauer	Geplanter Endtermin	Ist-End-Termin	Nacharbeit (Ja/Nein)
100	Berger	01.04.16	40	11.05.16	20.05.16	Yes
200	Müller	01.05.16	30	31.05.16	Open	No
300	Schmitz	01.02.16	10	11.02.16	11.02.16	Yes
400	Zeppelin	01.03.16	40	10.04.16	10.04.16	No
500	Meiner	01.06.16	30	01.07.16	Offen	Nein

Order data					Analysis date 15.06.2020		
Order no.	Customer	Start date	Plandau	Planned endter	Actual end date	Rework Y/N	Date deviation Y/N
100	Berger	01.04.2020	40	11.05.2020	20.05.2020	J	J
200	Müller	01.05.2020	30	31.05.2020		N	J
300	Schmitz	01.02.2020	10	11.02.2020	11.02.2020	J	N
400	Zeppelin	01.03.2020	40	10.04.2020	10.04.2020	N	N
500	Meier	01.06.2020	30	01.07.2020		N	J
5					3	3	2
(total)					(completed)	(without NA)	(without deviation)

Fig. 4.16 Determination of punctuality (according to Schmelzer and Sesselmann 2013)

Order data					Analysis date 15.06.2020		
Order no.	Customer	Start date	Plandau	Planned endter	Actual end date	Rework Y/N	Date deviation Y/N
100	Berger	01.04.2020	40	11.05.2020	20.05.2020	J	J
200	Müller	01.05.2020	30	31.05.2020		N	J
300	Schmitz	01.02.2020	10	11.02.2020	11.02.2020	J	N
400	Zeppelin	01.03.2020	40	10.04.2020	10.04.2020	N	N
500	Meier	01.06.2020	30	01.07.2020		N	J
5					3	3	2
(total)					(completed)	(without NA)	(without dev.)

Fig. 4.17 Determination of process quality (according to Schmelzer and Sesselmann 2013)

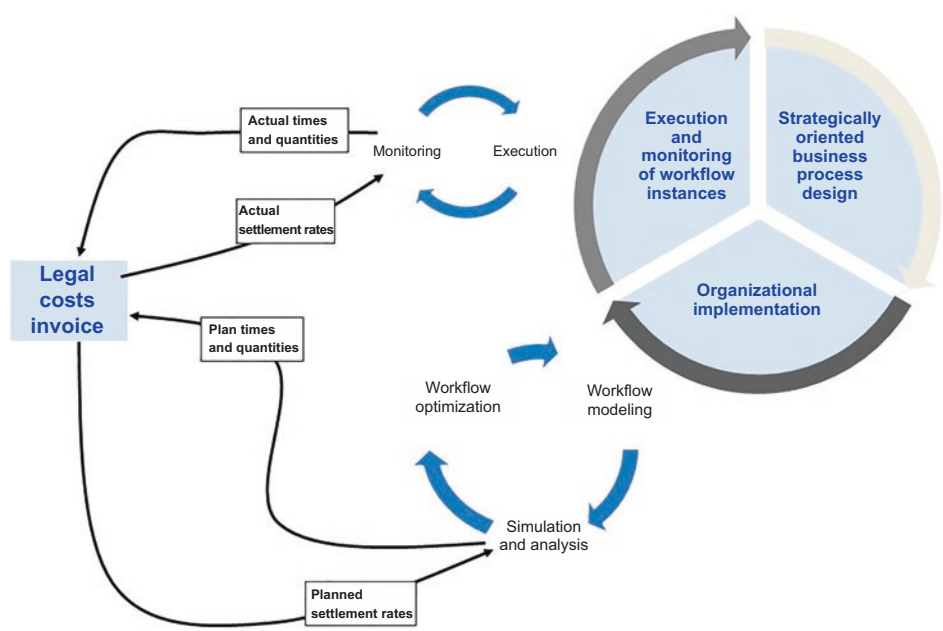


Fig. 4.18 Process costing

Calculation of process costs

Process cost calculation is based on the principle of absorption costing. For this purpose, the process cost rates from process costing are multiplied by the amount of utilization. The process cost rates can be divided into cost-specific components, such as personnel costs, energy costs, depreciation, interest and costs for the use of information technology. The determination of the quantitative utilization of resources by the process requires suitable reference values per process step.

Examples from procurement are “number of reminders” or “number of processed orders” (cf. Scheer 1998, p. 67). The cumulative process costs of the workflow steps result in the process costs of the workflow level, which can then be condensed to the business process level.

For process modeling, the above considerations result in the requirement that the modeling concept must be able to assign process cost rates to the process steps in the desired level of detail, i.e. the data model must be extended by attributes for modeling process cost rates.

4.6 Review Questions and Exercises

4.6.1 Questions

1. Explain possible structuring criteria for key figures.
2. What requirements must “good” process key figures meet?
3. Explain the concept of the “(Balanced-)Process-Scorecard” and the possible uses for process controlling.
4. What disadvantages does a control system such as the Process-Scorecard have?
5. Explain the use of process agreements in the context of process controlling.
6. How can you recognize a “good” key figure for process control?
7. Name some key figures for assessing the quality and efficiency of processes.

4.6.2 Exercises

4.6.2.1 Exercise Process Scorecard

Choose a main process for a fictitious trading company and identify suitable views for the process scorecard. Develop goals, key figures, target values and measures to achieve the main process.

4.6.2.2 Exercise Process Agreement

Create a process agreement from the perspective of the student service for enrolling students at a university. Applicants must first register online. They then upload copies of certificates, resumes and birth certificates. After being invited to enroll, applicants must

submit the originals of the documents. If the registration is successful, students receive a student ID and a ticket. According to the university's specifications, the process should be completed no later than one month after registration. Incorrect registrations, such as the admission of unqualified applicants, should not occur.

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Models simplify everyday work through abstraction.

Abstract

Models serve to simplify situations in reality. They abstract from unnecessary details and provide those involved in process management with a tool for documenting, analyzing and improving processes. The contribution presents basic questions of modeling and analyzing processes and then goes into modeling methods frequently used in practice: process map, process profile, tabular notation, swimlane diagrams, extended event-driven process chain (eEPK) as well as the Business Process Model and Notation. Finally, aspects of proper modeling are discussed and the presented methods are compared. Review questions and exercises ensure learning success.

5.1 Basic Questions of Modeling

5.1.1 Overview of Selected Modeling Concepts

Models simplify the view of complex reality. As an example, the planning of a train journey without local knowledge should be mentioned, for which i. d. R. Help would probably be required. For example, as a traveler, it would be possible to ask passers-by for directions or to use a timetable.

A timetable is a simplified model of reality that focuses on the goal of enabling interested users to navigate the traffic system. In Fig. 5.1 a timetable excerpt of the Cologne Transport Companies is shown, with which one can plan and carry out a train journey

Model



Fig. 5.1 Model of a train journey. (Image Source: Kölner Verkehrsbetriebe (KVB)(Ed.), City of Cologne)

in the city area. This “model” facilitates navigation by focusing on the essential aspects. These would be, in this context, for example, the two questions:

- How do I get from “A” to “B”?
- Which train do I have to take?

The symbols of the “timetable” model are standardized so that any user of different age groups can work with it without too much prior knowledge.

Business processes are—as already mentioned—often very complex and divided into work steps. In the past decades, therefore, various methods have been developed for the systematic representation of processes in order to reduce complexity.

As part of Business Reengineering and business process optimization, an analysis of the current and desired business processes, as well as their design and documentation, is carried out. For this purpose, business process models are created, which formally describe the business processes. Workflow models are derived from business process models by refinement. They serve the detailed specification of the business processes with the aim of execution by a workflow management system (WFMS).

Formal methods

Formal methods can be used to model processes. These are divided into script-based methods (script languages) and graphical methods (diagram languages).

Script languages allow the description of process models using a formal notation like programming languages. This makes it possible to achieve a very high level of precision in the model specification. However, the clarity of the process scripts is low and their interpretation requires detailed method knowledge, which makes their use in practice more difficult.

The diagram languages, which are much more intuitive than the script languages, can be divided into data flow, control flow and object-oriented approaches (cf. Fig. 5.2). In comparison to the script languages, they have established themselves more strongly in practice, especially where models are to be created in cooperation with application experts (e.g. employees from sales, accounting, production).

Data-oriented methods

Data flow-oriented methods do not describe the process itself, but the data flow, that is, the course of the data in the interaction of the individual activities. The process results only indirectly from the representations, whereby the sequence of process steps is difficult to read out of the diagrams. The importance of data flow-oriented methods has therefore decreased significantly in recent years. However, it is still necessary to take data aspects into account appropriately in process models. Data flow-oriented methods are, however, less expressive in terms of process management, as the process is not in the foreground of the modeling work (Meyer et al. 2011, p. 5).

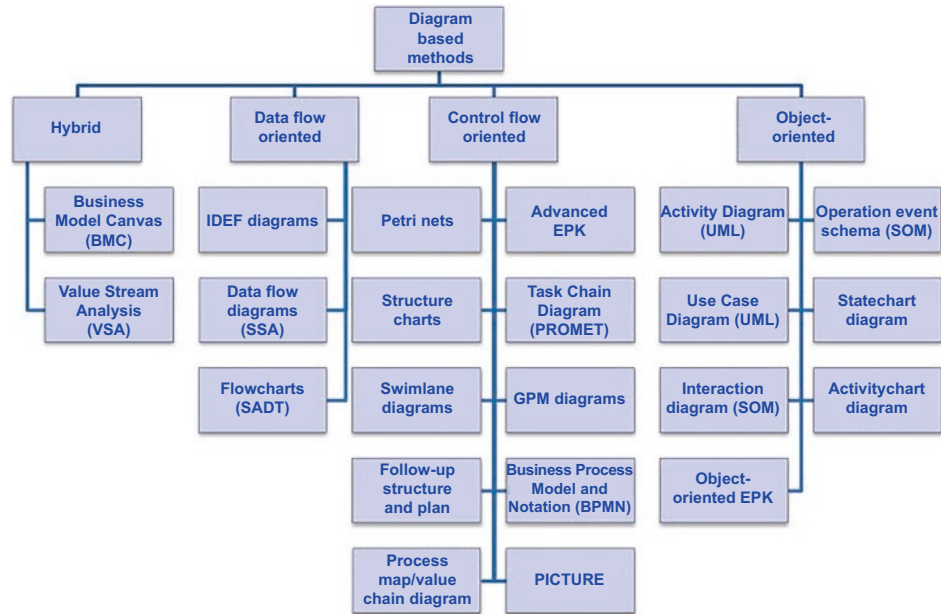


Fig. 5.2 Overview of selected modeling methods

Control flow-oriented methods

In control flow-oriented methods the sequence of activities is in the foreground, that is, the process modeling. In practice, process maps, swimlane diagrams, value chain diagrams (WKD), the extended event-driven process chain (eEPK), as well as the Business Process Modeling and Notation method (BPMN) have established themselves.

Object-oriented methods

The idea of integrating functions and data into so-called objects originates from software development. This has led to the development of object-oriented methods of modeling. In practice, the Unified Modeling Language (UML) with the Activity Diagram has established itself.

Hybrid methods

The hybrid methods include value stream analysis(WSA) and the Business Model Canvas(BMC). Value stream analysis is used in particular in the production and manufacturing sector to analyze processes. It is designed to identify waste resulting from unnecessary overproduction, transport routes, waiting times, overstocks, errors, and resulting rework (cf. Wagner and Lindner 2022, p. 4 et seq.). It is particularly suitable for analyzing processes in mass production, as it was developed for this purpose (cf. Wagner and Lindner 2022, p. 135). Due to this restriction in the application focus, the method is not further discussed in this book.

The still relatively young Business Model Canvas (cf. Osterwalder and Pigneur 2010) describes the central characteristics of the company business model above the process level. It is therefore very suitable as an introduction to process management, because the processes of a company are derived from the business model.

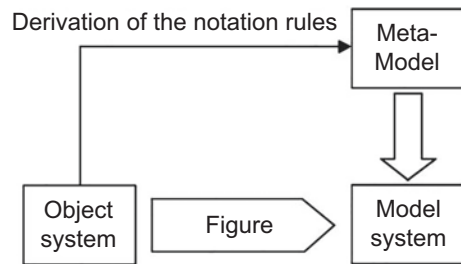
5.1.2 Terminology and Metamodel as Construction Features of Modeling Languages

Concept system

The task of a concept system is to delimit and categorize modeling-relevant phenomena and to name them by concepts (cf. Gehring 1998). Examples are the naming of information, activities, process relationships or assignment relationships. They are reflected in the notation of the modeling method and thus in the possibilities of expression. Business process models usually represent the following aspects (cf. Kurbel et al. 1997):

- **Process steps**, the activities required to create process outputs. Synonymous terms of an individual process step are process, task, function and work step.
- **Objects** are processed in process steps and exchanged between process steps. Examples are orders, complaints or offers. Objects are represented by information carriers of different presentation forms, such as e-mail, fax, document, etc. The forwarding of

Fig. 5.3 Metamodeling (see Gehring 1998)



objects is called object flow. Synonymous terms are information flow, data flow and document flow.

- **Dependencies** between process steps that are temporally, logically or technologically conditioned define the flow logic of a business process. Analogous terms are, for example, control flow and control flow.
- **Task carriers** carry out activities in process steps. Task carriers are, for example, processors, machines or programs. Alternative terms are department, organizational unit, function carrier, etc.

Meta-Model

Models are used to analyze and design real systems. They map an original or object system into a model system. Since a model should reflect the structure and behavior of an object system as faithfully as possible, special requirements must be placed on the mapping. The possibility of formal description of model systems makes it possible to introduce the higher-level modeling level of meta-modeling (see Gehring 1998) (see Fig. 5.3). A Meta-Model represents a whole class of model systems; each class element represents an instance of the meta-model here. Furthermore, notation rules are given for the creation of the model system. This allows the model system to be checked for completeness and consistency with the object system.

5.1.3 Process Modeling in Practice

Many companies use complex, historically grown and inadequately or not at all documented software systems. Cumbersome workflows and inefficient organizations force them to reorganize business processes and to develop or exchange software. The introduction of standard software for cost reduction can only lead to a rationalization success in connection with an analysis and redesign of the workflows. Especially larger organizations are therefore considering the establishment of an enterprise process model.

Companies

- Capture and documentation of business processes
- Weak points analysis of the overall organization

- Requirements definition of new information systems
- Selection and implementation of standard software
- Construction of a company process model

The customers of standard software providers need information about their functional scope when selecting products. Process models can be considered as an additional product component of the software and offer the customer an additional benefit in the context of carrying out use cases. These can be simplified by the data and process descriptions. Later, during use, the models can be used as a reference work.

Standard software providers

- Data and process models as product description
- Support of use cases at the customer
- Basis for individual further development (modifications)
- Comparison basis in the software selection process
- Training aid and reference work for the user

For consultants, the support of the customer in the reorganization of his workflows and structures is in the foreground. Another focus is the introduction support in the implementation of standard software or workflow management systems. In many cases, there is also the need to compensate for missing know-how at the customer.

Consulting companies

- Introduction of IT systems at customers
- Conducting vulnerability analyses
- Supporting consulting in organizational projects
- Conducting business reengineering projects

5.1.4 Case Study “Family Doctor’s Practice”

The following case study on the takeover of a family doctor’s practice forms the basis for continuous modeling examples. Although all information is designed to be realistic, it is not based on any specific case.

Basic data

A doctor plans to take over a general practice from a predecessor who has gone into retirement. He is considering what economic foundations he needs to consider here in order to generate a sustainable income (business model) and how the work in the practice is to take place later (business processes). For this purpose, he would like to make use of modern methods of business process management. The doctor’s practice is to treat patients with statutory health insurance, private health insurance and from abroad.

In addition to the usual “on-site services”, innovative online services are also to be offered, e.g. appointment scheduling, downloads, prescription orders or even an “online consultation hour”.

First analysis

A first analysis by the doctor resulted in a structured list of processes as well as first details of the two processes “Schedule appointment” and “Change bandage”. More processes must be collected.

- Control processes (serve the practice control)
 - Plan practice procedures,
 - Control personnel deployment,
 - Control finances
- Core processes (serve the visible performance for patients)
 - Perform treatments: schedule appointments, examination, treatment, documentation
 - Perform preventive examinations: schedule appointments, data entry, preventive examination, documentation
 - Administer vaccinations: online appointment scheduling, counseling, vaccination, issue vaccination passport
- Support processes (serve general support)
 - Procure materials and services,
 - Examine laboratory samples,
 - Provide IT support,
 - Clean and maintain the practice,
 - Book receipts, determine and pay salaries,
 - Write doctor’s letters.

Detail process “appointment”

The process “appointment” takes place on site in the practice or by telephone. It serves the preparation of examinations and treatments in the practice. The process can be described as follows.

Process goal: Appointment is made with patient.

Process steps (responsible persons in brackets)

- Greeting (assistant reception or telephone),
- clarify concerns (assistant),
- clarify resources / free times (assistant),
- appointment (assistant with patient)
- Patient farewell (assistant)

Result: Appointment is made/or termination of the process.

Participants (roles): Patient, assistant.

Detail process “change bandage”

The process “treatment” occurs in many variants, for example, appointment treatment, emergency treatment or also routine procedures such as “change bandage in the doctor’s office”. The process can be described as follows.

Process goal: New bandage, which is correctly positioned and protects the patient’s wound.

Process steps (responsible persons in brackets)

- Enter patient data (reception), wait (patient),
- Ask for complaints, remove bandage, examine and disinfect wound, apply bandage, check pressure/strength (assistant),
- if necessary, involve doctor (assistant), carry out examination and treatment (doctor)
- Document process (assistant)

Result: Wound is treated, bandage is applied and checked.

Participants (roles): Patient, reception, doctor, assistant.

5.2 Business Model Canvas (BMC)

5.2.1 Notation

The Business Model Canvas is a method for simple graphical modeling, structuring and optimizing business models. It was first developed in 2010 by Alexander Osterwalder as part of his dissertation (see Osterwalder and Pigneur 2010).

Its range of applications has expanded greatly. The BMC was originally developed to support the strategic accompaniment of startups, but is used by small and medium-sized enterprises and large corporations in all industries. For startups, it is used to move from a business idea to a business model and to further develop and realign the company. For established companies, it supports the analysis of the status quo, the identification of optimization potential and the adaptation to changed customer needs or market requirements (see Lukas 2018). Meanwhile, further areas of application have been opened, such as Data Science for the Business Model Canvas (see Neifer et al. 2020).

5.2.2 Modeling Example

The Business Model Canvas describes 9 aspects of service creation in a clear way: business partners, business activities, resources, value creation goals, customer relationships, distribution channels, customer segments, as well as costs and sources of income (see Maisch and Valdés 2022). In Fig. 5.4 you will find a small example for this which describes the business model of a doctor’s office on the basis of the case studies in Sect. 5.1.4.

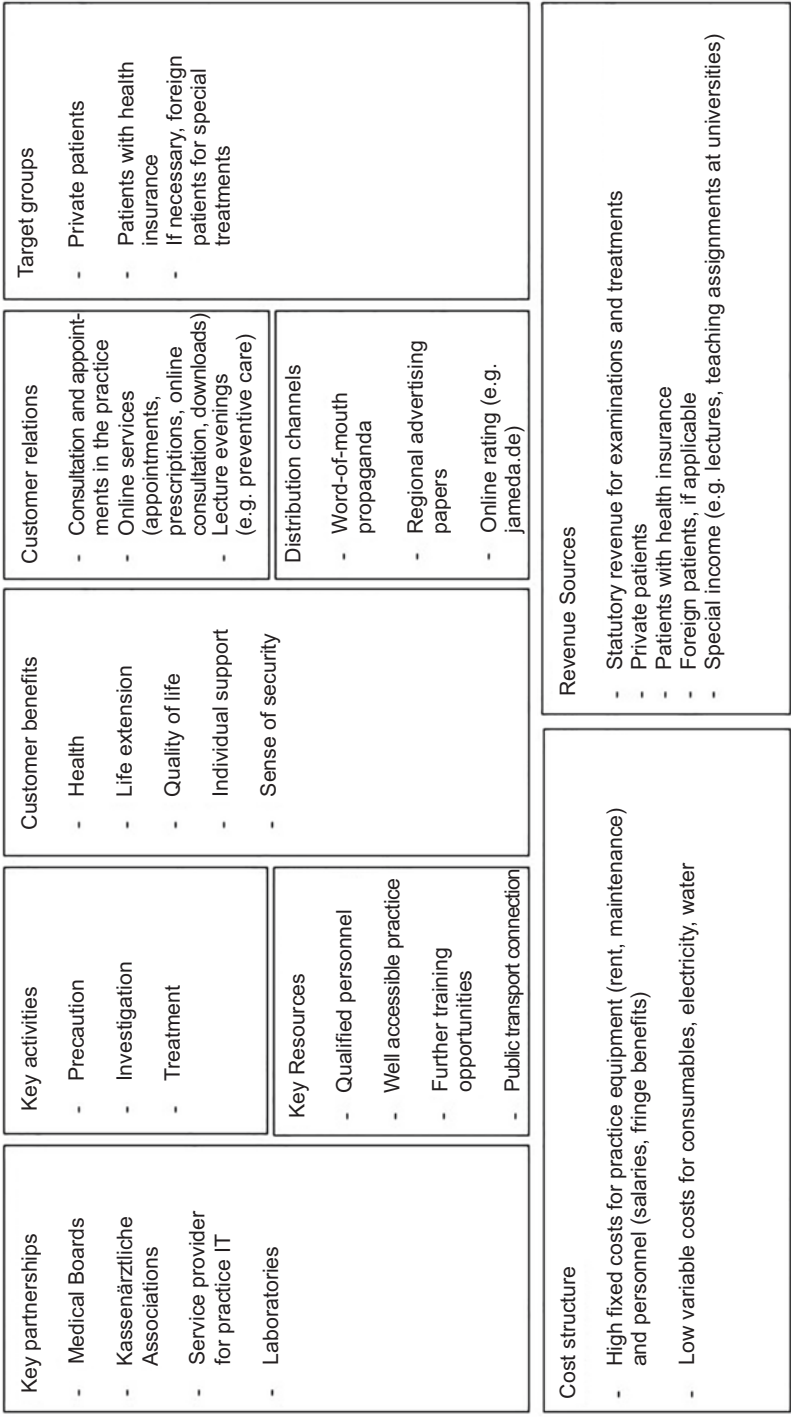


Fig. 5.4 Business Model Canvas modeling example family doctor's office

5.2.3 Assessment

The method helps to recognize relationships in the business model and to work out strengths and weaknesses. However, one moves easily on a very abstract level and neglects the derivation and implementation of measures (see Lukas 2018).

5.3 Process Map

5.3.1 Notation

Business processes are often differentiated depending on the proximity to the core business of a company (see, for example, Seidlmeier 2002, pp. 2 ff.). In order to show the essential business processes of a company in a clear way, process maps have established themselves, which show the essential business processes of a company. The purpose is to provide a rough overview of important workflows (processes) of a company. Target groups can be internal (management, employees) or external (suppliers, applicants). The processes shown are usually divided into control, core and support processes.

Control processes

Control processes are responsible for the integrative interaction of the business processes (e.g. strategy development, corporate planning, operational management). They are the entrepreneurial bracket for the value-creating and supporting processes.

Core business processes

Core business processes are business processes with a high value-added share. They are usually competition-critical and form the value-added process starting from the customer's wish, via procurement, storage, production, assembly and delivery.

Support processes

Support processes are business processes with no or only low value-added share. They are usually not competition-critical. Examples are accounting, cost accounting, reporting, human resources, canteen, fleet, information processing and law.

5.3.2 Modeling Examples

Fig. 5.4 shows an example of a possible notation for a process map. It was briefly mentioned in Sect. 1.3.4. The control processes are depicted in the header of the map, and the support processes in the lower area. The central element of a process map are the core processes of an organization, which are represented in the middle area with their essential process steps (Fig. 5.5).

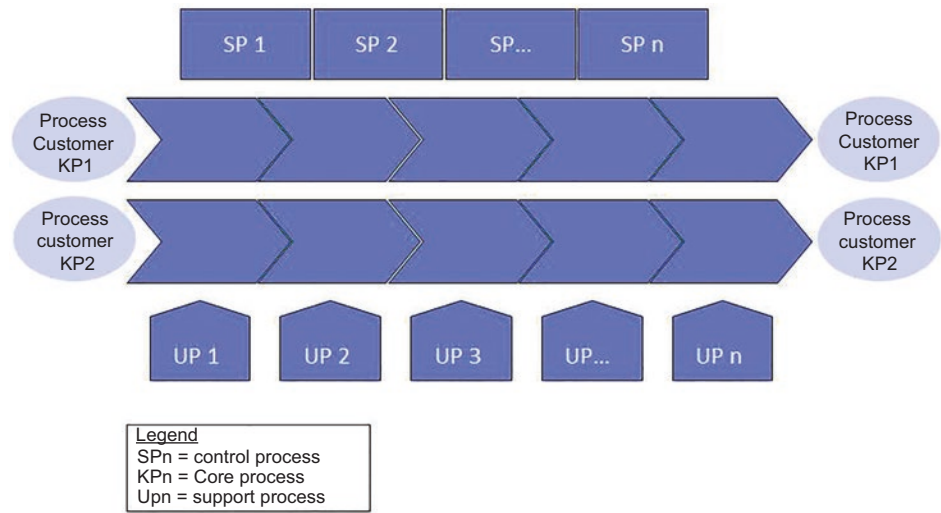


Fig. 5.5 Process map—notation

Fig. 5.6 shows an example of a process map based on the case study from Sect. 5.1.4. It shows three core processes (treatments, preventive examinations, vaccinations). In this example, the customer is represented by the patient.

Fig. 5.5 shows an example of a process map of a motor vehicle operation which is also already known from Sect. 1.3.4. The operation has two main processes, the sale of new or used vehicles, and the service process (repairs, maintenance, TÜV approval, etc.) (Fig. 5.7).

Another example can be seen in Fig. 5.6. The three core processes “new contract”, “contract change” and “damage settlement” describe the typical processes of an insurance company (Fig. 5.8).

Since the method for describing process maps is not standardized, many variations have developed in corporate practice. The previous examples are varied in practice by their own symbols or presentation techniques.

The modeling example for the family doctor’s practice (cf. Fig. 5.6) was modeled with the modeling tool “Bic Design” for illustration, which is used, for example, at the Bonn-Rhein-Sieg University of Applied Sciences in the context of courses (cf. Fig. 5.9). Each of these tools available on the market (cf. also Sect. 6.1.2) has its own graphical standards.

In practice, it is often difficult to decide whether a process should be included in the process map or not. Wagner and Lindner (2022) speak here of the “process worthiness”. The following questions can be used as orientation. If they can be answered predominantly with “yes”, the process should be included in the process map (cf. Wagner and Lindner 2022, p. 145):

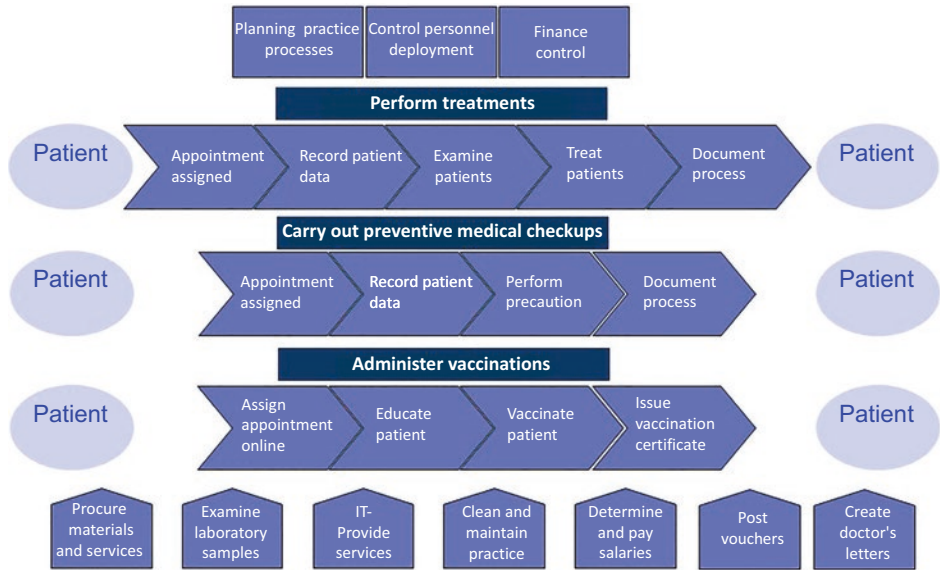


Fig. 5.6 Process map—example general practitioner's office

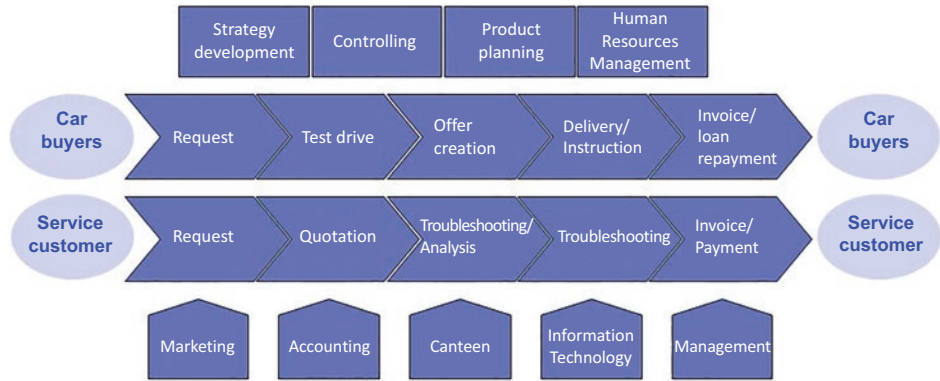


Fig. 5.7 Process map—Example car dealership

- Are different departments or areas affected by the process (high division of labor)?
- Are there interfaces to other processes?
- Are there interfaces to customer or customer processes?
- Can a process responsible be determined?
- Is the process often carried out?
- Are many resources bound by the process, which are also relevant for other processes?

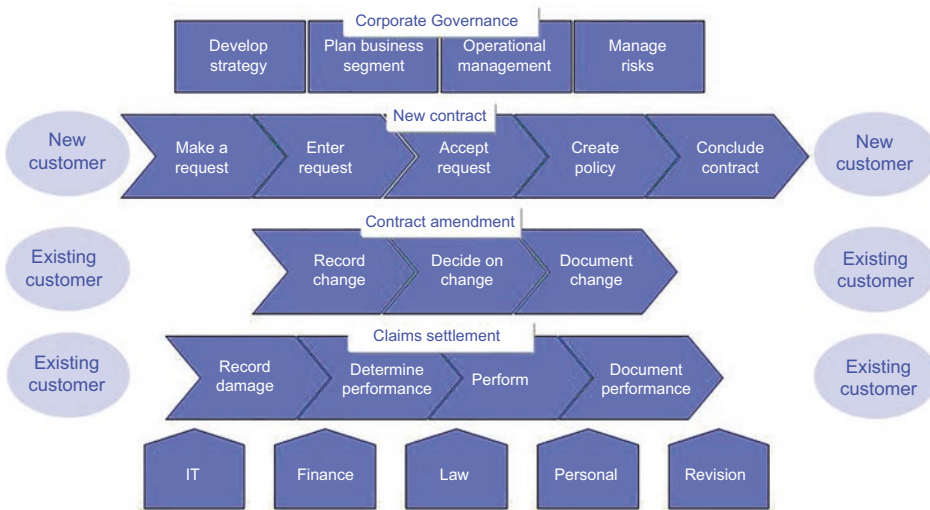


Fig. 5.8 Process map—Example insurance

5.3.3 Evaluation

The following properties characterize a “good” process map:

- **Clarity:** The representation should fit on one page if possible. The model section should describe the situation in a clear way, details and process variants are to be avoided.
- **Recognizability:** The type of company represented (e.g. doctor’s surgery, insurance, university) or part of a company (e.g. branch of a car manufacturer) should be clearly recognizable to an external observer.
- **Conciseness:** Memorable names for process steps. General descriptions without reference to the type of company are to be avoided (Not: “purchase-storage-sale”, but e.g. “procurement of food ...” for a food discounter)
- **Core processes:** Companies differ mainly in their core processes. Therefore, these should be highlighted in more detail in the process map with the central process steps.
- **Symbols:** Use of simple symbols, separated by process types (with legend)

Conclusion: A “good” process map can be recognized by the fact that a reader immediately gets an overview of the essential processes and thus the core business of the company.

The process map is a widely used tool for clear representation of overall context. With its help, a company can be represented in a comprehensible way with its essential processes (management processes, core processes, support processes).

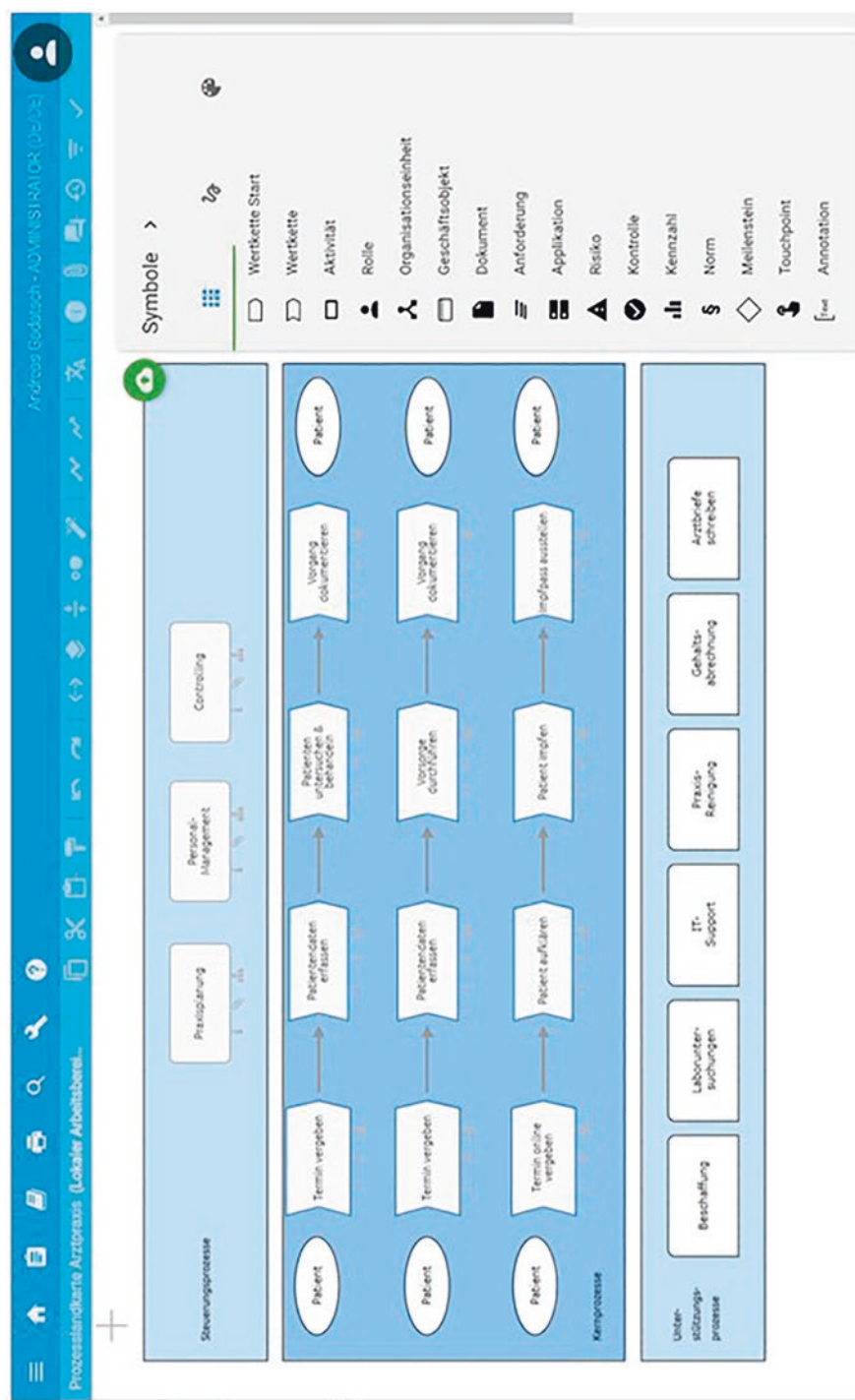


Fig. 5.9 Process map of family doctor's practice—modeled with Bic Design

The method can also be applied to sub-areas, such as sales, IT or a corporate unit. However, it cannot be used for detailed representations of processes or sub-processes.

The effort required for the creation and the passive use is minimal, as only a few symbols are used. In addition, this method is not standardized, which allows for the creation and use of own symbolism. On the other hand, the lack of standardization limits the comparison with reference models or representations of other companies.

5.4 Process Description

5.4.1 Notation

In addition to the process map, process descriptions describe the overall process and each process step through additional text information and, if necessary, further content, such as statistical indicators or an explanatory video. A uniform notation or a uniform description scope has not been developed so far. Important and frequently encountered contents of process descriptions are:

- process name,
- process designation,
- process responsible and other contacts,
- triggers and results of the process,
- additional explanations that go beyond the formal process models and
- key figures, such as number of processes per unit of time, number of employees, process control key figures.

5.4.2 Modeling Examples

A good example of a process specification is documented on the website of the Free University of Berlin. The university uses the process specifications for internal purposes and has published them freely accessible in addition to a process map published on the Internet. Fig. 5.7 shows a section from the interactive process specification “New programs set up” (Fig. 5.10).

5.4.3 Evaluation

The process specification supplements the process map for the detailed description of processes. Since not all aspects of a process can be accommodated in a graphic, the process specification offers a simple and generally comprehensible way to document details and special questions of a process appropriately. The content can be adapted to

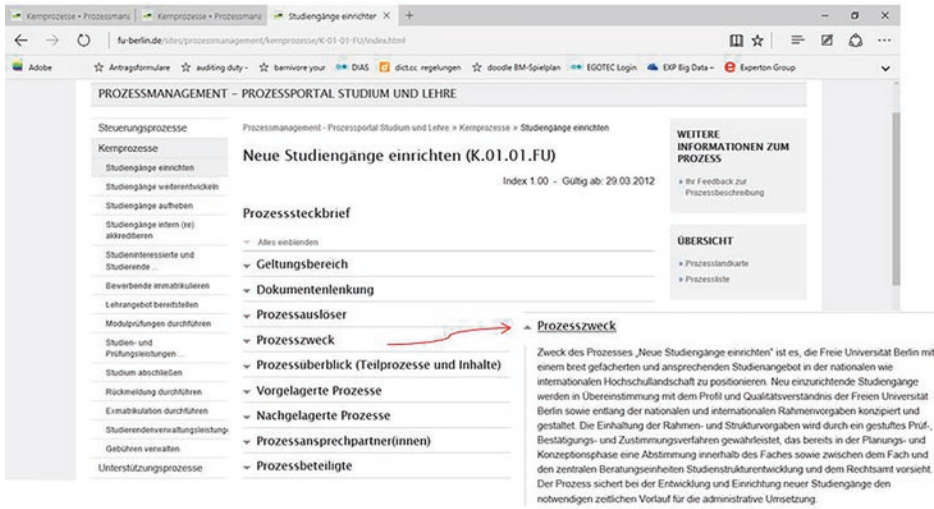


Fig. 5.10 Process specification Free University of Berlin (2015)

the requirements of the company. Ideally, a process specification is created for each process and sub-process and made available to employees and possibly external stakeholders (e.g. “Offer processing process for customers”). The contents are not standardized, so that own variants can be created without any problems. In practice, the intranet has proven to be a publication medium.

5.5 Tabular Process Modeling

5.5.1 Notation

Graphic modeling methods require the use of special software tools as part of a permanent application. If classic graphic programs are used, this will lead to a high level of creation and modification effort in the long term.

For “quick” process surveys, so-called “process survey forms” in tabular form are also used in practice. Even if the formal requirements for such concepts are not very high, the practical benefit is quite high. The easy comprehensibility and simple presentation of the content are aspects that are of high benefit for the initial survey or clear representation of the essential process elements.

A simple form for collecting process information is stored in Fig. 5.8. The notation of tabular models is not standardized and relatively simple. In the header area of the form, the process is discussed in general with some attributes such as “process name”, “date”, “creator”. The attribute “results” is important, which describes the general output of the process. In the lower area, the process steps are described sequentially, whereby for each

process step at least a designation, a responsible person, the necessary input (information, materials), the output (information, results) and the used software should be noted. It may make sense to also define a measure for process controlling (see Chap. 4) in order to be able to monitor the success of process execution (Fig. 5.11).

5.5.2 Modeling Examples

An example of a process that has been “modeled” in tabular form can be found in Fig. 5.9. It is the “appointment scheduling in a doctor’s office”. The table representation shows a process that takes place in the reception area of the doctor’s office and, in addition to the medical professional, has the patient as another process participant. The “number of patients” was chosen as the measure for process controlling. The overall output of the process can be modeled as an “agreed appointment” with the patient (Fig. 5.12).

The first line serves to identify the process (process name, date and creator). Then the process triggers (e.g. “patient has entered practice”) and the results (“patient will be discharged”) are briefly described. In addition to the process responsible (Process Owner), other participants and persons to be informed are noted. Then each process step is documented line by line. This is done by specifying the responsible party, the input (which information is used? e.g. health insurance card, doctor’s reports), the output (which

Process name		Date		Creator	
Trigger			Results		
Rollers		Description			
Process owner					
Involved					
To inform					
Process step	Responsible	Input	Output	IT deployment	Measured variable
Comments					

Fig. 5.11 Tabular process survey form

Process name: Appointment allocation		Date: 06.03.2013		Creator: A. Gadatsch	
Trigger: Patient calls or enters practice			Results: agreed date		
Rollers		Description			
Process owner		Med. FA			
Involved		Patient			
To inform		Laboratory if necessary			
Process step	Responsible	Input	Output	IT deployment	Measured variable
Welcome	Med. FA	-	-	-	-
Clarify the patient's concerns	Med. FA	Appointment request insurance card		Physician Practice Information System	Number of patients
Clarify resources / free dates	Med. FA	Scheduling overview Personnel deployment plan	Date	Physician Practice Information System	
Arrange appointment	Med. FA	-	-	-	-
Adoption	Med. FA	-	-	-	-
Comments: Comparable process for appointment changes. Not every patient may receive an appointment if concern in appropriate or lack of capacity availability.					

Fig. 5.12 Tabular process survey: “appointment scheduled”

information is produced?, e.g. prescription, referral), and the IT application (e.g. doctor’s office information system).

The tabular process representation can also be “converted” into graphical notation with many modeling tools at the push of a button, which we will discuss in more detail in Sect. 5.7 ff. In Fig. 5.13 we see a screen shot of the ARIS Toolset from Software AG, which shows the process “Appointment” as a table and in the modeling language EPK (see Sect. 5.7) (Fig. 5.14).

The process “Change bandage” from the case study is shown in Fig. 5.13 (Fig. 5.14).

5.5.3 Evaluation

The tabular process modeling is suitable for the quick recording of actual processes of simple and medium complexity. Possible uses in the context of target modeling are also conceivable. As soon as the processes are more complex, in particular if branching in the course is to be modeled, the method is less suitable.

Some tool manufacturers use tabular process recordings to generate “raw process models” (e.g. BPMN models Sect. 5.7, eEPK models Sect. 5.5.3) from them. This approach has the advantage that in the context of actual process recording with the employees of the specialist department, work can first be carried out with a simple method. Later, the work can be continued with a refined notation.

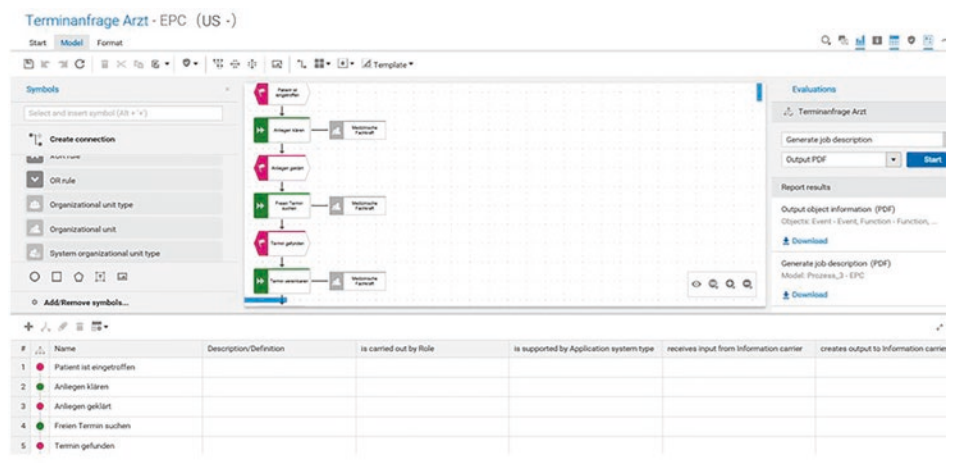


Fig. 5.13 Tabular process survey: “Assign appointment” shown alternatively as EPK model

Process name "Change dressing"		Date: 13.12.20xx		Creator: A. Gadatsch	
Trigger: Patient comes to practice			Results: Wound treated, dressing applied correctly		
Rollers		Description			
Process owner		Doctor			
Involved		Patient, assistant, doctor			
To inform		Health insurance			
Process step	Responsible	Input	Output	IT deployment	Measured variable
Record patient data	Reception / Patient	Insurance card		Medical practice-Information system	Number of patients
Waiting	Patient				
Examine wound	Assistant				
Investigate and treat complications if necessary	Doctor				
Apply bandage, Check pressure, strength	Assistant				
Document process	Assistant			Medical practice-Information system	Number of operations
Comments					

Fig. 5.14 Tabular process survey: Change dressing

5.6 Swimlane Diagram

5.6.1 Notation

Swimlane diagrams were developed by H. F. Binner at the beginning of the 1990s in order to represent sequences of activities in a simple way (cf. Binner 2000). The design

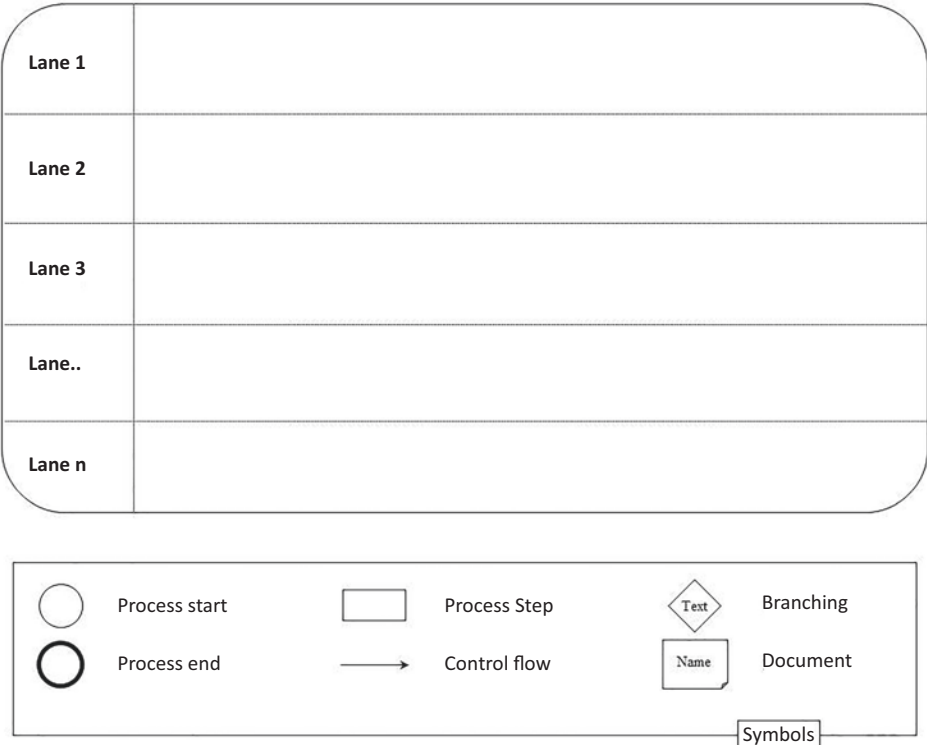


Fig. 5.15 Swimlane notation

is based on a swimming pool viewed from a bird’s eye perspective. The pool is the overall context, e.g. the company under consideration or a larger section, such as a department of the company. The swimming lanes correspond to the areas of responsibility of actors (e.g. a department) between which responsibility for a section of the process oscillates until the entire process is completed.

The representation has a certain similarity to the activity diagrams of the UML notation known from computer science or the task chain diagrams according to Österle (cf. Österle 1995). The notation has been further developed several times and can be expressed differently depending on the purpose (cf. e.g. Sharp and McDermott 2002, pp. 144 ff., 158 ff.). Organizational units are depicted as “swimming lanes” (lanes), activities (process steps) as rectangles, decisions as diamonds. This provides a good overview of rough processes with frequent department changes (cf. the example in Fig. 5.10).

In order to reduce complexity, the “happy path” is often only modeled, i.e. the standard course of events without special cases that occur in the normal case (Fischermanns 2013) (Fig. 5.15).

5.6.2 Modeling Examples

The “appointment” process was created with the freely available modeling tool “draw.io” (www.draw.io or diagrams.net) and is shown in Fig. 5.16. Since only two people are involved, two lanes are required for the relatively simple process.

The “change bandage” process is much more complex. It requires 4 lanes (registration, waiting area, assistant, doctor) and a branch (complications yes/no?). It is shown in Fig. 5.17.

Another example of a swimlane diagram can be seen in Fig. 5.11. It shows the simplified process of treatment in a hospital. The lanes represent the departments, such as administration, ward, X-ray, operating theatre and billing. The process begins in the top left-hand corner of the illustration with the capture of patient data. The patient is then examined and, depending on the result, X-rays are taken which then need to be evaluated. This is followed by the operation and finally the aftercare and discharge of the patient from the ward. Subsequent activities are the billing and monitoring of incoming payments (Fig. 5.18).

5.6.3 Assessment

The Swimlane method provides a good overview of processes and visualizes very clearly the change of department. This makes processes with many changes of staff very visible very quickly, which can be seen as an indicator of possible optimization. It does not require any tools and can be used simply on a blackboard or flipchart in workshops.

A disadvantage is the limited depth of detail and the low amount of information in the illustrations. The focus of the visualization is on the clear visualization of the control flow in the context of the participating organisational units, i.e. the sequence of individual activities and the organizational allocation. The notation uses very few, non-standardised

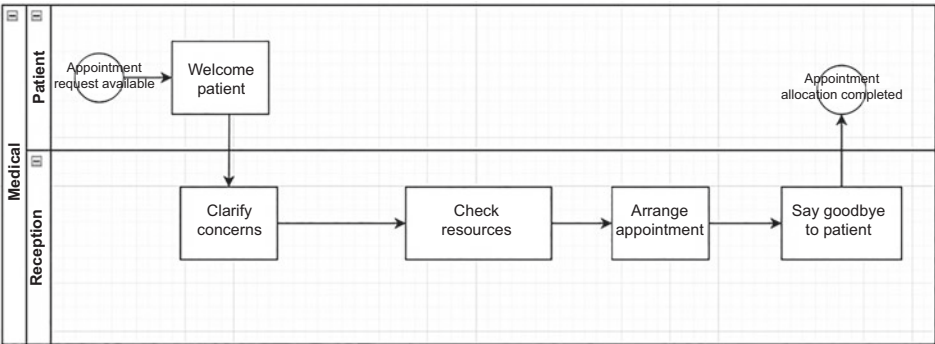


Fig. 5.16 Swimlane “appointment”—modeled with draw.io

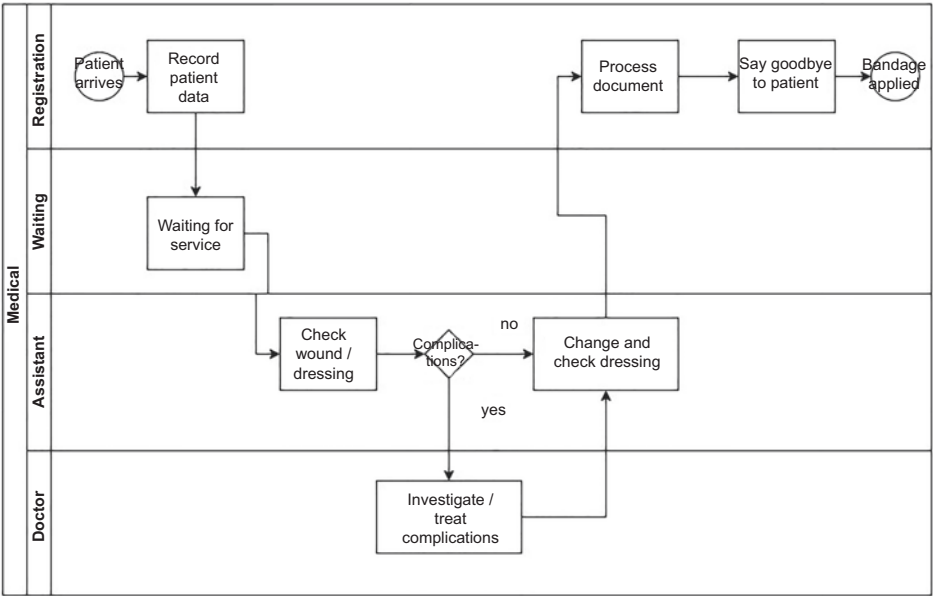


Fig. 5.17 Swimlane “change bandage”—modeled with [draw.io](#)

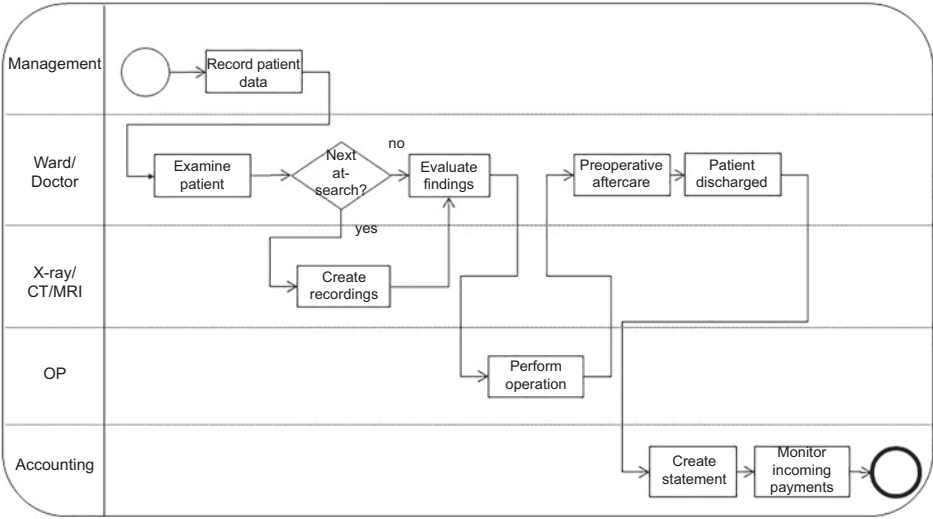


Fig. 5.18 Swimlane model of treatment in a hospital

elements and is very easy to learn. This makes the method also suitable for ad hoc use, e.g. on a flipchart in meetings. However, more extensive processes with many branches cannot be illustrated very well. Additional symbols need to be explained at least in a legend, but often lead to more confusing illustrations because of the limited space.

5.7 Event-Driven Process Chain (EPC)

5.7.1 Overview

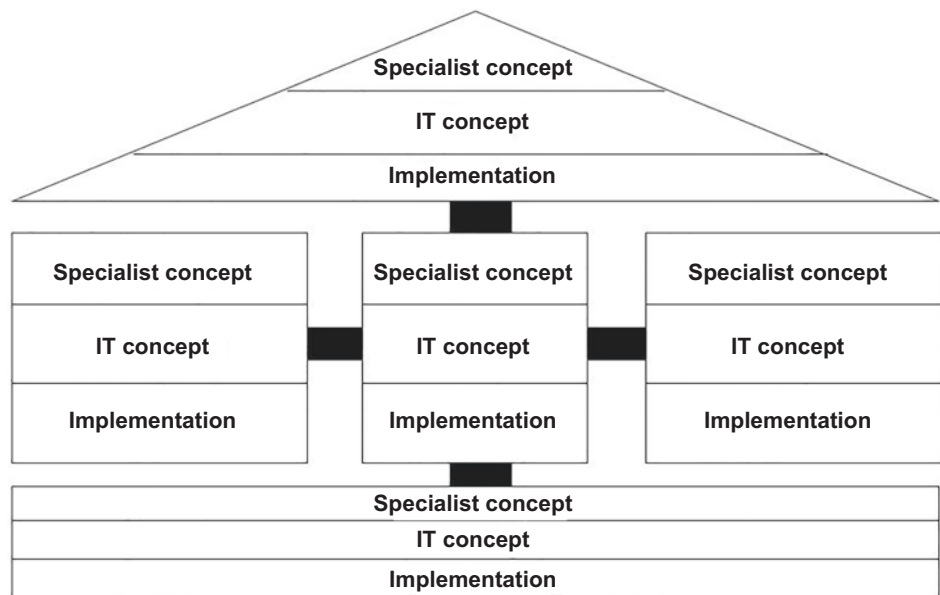
Up until the early 1990s, so-called data flow diagrams were used to depict which data “flows” between the organizational units of a company. They were used as aids in software development. The background for the then data-oriented approach was the comparatively limited storage space at the time. The programs for supporting workflows were designed so that the available storage space was used optimally. Process models for describing workflows were not yet common at that time. Against this background, the method of event-driven process chain (EPK) was developed (Hoffmann et al. 1992, p. 3) to bring the process into the foreground of modeling and thus the design of information systems.

The event-driven process chain (EPK) is a central part of the architecture for developing and describing information systems ARIS (Architecture Integrated Information Systems, see Scheer 1998) developed by A.-W. Scheer at the University of Saarland as well as the modeling concepts anchored in the architecture developed in 1991 by Keller, Nüttgens and Scheer (see Keller et al. 1992).

- The previously cited original article (Keller et al. 1992) is still available on the institute’s website and is expressly recommended to interested readers as reading material. The last researched link is: https://www.uni-saarland.de/file-admin/user_upload/Fachrichtungen/fr13_BWL/professuren/PDF/heft89.pdf.

The modeling approach of the EPK has established itself in practice very quickly as the leading semi-formal method for modeling business processes. One reason for this was that the EPK was used by SAP AG, Walldorf, to document its successful ERP system “R/3” (Scheer 1998, p. 125). This had the consequence of a rapid spread of the EPK method due to the success of the SAP software (Rump 1999, p. 61). An early description of the method can be found in Keller and Teufel (1997).

The notation of the EPK method has been modified many times and published in various variants. There is no uniform standardized version, as for example in the BPMN 2.0 method (see Sect. 5.7) (see Jannaber et al. 2016; Riehle et al. 2016 for this). The author therefore relies to a large extent on the original version by Keller, Nüttgens and Scheer (see Keller et al. 1992) as well as the variants common in practice, which were set by various modeling tools.



ARIS architecture ("ARIS house") according to A.-W. Scheer

Fig. 5.19 ARIS house (Scheer 1998)

Classification into the ARIS architecture according to A.-W. Scheer

The ARIS architecture developed by August-Wilhelm Scheer (University of Saarbrücken) differentiates between data, control, function, organization and performance views for the holistic specification of information systems. In addition, the project phases of conceptual design, IT concept and implementation are distinguished (cf. Fig. 5.12). ARIS is a general reference framework for business process modeling. It provides level- and view-specific modeling and implementation methods (Scheer 1998, p. 1) (Fig. 5.19).

Modeling phases

ARIS is designed as a method-neutral approach model and accompanies the way from the problem statement to the executable program. It knows three consecutive modeling phases: Business Concept, IT Concept and Implementation. The starting point of the modeling is an informally described business problem, which is successively refined up to the implementation. In addition to methods for modeling, numerous software tools (e.g. the products "ARIS Business Architect" and "ARIS Express" from Software AG, Darmstadt) are available to support the practical implementation. ARIS is adaptable to current developments due to its generic structure and is still considered the leading and established framework concept of business informatics (cf. Figs. 5.13 and 5.20).

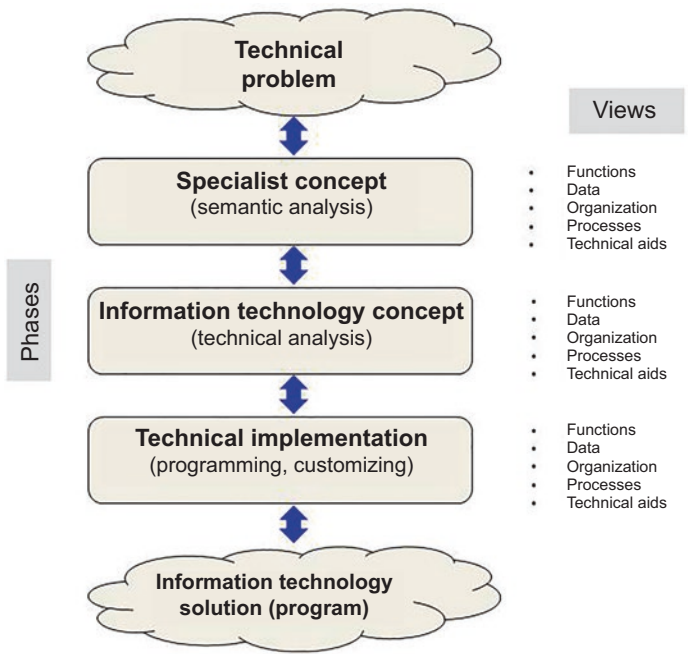


Fig. 5.20 ARIS—From problem to program (Scheer 1998)

The **business concept** serves the formal representation of the business problem so that it can be implemented in information technology solutions. The business concept is of a long-term nature, as it is the content-bearing element of the business application concept.

The **information technology concept** (IT concept, formerly known as data processing concept or DV concept) serves the adaptation of the business concept to requirements for technical implementation in a general form that is independent of the implementation. The business concept and the IT concept are only loosely coupled here.

The **implementation** is the implementation of the IT concept into concrete software and hardware components. It describes the computer-aided realization of the business concept. The ARIS concept is suitable for both the individual development of software and the introduction process of standard software (cf. e.g. also Kirchmer 1996, p. 66 ff.).

Modeling views

ARIS distinguishes four secondary views, the organizational view, the data view, the function view and the performance view. The integrating central view is the control view.

The **Organizational view** describes the organizational structure of a company. For this purpose, organizational charts are used, which map the hierarchical relationships.

The **Data view** shows the information objects relevant for modeling and their relationships to each other. For this purpose, extended entity-relationship diagrams are used.

The **Function view** comprises structured operational activities. For this purpose, function trees are used, which map the relevant business functions and their relationships to each other at different aggregation levels.

The **Performance view** describes the products of a company, i.e. the tangible and intangible services including the money flows (Scheer 1998, p. 93 ff.). The description is made using a product model.

The **Control view** represents the business processes of a company. It integrates the partial views of the ARIS concept and uses, for example, the extended event-driven process chain (eEPK) to describe business processes.

ARIS as a method for software development

Fig. 5.14 shows the classification of the tasks that arise in individual software development or the introduction of standard software into the ARIS concept, as well as the employee groups primarily used (employees of the department, employees of the IT department with a focus on organization or software development). All ARIS phases are therefore to be carried out and affected to a different extent (Fig. 5.21).

In **individual software development**, the business requirements are first recorded and implemented in the form of a business concept. This is followed by the technical design of the planned information system and its implementation, testing and acceptance.

In the case of the **introduction of standard software**, all activities focus on the business concept level after the decision for the standard software has been made, as the software is already finished. The decision-making process “Make or Buy” is largely

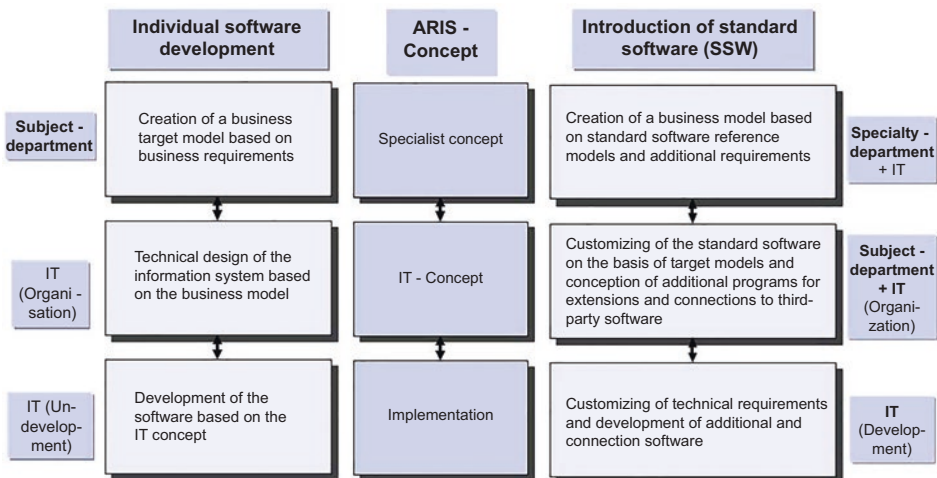


Fig. 5.21 ARIS as a method for software introduction

preceded by the ARIS concept. Except for additional programs for functional extensions and interface programs to “foreign systems”, the development work of the software manufacturer can be taken over. Based on reference models, which document the scope of the standard software, a business target model is created. Of particular importance here are the target process models in the form of EPC models, which will be dealt with in more detail. As part of the IT concept and implementation, customizing activities take place, i.e. the business model is anchored in the standard software system in the form of parameters. In addition, additional programs (so-called Add Ons) must be designed and programmed.

5.7.2 Basic Notation (EPC)

5.7.2.1 Events and Functions

The basic notation of the EPC method describes the course of the business process with only a few basic symbols. The starting point of every process is an event. It describes the question of what triggers the process. This can be, for example, the input of an order by fax. If necessary, several events may be required. For example, the payment of a life insurance policy will only take place if several conditions are met at the same time. After the triggering event, a function to be fulfilled is executed.

The four basic elements of the EPC are:

- The **function** that changes the state of objects,
- the **event** that triggers state changes of objects,
- the **edge** that links functions and events and
- the **connector** that connects functions and events to a process.

Function

Functions describe transformation processes of information objects to achieve corporate goals. They can be described on different levels. A process or a process chain is therefore a comprehensive procedure (e.g. spare parts sales). A function is a complex activity that can be further subdivided and directly enters a process (e.g. order processing). The activity described by the function symbol is carried out by actors (people or software).

A subfunction is an activity that can be decomposed into further subfunctions or elementary functions and enters a higher-level function (e.g. order checking). Elementary functions are activities that cannot or should not be further decomposed.

A criterion for the maximum meaningful decomposition of processes is the meaningful closed processing of the function at one workstation (e.g. material availability check). The representation of functions is carried out as a rectangle with rounded corners (cf. Fig. 5.15).

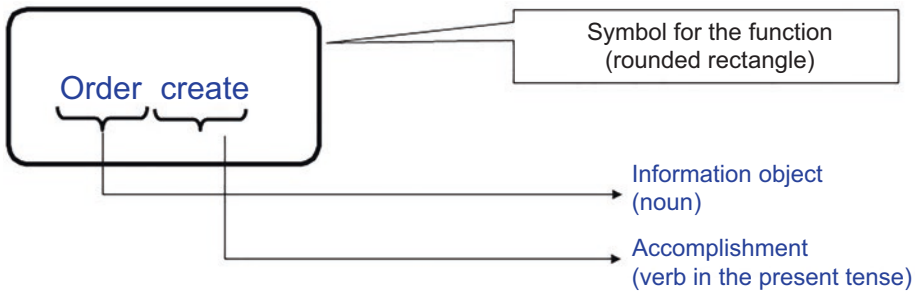


Fig. 5.22 EPC notation “function”

The function is a so-called “active” object type of the EPC, which maps a task that is carried out by people or systems. Functions can make decisions. The function refers to one or more information objects and an activity that changes the information. For this reason, the name of the EPC is composed of an information object (noun) and a description of the task (verb).

Examples of functions are, for example, “create order”, “check order”, “evaluate employee”, “create calculation”, “book invoice” (Fig. 5.22).

Event

Events are passive object types, i.e. they only represent states and cannot make decisions. They trigger functions and are in turn results of already executed functions. Events can occur within (e.g. “Candidate was rejected”) and outside the company (e.g. “Application has been created”). The processing of an object changes its state. For example, an order of a customer is supplemented with relevant order characteristics such as the customer number, material numbers, etc. Events describe a state that has occurred, i.e. they describe the object that has experienced a change in state (cf. Hoffmann et al. 1992, p. 5). Events are represented as hexagons (cf. Fig. 5.16). The name of an event consists of an information object (noun) of the underlying data model and a verb in the perfect tense, i.e. a state that has occurred. Examples of events are “Credit limit has been exceeded”, “Order has been received”, “Offer has been created” (Fig. 5.23).

5.7.2.2 Basic Modeling Rules

An EPC begins and ends with an event, with a process-initiating event being described as a start event and a process-finalizing event as an end event. Subsequent processes can be triggered by end events of a previous process, i.e. an end event can indeed be a process-initiating start event in another process.

A simple example of an EPC is shown in Fig. 5.17 on the left, it shows the sequence of events and functions (each only with placeholder names) and on the right the process for processing applications in a greatly simplified form (Fig. 5.24).

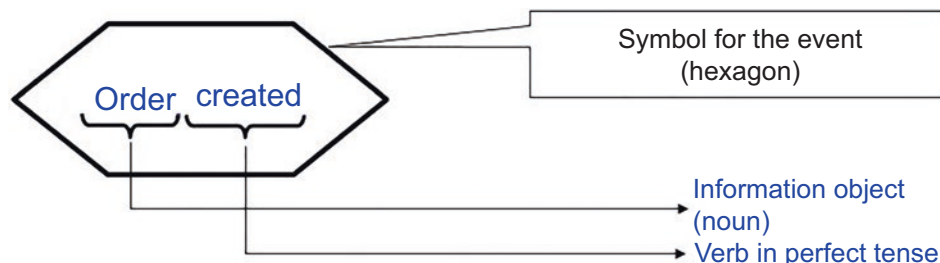


Fig. 5.23 EPK notation “event”

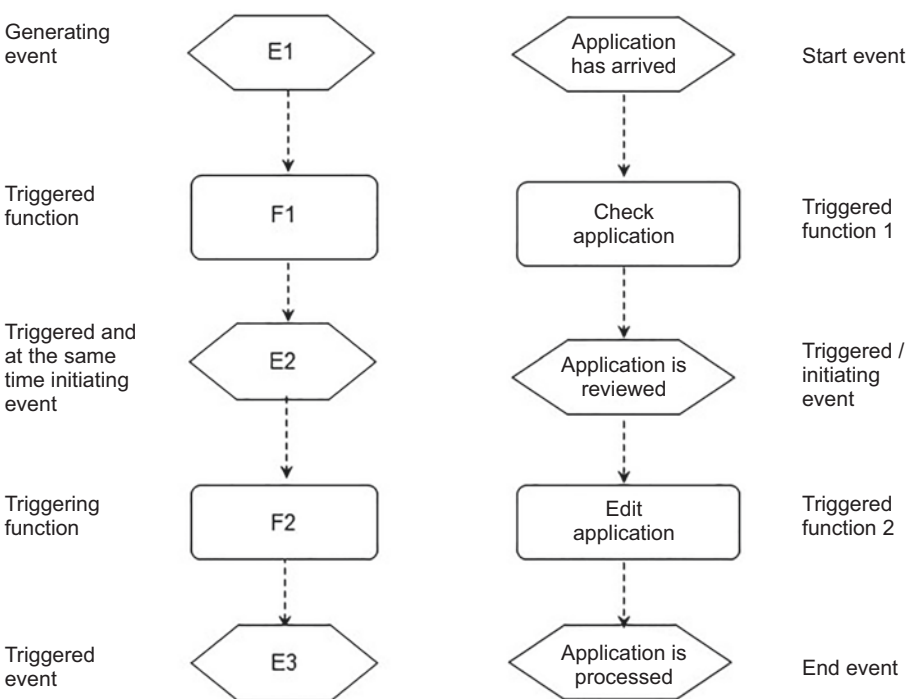


Fig. 5.24 EPC notation “Simple Example”

5.7.2.3 Connectors

After the basic structure of the EPC has been introduced, the question arises as to how it can be further refined. In practice, functions can be triggered by more than one event and can also trigger several events. For example, the event “customer is creditworthy” depends on several preconditions that must be checked by means of several functions. In order to represent such constructs using the event-driven process chain, three logical connectors are used: the conjunction (“and” connection), the disjunction (“exclusive or” connection) and the adjunction (“inclusive or” “connection”) (see Fig. 5.18 and 5.25).

	And	"conjunction", "both and" Example: A and B
	Or	"adjunction", "inclusives or", "at least one" example: either (A or B) or (A and B)"
	XOR	"disjunction", "exclusive or", "either or" example: (A or B), but not (A and B)

Fig. 5.25 EPC notation “connectors”

Fig. 5.19 shows a schematic representation of an EPC with the “XOR connector”. The left column shows the formal syntax of a so-called XOR split (branching of the process with XOR) with subsequent XOR join (merging of the process with XOR). The right column shows a business example. The concrete process can either follow the left path from E1 to E2 via E3 and then to E6 or the other (right) path from E1 to E4 to E5 and then finally to E6. With the example on the right-hand side of the applicant’s examination, the situation can be somewhat more easily understood, the applicant either receives a rejection or an employment contract is offered. Both options exclude each other.

Important: It is not necessarily required that the process is closed again using an XOR join. The EPK shown could also end with two events (E3 and E5 or “Cand. is rejected” and “Contract is offered”). However, if the join is carried out, it is important to use the same operator. For example, if an AND operation was used instead of the XOR join in front of function “F4”, the process would “wait forever” because both events cannot occur (Fig. 5.26).

Fig. 5.20 shows a schematic representation of an EPK with the “AND connector”. The left column shows the modeling schema again with the basic notation using the AND connector. In the right column you can see a corresponding business example. In this case, the process is split into two strands after function F1 is carried out and brought back together before function F4. The business example shown on the right shows the area of application. After the order has been processed, two events are certain: The order has been checked and the order data has been recorded. This means that two functions can be carried out in parallel, namely “Order planning” and “Order confirmation”. If both functions are finished (if the AND connector waits), work on the function “Assemble parts” can begin. If this is done, the event “Parts are assembled” occurs. This could in turn trigger further process steps, but this has been omitted here for the sake of simplicity.

Important: It is also important here that the process is closed with the same connector. If something were to stand in front of function F4 (or “assemble parts”) an XOR

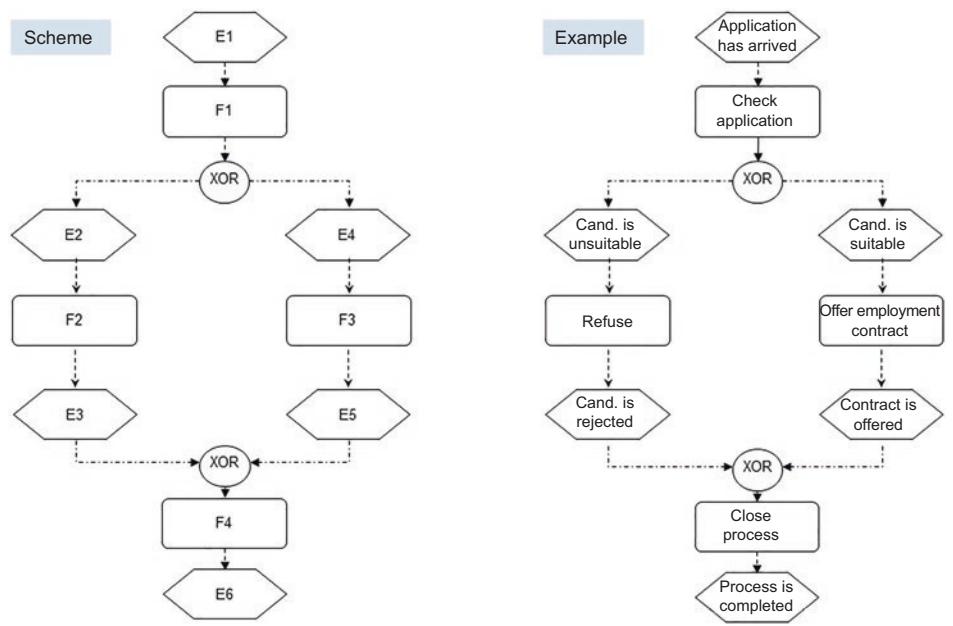


Fig. 5.26 EPK notation “Example of the use of the XOR connector”

connector, then F4 or “assemble parts” would be triggered immediately if one of the events E3 or E5 or “order is planned” or “order is confirmed” occurs. This could result in an order being confirmed that is not yet planned (Fig. 5.27).

In Fig. 5.21 a formal (left column) and intuitive (right column) representation of an EPK with the “OR connector” is visible. The possible processes from a formal perspective (left column) are:

1. Variant: (left path) E1-F1-E2-F2-E3-F4-E6
2. Variant: (left path) E2-F1-E4-F3-E5-F4-E6
3. Variant: (both paths): E1-F1-E2/E4-F2/F3-E3/E5-F4-E6

When looking at the right column, the use of the OR connector becomes more apparent. After the guest has chosen his drinks, he either drinks only beer (left path), only water (right path) or beer and water together (both paths). After he has drunk his drink or both drinks, the bill can be paid.

Important: It is also important to close the split path with the same connector, otherwise error situations could occur (Fig. 5.28).

A more complex modeling example can be found in Fig. 5.22. It depicts the following situation with the previously known EPK notation:

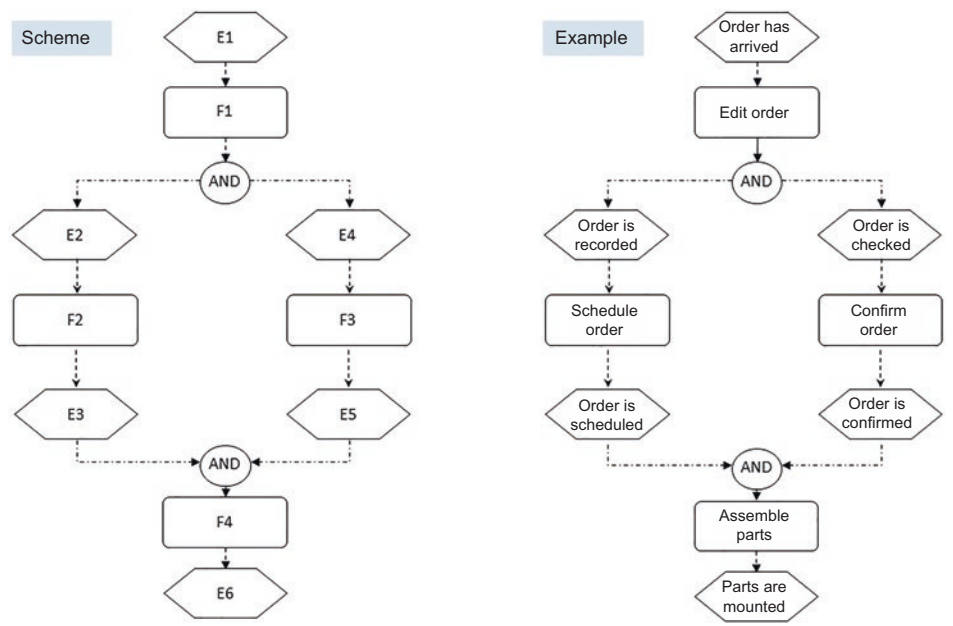


Fig. 5.27 EPK notation “Example of the use of the AND connector”

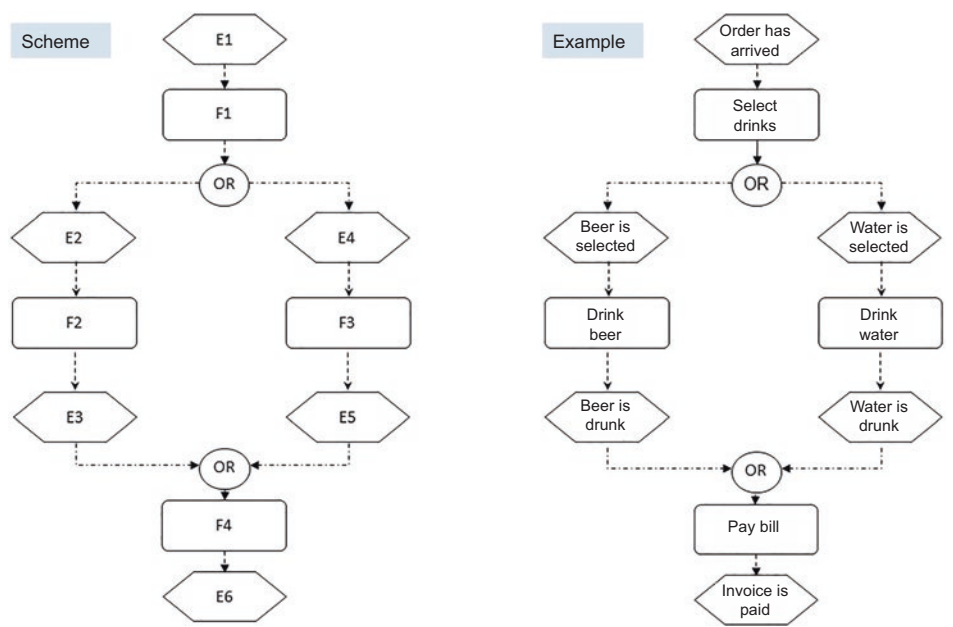


Fig. 5.28 EPK notation “Example of the use of the OR connector”

- After the customer has returned the vehicle, the condition of the car is checked. If at least one defect is present, a list of defects is first created.
- If the vehicle is damaged, it is repaired.
- If the fuel tank is not filled, the vehicle is refueled.
- If the vehicle is not completely clean, it is cleaned.
- If the tire pressure is not correct, air is filled or released.
- Then the vehicle is parked in the parking lot. It can now be rented again (Fig. 5.29).

5.7.2.4 Special Modeling Aspects

Although the basic symbols of the EPK introduced so far are quite simple, their application leads to some special cases. So the standard rule of EPK modeling is: Events alternate with functions, functions may not follow functions. However, there are also special cases.

Optional Events

For the specification of processes, nested connectors or optional events are also possible. Figure 5.23 shows that events can also follow events (right column) if this creates more clarity or is useful for organizational reasons.

This can be best explained using the business situation depicted at the bottom of the representation: As soon as the work permit and the certificate are available and the work experience is sufficient, an employment contract can be offered to the applicant (left column). To specify the decision, the event “person is suitable” can also be recorded (right column) to precisely capture this situation. Both models are equivalent but depicted with different levels of detail. The upper part of the representation shows the model schematically without going into specific facts (Fig. 5.30).

Nested Connectors

Not only can events follow events if required. Connectors can also be used one after the other. In Fig. 5.24 in the upper left part of the display, an event E3 is shown which leads into the AND connector. In the upper right part, an alternative representation of the situation is shown which does without E3. In this case, the XOR connector is followed by the AND connector before F1. The lower representation of the formal model should be somewhat more intuitive. If “Paid in advance” or there is a “guarantee”, the customer is considered to be solid (“good credit”) and the goods can be delivered. This event can also be omitted, as can be seen on the right side. It only serves to clarify the situation (Fig. 5.31).

5.7.2.5 Types of Linkage of EPK

Using the connectors presented, two types of linkage of functions and events can be distinguished. With **event linkage**, two or more events are linked to a function by means of a connector. Depending on whether triggering or generated events are involved, the linkage can be further subdivided into linkage of triggering or generated events (cf. also

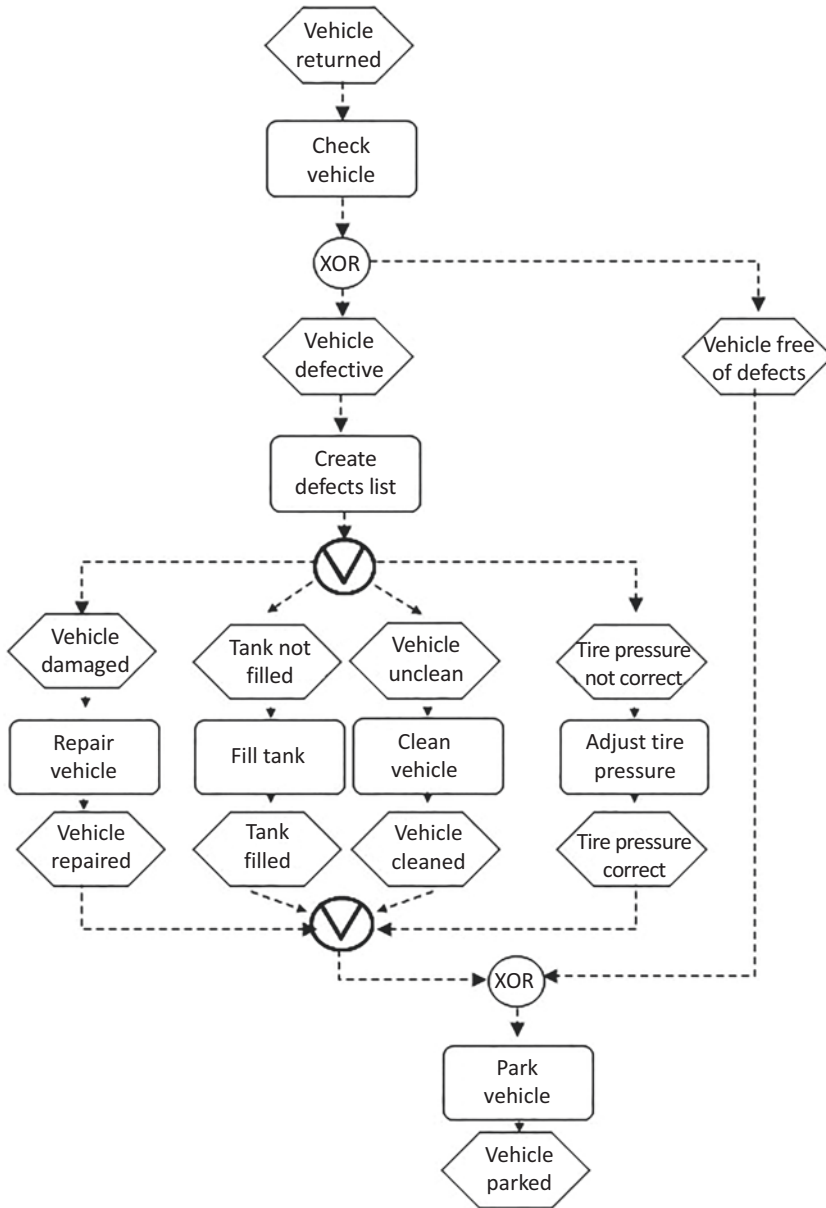


Fig. 5.29 EPK modeling example “defect processing”

Hoffmann et al. 1992, p. 12). With **function linkage**, two or more functions are linked to an event by means of a connector. Depending on whether triggering or generated functions are involved, the linkage can be differentiated in analogy to event linkage into linkage of functions with a triggering or generated event.

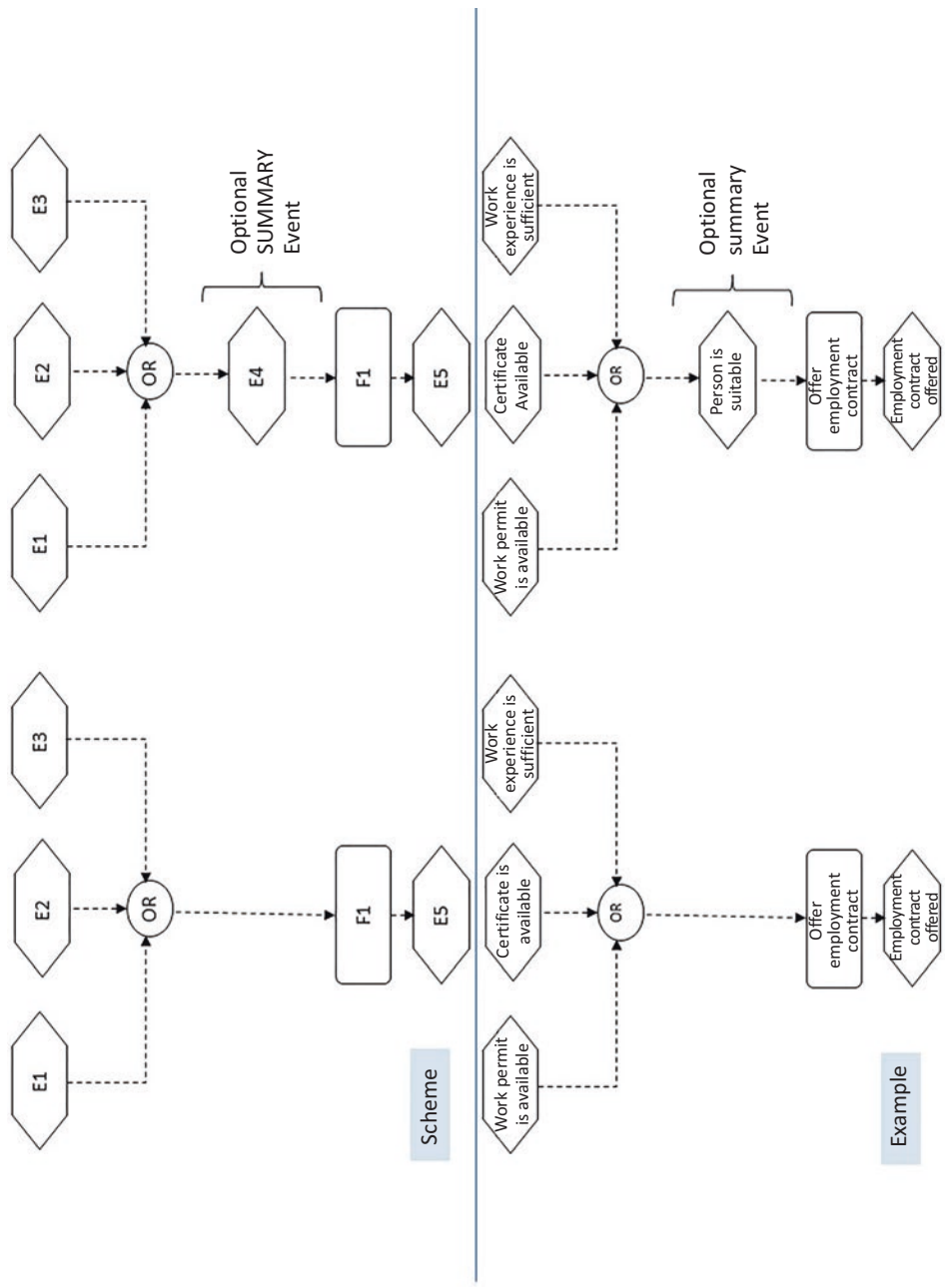


Fig. 5.30 EPK—Optional Events (Schema and Example)

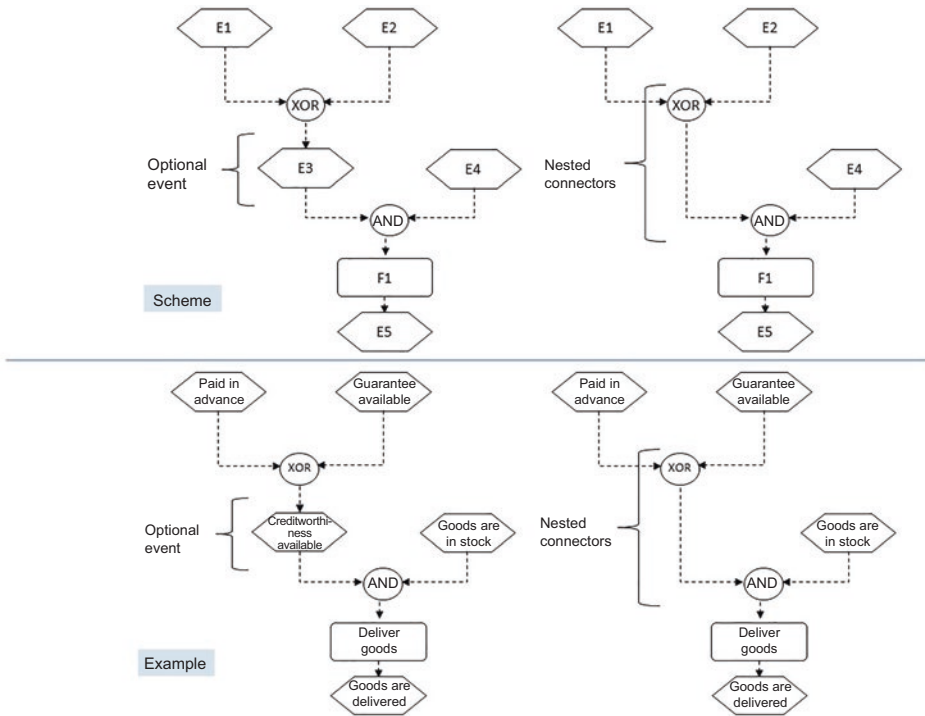


Fig. 5.31 EPK—Nested Connectors (Schema and Example)

All combinations are possible except the following special cases: The function link with a triggering event is only possible via an “AND” link, since events cannot make decisions as passive model elements. The “OR” and the “XOR” link are not allowed here. The conceivable case groups are shown in Fig. 5.25 (see also Hoffmann et al. 1992, p. 12 (5.32)).

Event link: Linking of triggering events with a function First, case group 1 “Linking of triggering events with a function” is presented. The common feature of this case group is the initiation of a function by one or more events as an input requirement.

- The function in case 1a is triggered when all events have occurred. Example: If the applicant meets the conditions A, B and C, he is invited to the job interview.
- The function in case 1b is triggered when at least one event has occurred. Example: If the applicant meets one or more of the conditions A, B or C, he is rejected.
- The function in case 1c is triggered when exactly one of the alternative possible events has occurred. Example: If the applicant meets one of the conditions A or B or C, he is rejected.

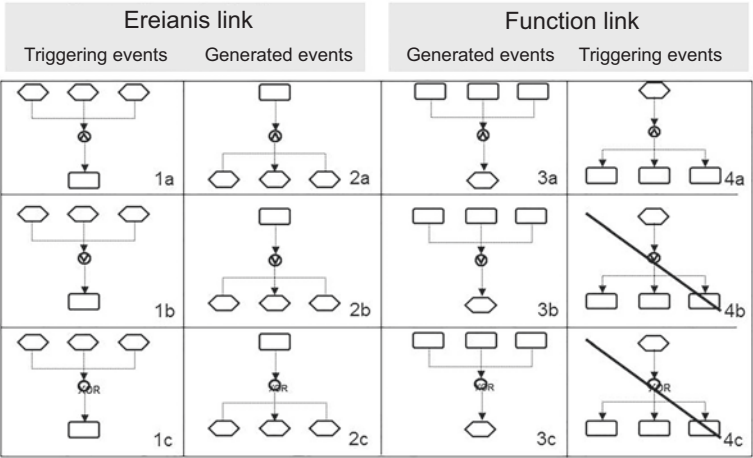


Fig. 5.32 EPK link types (see, for example, Hoffmann et al. 1992, p. 12)

Event linkage: Linking of generated events with a function Case group 2 “Linking of generated events with a function” explains the generation of one or more events after the execution of a function.

- After execution of the function in case 2a, all events are generated. Example: If the order was created, the master data is up-to-date, the order was checked, etc.
- After execution of the function in case 2b, at least one event is generated. If the order was created, at least one of the events A, B or C is generated.
- After execution of the function in case 2c, exactly one of the alternative events occurs.

Function linkage: Linking of several generating functions with an event The function linkage connects functions with generated or triggered events.

Case group 3 “Linking of several generating functions with an event” describes the generation of an event after the execution of one or more functions as an input requirement.

- The event in case 3a is generated if all functions have been executed. Example: The order is “released” if the order data was previously recorded and the credit limit was checked.
- The event in case 3b is generated if at least one function has been executed.
- The event in case 3c is generated if exactly one of the alternative functions has been executed.

Function Linkage: Linking of functions with a triggering event Case group 4 “linking of functions with a triggering event” is the generation of one or more functions by a triggering event.

Since events are passive model components and therefore cannot decide about the selection of relevant functions, only the conjunction “AND” (“AND” linkage) is allowed. When the event occurs, all functions are started.

Case 4b is not allowed because the event cannot make a decision about the selection of functions as a passive object type. Case 4c is also not allowed for the reason.

5.7.2.6 Modeling Rules of the Elementary EPK Notation

Modeling work is usually done in a division of labor. Before a modeling project, modeling rules are usually agreed in practice in order to achieve a constant quality and comparability of the models. The following modeling rules for the EPK (cf. Seidlmeier 2002, p. 78) are common:

- Each EPK begins and ends with one or more events.
- Events and functions alternate during principle. Connectors (see below) describe branches.
- In and out of functions, only one control flow edge runs.
- No object is in the model without an edge.
- An edge connects exactly two different objects.
- After an event, an OR or XOR connector may not be present in principle (exceptions: loop constructions, summary of events to higher-level events).
- Paths branched by connectors are brought together again by similar connectors.
- If multiple paths are reconnected with a connector, the connector may only have one outgoing edge.
- Direct connections of connectors are allowed.

5.7.2.7 Exercises for the Basic Notation

The symbols proposed by Keller et al. (1992) for “function”, “event” and the connectors “AND”, “OR”, “XOR” are partly displayed in software tools (e.g. ARIS Express by Software AG) in different colors and graphics. In Fig. 5.26 you will find an example for the use of the XOR connector.

Exercise for Fig. 5.26

Which of the following statements are true for the EPC model in Fig. 5.26?

- a) E2 can only occur after F1 has been completed.
- b) F1 is always followed by exactly one of the events E1, E2 or E3.
- c) F1 is followed by the events E1, E2 and E3 (Fig. 5.33).

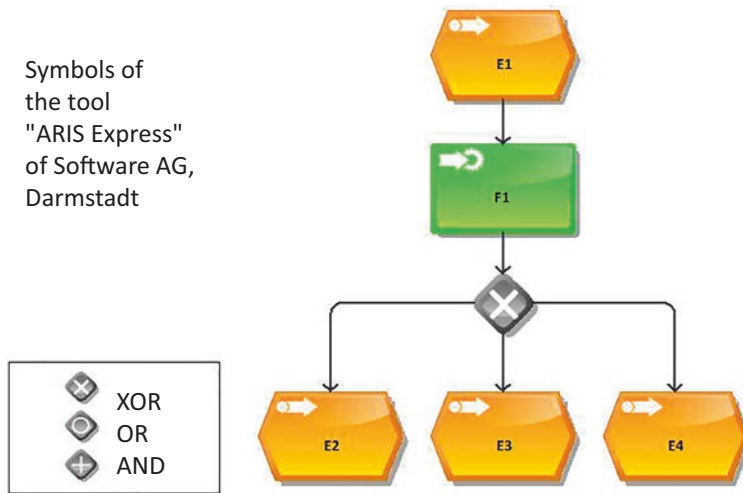


Fig. 5.33 EPC example 1 with ARIS Express (Software AG, Darmstadt)

Solution to the statements in Fig. 5.26:

- a) E2 can only occur when F1 is completed: True.
- b) F1 is always followed by exactly one of the events E1, E2 or E3: True.
- c) F1 is followed by the events E1, E2 and E3: False, only one of the three events is possible.

Exercise for Fig. 5.27

Another EPK model is shown in Fig. 5.27.

Which of the statements are true?

- a) F1 is executed if E1 or E2 occurs.
- b) If E1, E2 and E3 occur simultaneously, F1 is executed.
- c) If F1 has been executed, it can be concluded with certainty that E2 has occurred (Fig. 5.34).

Solution to the statements in Fig. 5.27

- a) F1 is executed if E1 or E2 occurs: False, in addition, because of the "AND" connector, E3 must also occur.
- b) If E1, E2 and E3 occur simultaneously, F1 is executed: True.
- c) If F1 has been executed, it can be concluded with certainty that E2 has occurred: False, the combination of E1 and E3 is also conceivable.

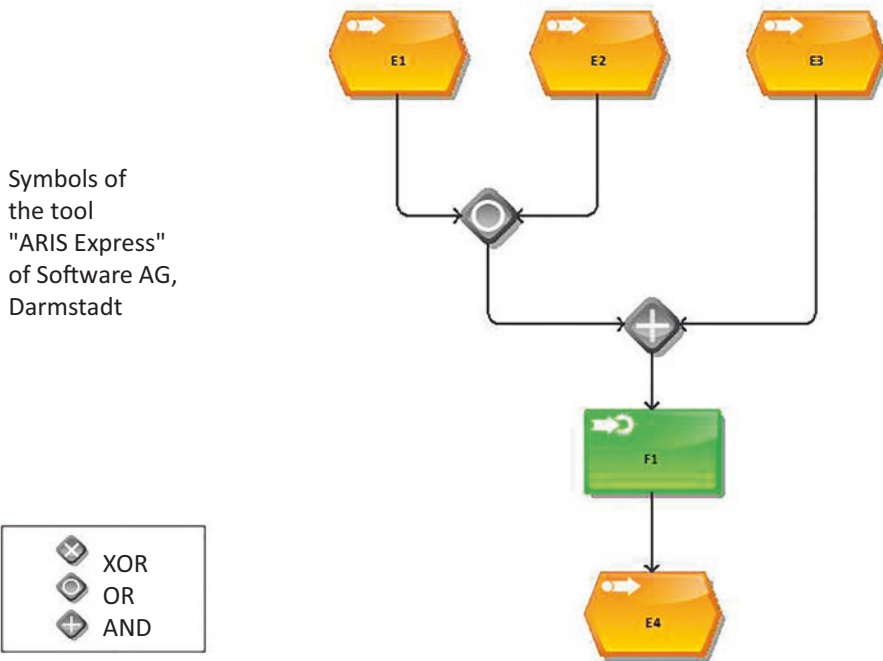


Fig. 5.34 Modeling example 2 with ARIS-Express (Software AG, Darmstadt)

Exercise for Fig. 5.28

Another EPC model is shown in Fig. 5.28.

Which of the statements are true?

- Function F1 is executed if events E1, E2 and E3 are activated at the same time.
- If function F2 has been executed, events E4, E5 and E7 have occurred before (Fig. 5.35).

Solution to the statements in Fig. 5.28

- Function F1 is executed if events E1, E2 and E3 are activated at the same time: Correct, but other combinations (e.g. only E1 or E1 with E3) are also conceivable.
- If function F2 has been executed, events E4, E5 and E7 have occurred before: Correct, but E6 has also occurred (because of the AND connector).

5.7.3 Extended Event-Driven Process Chain (eEPK)

5.7.3.1 Need for Extensions

The notation of the EPK introduced so far is not sufficient to create meaningful models for practical use. The method is only able to provide sufficiently detailed models for rep-

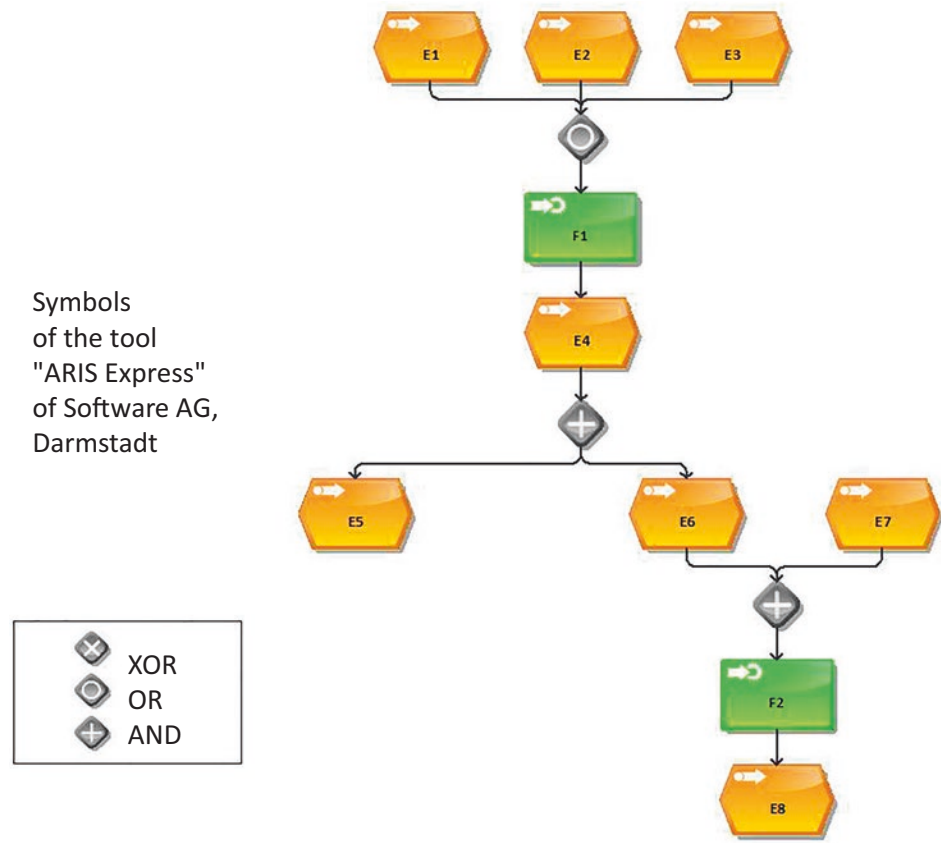


Fig. 5.35 Modeling example 3 with ARIS-Express (Software AG, Darmstadt)

representations of the flow logic. However, the use in practice requires more depth of detail, especially when introducing and revising information systems.

The EPK method has therefore been extended by several elements, the most important of which are the “organizational unit”; the “information object”, the “application system” and the “data flow”.

The **organizational unit** is used to describe the people, roles, positions, departments or external partners involved in a process, such as customers in the sales process, applicants in the acquisition of employees.

The **information object** maps the information (input and output) processed by the process, which are described in more detail in the data view (ERM model).

The **application system** is used to represent the information processing support of business processes.

The **data flow** is used to link function and information object and shows whether a function uses, changes or generates data.

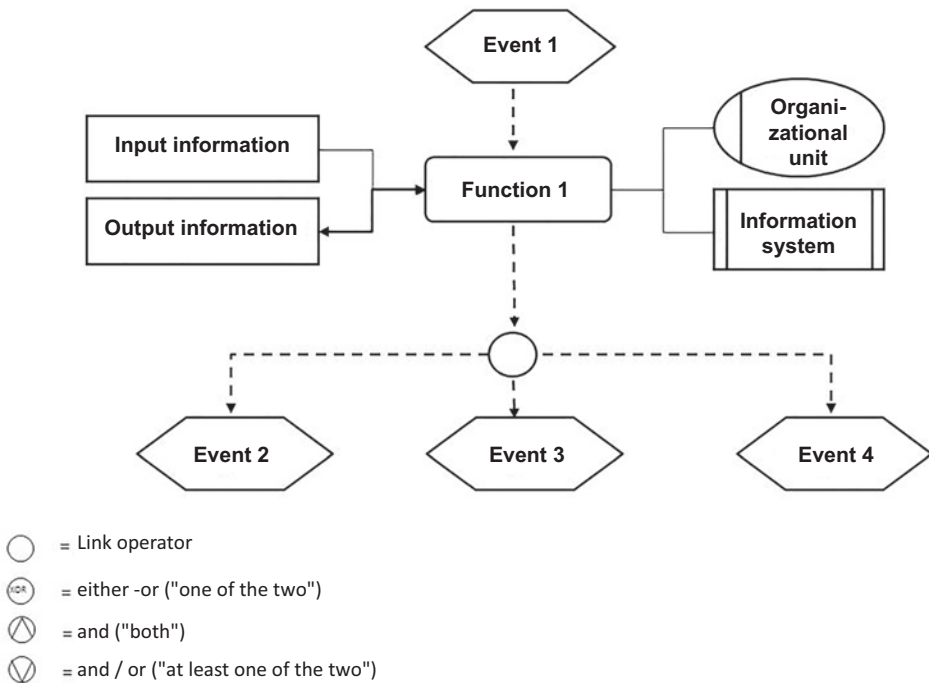


Fig. 5.36 Modeling elements of the eEPK according to Keller et al. 1992

Concept system

The complete concept system and the derived original notation according to Keller et al. (1992) is referred to as “Extended event-driven process chain” (short “eEPK”) (cf. Figs. 5.29 and 5.36).

The semantics of the symbols of the extended event-driven process chain is explained in Fig. 5.30 (Fig. 5.37).

Verbal and symbolic modeling

The connection between the verbal process description and the symbols of the eEPK is shown in Fig. 5.31. The respective bold terms of the textual process descriptions are assigned to the eEPK symbols (Fig. 5.38).

5.7.3.2 eEPK notation

The complete notation of the eEPK is summarized in Fig. 5.32 (see also Keller and Teufel 1997, pp. 166 ff.). The symbols can be divided into different categories: Event nodes (representation of events), activity nodes (representation of activities), condition nodes (representations of conditions that decide on the further course of work), organization nodes (representation of the participating organizational units), control flow edge

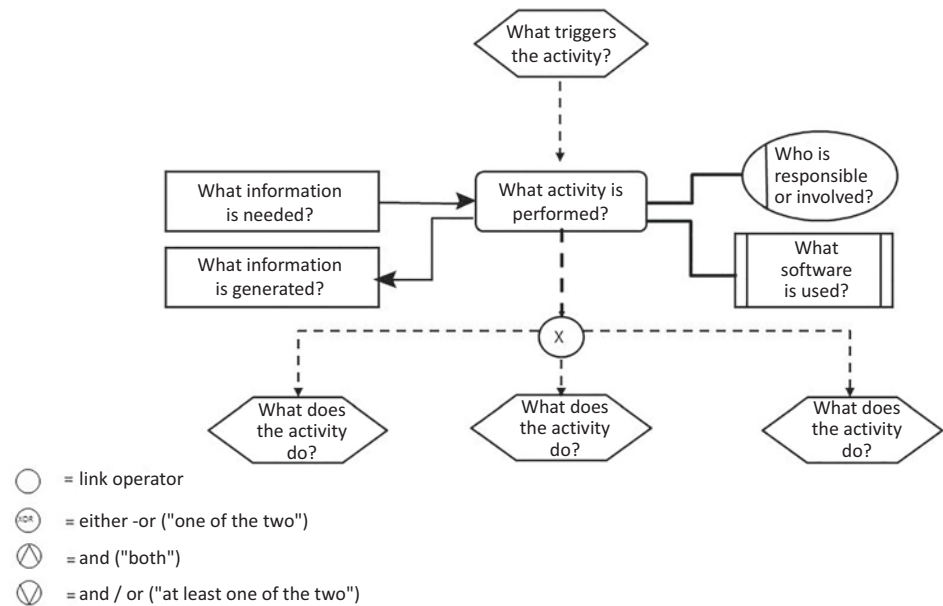


Fig. 5.37 Semantics of the eEPK

Process description and mapping of modeling elements to the eEPK:

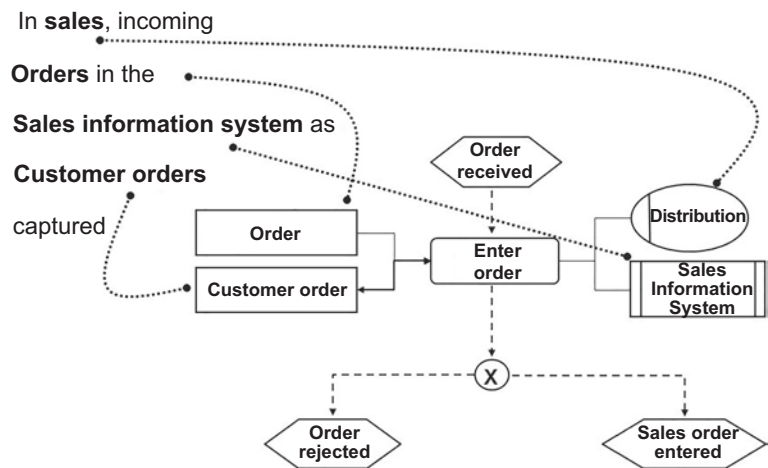


Fig. 5.38 Process description and assignment of modeling elements to the eEPK

(representation of the sequence of activities), data flow edge (representation of input and output relationships between information objects and functions) and assignment relationship edge (assignment of the organizational units involved in a function) (Fig. 5.39).

Symbol	Naming	Meaning	Edge/Node Type
	Event	Description of an occurred state, on which the further course of the process depends	Event node
	Function	Description of the transformation from an input state to an output state.	Activity node
	"exclusive or"	Logic operation operators describe the logical combination of events and functions	Condition node
	"or"		
	"and"		
	Organizational unit	Description of the outline structure of a company	Organization node
	Information object	Illustration of real world objects	Activity node
	Application system	Application systems for process support (e.g. SAP ERP)	Activity node
	Control flow	Temporal-logical connection of events and functions	Control flow edge
	Data flow	Description whether a function is read, written or changed.	Data flow edge
	Assignment	Allocation of resources/organizational units	Allocation relationship edge

Fig. 5.39 Notation elements of the eEPK

An example of the process “car rental contract conclusion” to be sketched below is shown in Fig. 5.33:

- After the customer’s arrival at the car rental center, the customer advisor enters the driver’s license data and additional information (insurance coverage) into the ERP system. If necessary, further drivers will be recorded who must hold a valid driver’s license. The driver’s license data of the other drivers will also be recorded.
- The customer can choose full insurance, partial insurance or no protection. If necessary, the customer advisor will take over the insurance coverage and the desired own contribution.
- The customer hands over his credit card to the customer advisor. The credit card is checked using the credit card billing system. The card data is transferred to the customer data of the ERP system. The rental contract is then printed out and handed over to the customer (Fig. 5.40).

5.7.3.3 Modeling Examples

The following two modeling examples are presented using the modeling tool “ARIS Express” from Software AG (Darmstadt). The slightly modified symbols can be seen in Fig. 5.34 (Fig. 5.41).

In Fig. 5.35 the following business process “creation of offers” is depicted with the “eEPK method”.

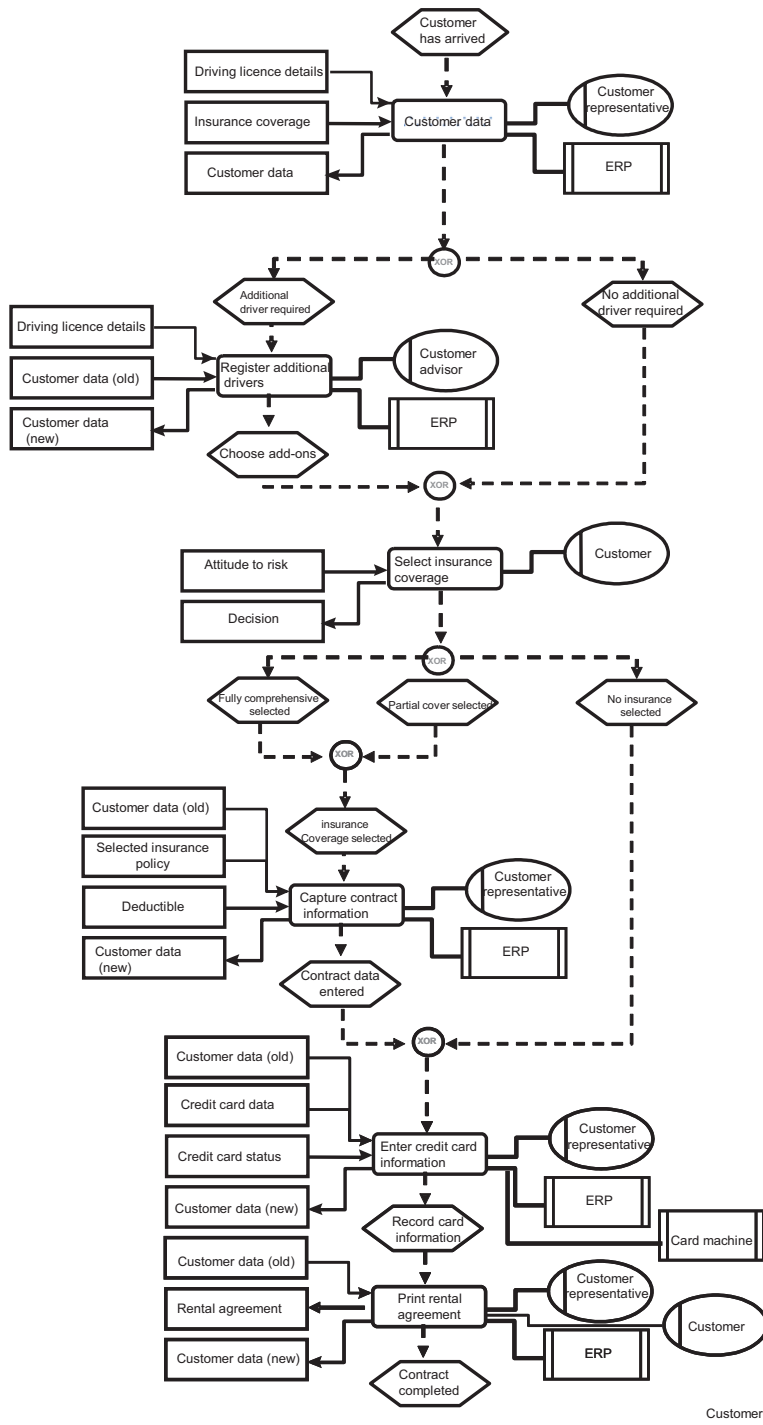


Fig. 5.40 eEPK example model “contract conclusion”

Standard symbol	ARIS Express Icon	Naming	Example
		Event	Customer has rental request
		Function	Conclude rental agreement
		"exclusive or"	
		"or"	
		"and"	
		Organizational unit	Train acceptance
		Information object (entity)	Rental agreement
		Application system (IT system)	MS Word, SAP ERP

Fig. 5.41 Notation elements of the eEPK of the tool “ARIS Express”

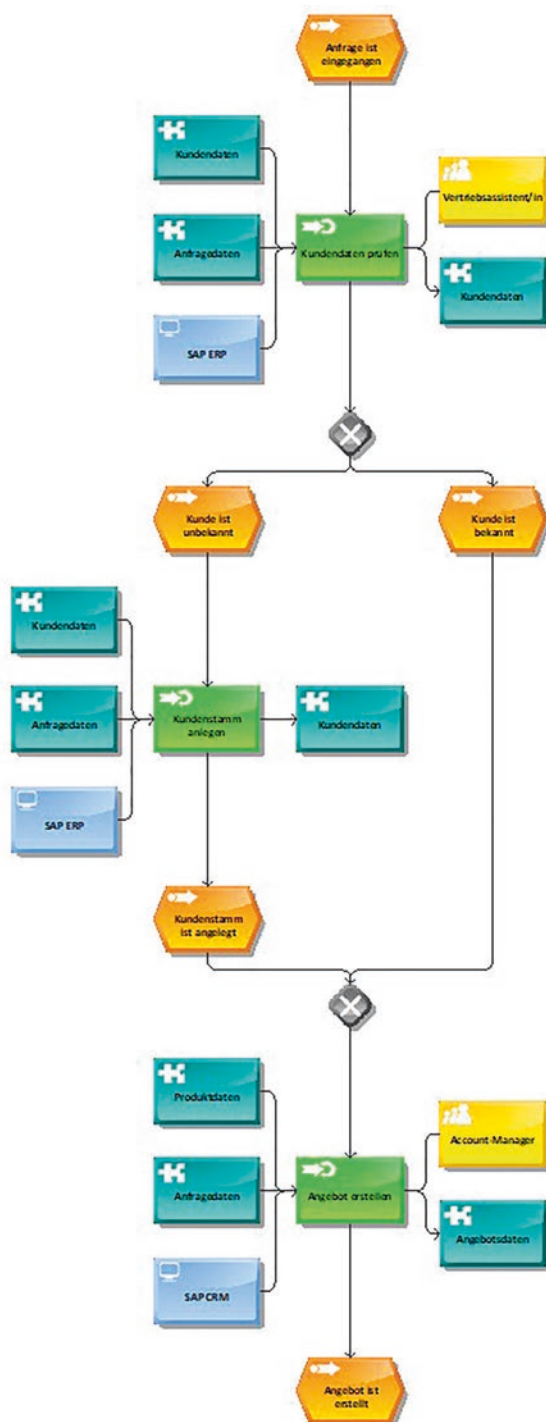
- After receiving the customer inquiry, the sales assistant checks using the “SAP ERP” system whether the customer is known. For this purpose, customer data and inquiry data are used.
- For new customers, a customer master record is created using “SAP ERP”.
- Then, using inquiry data and product data, an offer is created in “SAP CRM” by the account manager (Fig. 5.42).

The process “change bandage” was selected from the known case study on the takeover of the family doctor’s practice and modeled as an eEPK model with the Bic Design tool. The tool uses its own symbols, mostly in rectangular form. The entire model can be seen to the right in the representation and a somewhat more readable model section is shown in the upper left (cf. Fig. 5.43).

5.7.3.4 Evaluation of the eEPK

The eEPK has a fixed place as a traditional method in the German-speaking world. It is present above all where large and complex information systems are developed and maintained. It is embedded in the classical ARIS architecture and used by many companies. Due to its few symbols, it can be learned relatively quickly, which is also practiced at many universities and vocational schools as part of the economic informatics training. The focus is on the representation of the process logic taking into account participants (organizational units), data and information systems. Since it requires a lot of “space” for the models due to the many model elements, the use of tools is required to manage the models and, if necessary, to display details using the “zoom function” or the like.

Fig. 5.42 eEPK example
offer processing (with ARIS
Express)



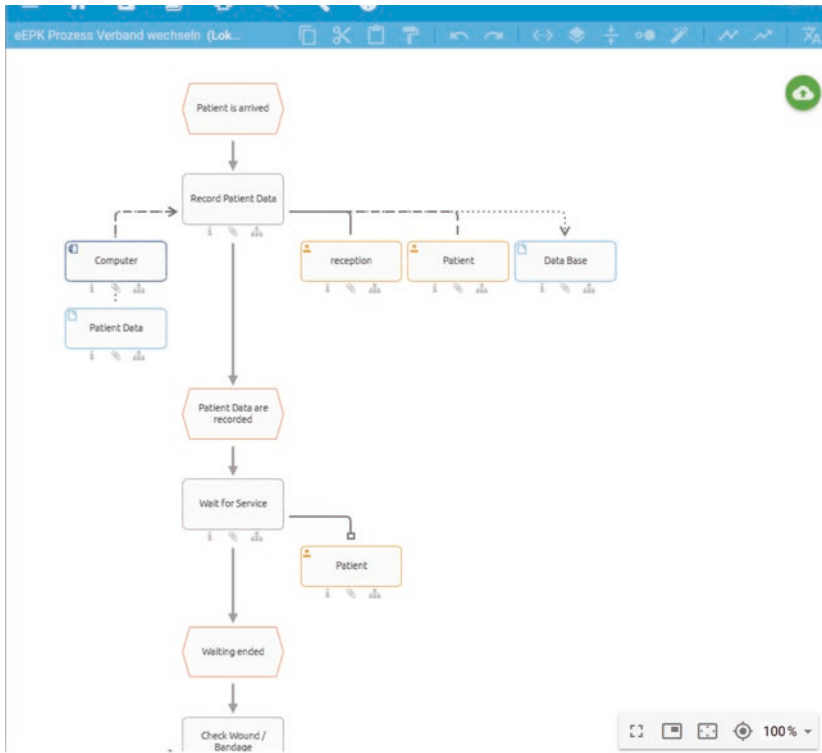


Fig. 5.43 eEPK model “change bandage”—modeled with Bic Design

5.8 Business Process and Model Notation (BPMN)

5.8.1 Overview

The Business Process Model and Notation (BPMN) is a relatively young and internationally standardized method for representing and executing processes. The current version BPMN 2.0 is the result of a long development. Important milestones of this evolution are shown in Table 5.1. The main new features of version 2.0 were mainly a significant expansion of the language scope and the introduction of executable elements (Spath et al. 2010, p. 16). The current version can be found on the OMG website (www.omg.org)

The spread of BPMN has increased significantly since the release of Version 2.0. In the analysis of Minonne et al. (2011), BPMN was already ahead of the previously leading modeling method eEPK by two percentage points with 49% in 2011. In a survey of the universities Bonn-Rhein-Sieg and Koblenz among management and specialist staff of IT management, the topic “BPMN” was positioned on place 5 of the current topics in

Table 5.1 Milestones in the development of BPMN 2.0

2002	Development of the method by IBM employee Stephen A. White
2005	Takeover of further development by the Object Management Group (www.omg.org)
2009	Presentation of Version 1.2
2010	Presentation of Version 2.0

BPMN 2.0

IT management (cf. Komus et al. 2016). In Switzerland, BPMN has been the standard method for companies and authorities for a long time (cf. eCH 2011). BPMN provides a very comprehensive notation with which technical and technical aspects can be mapped, which sets it apart from other methods.

5.8.2 Basic Notation

The essential symbols of the notation have been known for a long time and are based on the usual swimlane diagrams (cf. Sect. 5.5). The central elements of the BPMN notation can be seen in Fig. 3.16 (cf. White 2010). The symbols are also easy to understand for inexperienced users: rectangles describe activities, circles different event types (e.g. start or end), the diamonds known from flowcharts specify possible decisions in the process and edges the control and message flow.

The distinction between message and control flowallows to represent related processes and in addition to model the message flow when crossing organizational boundaries. In addition, special symbols are available for gateways (decisions), events (events), textual explanations, etc. (cf. Figs. 5.36 and 5.44).

A simple modeling example with the basic BPMN notation is shown in Fig. 5.37 (Fig. 5.45).

5.8.3 Activities

BPMN knows the basic form of an activity and numerous specializations (e.g. transaction, manual activity, call activity). Because of the large number of symbols and the complex notation, software tools are necessary for the use of BPMN. However, the scope of the specializations supported by the tools and their graphical expression do not always correspond to the original OMG specifications (OMG 2011, pp. 151 ff.). The examples in the book were carried out with the tool “ARIS Business Architect”, version 9.7 of Software AG (Darmstadt). They can also be created with the free tool “ARIS Express” of the same manufacturer.

Symbol	Naming	Meaning
	Activity (atomic)	An activity describes a process that is executed by the company. It can be atomic (task) or composite, i.e. contain subprocesses.
	Activity (with sub-processes)	
	Start event	Events are events that occur during a process. They can be triggering or the result of an activity. There are three basic types (start, intermediate and end) and special cases.
	Intermediate event	
	End events	
	Decision (Gateway)	Gateways are synchronization points in the process flow. They decide on the further course of the process. There are several gateway types: XOR, OR, AND and event-based decision.
	Control flow (Sequence flow)	The control flow describes the timing of the activities in the process
	Message flow	The message flow describes the exchange of messages between two objects (activities, events or decisions).
	Connection (Association)	The connection indicates that data, texts or other objects are connected to the control flow, e.g. input or output of an activity.
	Data Object	The data object indicates which information/data are required as input or output of an activity
Name		

Fig. 5.44 Basic notation elements of BPMN (cf. White 2010)

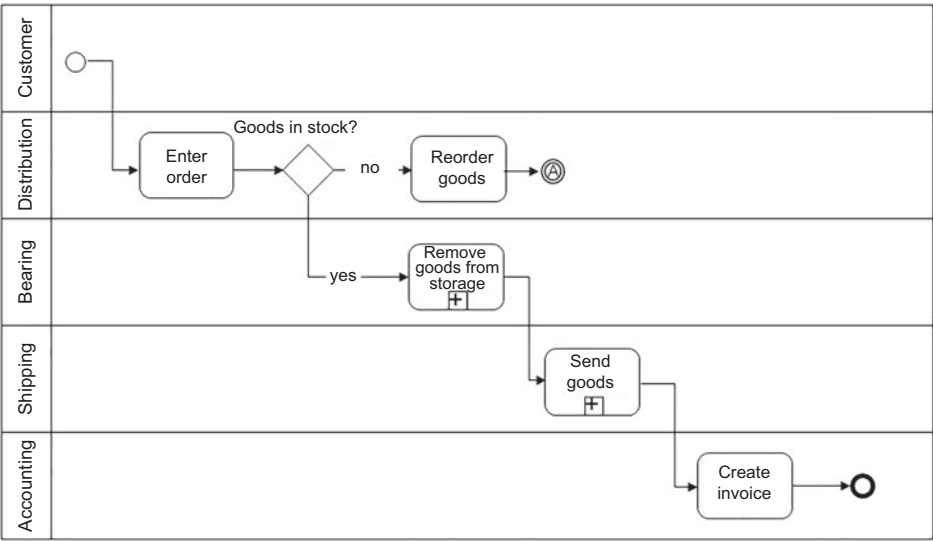


Fig. 5.45 Simple notation example with BPMN (cf. White 2010)

In Fig. 5.38 an example taken from Seidlmeier (2015, p. 170) is presented. It contains three manual activities (“Check invoice”, “Request invoice correction” and “Pay invoice”) (Fig. 5.46).

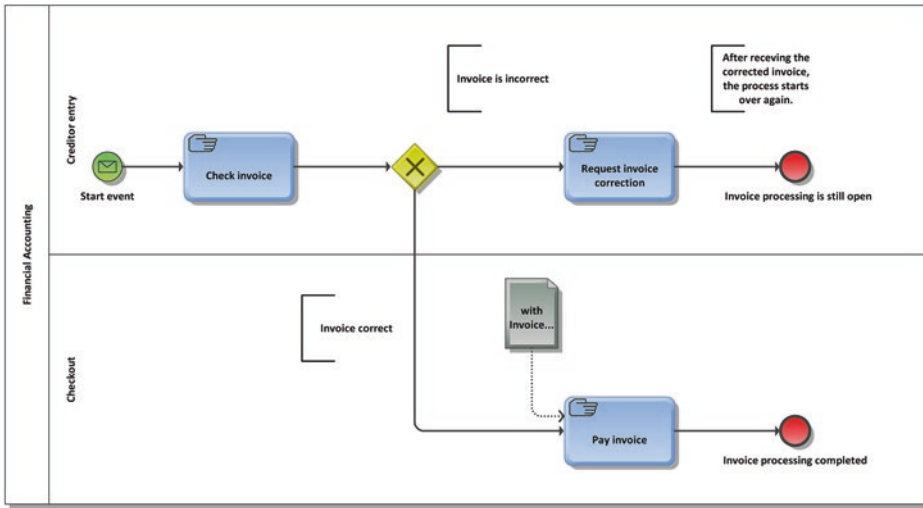


Fig. 5.46 BPMN—Example of activities. (Taken from Seidlmeier 2015, modeled with ARIS 9.7)

The control flow is represented by arrows and can be further specified by annotations (e.g. “account ok”). Compared to other modeling methods, BPMN offers the possibility to mark the “standard flow”, which usually occurs, in the model. In Fig. 5.39 a process from finance is shown, which uses this possibility.

After the activity “instruct payment”, the activity “settle payment immediately” is usually carried out for payments up to 10,000 €. This path of the control flow was marked as the “standard sequence flow”. The two other paths concern larger payments, for which separate paths are provided (Fig. 5.47).

5.8.4 Pools and Lanes

Since BPMN is based on the swimlane method, pools and lanes play a special role (see the schematic representation in Fig. 5.40). A pool represents a separate process. A lane describes details of a process within a pool, differentiated by organizational units, roles or IT systems (see OMG, pp. 109 ff.) (Fig. 5.48).

Fig. 5.41 shows a process divided into organizational units, which was taken from Allweyer (2015, p. 22). The division into organizational units is the most commonly used method in practice, according to the author’s perception (Fig. 5.49).

Between Pools (independent processes), messages can be exchanged that have an impact on each other process. As the application process in Fig. 5.42 shows, the processes taking place in the “Applicant” pool and in the “Company” pool influence each other. They each represent the view of the process participant on the same process (Fig. 5.50).

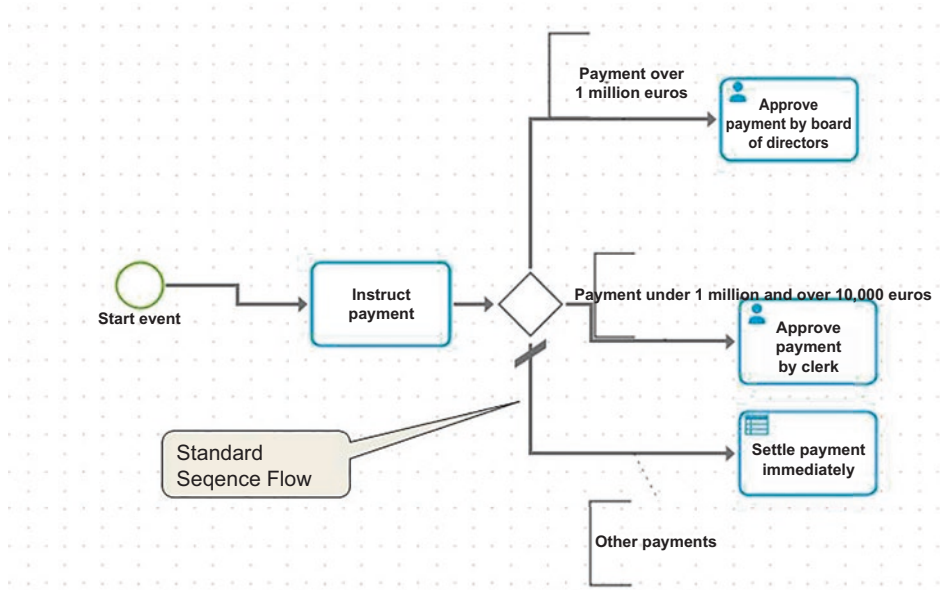


Fig. 5.47 BPMN—Example of standard sequence flow

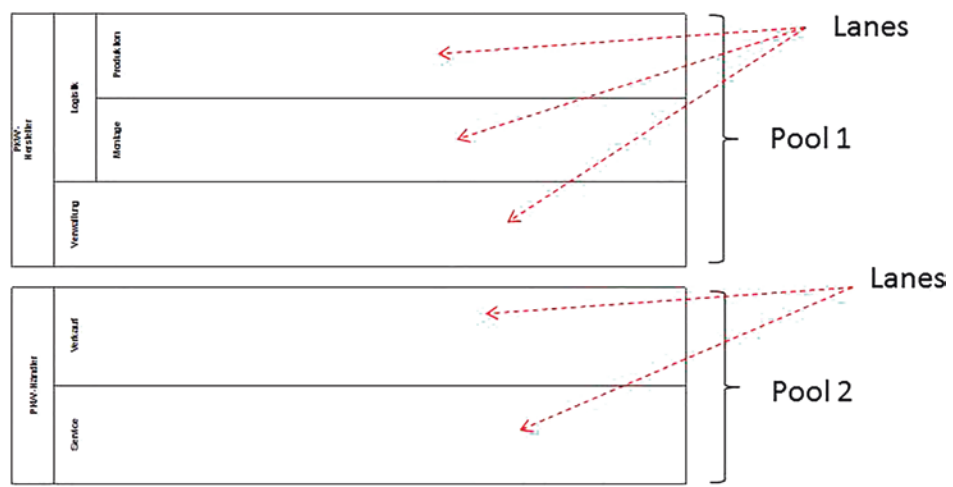


Fig. 5.48 BPMN—Pools and Lanes

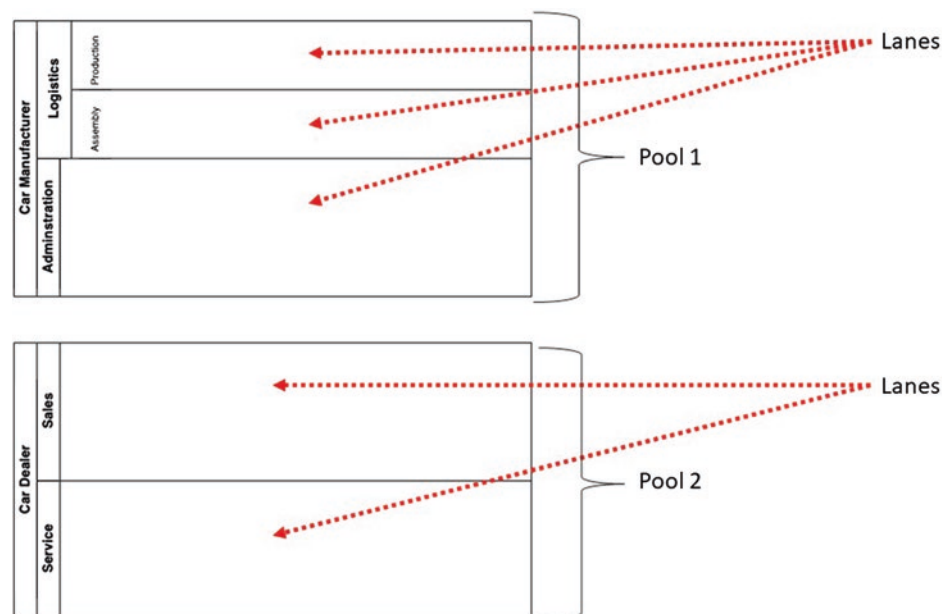


Fig. 5.49 BPMN—Pool with Lanes according to organizational units. (Taken from Allweyer 2015, p. 22)

5.8.5 Gateways

Gateways are used to represent possible branches (SPLIT) or mergers of paths in processes (cf. OMG 2011, pp. 287 ff.). They therefore form different variants that a concrete process can follow in a process model. From the eEPK method (cf. Sect. 5.6.2) the AND, XOR and OR connector are known, they are also used in the context of the BPMN notation. In addition, BPMN knows further variants, which are only briefly introduced here.

Exclusive Gateway (“XOR” Gateway)

The “Exclusive Gateway” corresponds to the XOR connector of the eEPK method. A path from several possibilities (selection 1 from n) for the further course of events (SPLIT) or the merging of several paths (JOIN) is selected. For the representation (see Fig. 5.43) two symbols are provided, which are not always supported together by tools (Fig. 5.51).

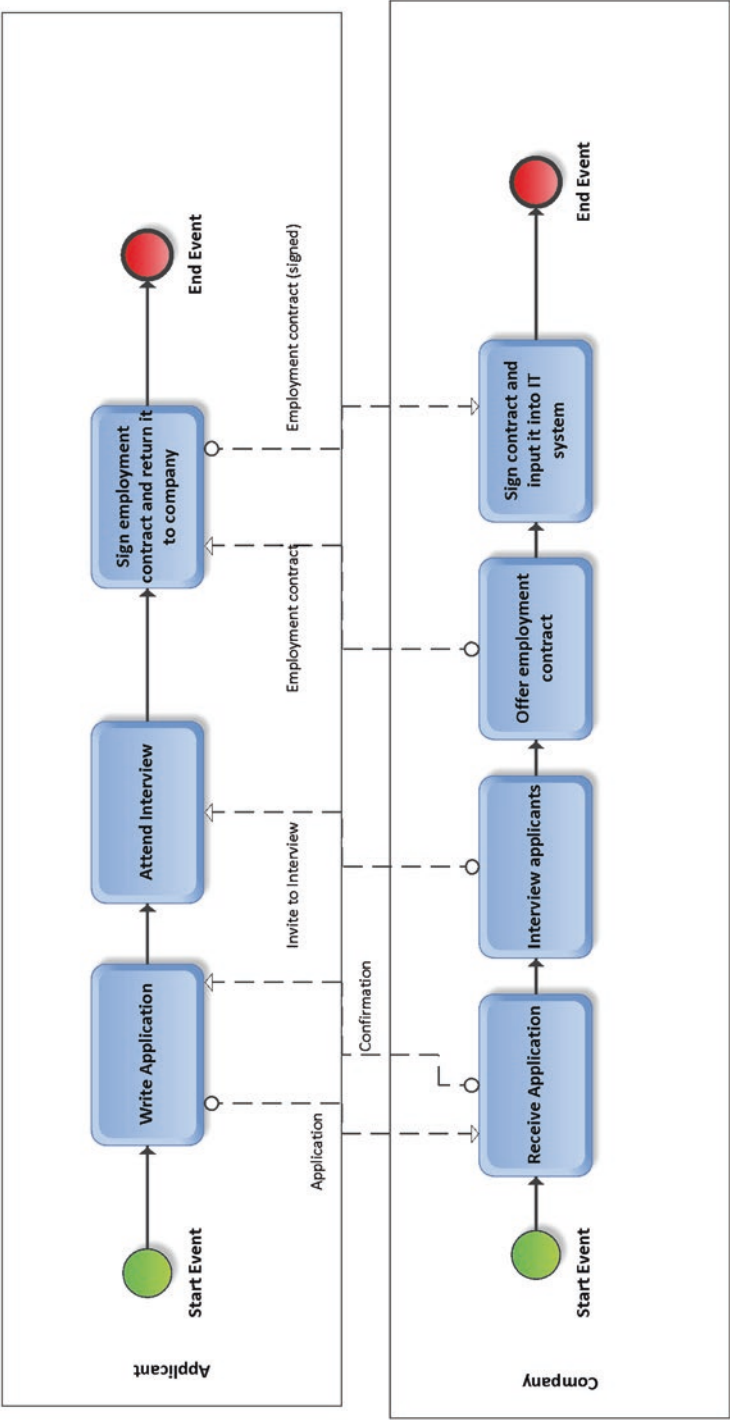


Fig. 5.50 BPMN—Message flow between Pools. (Simplified representation according to Allweyer 2015, p. 51)

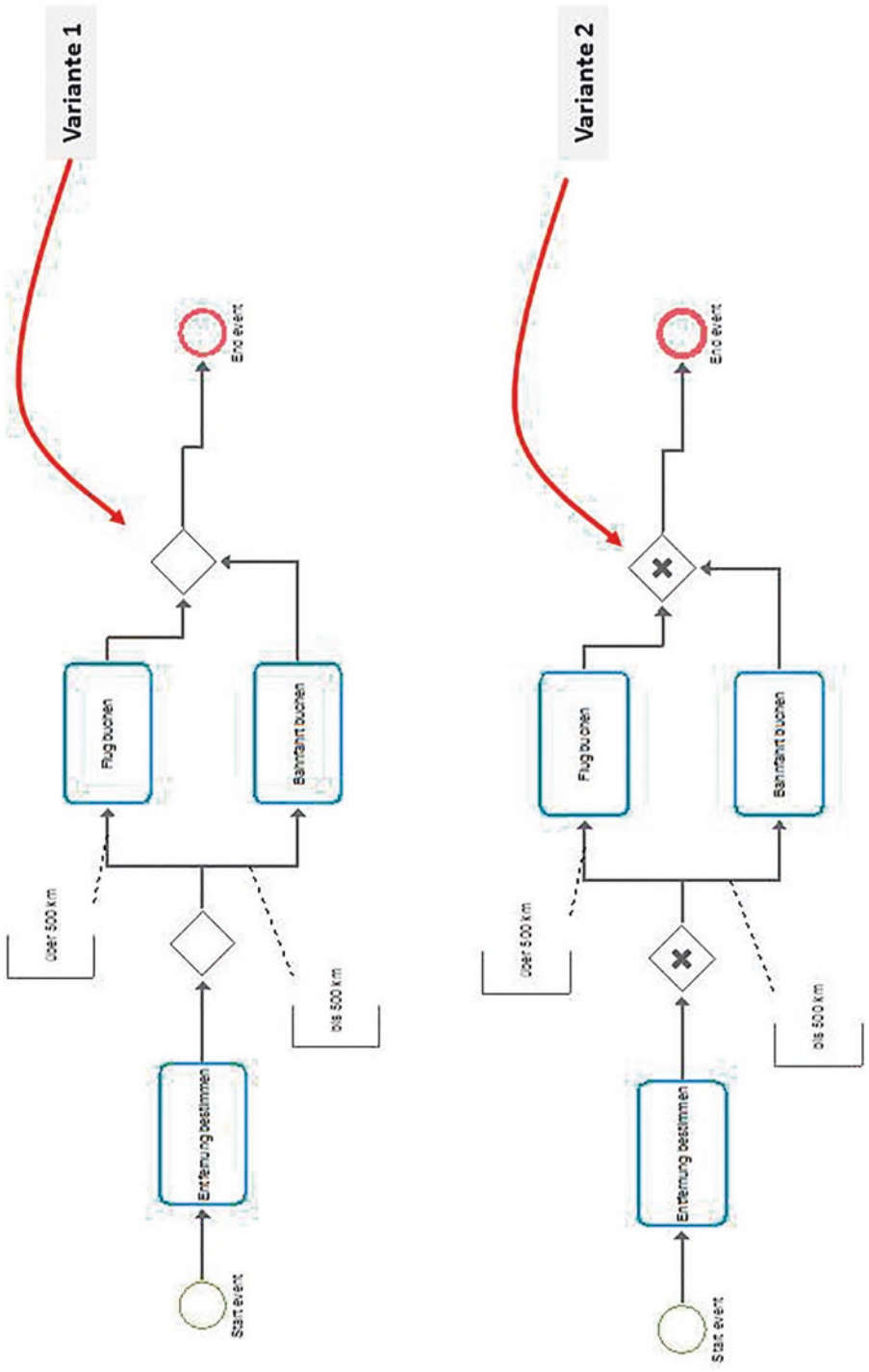


Fig. 5.51 BPMN—Exclusive Gateway. (XOR Gateway, taken from Allweyer, T.: BPMN 2.0, 3rd edition, Norderstedt 2015, p. 24)

Parallel Gateway (“AND” Gateway)

The “Parallel Gateway” (see Fig. 5.44) corresponds to the UND connector (selection n from n) of the eEPK method. The process is continued in all paths (SPLIT) or it is waited for all incoming path events (JOIN) until it is continued (Fig. 5.52).

Inclusive Gateway (“OR” Gateway)

In the inclusivegateway (cf. Fig. 5.45) one or more paths are selected. It corresponds to the “OR” connector of the EPK method (selection x from n , $x = 1, \dots, n$). In the process shown, after the “media for job advertisement” process step, several media can be selected, for example only the “homepage”, the “newspaper and the homepage” or any other combination (Fig. 5.53).

Complex Gateway

The complexGateway applies any (complex) rules. It is used when the classic gateways (“XOR”, “AND”, “OR”) cannot or only very unclearly map a situation. In practice,

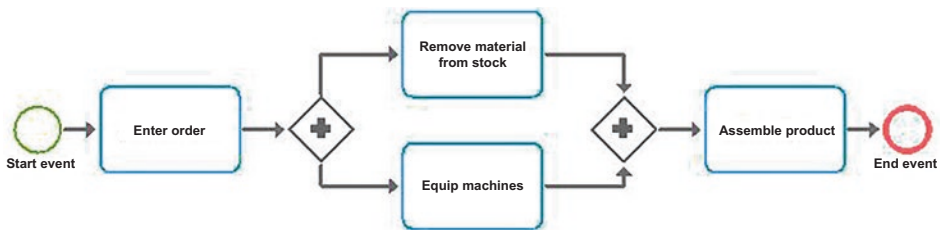


Fig. 5.52 BPMN—Parallel Gateway (AND Gateway)

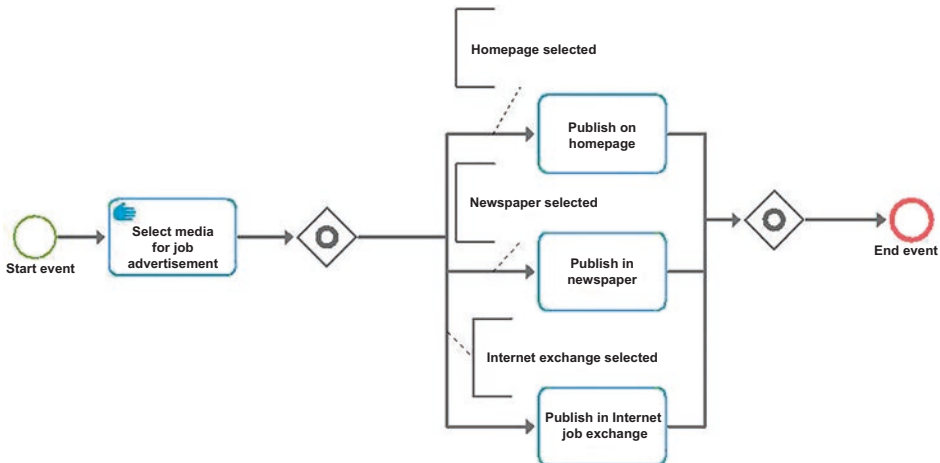


Fig. 5.53 BPMN—Inclusive Gateway. (OR Gateway, taken from Allweyer, T.: BPMN 2.0, 3rd ed., Norderstedt 2015, p. 32)

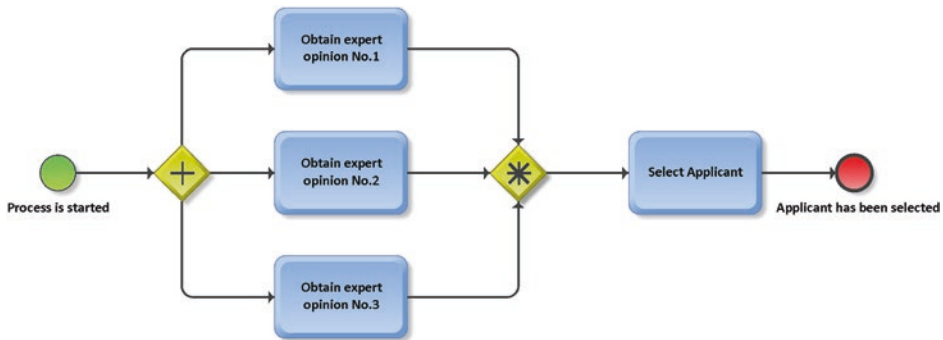


Fig. 5.54 Complex gateway. (Taken from Allweyer, T.: BPMN 2.0, 3rd ed., Norderstedt 2015, p. 37)

however, the complex gateway is rarely used because the technical implementation is difficult to realize. However, it can be used for expert modeling. In the process step “Select applicant” in Fig. 5.46 a rule is applied which is not specified in more detail in the model, in which the contents of the expert opinions on the applicant play a role (Fig. 5.54).

5.8.6 Data

BPMN is a modeling language for processes, i.e. the focus of modeling is on the control flow (sequence of steps) and the message flow between different process steps. In principle, modeling of data can be dispensed with if all data required in the pool is available.

An example of this is a pool whose process is fully supported by a BPM system (e.g. SAP ERP). If data is passed on between process steps (tasks) because, for example, different information systems are used, data objects have to be modeled. Example: An invoice is recorded in the sales system and transmitted to the customer. The invoice data is then electronically transmitted to an accounting system and booked there as a debtor’s account.

BPMN provides various symbols for data modeling (see OMG 2011, pp. 203 ff.).

- **Data storage** is used permanently by several process steps,
- **Data objects** is generated or processed by process steps.

In Fig. 5.47 data objects are shown as input or output of process steps. The “application” is input for the step “check application”, the “notice” is output from “grant notice” (Fig. 5.55).

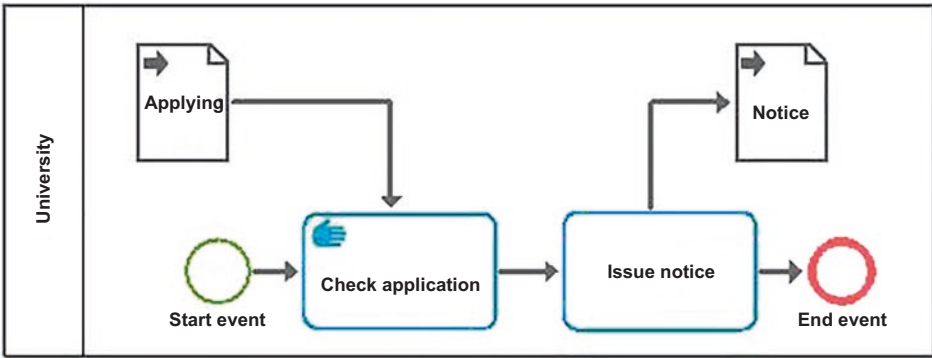


Fig. 5.55 BPMN—Data objects as input or output of process steps

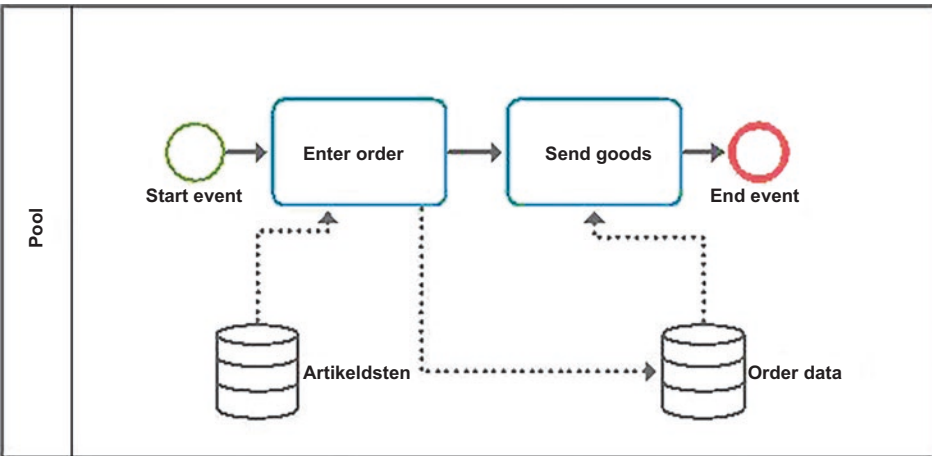


Fig. 5.56 BPMN—Data storage for multiple steps

Fig. 5.48 shows data storage, which was modeled as data objects in order to access multiple steps. The article data is read by the process step (task) “create offer”. The process step “enter order” writes to the data store “order data”, which data is read by the process step “ship goods” (Fig. 5.56).

5.8.7 Events

Start, intermediate and end events

BPMN knows numerous variations of events. The OMG distinguishes between start, intermediate and end events, which in turn occur in different variants (see [OMG 2011](#),

pp. 287 ff. and Fig. 5.49). Events can be used in the undefined state (standard) or defined for special situations. Start and end events must be, intermediate events can be modeled. They are only necessary if they have to be reacted to inside or outside the process. Otherwise, they only indicate a state within the process (Fig. 5.57).

The use of special events as start, intermediate and end event is shown in Fig. 5.50 using the process “Baking a cake in the oven”. The “Condition Event” comes into play when the oven temperature has reached a certain degree. Only then can the process continue. The “Time Event” provides a pause in the process so that the cake can remain in the oven (bake). The “Signal Event” at the end of the process indicates that the cake is ready and can be consumed (Fig. 5.58).

Modeling of messages

In BPMN, messages are also counted as events. Messages connect process steps that are dependent on each other. Fig. 5.51 shows the use of message flows using the example process “Cancellation of process parts” (taken from Allweyer 2015, p. 37). For an order,

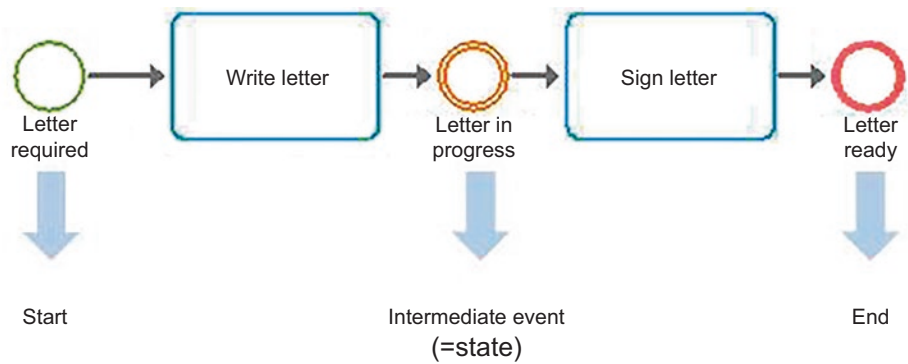


Fig. 5.57 BPMN—Standard events

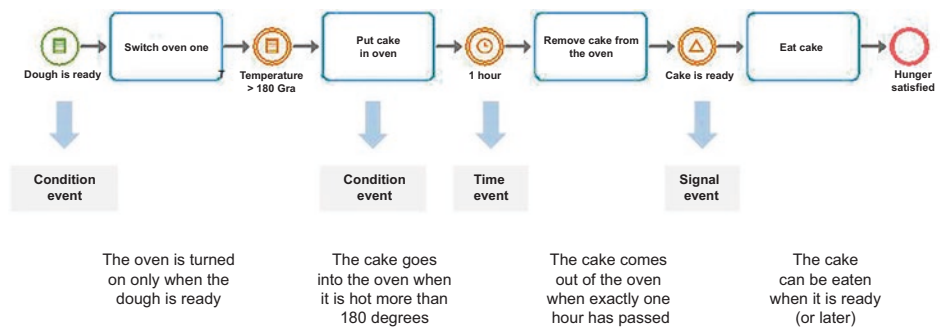


Fig. 5.58 BPMN—Special events

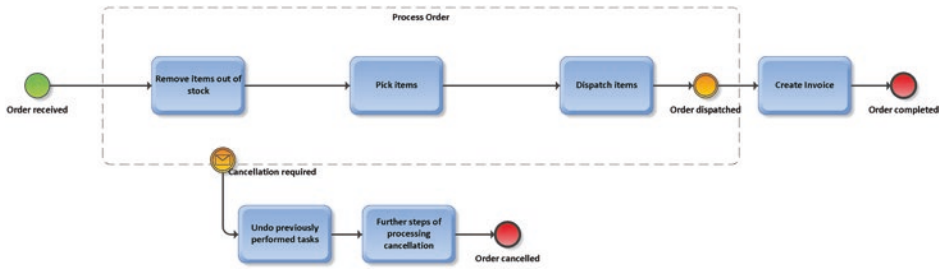


Fig. 5.59 BPMN—Use of messages to represent dependencies (cf. Allweyer 2015, p. 37)

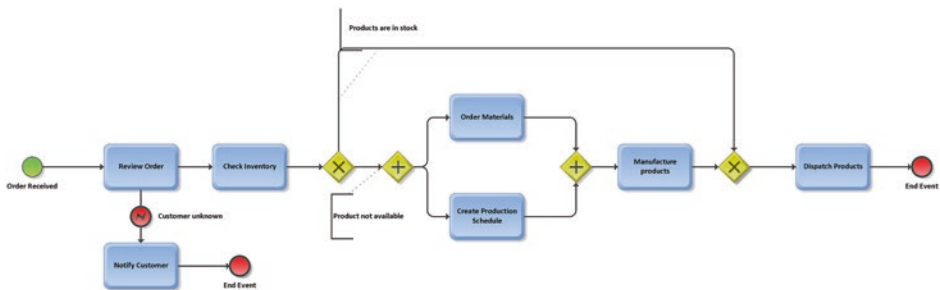


Fig. 5.60 BPMN—Error events (GI 2010)

material has already been stored; however, the further steps (e.g. commissioning of goods, shipping of goods) have not yet been started. Due to a miscalculation, the wrong order was processed. The process must be canceled (Fig. 5.59).

Modeling of error situations

Another application for special events are error situations. In the example of Fig. 5.52 an excerpt from a production planning is described: The customer specified on the order does not exist. The process cannot take place and must therefore be aborted (Fig. 5.60).

Multiple events

The modeling of Multiple events is in other modeling languages (e.g. eEPK with the “OR” connector) partly very expensive. In the present example of an application process in Fig. 5.53 one of the events must occur so that the application can be checked:

- Application as letter,
- Application as email or
- Application by telephone (Fig. 5.61).

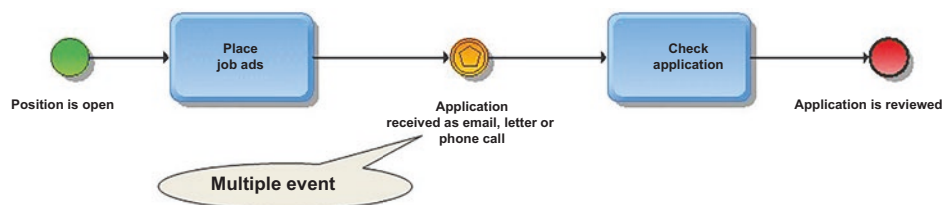


Fig. 5.61 BPMN—Multiple events

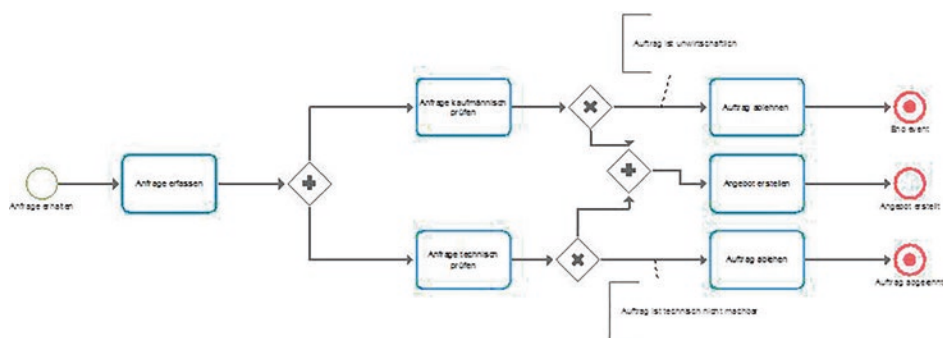


Fig. 5.62 BPMN—Termination of processes

Termination of processes

The termination (termination) of concurrent processes can be described in detail using BPMN. This is particularly interesting when events occur in the course of time that require already running parallel processes to be aborted. In the example in Fig. 5.54, the process ends as soon as it is not technically possible or the economic conditions are not given. Running process parts are aborted. For example, within three days, the technical manager could determine that the customer’s request is not realizable, and the ongoing commercial review would not be continued (Fig. 5.62).

5.8.8 Modeling Examples

The well-known process “Schedule appointment” from the case study of the family doctor’s office can be seen as a BPMN model in Fig. 5.46 using the Bic Design tool. In addition to start, intermediate and end events, activities as well as information objects (insurance card, appointment slip) and information systems (doctor’s office information system) have been modeled (Fig. 5.63).

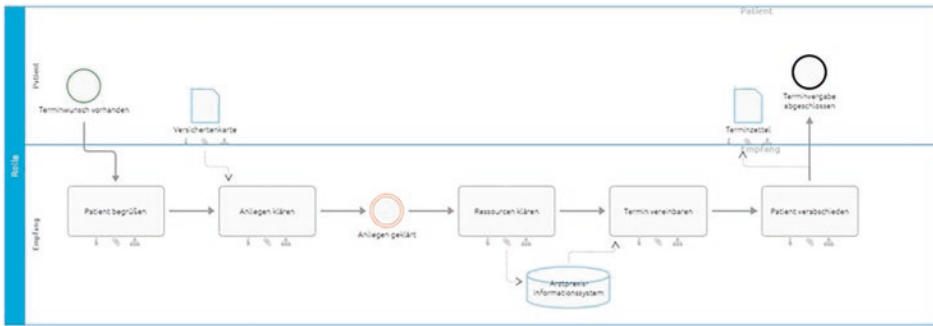


Fig. 5.63 BPMN model of the process “Schedule appointment”—modeled with Bic Design

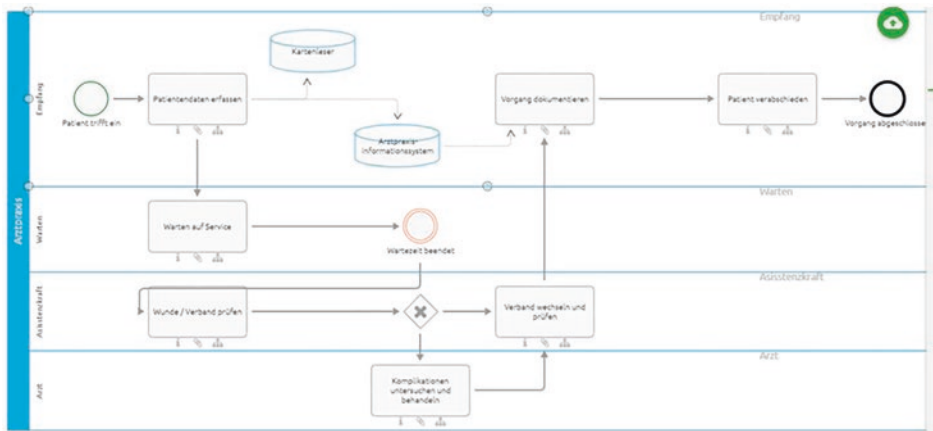


Fig. 5.64 BPMN model of the process “change bandage”—modeled with Bic Design

The somewhat more complex process “change bandage” in Fig. 5.64, also modeled with Bic Design, contains numerous details that can be represented with BPMN (information systems, branches, multiple lanes).

The example modeled with the “ARIS Express” tool in Fig. 5.55 was taken from Allweyer (2015, p. 32) and modified:

- Applicants write applications and send them by post or email to a company.
- First, the company confirms the receipt of the application and then checks it.
- Interesting applicants are invited by the company for an interview and receive an invitation with date and time, while unsuitable applicants receive a rejection.
- The invited applicants confirm the proposed appointment and take part in the interview.
- After the interview, the process is completed (Fig. 5.65).

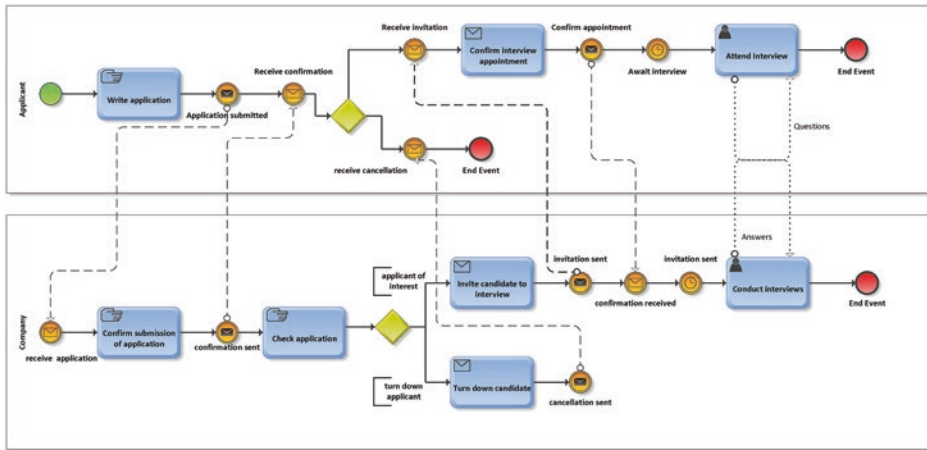


Fig. 5.65 BPMN—modeling example application. (Taken and modified from Allweyer 2015, p. 32)

5.8.9 Assessment

In comparison to the eEPK method, the better comprehensibility of the BPMN method is highlighted, since the simple basic symbols and the use of pools and lanes also make the process understandable for inexperienced users (cf. Krems 2016). A disadvantage is the very high level of effort required for familiarization in comparison to other methods when using the full notation.

If the method is only used for documentation, a selection must be made from the over 100 symbols. This can be done, if necessary, via a company-specific modeling handbook. The BPMN method can only unfold its full effect if it is not only used for documentation and business modeling, but also for technical modeling and execution of the models.

5.9 Simulation of Processes

5.9.1 Goals of Process Simulation

Process simulation is considered to be the central tool of process management for quantitatively analyzing processes and identifying improvement potentials (cf. the comprehensive literature review by Rosenthal et al. 2021). With simulation, i.e. the reproduction of reality in a model in order to experiment with it, three goals are pursued: the verification of the formal correctness of the process models, their realism and the evaluation of alternative process models.

1. Goal: Checking the executable nature of process models

The first goal relates to the verification of process models with regard to formal correctness and consistency. To achieve this goal, it is necessary to use the syntax provided for modeling, to take into account the semantics underlying it and to create an executable workflow model. The executable nature of a workflow model can be checked using a simulation tool. Achieving this goal does not yet allow any statements to be made about the content of the workflow models checked, but only about the question of whether the workflow model examined is executable and can also serve as a basis for execution by a WFMS.

2. Goal: Validation of the realism of process models

An important prerequisite for the application of the simulation is that the simulation model maps the reality in such a way that this section of reality is sufficiently reflected for the simulation goals (cf. Klügl 2006, p. 412). The second goal relates to the clarification of the factual correctness, i.e. the answer to the question to what extent a process model adequately maps reality. One way to validate a process model is that the results of the simulation experiments are based on an actual model, such as information on average cycle times, average capacity utilization, etc. with different observations in reality are compared. To validate an actual model, it is therefore necessary to have relevant actual data from reality that can be contrasted with the simulation results of the actual model.

3. Goal: Evaluation of alternative process models

The third goal is to provide information for qualitative process improvement. It is about clarifying the question to what extent alternative target models are suitable for improving the degree of achievement of process objectives, such as cycle times, capacity utilization or process costs.

In Fig. 5.56 the relationship between reality, model types (actual model, target model) and simulation goals is illustrated. Starting from reality, an actual model is created by its mapping. First, the formal correctness and consistency of this model is to be checked (1st goal) by checking the executable of the actual model. Subsequently, the actual model can be validated for its realism (2nd goal), provided that the formal correctness is given. For this purpose, simulation experiments can be carried out with the actual model and the results can be compared with the observable values of reality. After that, the actual model can be improved by, if necessary, creating several alternative target models and checking their executable again (1st goal). Thereafter, the target models are evaluated with regard to the achievement of the process objectives (Fig. 5.66).

5.9.2 Analysis Variables

The Fig. 5.67 shows a structure of analysis variables of process simulation, which serve to answer the questions listed above. Therefore, in process-related and resource-related

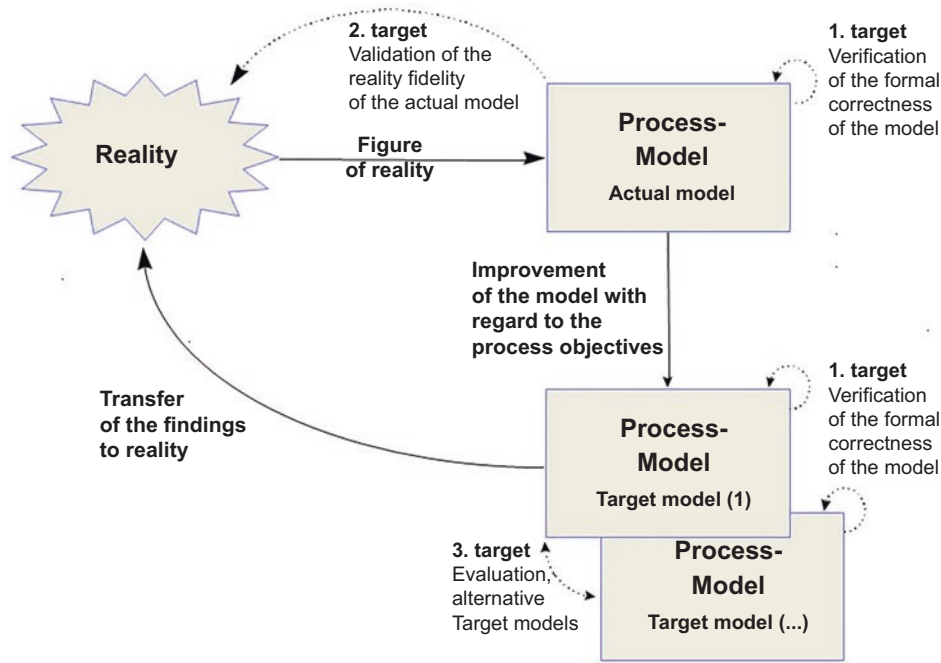


Fig. 5.66 Goals of process simulation

analysis variables is to be differentiated, which in turn can be divided into time-, value- and quantity-oriented variables.

With the help of process-related analysis variables can be generated in the context of a simulation run instances in terms of their process behavior be evaluated. The processing time of an instance describes the duration of execution from the beginning of instantiation of the simulated instance to the completion of the last process step. The processing time is often higher than the execution time, because, for example, due to insufficient resources waiting times can occur. The evaluation of the process execution with cost rates for the temporal use of resources results in the process costs for the execution of the instance.

Resource-related analysis variables consider the generated in the context of the simulation instances from the perspective of the required resources, ie in essence the personnel activity carriers (processor), but also the used computer resources (programs). Operating times and waiting times provide information about the utilization of resources, which, with the process cost rates evaluated, the utility or idle costs result. Downtime occurs due to unplanned non-use of resources (eg illness, prevention) and should be included in the idle costs. The processed or still to be processed by the used resources objects are described by object variables. The object input refers to the generated for a simulation run instances, the object output the actually processed during the simulation

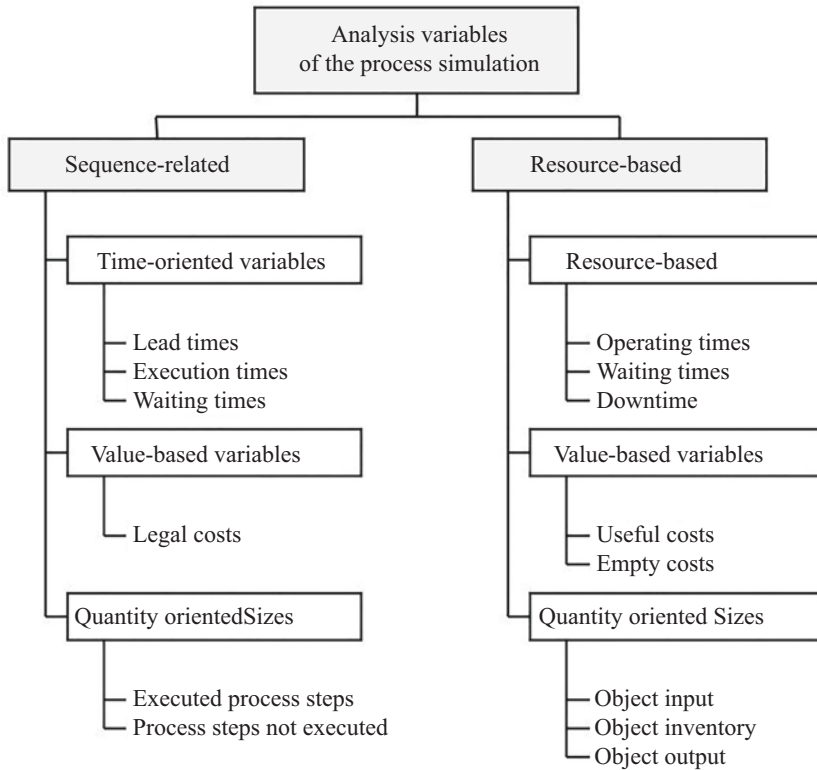


Fig. 5.67 Analysis variables of process simulation

run instances. The still in progress after the end of the simulation run objects identify the still unprocessed work in progress of a resource (Fig. 5.67).

5.9.3 Carrying Out a Simulation Study

The process of a simulation study can be carried out in seven process steps, it is described below using the example of the order planning.

1. STEP: SETTING GOALS

Before the investigation begins, the goals must be quantitatively defined. A suitable goal is, for example, the minimization of cycle times in engine assembly by the selection of suitable priority rules for the processing of workshop orders based on simulations.

2. STEP: INFORMATION ACQUISITION

This step serves to capture the relevant basic data including a plausibility and completeness check. In addition to processing and assembly times, output quantities, capacities of processing and assembly stations, as well as disturbance variables such as downtime or absenteeism, are to be recorded. The data is to be classified and condensed in such a way that an assignment to processing and assembly stations is possible.

3. STEP: MODEL FORMATION

For the execution of the test series, information is required as to what is to happen to the data. These are here processing sequences of the processing and assembly stations as well as the priority rules to be investigated.

4. STEP: IMPLEMENTATION

This includes the concrete capture of the model in the simulator, i.e. the data entry of the model in the computer. The level of detail depends on the specification of the goal.

5. STEP: VALIDATION

The model is to be checked to see if it still corresponds to the reality to be depicted. Preliminary simulation runs can be used for this purpose, which are then compared with already known results.

6. STEP: EXPERIMENTING (SIMULATION)

This is the actual simulation phase. The user systematically plays through various test series with varying parameters. The test parameters and results are to be documented. For example, several simulation runs can be carried out with different planning periods, capacities or buffer capacities.

7. STEP: RESULT ANALYSIS AND EVALUATION

The simulation software often only provides numerical results that need to be graphically processed for analysis. The evaluation can lead to changes in the model and to new test series.

It is important that during each simulation run, not multiple parameters (e.g. simulation duration and capacity data) are changed at the same time. Otherwise, the causes of parameter changes cannot be assigned to the effects of the simulation results.

5.10 Principles of Proper Modeling

Modeling content must not only be error-free, but also target group-oriented. For this purpose, the “Principles of Proper Modeling (GoM)” have been developed, whose term is based on the “Principles of Proper Accounting (GoB)” of accounting (cf. Becker et al. 1995; Scheer 1998, p. 198 ff.). The GoM include rules in the form of principles in order

to create high-quality and error-free models: Principle of correctness, principle of relevance, principle of economy, principle of clarity, principle of comparability and principle of systematic structure (Scheer 1998, p. 198 ff.).

Principle of correctness

A model is then **correct**, if it is syntactically and semantically correct, i.e. the notation is applied correctly and the model is behaviorally congruent with reality.

Principle of relevance,

A model is then **relevant**, if only those parts of reality are mapped in the model that are required for the purpose of the model.

Principle of economy

A model is then **economical** if the effort for the creation corresponds to the expected benefit. In practice, expensively created and complex actual models often violate this principle if too many unnecessary details and variants are created which can not be used for the target concept, or quickly become outdated.

Example of uneconomical process modeling

An international corporation has all actual processes from the top process at corporate level modeled down to the elementary level in all companies. However, it only uses the models for documentation and not for optimization. In addition, the models' actuality can not be guaranteed, which results in the fact that a few months after the actual data collection, reality looks completely different than the modeled processes. ◀

Principle of clarity

A model is then **clearly designed** if it is understandable for the recipient. In addition, models are to be appropriately structured into sub-models in order not to lose overview. There are overview models for management, detail models for clerks, technical models for workflow developers.

Principle of comparability

A model is **comparable** if the modeling languages used (eEPK, BPMN, etc.) can be traced back to comparable metamodels, i.e. have the same structure.

Example of comparable models

An "XOR" connector of the eEPK is comparable to the "XOR" connector in BPMN. On the other hand, there are some notation elements in BPMN that are missing in the eEPK (eg message flow). For this reason, models created with eEPK or BPMN are only partially comparable. ◀

Principle of systematic structure

A model is then **systematic** if different views (eg data view, organization view) can be integrated into an overall view (eg process view) and consistently modeled.

5.11 Selected Modeling Methods Compared

In practice, simple, non-formalized flow diagrams (63%), the BPMN 2.0 methods (49%), and the classical eEPK (47%) are used to a greater extent, with UML (20%) being used to a lesser extent (Minonne et al. 2011, p. 30). A flow diagram can be used to summarize all non-formalized methods composed of any number of diagrams (circles, rectangles, text, arrows, etc.). They are often used in conjunction with graphic programs. The high level of use can be explained by the low effort and simplicity of the application.

For ad hoc situations in conversation, the methods are quite useful for visualizing relationships quickly. For professional process management, the aforementioned other methods are usually used in later project phases when it is determined that several people need to access the process diagrams remotely and that the change effort is too high.

The variety of notations presented in this book is shown in Fig. 5.58. The value chain (WKD) is the simplest method. It only has rudimentary representation options. The swimlane method has additional notation elements for representing organizational aspects (departments = lanes) and to a certain extent process-related elements (yes/no decisions) compared to the WKD method. The eEPK method is able to integrate data structures and information systems in addition. The complex BPMN (here only in the basic notation) has the most comprehensive notation, as it covers both business process modeling and information technology implementation (workflow). UML plays a special role, as it was designed for software development and is therefore focused on the fine modeling of processes as a specification for programming. It is more of a tool for software architects than for process modelers. For this reason, a representation of UML was omitted here, interested readers can look up the method, for example, in van Randen et al. (2016) (Fig. 5.68).

When methods are compared with regard to the target group, modeling depth, standardization and dissemination as well as availability of tools, complexity of the method and the necessary training requirements, the picture in Fig. 5.59 shows. The potential areas of application result from the complexity. Simple methods, such as WKD, are more aimed at management and the specialist department, complex methods more at the IT departments. The amount of training correlates with the complexity of the method, i.e. the number of modeling elements (Fig. 5.69).

WKD, eEPK and Swimlane are not standardized. This is surprising, especially in the case of the eEPK method, which was developed as early as 1992. However, a standardization offensive started in 2016 apparently “came to nothing” (cf. Laue et al. 2021, p. 75). This makes it difficult to use in the company, as a modeling manual must fix the conventions beforehand. The eEPK is also more of a “German method”, it is used mainly in the German-speaking world.

Method	Event / State	Function / Process	Organization	Software	Data	Branch / Connector	Control, data, message flow	
WKD	 Start Process	 Process					 Control flow	
eEPK	 Event	 Function	 and more	 Application system	 Information	 AND XOR OR	 Control flow	 Data flow
Swim-lane		 Process step	 Lanes		 Document	 Rule	 Control flow	
BPMN	 Start-, Intermediate, Final event	 Activity	 Lane		 Document	 AND XOR OR Event Complex	 Control flow	 News flows
UML Activity Diagram	 Start	 End	 Activity				 Control flow	

Fig. 5.68 Modeling methods compared—notation

Method	Main target-Group	Modelling-rung depth	Standardi-sation	Spread	Tools	Complexity	Training needs
WKD	Management	Rough models	no		yes	Very low	minimal
eEPK	IT/ specialist department	Detail models	no	DACH countries	yes	high	high
Swim-lane	IT/ specialist department	Rough models	no	international	yes	very low	minimal
BPMN	IT/ specialist department	Detail models	OMG	international	yes	very high	very high
UML Activity Diagram	IT	Detail models	OMG	international	yes	high	medium

Fig. 5.69 Modeling methods in comparison—characteristics

5.12 Review Questions and Exercises

5.12.1 Questions

- Compare the modeling approaches of the “eEPK” and “BPMN” methods
- Explain the idea of the concept of the principles of proper modeling
- Name two principles of proper modeling of your choice and give a suitable example of how the principle can be violated

5.12.2 Exercise in Process Modeling “Treatment in the Hospital”

- **Task:** Model the business process “treatment in the hospital” with the eEPK or BPMN method.
- **Data:** The data of the patients referred to the hospital by a doctor are recorded by the hospital administration. Here, the patient data are transferred from the health insurance card and the admission form to the hospital information system (KIS). Subsequently, the examination and treatment are carried out by a doctor. For emergency patients, the capture of the patient data is only carried out after the treatment. The doctor is supported by an assistant, who records the diagnosis and treatment data as well as the prescribed medications in the KIS. Finally, the patient is discharged by the doctor. For this purpose, the patient receives a doctor’s letter.

5.12.3 Exercise in Process Modeling “Apply for Business Trip”

- **Task:** Model the business process “apply for business trip” with the eEPK or BPMN method.
- **Data:** Before the business trip, the applicant fills out the paper form “business trip application”. This is forwarded to the superior. The superior returns the signed application to the applicant. Occasionally it happens that the application is rejected (e.g. no budget left). In such a case, the application is returned to the applicant. If the applicant needs an advance, he fills out another form “advance application”, which he sends together with the business trip application to the travel office. The travel office then transfers the money to the applicant. For this purpose, it uses a travel processing software. After returning from the business trip, the applicant fills out a third form “account” and sends it to the travel office. The travel office then processes the account. If questions arise, these are clarified by telephone with the applicant. For the payment, the travel office uses the travel processing software again.

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Processes and IT belong together

Abstract

Process management is often associated with IT tools. First and foremost, process management is a method of better understanding and continuously improving work in the company. However, due to the complexity and variety of relationships, IT tools are required to document processes and also support them in operational operation. The contribution deals comprehensively with the possible IT support and presents the tools common in practice for process modeling and analysis, workflow management systems, enterprise resource planning systems, etc. Finally, the article addresses current aspects such as digitization, big data, cloud computing and Industry 4.0 with regard to the linking points to process management. Review questions and a case study support the learning process.

6.1 Tools for Modeling, Analyzing and Designing Processes (BPM-Tools)

6.1.1 Objectives and Concept

The software market for tools to support basic business process management has been characterized for years by a large number of manufacturers and a variety of products. Basically, tasks of visualization, modeling, simulation, process execution (workflow management) and system development (computer aided software engineering) can be

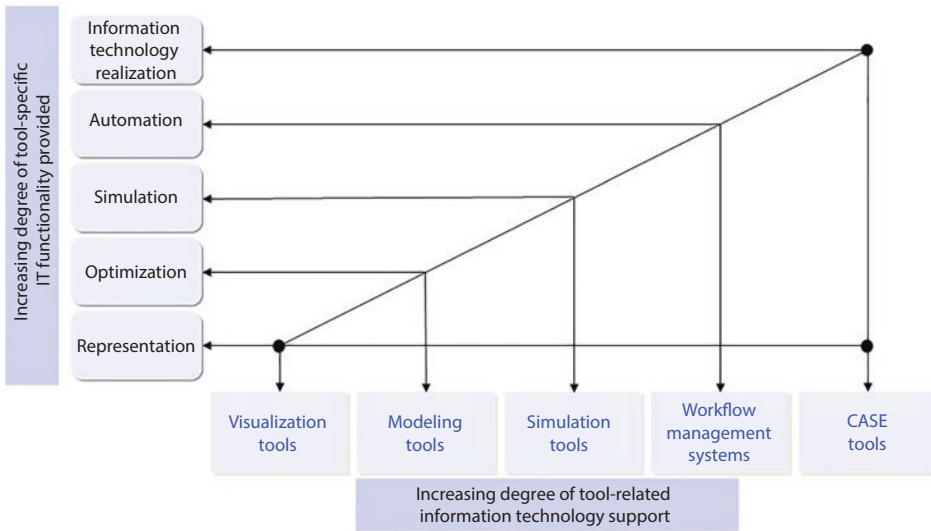


Fig. 6.1 Forms of tool support (Nägele and Schreiner 2002)

distinguished, which are covered by different product categories (Fig. 6.1). Tools with this functionality are also referred to as “BPM-Tool”/“BPM-Tool”.

► **BPM-Tool** A BPM-Tool is a software system that supports the modeling, analysis and documentation, possibly also the simulation of processes. The use is on standard hardware (in particular laptops, desktops and partly on tablet computers) using file or database systems for storing the model data. Powerful BPM-Tools offer in addition to the modeling still possibilities for the review of the models (syntax and consistency).

The graphical preparation and documentation of processes is provided by many tools in different form and quality. The range extends from the pure “painting program” to the database-based modeling tool, which supports a variety of methods.

The modeling and simulation of processes is a domain of products specialized hereon. The automation of processes is carried out by workflow management systems, which are often referred to as “business process management systems” or “BPMs”.

So-called CASE-Tools (CASE stands for Computer Aided Software Engineering) support the development and testing of information systems, i.e. the process of providing information systems. For many of the mentioned product categories, free products are available that can be used at least for teaching and learning purposes and with limitations in practice process management.

The selection of a suitable tool can be done in two steps using company-specific criteria. The first step is to clarify whether a pure documentation tool is needed or whether support for process execution is required.

Criteria for documentation tools

Concerning documentation, the tools must cover functions for modeling and representing processes. In particular, the two best-known modeling standards BPMN and eEPK are of importance here. In addition, attention should be paid to the possibility of importing and exporting model data. As general important criteria, the usability, i.e. the simplest and most intuitive operation of the tools, also applies.

Criteria for process execution

BPM-Tools with an execution component compete with enterprise resource management systems as process-supporting systems. Therefore, it is important to choose products that support a high stability of IT operations and at the same time are flexible enough to react to changes without changing the program.

6.1.2 Selected Modeling Tools

The market for modeling tools is very comprehensive. The number of products offered worldwide is probably in the lower three-digit range, as the website www.bpmn.org lists numerous tools that support the BPMN notation (cf. Allweyer 2014, p. 19). Meanwhile, in addition to the classical on-site installation of purchased licenses on own hardware (on premise), various cloud-based usage models are offered. They range from the public cloud solution to the self-operated cloud in the company's own data center (on-premise cloud in the company's own data center). For first "attempts" with modeling tools, the public cloud solutions are usually sufficient, but they are often not suitable for productive use. However, the manufacturers usually offer paid versions with more extensive functionality. Here, in particular, aspects of data protection, operational safety and questions of administration come into play.

The Table 6.1 gives a short overview with a short description of some selected tools for process modeling.

6.2 Tools for the Control, Automation and Machine Analysis of Processes**6.2.1 Workflow Management Systems (WFMS)**

Process management is more than just the graphical modeling of processes. This aspect is often forgotten in practice. Processes must also be executed and controlled in everyday life. For this purpose, technical support is required that goes far beyond the graphical modeling and representation of processes. Workflow management systems (WFMS) play a key role here. They support the modeling, simulation and, above all, the execution and monitoring of business processes at the workflow level of detail.

Table 6.1 Overview of selected tools for business process management

Tool	Manufacturer	Description
Adonis Community Edition	BOC	Modeling with BPMN and other notations such as process map. Limited free version
ARIS Business Architect	Software AG, formerly IDS Scheer	Database-based modeling with a very large number (>100) of notations such as eEPK, BPMN, process map, in addition data modeling, function modeling, etc. concepts
ARIS Express	Software AG, formerly IDS Scheer	File-oriented modeling with a limited selection of notations such as eEPK, BPMN, process map, in addition data modeling
BIC Design	GBTEC	Modeling of different notations such as eEPK, BPMN
Bizagi Modeler	Bizagi	Part of the Bizagi Suite, which also supports process execution (Bizagi Studio and Engine)
Blueworks Life	IBM	Modeling tool of the workflow management system “IBM Business Process Management”
iGrafx	iGrafx	Modeling of BPMN and other flowcharts (including the relatively rare IDEF0 diagrams)
Innovator	MID	Modeling of BPNN and other notations (including the relatively rare IDEF0 diagrams)
Signavio Process Editor	Signavio	Tool specifically developed for BPMN, which also supports EPK and value chains
TIBCO Business Studio	TIBCO	Modeling component of the TIBCO workflow system

Need for WFMS

The use of WFMS is not useful for all business processes. The process to be supported by a WFMS must be at least partially automatable and should take place regularly. A typical example is order processing in the insurance industry. One-time processes are not useful to support by WFMS. The higher the proportion of repetitive activities, the more useful WFMS are. The complexity (structure) of the processes can vary, however. WFMS are more likely to be useful for highly structured processes because they describe and document the process logic in the form of workflow models. But also simple, less complex processes that run several times a day are suitable for support by WFMS. Examples of this are processes of application processing. Simple processes that are carried out only 1–2 times a month, however, are less common.

In recent literature, WFMS are also referred to as process management systems (PMS) or business process management systems (BPMS). Dadam et al. (2011, p. 364) distinguish in this context form-oriented, document-oriented and service-oriented process management systems (PMS).

Form-oriented PMS are used to display the contents of database tables. Document-oriented PMS support the display and editing of documents, such as incoming invoices.

Service-oriented PMS can execute any service for each process step, which in addition to the two mentioned tasks also includes the call of external applications (e.g. an ERP system).

► **Definition: Workflow Management System** A workflow management system is an application-independent software system that belongs to the middleware area and supports the modeling, execution, and monitoring of workflows, as well as additional functions such as workflow simulation and analysis; in particular, it is able to interpret (semi-)formal workflow specifications, trigger the execution of process steps by the intended activity performers—employees or application programs—and, if necessary, provide the necessary work instructions, tools, application programs, information and documents (Gehring 1998).

The basic operation of a workflow management system is shown in Fig. 6.2. A workflow consisting of several workflow steps (here “order processing”) is supported by different people to a certain extent, but also by different applications. Partially automated workflow steps with personal interventions, but also a fully automated workflow step are shown. The applications are supported to a certain extent by means of classical office products, but also by means of ERP systems or self-developed database solutions.

Areas of application

WFMS can be used for any type of workflow in principle. The focus of application is currently mainly on commercial and administrative business processes or office processes, while production-technical processes, for example, are supported by production

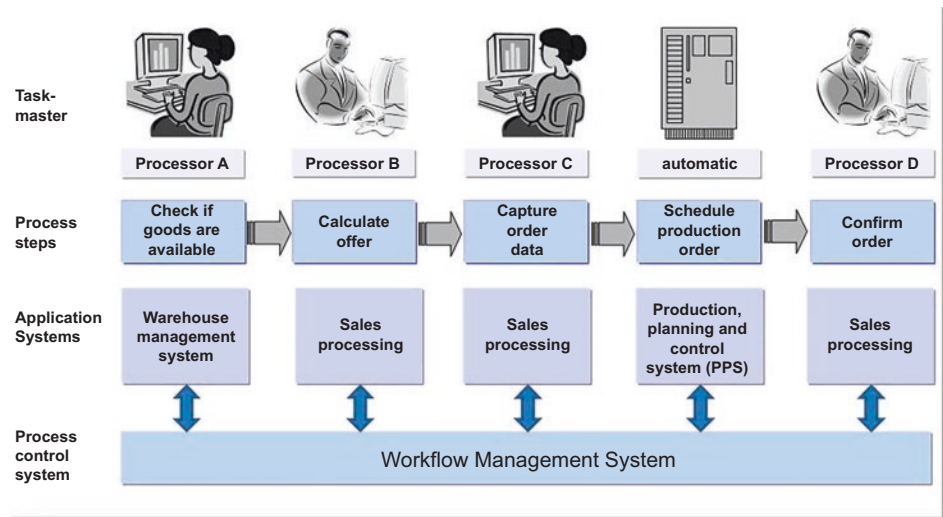


Fig. 6.2 Basic principle of a workflow management system

planning and control systems and production control systems. However, there are several approaches that aim at a merger of these two system areas, which are still separate, in order to enable a continuous IT support for administrative and production processes (cf. Loos 1997 or Lassen and Lücke 2003). The high similarity of the basic functionality of workflow management systems and production planning and control systems makes it possible to transfer classical, mature PPS methods, e.g. capacity planning and balancing, process termination or a load-oriented role resolution, to workflow management. Lassen and Lücke concluded in their study that an integration of PPS systems with WFMS leads to an improvement in the planning and control of business processes, since integration deficits in order processing can be reduced (cf. Lassen and Lücke 2003, p. 20, extended).

Selected WFMS

The market for workflow management systems or BPM suites is as comprehensive as that for pure modeling tools. The study published by Adam et al. (2014) at the Fraunhofer Institute for Experimental Software Engineering (IESE) in Kaiserslautern shows the central results of a comprehensive market analysis. As part of the study, 20 BPM suites were analyzed and evaluated. The aim of the study was to compare a selection of relevant tools in terms of their power and comfort from the perspective of BPM experts and BPM users. The weighted degree of fulfillment was evaluated in relation to a standardized requirements catalog that had been developed beforehand, the power of the software in terms of adaptability with standard tools and the comfort of the provided functionality (Adam et al. 2014, p. 6).

The products analyzed in the study are listed in Table 6.2. It is concluded that all products have a high power. Regarding comfort, the study showed that only very few products offer a high comfort in all individual aspects, the spread is much higher. The product of the company Bizagi receives the best overall rating. At the same time, it has the highest comfort of all the tools examined. The product of the company SoftProject received the best rating for power (Adam et al. 2014, pp. 124–125).

- Workgroup computing, which goes back as far as Krcmar (1993), can be distinguished from workflow management as a special case. Workgroup computing, or more recently “social workflow”, aims to replace internal email communication and support internal company communication with social web functionality (like “Facebook”). The functionality includes the following processes

- Common editing of documents and projects
- Chat, audio/video conference, webinar, newsfeeds,
- Tagging/liking/following of files and people,
- Commenting on messages,
- Wikis, group calendar, surveys.

Table 6.2 BPM suites of the Fraunhofer market analysis (Adam et al. 2014)

Tool	Manufacturer	Description
AgilePoint iBPMS	AgilePoint Inc	Owner-managed US company, strong integration with Microsoft products
Agito BPM	Agito GmbH	Small Berlin company, only on the market since 2011
Appian	Appian Software GmbH	Family-run US company, product has been awarded several times
Appway Plattform	Appway Numcom Software AG	Swiss company, product has been awarded several times
Axon.ivy BPM Suite	AXON IVY AG	Swiss company, spin-off of Landis + Gyr and ETH Zurich, product has been awarded several times in highest categories
Bizagi Suite	Bizagi Ltd	Company founded in Colombia, today headquartered in the UK, 2014 multiple times as “Finalist” for BPM awarded
DHC Vision	DHC Business Solutions GmbH & Co. KG	Saarbrücker company, focus is on process support and regulatory requirements
@enterprise	Groiss Informatics GmbH	Austrian company, was awarded by the Swiss Railway in 2013
HCM VDoc Process	HCM Customer Management GmbH	Stuttgart company, on the market since 2000, product received the “Best of Industrie-IT” award in 2014
IBM BPM	IBM Deutschland GmbH	International IT corporation, product received multiple international awards from analysts
BPM inspire	Inspire Technologies GmbH	German company, on the market since 2008, medium-sized company prize and TÜV certification of the product
JobRouter	JobRouter AG	Mannheim company, on the market since 1993, focus on providing a development platform, Innovation Prize of the Mittelstand
K2 blackpearl	K2 Northern Europe GmbH	South African company with several thousand customers worldwide, numerous international awards
Metasonic® Suite	Metasonic GmbH	German company from Pfaffenhofen, on the market since 2004
ORACLE BPM Suite	ORACLE Deutschland B.V. & Co. KG	International IT corporation, comprehensive platform for BPM based on various standards (BPMN, BPEL, etc.)

(continued)

Table 6.2 (continued)

Tool	Manufacturer	Description
FireStart	PROLOGICS IT GmbH	Austrian company, on the market since 2006, Look and Feel of Microsoft applications
X4 BPM Suite	SoftProject GmbH	Ettlingen software house, on the market since 2000, numerous adapters for third party software integration
T!M—Task !n Motion 4.0	T!M Solutions GmbH	Freising-based company, on the market since 2007 market, several recent awards of the product

Current typical products with a specification of the core functionality are, for example:

- Jira (workflows, tickets),
- Trello (Kanban board, project management for smaller teams),
- Asana (like Trello, but for larger teams and more functionality)
- MS Teams (collaboration platform).

6.2.2 Robotic Process Automation (RPA)

The automated processing of process steps is the subject of many information technologies that have been in use for a long time. Many organizations are facing enormous cost pressure and need to optimize processes. Often, repetitive, long-established processes are in the focus, which cause high personnel costs through manual work and data entry in connection with older information systems. The current shortage of skilled workers exacerbates this effect.

As a possible instrument for process acceleration, software robots (Robot Process Automation or “RPA” or also “bots”) can be used, which automatically carry out the tasks of the persons involved. The human input on the user interface (GUI) of the application is simulated by a program (i.e. a virtual employee), but not changed in content or in any other way. In research literature, Robot Process Automation is currently considered “The newest technological ‘star’ on the firmament of BPM research” (Reijers 2021, p. 4).

If the application provides an so-called “Application Programming Interface” (API), it can also be directly controlled by a workflow engine. However, it is usual for the robot to access the normal user interface, which is also used by the “real” employee.

► **Definition “RPA”** RPA is a technology that replaces humans in the processing of simple, repetitive IT tasks without changing the technologies used. RPA programs access the same interfaces as the human processor.

RPA programs are “rule-based software robots” for repetitive tasks with typical applications such as automated data entry, integration of systems without programmed interface and process control (ensuring desired process = actual process) (cf. van der Aalst et al. 2018). The widespread use of RPA applications shows that there is enormous potential and interest for this. 60% of companies in the DACH region want to have between ten and 25 business processes carried out by so-called software robots by 2020 (cf. Freund 2019).

► **Definition “bot”** The bot is a virtual operator or a virtual workforce that performs pre-determined and routinely performed business processes. It simulates the human input on the user interface (GUI) of an application.

Typical RPA processes that can often be found in practice are, for example, travel expense accounting, termination processing, onboarding of new employees, invoice processing, order processing or vacation application processing. But also interactive processes, such as conducting customer conversations via voice input and output, are conceivable (Czarnecki et al. 2019, p. 797).

RPA tools can perform the same operations on the desktop as a human user, e.g. mouse movements and actions, input of characters, copying or changing of displayed values, clicking on buttons, log-in and log-out operations. Receiving and sending of emails (Schiklang et al. 2019, p. 4). The interaction between the RPA software and an application is shown in Fig. 6.3. Steps 1–4 simulate the human operator. If information is missing in the process flow, the RPA software informs a previously set user group (e.g. by email).

An example of a process supported by RPA technology is shown in Fig. 6.4. After logging in to various systems, the employee copies the necessary information, switches the window to the target application and enters the information in the “invoice position”

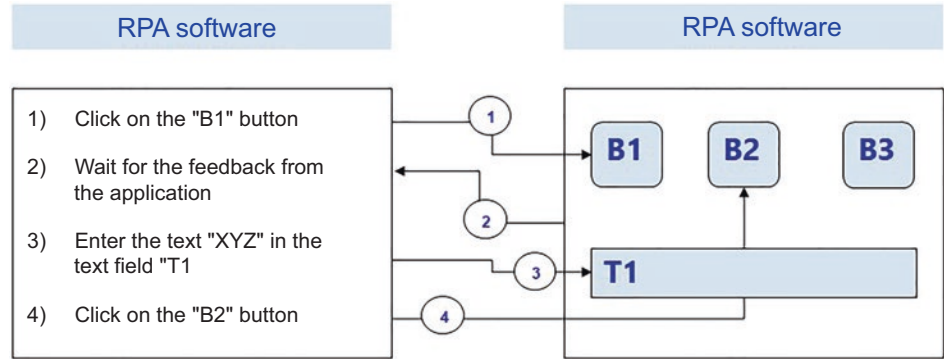


Fig. 6.3 Interaction between RPA software and application. (Adapted from Häuser and Schmidt 2018)

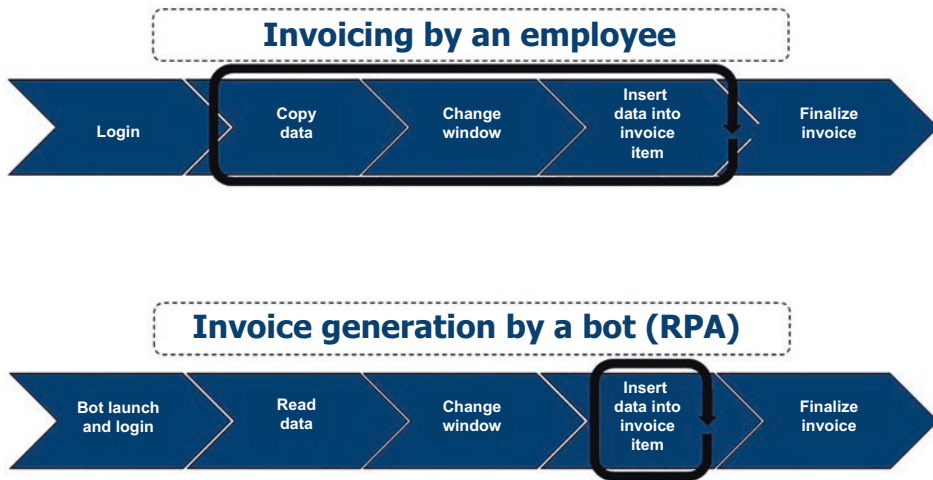


Fig. 6.4 RPA example process invoice processing. (Adapted from Baumbach et al. 2016)

mask. This process part is repeated as often as there are invoice positions. After starting the bot, this logs in to the various systems and reads the data necessary for all invoice positions. Then the bot switches the window to the target application and enters all information in the “invoice position” mask for all positions.

The achievable cost advantages using RPA technologies are enormous. If the costs of a German full-time employee are set at 100%, only 35% are incurred for a comparable employee in a low-wage country. The cost share of a bot when using RPA technology is only 10% (cf. Hermann et al., p. 30).

Requirements for RPA

Not all processes are suitable for the use of RPA technologies. Important requirements that a process or sub-process must meet are (cf. Deloitte 2017):

- Rule-based processing of process steps,
- Processing of structured input data,
- Repeated execution of the process,
- Medium to high transaction volume,
- Stable process flow, low changeability.

Potential risks

The use of an RPA software does not require a revision of the processes, so there is the risk of “cementing existing processes”. Instead of redesigning the processes (business reengineering), they are automated by the robot and thus used for several more years. RPA can be easily used in the department as “shadow IT”, like Excel macros and similar

tools. This often leaves the information and process management without control over the implementation. The tools are often said to be inflexible and prone to errors. Special cases in workflows and changes in the interface can lead to error situations in the processing of the processes.

RPA and BPM

The Table 6.3 shows the main differences between an RPA solution and Business Process Management (BPM) (cf. Williams 2017).

The increasing use of RPA technology raises a number of questions that need to be regulated. For example, it needs to be clarified who is responsible for the selection of RPA-relevant processes, who develops, tests and operates the RPA programs. In addition, tasks such as error analysis and quality improvement need to be clarified. Such questions have led to a demand for “RPA governance” (see Petersen and Schröder 2020), which is intended to regulate these aspects. For this purpose, specific roles are to be defined depending on the company’s requirements, such as “RPA Program Owner” or “RPA Developer” (see Petersen and Schröder 2020).

6.2.3 Process Mining

The term Process Mining goes back to the work of Will van der Aalst (see, for example, van der Aalst 2011). This refers to a technology that generates a process analysis based on real process data and makes it available using visualization techniques. Fig. 6.5 visualizes the overlap between “Process Science”, i.e. the analysis and modeling of

Table 6.3 Comparison of RPA and BPM

Robotik Process Automation (RPA)	Business Process Management (BPM)
RPA carries out the processes instead of an employee	BPM is a method for process optimization
The focus is on replacing human work with robot software	The focus is on process improvement (in terms of content, time, finances) through restructuring
RPA does not change the process, only the interface on the top level (human-machine) is changed	BPM changes the process, RPA can be used as a partial solution for individual process steps
RPA is a software technology	BPM is a management approach
RPA is a fast and inexpensive transitional solution that can be used instead of BPM (“Quick Wins”)	BPM offers the opportunity for a fundamental change in processes, but is more expensive
Frequent changes in the process hinder RPA	Frequent changes in the process are implemented by BPM

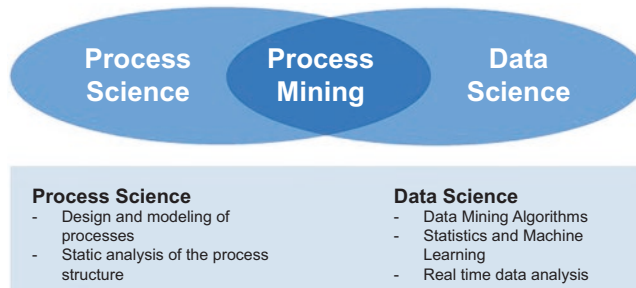


Fig. 6.5 Process Mining as the intersection of Process Science and Data Science

processes, and “Data Science”, the application of statistical methods to data, which are referred to as “Process Mining”.

A “Process Miner” analyzes the data generated by processes (newly created customer master records, newly recorded customer orders, changed master data or orders, etc.) and assigns them to processes. This makes it possible to compare target process models with actual models that reflect reality. Data sources, for example, include the log data of ERP systems, workflow management systems, or other databases that are supplied with data by the processes (e.g. order data, customer data in an order process). The deviations of the actual processes can, for example, be compared with the target processes to research the reasons why the processes take place differently in reality than planned (or modeled).

Possible reasons for deviations of actual and desired processes can be found in the following causes:

- Repetition of process steps,
- Unplanned rework,
- Redirections in the process flow (e.g. unnecessary process steps without added value for the process result),
- Delays of process steps.
- Lost items (e.g. customer request is not processed, it “gets lost”),
- Ping Pong (process flow takes place back and forth between different locations without creating added value)
- Loops (repeated processing, e.g. an order is recorded, checked and changed again),
- Crime (e.g. unauthorized creation of a debtor master record and booking of an invoice on this debtor, which is then paid automatically),
- Ineffectiveness (e.g. request is not processed to the last step).

Situation in practice A study by the FH Münster showed that the penetration in practice is still very heterogeneous: Only 5% of companies use process mining and 7% are in the test or implementation phase (see IPD 2021). What is interesting about this study is that

39% of companies say that process mining is interesting, but they do not pursue it for various reasons (see IPD 2021).

Fig. 6.5 shows the basic idea of process mining. The process modeled in the upper area is often not lived. Employees omit individual process steps, they need more time or incur higher costs than planned, and, for example, using shadow IT (Excel macros, etc.), they add additional process steps.

Example The Fig. 6.6 shows an example with a small process model and fictitious log data of the ERP system. It contains the process data of four orders (100, 101, 102, 103), which run through the process differently.

The Fig. 6.7 shows the course of the orders 100, 101 and 102. With order 100, the customer’s creditworthiness is first checked, the order is placed and then the goods are shipped. With order 101, the customer does not have a good credit rating, so the order is rejected. With this, the process is ended. With order 102, the order is rejected after the creditworthiness check. Nevertheless, a shipment of the goods takes place, which should not have happened according to the process model. From the log data, this situation is indeed apparent, but only with difficulty. In practice, this can have different causes, e.g. processing errors, fraud, software errors. The course of order 103 (normal order) corresponds to the course of order 100. Therefore, a visualization was dispensed with.

Fields of application The range of applications of Process Mining is large. For example, an increasing number of auditors are using this technology to carry out an automated

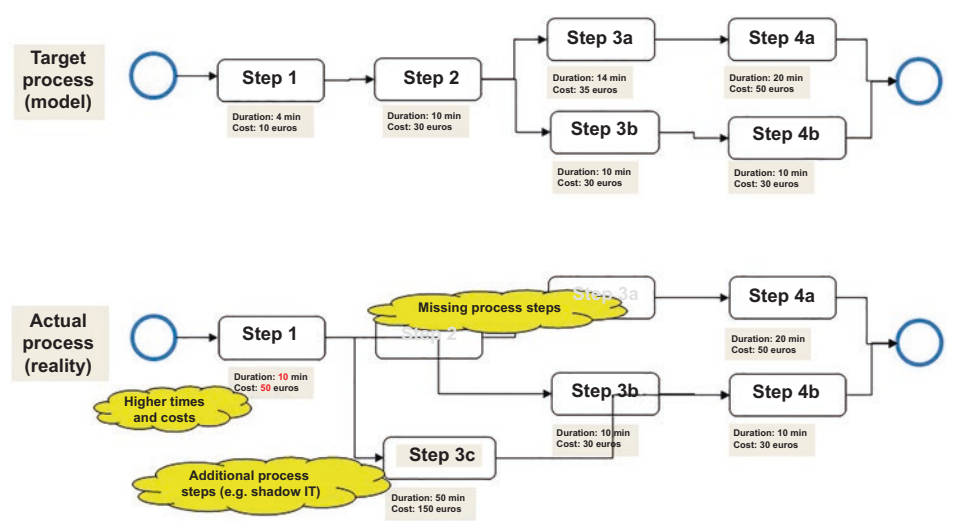
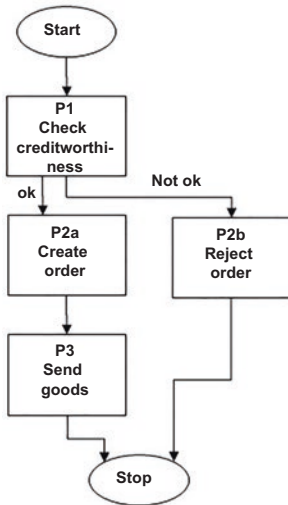


Fig. 6.6 Process Mining—Target and actual processes in contradiction

Process model



Log data

ID	Order	Customer	Process ID	Process name
1	100	10	PI	Check creditworthiness
2	100	10	P2a	Create order
3	100	10	P3	Send goods
4	101	20	PI	Check creditworthiness
5	101	20	P2b	Reject order
6	102	30	PI	Check creditworthiness
7	102	30	P2b	Reject order
8	102	30	P3	Send goods
9	103	10	P1	Check creditworthiness
10	103	10	P2a	Create order
11	103	10	P3	Send goods
...				

Fig. 6.7 Example: Process model with log data

analysis of complete business processes. This makes it possible to detect and assess irregularities in transactions (cf. Bruckner 2019, p. 6).

In the meantime, there is a range of commercial software tools that have taken up van der Aalst’s ideas and implemented them in products. The following is an alphabetical selection of current tools (cf. the interactive product overview of the consulting firm Gartner, Gartner 2021).

- ARIS Process Mining SaaS, Software AG,
- Celonis Process Mining, Celonis,
- Fluxicon Disco, Fluxicon,
- Signavio Process Intelligence, Signavio,
- UiPath Process Mining, UiPath (formerly Process Gold).

Process Mining in the Mittelstand

The Bonn-Rhein-Sieg University of Applied Sciences, together with the Cologne consulting firm BPM&O GmbH, conducted a survey among several medium-sized companies (SMEs) in order to analyze the state of the art in terms of possible applications in SMEs (cf. Tsingeni et al. 2022). Three companies were surveyed: a bank, a toy manufacturer and a food company (cf. Tsingeni et al. 2022).

The study showed that many SMEs are interested in Process Mining but have not yet been able to undertake such a project for various reasons (e.g. lack of resources). Important prerequisites for the successful introduction of Process Mining are:

- Process culture with understanding of process management (thinking and acting in processes, assignment of responsibility for processes),
- Selection of suitable processes (often end-to-end processes are distributed across several systems, making it difficult to obtain a clear process identification),
- Definition of objectives to be achieved with Process Mining (e.g. identification and elimination of weaknesses in processes) (cf. Tsingeni et al. [2022](#)).

Conclusion from the perspective of process management

Process Mining can support process management in that it can detect conscious or unconscious errors in processes and identify weaknesses in implementation. It represents a bottom-up approach, which, in contrast to the classical top-down approach of process management, starts from the actual data. For controllers, auditors and compliance officers, Process Mining offers the opportunity to support process controlling and actual analysis.

6.3 Tools for Professional Process Support

6.3.1 Standard Software Versus Individual Software

A key question in business informatics is the procurement of the application software required to support business processes. For process management, it is important whether ready-made standard software is used, which was developed for many companies, or whether software specially made for the company is used.

Trend towards standard software

In view of current trends, such as cloud computing, the use of standard software is gaining increasing importance. The classic development of software with own employees, which may be supported by specialized consultants, is becoming increasingly rare. However, it still dominates in some industries (insurance, banking), in which the range of standard software is relatively low. The classic purchase of standard software with subsequent implementation by own employees and consultants has been increasing for years in areas with high software availability (e.g. manufacturing industry, mechanical engineering, trade).

In-house development as the domain of small and medium-sized enterprises

The classic in-house development by an external software house commissioned (third-party development) is typically found in small and medium-sized enterprises, which

usually have only limited development resources, but also in larger companies for specific applications. Since standard software requires comparatively high investments in employees, hardware and other resources, the rental model for standard software has found widespread use in recent years. This eliminates the purchase of the standard software, as this is carried out by a provider specialized in this. The provider has the know-how for the introduction and implementation support, the user pays for the use of the software.

Example from the insurance industry

In the insurance industry, the proportion of software developed in-house is still relatively high compared to the use of standard software. At a panel discussion with IT managers, the owner of a software house that mainly produces custom software expressed the following opinion: “One of our customers in the field of private health insurance has developed a software solution for assessing the risk of damage itself. Input for this software are the applicant’s pre-existing conditions as well as other data such as place of residence, profession, etc. As a result, the program delivers a probability of damage and a possible risk surcharge or, in extreme cases, a rejection recommendation. This system incorporates decades of experience that are very specific to the company. We cannot imagine that such tasks will be solvable by standard software in the foreseeable future.”

The representative of a large standard software provider who was also present was not in agreement with this. His response was: “The industrial use of standard software is not limited to accounting and warehouse management. About 25–30 years ago, supporters of custom development argued that the core business of an industrial company was so specific that it would be hardly possible to produce production planning and control software for the mass market. Today you will find software from the market leader or a competitor in almost every company in the field of manufacturing and production. In this case, insurers can benefit from this. I can also imagine that standard software providers offer software solutions in the form of frameworks for the core business of insurance companies, which can be supplemented with own modules for specific requirements. Such a module can be, for example, the rating or the risk assessment.”

There are also similar examples in the manufacturing industry in the field of intersection optimization. Providers of standard software offer their customers the opportunity to integrate their own modules for intersection optimization at defined points.

**Use of custom software for process support**

- **Customized Solution**

The development and use of individual software has the advantage of a tailored solution that may not require organizational adaptation. With custom development, the

user specifies the desired requirements and implements them in the form of a technical solution. No adaptation of the organization is required because the wishes are taken into consideration. Depending on the company, this can be seen as an advantage. However, the problem is that the “installation” of custom developments can lead to complex application architectures. Older and newer generations of software systems are interconnected, with the result that maintenance options for future developments are made more difficult. It is critical in this context to note that the use of standard software can also “enforce” long overdue organizational changes or at least induce them. Therefore, overdue organizational changes can be more easily delayed using custom software than using standard software.

- **Independence**

The advantage of developing software in-house is that no dependency relationship is established with a software supplier, which usually lasts for several years, often for decades. This bond is much stronger than, for example, the bond with a manufacturer of computer hardware, because the exchange of software causes much more effort than the exchange of hardware.

- **Strategic Aspects**

Companies need unique selling points to be able to compete in the long term and to ensure customer loyalty. They want to present themselves individually to the outside world to differentiate themselves from competitors. A homogeneous support of business processes using standard software is a reason for many companies to develop custom software. With this instrument they have the possibility to create specific competitive advantages in selected process areas. Possible candidates here are product- or market-related process areas with an innovative character (e.g. product development) or systems that are visible to the outside, such as sales or systems for the design of the Internet presence. In the classic business areas, which mainly cover inward-looking business processes, such as accounting, cost accounting, controlling, human resources, but also logistics and materials management, the scope of the functionality and the quality of available standard software packages are so high that custom development is hardly an option anymore.

- **Costs difficult to Plan**

The development of individual software requires a high financial expenditure and is very risky because of the many unknown influencing factors. Again and again it can be seen in practice that, despite modern development methods, the documentation of the delivered programs is often insufficient or not available in individual cases—often for time reasons.

- **Dependent on employees**

The independence of software suppliers is exchanged for a dependence on key employees in software development in individual development. IT departments are often restricted in their performance in the event of vacation or illness of certain employees. However, this disadvantage can be increasingly reduced, but not excluded, using modern methods of software engineering and quality assurance. Ultimately,

the dependence on individual IT employees is a risky problem, especially for smaller companies.

Use of standard software for process support

- **Purchase of know-how**

For many companies, the use of standard application software (short: standard software) is an alternative to modernize their workflows. With the standard software they not only buy software, but also predefined business processes, which still have to be adapted to the needs of their company. Alternatively, the organizational structure and workflow of the company must be changed if the software “does not fit the processes”. The latter is often the case in practice.

- **Cost advantages**

The purchase costs are lower in comparison to the costs for an individual development, since the development costs of the manufacturer are spread over a larger number of customers. It should be noted that, however, the purchase costs are not the dominant cost factor for the introduction of standard software. With the introduction of standard software, costs for the conversion of hardware are often also incurred. Sometimes the procurement of additional software, e.g. a certain database management system or faster terminals is necessary.

- **Current software**

The competition among manufacturers of standard software leads to the constant improvement of the products, and as a rule, the current business knowledge is represented in the form of programs. The customers of the standard software providers therefore benefit from the permanent further development of the software and can usually also use current standards of the software market. An example from recent times is the further development of business standard software for Internet functionality.

- **High functionality**

In contrast to earlier years, business standard software today has a very comprehensive functionality that usually covers the usual requirements of a function area (e.g. finance) in full. Not infrequently, some functions are not used when first introduced because of the wealth of functions but used at later times. Standard software is often used internationally. Modern software packages can be used for all common languages of the world, including Japanese, Chinese, Arabic and Russian characters. This makes it possible to work with the same standard software—on the same data sets—worldwide. Every employee can operate the system in his individual language. Business standard software is available hardware- and software-independent for different industries. Early generations of standard software offered little or no opportunity for customer customization. Current software products can be customized to the needs of the users within the scope of the offered functionality via so-called customizing. The number of “screws”, i.e. the possible parameterizations, are so complex that no general statements are possible. In principle, it is possible to adapt such standard software products to the requirements in the respective company to a large extent,

although nevertheless adjustments of the organization and the business processes must be expected.

- **Organizational Changes**

The use of standard software is often justified with strategic aspects. The decision to use standard software, often also for a specific manufacturer, is usually made at the top management level. This can make sense for a suitable motivation, for example if it is desired to use the software introduction project to justify and implement necessary organizational changes. This situation is often encountered in practice, because in individual software development, the current organizational structure and the current business processes are often used as the basis for target concepts. In this case, the use of standard software can lead to the desired “reflection” on the advantages of the current organizational structure and the business processes. Such projects then have the character of business reengineering projects, and the use of standard software serves as a tool, i.e. as an enabler for the revision of business processes. In addition, the responsible employees of the specialist departments are involved in the project responsibility at an early stage, since a functional software system already exists and the adaptation to the operational requirements is a main economic task. This ensures that the future software solution corresponds more to the requirements than is often the case with in-house development, where employees of the specialist department are only marginally involved during the development work.

- **Lower Follow-up Costs**

The essential strategic advantage of using standard software lies in the lower follow-up costs for extensions of the installed system. The extension of the functionality of self-created applications often exceeds the effort of the original introduction project, but often leads to complete (maintenance) projects. With the use of standard software, functional extensions by activating the required functions and their parameterization (customizing) are always possible. However, this only applies if the functional extensions remain within the standard scope of the software. If requirements must be met that go beyond this, additional development (so-called add-ons) or even changes in the source code (so-called modifications) are required, which have the character of in-house development.

- **Strategic aspects**

However, the strategic aspects are not always decisive or correct. The use of business standard software in the field of internal administrative business processes, which have a cross-sectional character, such as accounting and personnel, is less critical. The selection must be more careful in core business processes that provide essential value contributions to the company. In part, no competitive differentiating features can be implemented with standard software here. The decision for a manufacturer and his product inevitably leads to a certain, partly also high, dependence, since usually only a small influence, mostly only indirectly via user groups etc. on the product policy and further development is possible. The dependency is all the heavier if standard software is used for strategically relevant process fields in companies. Process fields with a high

degree of standardization, such as finance or office communication (word processing, e-mail, etc.), are less critical. The practice of modification (change of the source code) common in previous software generations is often not possible for technical reasons (no delivery of the source code by the manufacturer) or too time-consuming in the long term. The latter is increasingly the case as changes to the source code are no longer realizable with reasonable effort when switching to release several times a year.

- **Training**

The introduction and operation of standard software requires a one-time training and usually also consulting effort by the manufacturer or specialized consulting companies as part of the initial introduction, but also permanently during operation.

- **Expensive special personnel**

The required special personnel is usually expensive and difficult to obtain. By early involvement and qualification of employees who later act as coaches in the company, this disadvantage can be mitigated with adequate project management.

- **Changed requirements for personnel**

It must also be considered that, when using standard software in practice, changes in the personnel structure of the IT department will occur in the medium to long term, making it more difficult to switch from standard software back to individual development. The reason for this is the significant reduction in personnel capacities for application development, as this is only required for the development and adaptation of extensions (add-ons). Instead, personnel resources are built up with business-economic and organizational skills, i.e. the introduction of standard software is associated with a change in the IT department from development to internal company consulting.

6.3.2 Enterprise Resource-Planning Systems (ERP Systems)

Objectives and terms

ERP systems (ERP = Enterprise Resource Planning) were of great importance for the development of process management and are still a central component in the implementation of processes. Until their development and introduction in the 1980s, dedicated systems were used for individual business functions for the most part. This led to the fact that companies sometimes even operated systems for warehouse management, sales, accounting, personnel, etc. on their own hardware, which managed the necessary data themselves. This resulted in high data redundancy and brought with it a multitude of interface programs to ensure mutual data exchange, which was often no longer controllable.

► **Definition of ERP system** An ERP system is a software system that is modularly structured and in which several standard business applications are integrated through a common database. Typical business applications are finance and controlling, production planning and control, purchasing and logistics, sales and shipping, and personnel. Each

business application is represented by a module consisting of several individual programs. The focus of an ERP system is on process support within a company, rather than on the support of inter-company business processes with suppliers and sub-suppliers. Individual adaptation to different needs is achieved through customizing.

Characteristics

ERP systems are characterized above all by the following features: data integration, process integration, operational functionality, a uniform development concept, layer architecture, and transaction orientation (cf. Table 6.4).

Data Integration

A key feature of integrated standard software is the shared use of data. As an example, sales data can be mentioned. Customer master records are usually created originally

Table 6.4 Features of ERP systems

Feature	Short description	Example
Data integration	The software modules of the system use data together	A sales and accounting module each use customer master data
Process integration	Cross-departmental business processes are supported by multiple involved software modules	Customer order processing is supported from the customer inquiry, over production to the delivery and payment of the goods by means of several software modules (sales processing, production planning, shipping, finance)
Operational functionality	Support of operational tasks of a company for the processing of business transactions	Order processing, production planning, customer accounting, incoming invoices, salary statements
Uniform development concept	Software modules use a common repository and are based on uniform development standards	Same screen mask layout, similar error messages
Layer architecture	Software architecture to support processing distributed over multiple departments and locations, possibly also countries	Client/server architecture to realize decentralized access to data and functions
Transaction orientation	Online processing of business transactions and storage of data in databases	Creating a customer order, booking an incoming invoice
Tenant capability	Separation of the logical view (company, area) from the technical view	On an SAP® system, several legally separate companies can be invoiced. Users and data are completely separated

Characterization of ERP systems

by employees in sales. Here tasks such as assigning a customer number, customer name, address, sales data, etc. fall. The accounts receivable clerk can pick up these information available in the ERP system and extend them with specific information of the accounting (e.g. credit limit, co-account, payment terms). Both employees access the same data sets. The data integration is particularly noticeable in the “posting” of business transactions in all activated components of the standard software. If a company uses, for example, an integrated application system with the sub-functions logistics / materials management, production planning and accounting, a goods receipt booking of a raw material necessary for production control triggers the following activities:

- Update of the quantitative inventory levels in logistics and materials management,
- Triggering a production order that is waiting for this material,
- Increasing inventory values in accounting. ◀

Process Integration Of particular importance for business process management is the question of the continuity of process support, i.e. the waiver of employee, department or software change. Only a continuous connection of several application components to a business process allows to dispense with interfaces to a large extent and to capture data only once, at the place of origin, and to further process them in all components. Integrated databases require the plausibility check of all data already during the input into the system. So even with a quantitative goods receipt booking the accounting relevant data fields have to be captured and checked. For example, it must be determined whether a cost center to be charged with the goods receipt even exists. In such an application case, the data of the business transaction must also be forwarded to all affected application components, i.e. “posted”.

Only the consistent, interface-free connection of individual functions, such as logistics, accounting or sales, to an overall system makes an integrated standard software. All system components access common databases here. The “integration” of various modules via batch interfaces only represents the data supply of the individual components and cannot be referred to as integration. “Batch programs” run without user intervention. Another, somewhat outdated, term for this is “batch processing”. Dialog programs, in contrast to batch programs, require user interaction, such as the capture of customer data on the screen. Data that is required for non-integrated programs for different purposes is exchanged via interface programs. With high data volumes, this is often done in the form of batch programs. This is the case, for example, when transferring customer master data from a sales system to a financial accounting system.

It is decisive whether process chains can be mapped across all modules of the standard software. For example, order processing must be implemented and executed seamlessly without media breaks in the standard software system according to the actual processing flow through the sales, logistics, production and shipping functions. In the

background, the administrative functions such as accounting and controlling must be supplied with the necessary information.

The process overview of the primary process procurement in Fig. 6.8 shows how, starting from purchase requisitions (BANF), logistics oblige data are carried forward in controlling (e.g. on an order or a cost center). Purchase requisitions are requests to the purchasing department to procure certain materials or services. They can, for example, be created by the demand creator (e.g. head of the cost center) in the Materials Management module. Under an oblige, commitments are to be understood which arise from contracts or dispositions and have not yet been recorded in accounting. A purchase oblige is triggered by the purchase order in purchasing. The goods receipt leads to preliminary actual values in the controlling reports, which are replaced by the later invoice receipt from the “actual actual”. The goods receipt is reflected in the material accounts of the general ledger in the secondary process finance. The invoice receipt is also booked in the creditors’ subsidiary ledger and in the main ledger via the co-booking technique in the Materials Management module.

Fig. 6.9 shows the core process of Distribution, which acts as a data provider for the involved cross-sectional processes. The order entry in the Sales module triggers the credit limit check in the Accounts Receivable module. At the same time, the relevant controlling objects (e.g. order) are updated and enter forecast analyses in the Sales Controlling module. The booking of the goods issue in Sales in turn triggers the billing. If standard business software is used here, this triggers the booking of the invoice, which is updated in the subsidiary ledger Accounts Receivable and via the subsidiary ledger technique in the main ledger. Depending on the customer’s behavior, the dunning or payment processing follows in the Finance area. In the process overview, the consideration

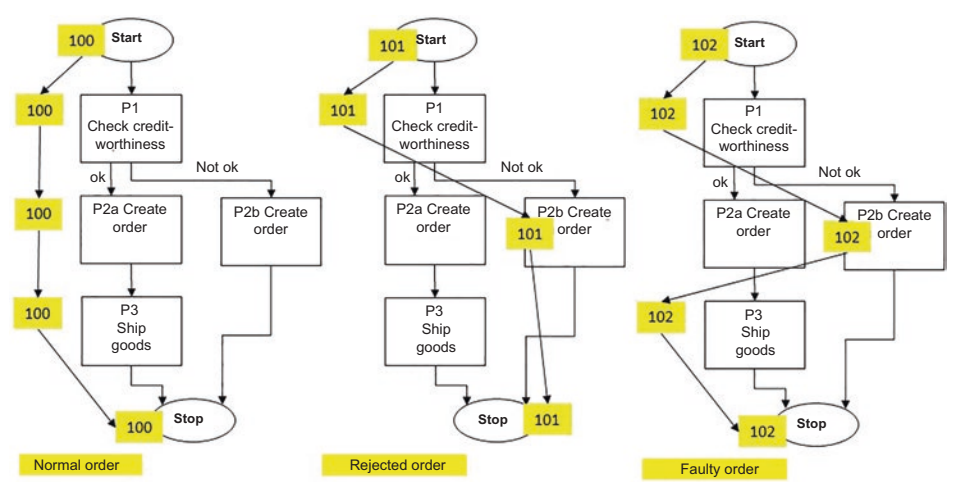


Fig. 6.8 Example of process flows in visualized form based on log data

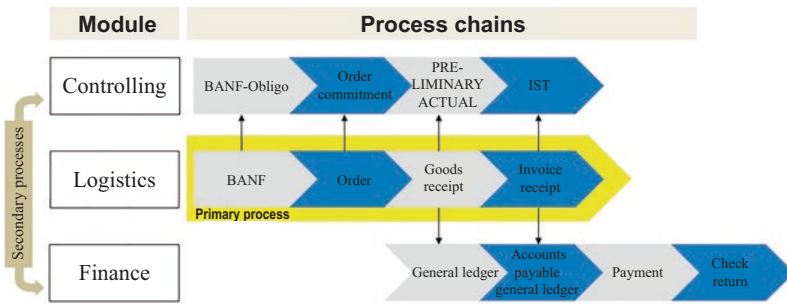


Fig. 6.9 Process integration using the example of purchasing logistics

of the payment-relevant processes in the context of financial disposition (Treasury) was neglected for the sake of simplicity.

Operative functionality ERP systems support functions that are necessary for the operational processing of the regular business transactions of a company. Examples are the capture of orders, orders, execution of payroll and salary calculation, etc. They thus differ from management information systems, which can be used to support the analysis of data, e.g. for customer sales analysis.

Unified development concept Individual independently designed partial functions cannot be integrated into a complete system so that it can meet the above requirements. Integrated standard software systems are therefore based on a unified development concept. In the form of a layer model, a basic system with cross-cutting, necessary “services” for all partial functions is designed on a lower level. In addition, uniform standards are used in integrated systems, such as a uniform screen and print output layout, used database systems or database system interfaces and the use of open interfaces (e.g. TCP/IP).

Layer architecture ERP systems are not single-user systems, such as a word processing program that is fully installed and used on a single workstation. They support business functions that are usually required by several employees in different departments and also at different locations. For this reason, a layered architecture is necessary, which is usually realized in the form of the client/server principle with a separation of presentation, processing and data storage.

Transaction orientation The support of operational business cases requires the change of data using online transactions. ERP systems work transaction-oriented, i.e. they provide several transactions to support business processes (e.g. transaction for creating a customer order, for capturing an order, for changing an employee master record, etc.).

Multi-tenancy In addition, multi-tenancy is often added to many standard ERP systems. This refers to the ability to account for several companies from a business point of view completely independently of each other in a single technical installation. General settings that apply to all companies are limited to a few general aspects (e.g. calendar entries). In addition, each company can be individually configured and invoiced on the installation.

6.3.3 Economic Viability of Standard Software

Often, expectations are associated with the introduction of standard software, which should preserve the competitiveness of the company and thus secure it. In addition, it is assumed that there will be lower costs than with individual software. In addition to the acquisition costs for the standard software, however, other types of costs are significantly more important when introducing standard software.

Economic viability of standard software using the example of SAP ERP® (cf. Buxmann and König 2000)

Because of the high demand for product-specific know-how, the introduction of SAP® systems is usually associated with the use of external consultants. Consultants are usually used in all project phases, in the often associated with the introduction of SAP® reorganization of business processes (business reengineering), but above all in the customization of the system, user training, the implementation of extensions (especially custom development) and very often over long periods of time in the introduction of support for the productive system. Consequently, consultant costs, as the above empirical study shows, are at the top of the cost categories associated with the introduction of SAP® systems.

- Costs for external consultants,
- Costs for the purchase or expansion of hardware and system software,
- Costs for the release of own employees for the introduction project,
- Acquisition and maintenance costs for the standard software,
- Costs for training measures.

Despite the enormous costs, the success of the SAP® system shows that there are significant potential benefits to offset the effort, which can justify their use. The key benefit categories are:

- Better planning, control and monitoring of business processes,
- Uniform and consistent data basis,
- Improved flexibility about adaptation of information systems and business processes to changed requirements,

- Reduction of process times of business processes,
- Qualitative improvement of business processes. ◀

Martin et al. (2002) distinguishes four categories of benefits of the use of ERP systems:

- Process efficiency (business processes),
- Market efficiency (customer and market orientation),
- Resource efficiency (productivity and efficiency) and
- Delegation efficiency (information retrieval efficiency).

Under process efficiency they understand the ability of a company to improve business processes in the categories of cost, quality and time. Examples are the reduction of processing times for orders or an increase in delivery punctuality. Market efficiency is the improved use of opportunities on the sales and procurement markets through coordinated action towards customers or suppliers. This can be done, for example, on the procurement side by demand bundling or on the sales side by improved products and services. Resource efficiency is the improvement of productivity and efficiency, i.e. the optimized use of resources in the form of people, facilities, machines and capital. Examples of this are improved capacity utilization, reduced inventory levels or reduced number of employees required. The delegation efficiency measures the increase in the use of the problem-solving potential of hierarchically superior units. Examples of this can be a higher speed and quality of information processing through IT-supported reports and analyses. Often, global analyses for an entire corporation are possible without merging and consolidating various data sets, thanks to access to the same database.

6.4 Introduction Processes for Standard Software

6.4.1 Connection to Process Management

Projects for the introduction or update of standard software also involve aspects of process management, as workflows have to be adapted, can be discontinued or new ones added. The introduction or update of standard software often poses new challenges for employees in the companies concerned. In addition to technical and business questions, new requirements are also placed on the cooperation of employees within and between the departments of the company concerned, as integrated software systems do not know any departmental boundaries. The introduction of a business standard software, in particular ERP systems, represents a massive intervention in an ordering system that cannot be coped with without conflicts (Maucher 2001, p. 23). The use of standard software therefore changes the processes in the company. Therefore, the introduction phase of the software has to be planned carefully, as the changes here become effective within a short time.

According to Heilmann et al. (2003, p. 283), there are three basic scenarios for the introduction of new information systems:

- *Complete replacement of the old system* by a new application software In this case, individual elements may be standardized, but the share of individual development of the new software dominates.
- *Complete replacement of the old system* by a standard software: Here the old system is completely replaced, there is no significant share of individual development (e.g. complete introduction of an ERP system)
- *Porting of the old system* to a new technical platform (1:1 migration): Here, no change in the functional scope is sought, but the old application is to continue in a new system environment. So it's just a change of technology (e.g. transformation of a system into a private cloud).

For the introduction of a business standard software, such as SAP ERP®, there are two basic strategies: the “Big-Bang-Strategy”, i.e. the date-related exchange of the system in one go, or the “Step-by-step Strategy”, i.e. the gradual transfer of processes to a new system. Mauterer (2002, p. 23) also speaks of SmallBangs here.

6.4.2 Big Bang

In the Big Bang there is the possibility to carry this out for the entire company or, in the case of a decentralized organizational form, successively after the definition of a master system, for decentralized units (e.g. countries or regional branches) as so-called roll-out. With the successive strategy, criteria for the definition of the sequence of steps must be defined, usually one differentiates the department-related or function-oriented conversion and the market-oriented or process-related conversion of the system.

The Big Bang strategy is a theoretically optimal solution, since no interface problems occur and from the beginning an integrated software solution covering the entire process is available. There are also no transitional problems with double work in the old and new system, and there is also no risk of data inconsistencies, as old data before the cut-off date and new data after the cut-off date can be strictly distinguished.

The biggest disadvantage is the extremely high project risk, which can endanger the existence of the company in the event of a total failure of the new system. Examples from practice show that this can occur. To reduce project risks to a minimum, extensive tests and rollback scenarios are necessary. A Big Bang requires very high demands on project management and requires a concentrated use of personnel resources (department and IT department and usually also external consultants) within a very narrowly defined time frame.

6.4.3 Roll-Out

To mitigate the disadvantages of the Big Bang, companies with a decentralized organization (e.g. regional branches with a similar structure, locations in several countries) have the possibility of a Roll-Out. First, a central master system with common processes is defined and then successively rolled out, i.e. distributed to the regional units. If necessary, rolled out systems are locally adapted before they are put into productive operation.

The pursuit of a Roll-Out-Strategie leads to significantly lower risks, as the experiences of the first projects can be used for follow-up projects and only parts of the company (e.g. a branch) are affected in the event of problems. The use of resources can also be significantly stretched over time.

Unfortunately, a roll-out is not always possible. A decentralized organization with a manageable level of complexity is necessary. Are the local organizations so large that no local Big Bang can be carried out here, the successive strategy must be used. Further disadvantages are that only after the completion of the entire roll-out, which can take years depending on the size of the company, an integrated system is available.

6.4.4 Step-by-Step Function-Oriented Introduction

In many companies, terms such as “accounting system”, “warehouse management system” or “sales system” are known. The terms refer to a functional division of labor and document the function-supporting systems developed for this purpose. In such an architecture, it is quite common to proceed “functionally” when changing systems, i.e. the “old” warehouse management system is replaced by a new warehouse management system.

The function-oriented approach gradually releases individual functions or function areas (accounting, warehousing, ...) from the old system by, for example, new software and temporarily connects the two “worlds” by interfaces. For a transitional period, processes are therefore supported by “old software” and “new software” in parallel. The advantage over Big Bang strategies is the shorter individual project duration, as the overall project can be divided into several independent sub-projects. The individual projects are easier to handle, so the overall project risk is reduced.

On the other hand, disadvantages arise which are caused by the interface problem. The effort for the implementation of the interfaces is enormous. For the transitional period, which can easily last for several years, no integrated overall system is available. Where no interfaces are implemented (can be), there is a high manual effort by the affected employees. In addition, there is the risk of inconsistencies due to data redundancies, as the “old world” and the new ERP system must be supplied with data.

6.4.5 Step-by-Step Process-Oriented Introduction

Process orientation has established itself as a paradigm of corporate governance since the 1990s and has been established in many companies. As an alternative to the traditional functional approach to the introduction of standard software, a strategy following this paradigm is therefore recommended. The individual steps of the migration are carried out here according to market-oriented criteria. This means that individual process chains are completely separated from the old system and immediately supported throughout by the new ERP system. As a rule, the primary processes are converted successively, and the cross-sectional processes (accounting/personnel) are planned in block at the beginning or end of the project. A prerequisite for such a procedure is that the individual processes can also be separated out organizationally and operated separately.

In general, the same advantages apply to this approach as to the function-oriented introduction of standard software. However, the project risk is much lower, as the sub-processes are autonomous, and the sequence of the individual projects can be controlled according to the risk for the company. For example, less critical processes can be switched first. Later, when experience has been gained and the project team is “fit”, other processes can be followed up. As a rule, it is advisable to switch the replacement business and then the new business. This ensures that experience can be gained not with the core business, the sale of new products, but with the downstream replacement parts business.

The effort for project implementation is also lower, as fewer interfaces must be supplied due to the continuous process support. As a rule, only interfaces to cross-sectional processes such as accounting and personnel and possibly shared master data (e.g. material master, customer master) remain. In addition, the interfaces are more constant during the system changeover than in the function-oriented changeover.

In general, the disadvantages of the function-oriented changeover apply. In addition, no further negative aspects are to be recorded, possibly certain redundancies have to be accepted in the master data maintenance.

6.4.6 Strategic Portfolio

The strategic portfolio in Fig. 6.10 arranges the presented action alternatives according to the particularly important decision criteria “project risk” and “effort” in a portfolio representation. Here, the effort for the realization and dismantling of interfaces is placed in the foreground, as it is the most essential distinguishing feature.

If the essential aspects are compared, there are many advantages for the successive process-oriented approach, but hardly any disadvantages worth mentioning for the existence of the company. In principle, a case-related examination of the decision basis is required, as the numerous conditions must be met and external decision parameters, such as time pressure, corporate policy guidelines, etc., must be taken into account.

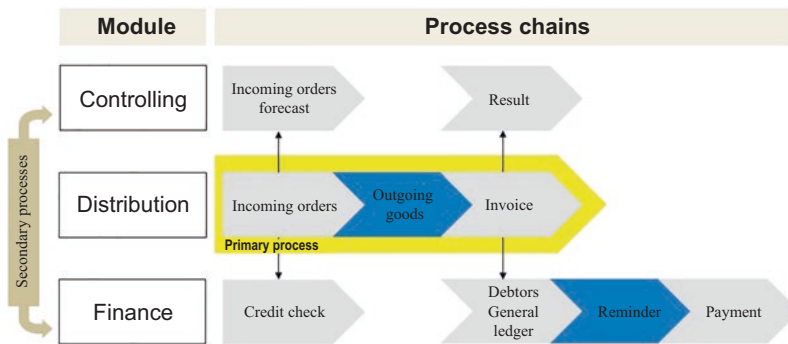


Fig. 6.10 Process integration using the example of sales logistics

6.4.7 Practical example SAP S/4 HANA

The provider of ERP systems for medium and large companies, SAP SE, offers its product “SAP S/4 HANA” in three operating variants (cf. Brugger et al. 2021, p. 107):

- On-Premises Version: (Installation at the customer, maximum functionality, adaptation of source code possible),
- Cloud Version: (Installation, operation and maintenance in the public cloud of SAP, reduced functionality, only standard processes),
- Hybrid Version: (Installation on a private cloud with own hardware, Infrastructure as a Service provided by SAP).

In the on-premises version, the customer purchases a software license and operates the software on the company’s own hardware at its own responsibility. Maintenance and administration are carried out and responsibility by own personnel (see Brugger et al. 2021, p. 108). The customer retains full control over the type and extent of the use of data, applications and processes. In the cloud version, only standard processes can be configured and used, the release changes are periodic (see Brugger et al. 2021, p. 109). The customer can concentrate on using the software but has to make do with a smaller range of functions and is no longer as free in his decisions as in the on-premises version. The advantage of the hybrid version for the customer is that he does not have to share the hardware resources with other companies as in the cloud version and has more control over the use.

For the introduction, a radical approach (greenfield) or a moderate step-by-step approach (brownfield) is proposed, which is suitable depending on the interests of the companies (see in detail Brugger et al. 2021, pp. 114–116).

With the radical Greenfield approach one can speak of business reengineering, there is a complete re-implementation of the system, which requires a comprehensive requirements and process analysis. The company must orient itself to the SAP standard processes in terms of its processes. This is also a prerequisite for the use of the public cloud

version, so that the maximum use of the existing possibilities of SAP S/4 HANA is possible. Existing structures and processes often must be changed (see Brugger et al. 2021, pp. 114–116).

With the Brownfield approach one can rather speak of business process optimization. There is a technical upgrade of existing components with migration of databases, which only requires a selective process analysis. Therefore, only a partial adaptation of the business processes is required. This variant makes sense for companies that do not want to use the cloud variant. Therefore, existing structures and processes can be largely preserved (see Brugger et al. 2021, pp. 114–116).

6.5 Effects of Current Technologies on Process Management

6.5.1 Digitalization

Currently, numerous discussions are being held under the term “digitalization”, i.e. the electronic planning, control and execution of business processes. However, the term cannot be considered independently of trends such as “big data”, “cloud computing”, “Industry 4.0” or “Internet of Things” and “Social Web” because the mentioned topics are strongly interlinked (cf. Fig. 6.11).

Digitalization and the associated technologies offer a high growth potential if management is willing to invest in innovative new business models and processes (cf. Gadatsch 2016, p. 63). The long-standing path of digitalization of processes is increasing and reaches examples of applications that were previously unthinkable. For example,

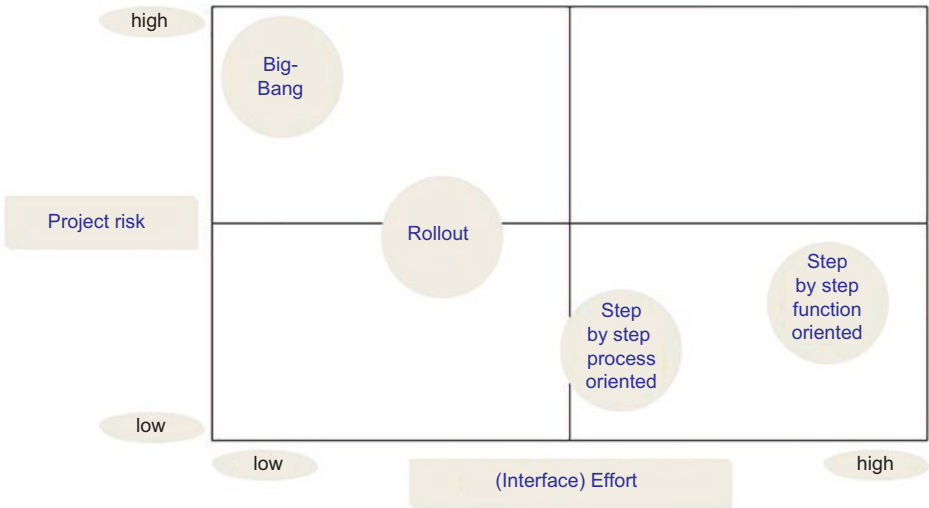


Fig. 6.11 Strategies for the introduction of SSW

Werth, Greff and Scheer sketch a digitalized consulting process under the label “Consulting 4.0”, which, unlike today, where the digitalization of many consulting companies ends at the contact form, includes the entire consulting process from problem identification, analysis, problem solving and implementation (cf. Werth et al. 2016).

Homo Digitalis

Digitalization not only affects workflows in companies, but also the people themselves. The futurologist Markowetz speaks of the “Homo Digitalis”, i.e. a new generation of people who cannot cope with everyday life without digital networking (cf. Markowetz 2015, p. 15). They are permanently connected to the Internet and base their actions on the recommendations of digital helpers. This aspect will change the business processes of the future in a sustainable way.

Effects on human work

The use of new technologies comes at the same time as a drastic change in paradigm in the world of work (cf. Fig. 6.12). Many companies no longer see the sole focus on profit as a meaningful corporate purpose but are looking for more sustainable corporate visions. The hierarchy as a steering instrument has long since ceased to be effective in many cases and is replaced by network-oriented structures that place the team concept in the foreground. Reciprocal knowledge transfer replaces centralized, control-oriented behavior of management. The already mentioned agile methods replace planning-oriented methods of corporate governance and subordinate levels (here process management). Transparency replaces discretion and tries to replace the long-standing and here also discussed silo thinking in companies (Fig. 6.13).

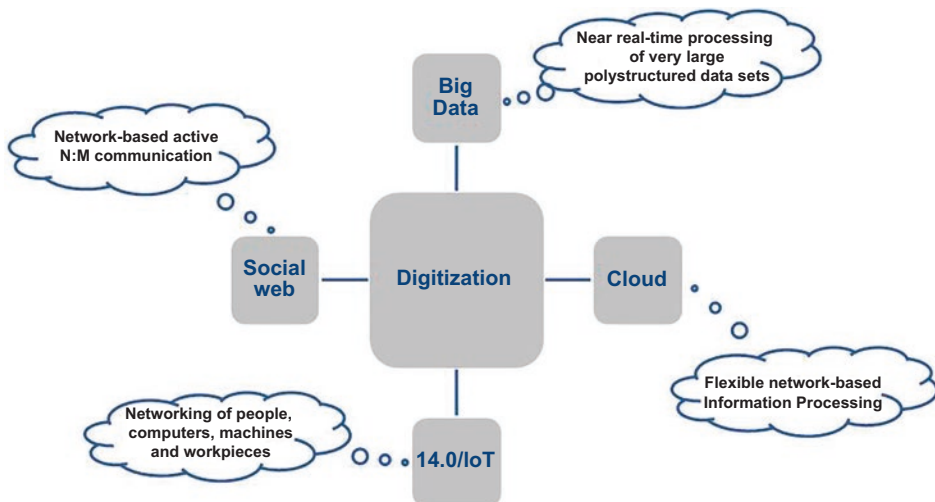


Fig. 6.12 IT megatrends

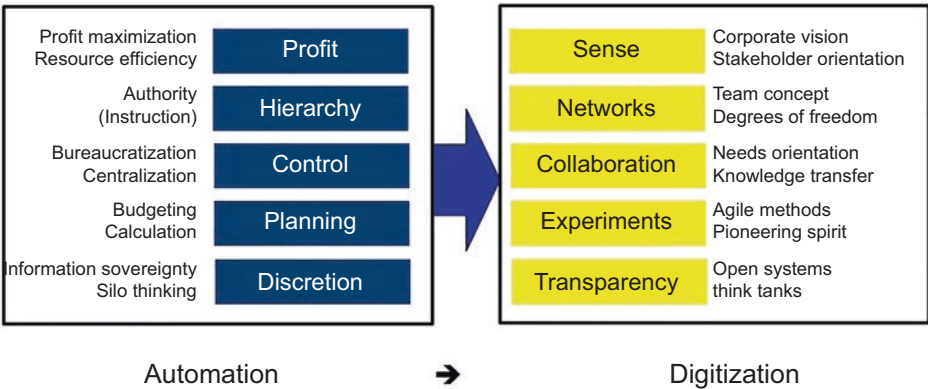


Fig. 6.13 Paradigm shift in the world of work

Digitization leads to a changed profile of requirements for the population (cf. Fig. 6.12). The demand for highly qualified employees is increasing, less qualified persons must be re-qualified. This also applies to occupations that are generally not necessarily classified as low-skilled but work mainly based on rules. Digital tools make it possible to simulate and automate rule-based work and the processes associated with it.

Effects on selected professions (here: tax consultant, auditor)

Many firms of tax consultants and auditors are probably not yet prepared for this scenario: The (future) largest provider of auditing & tax consulting in the world probably does not own a firm, does not employ its own tax consultants and auditors and works exclusively digital based on a platform similar to the known models of Uber, Airbnb or Facebook.

Activities like “maintaining bookkeeping”, “preparing and checking financial statements and annual reports”, “preparing and checking tax returns” are rather among the losers, i.e. these activities are carried out digital by robot software. Activities like declaration advice and design advice are less rule-based and therefore not as affected by digitalization (cf. Fig. 6.14).

The situation is similar with auditors, but not as critical. The share of advisory activities is higher here than in tax consulting professions and therefore less susceptible to replacement by robots and similar technologies. Nevertheless, a number of rule-based tasks such as the classical audit tasks are threatened by digitization (cf. Fig. 6.15). Consulting and expert activities as well as fiduciary administration can be supported by systems but will not disappear.

Effects on selected processes (here: controlling processes)

Controlling means, in a nutshell, planning, steering and control. This activity requires comprehensive methodological knowledge. Below are some examples from controlling

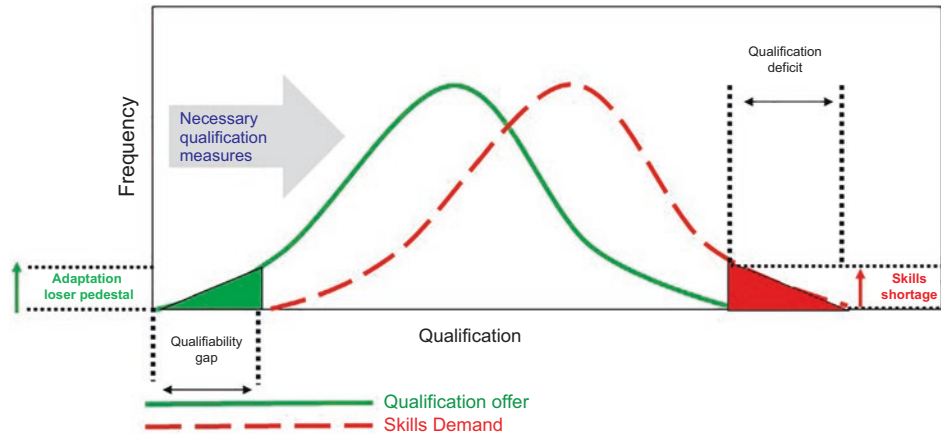


Fig. 6.14 Effects of digitalization on human work

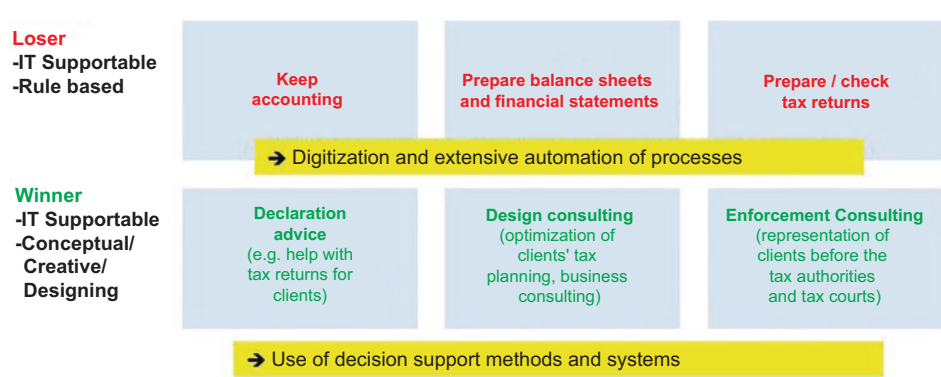


Fig. 6.15 Effects of new technologies on the profession of “tax consultant”

which show that there is still potential for new technologies to be used to support them (taken and modified from Gadatsch 2016, p. 65).

- *Public financial controlling*: The US state of North Carolina was able to use Big Data to combat billing fraud and identify suspicious claims totaling \$200 million.
- *Public customer controlling*: In France, a city evaluates social media posts to identify and prioritize the needs of its citizens.
- *Production controlling*: The analysis of machine parts in operation makes it possible to create dynamic preventive maintenance plans
- *Production controlling*: In this sector, numerous solutions have already been implemented in practice. The software provider “Blue Yonder” reports on a predictive maintenance solution. This allows its software to detect, based on systematically

evaluated machine data, which plants worldwide could have technical problems soon (cf. Blue Yonder 2015).

- *Sales controlling*: Here, an improved analysis of customer behavior, the prediction of customers who are about to leave, and the real-time analysis of the effectiveness of advertising campaigns are possible.
- *IT controlling*: The prediction of system failures and disruptions or clusters of user requests can help improve the stability of information systems and, as a result, simplify IT personnel planning.
- *Financial controlling*: A classic application is the detection of fraud in payment transactions, preferably in real time. The algorithms developed by financial and credit card companies can also be applied to internal payment flows.
- *Personnel controlling*: The shortage of well-trained personnel can be alleviated by the early detection of employees and employees who are willing to leave if timely countermeasures can be taken.

6.5.2 Big Data

Interest in Big Data is steadily increasing (see Google Trends 2015). At the beginning, mainly technical aspects such as storage technologies (e.g. *In-Memory*) and database types (e.g. *No-SQL databases*) were in the foreground. Meanwhile, business applications such as the development of new strategies and business models as well as the optimization of business processes are being discussed.

Origin of the term

The authorship of the term cannot be clarified conclusively (see Klein et al. 2013). Basically, Big Data can be seen as a normal further development of the analysis and use of data. For this purpose, the terms “Decision Support”, “Executive Support”, “Online Analytical Processing”, “Business Intelligence and Analytics” have been used among other things in the past 30 years. Since about 2010, the term Big Data is increasingly used. Often, the so-called “three Vs” volume, velocity and variety of the analyst and consulting firm Gartner are referred to for description (see Beyer and Laney 2012). Big Data is characterized by a high volume of data (Data Volume), enormous speeds of information processing (data velocity) and the variety of data that can be processed (data variety). These features were later supplemented by the aspects of value and validity (see Bachmann et al. 2014, p. 23). Other authors have added the aspect of truthfulness or credibility (veracity) (see Beyer and Laney 2012).

Definition Big Data

One of the first definitions comes from Doug Laney, who defines Big Data 2001 as “data sets that are larger than usual” (cf. Beyer and Laney 2012) has. The approach of using poly structured data makes it clear that not only structured data from ERP systems and

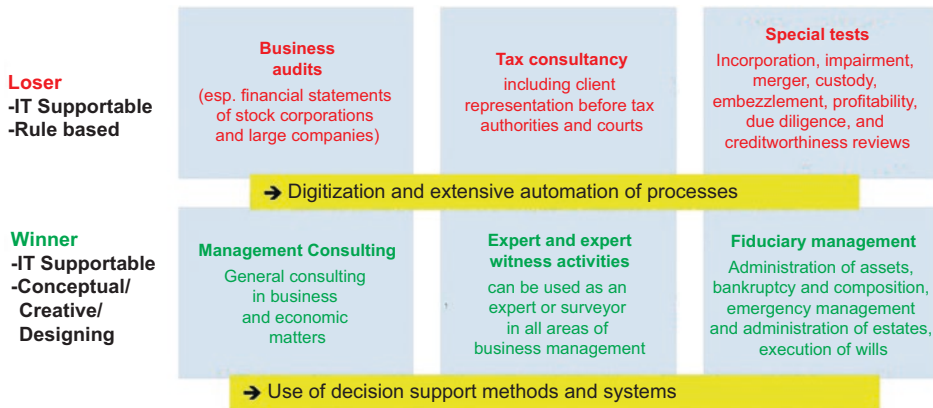


Fig. 6.16 Effects of new technologies on the profession “auditor”

other sources, but also semi- or unstructured data such as videos, images or free text can be used (cf. Fig. 6.16).

A practice-oriented and management-usable definition comes from BITKOM: “Big Data is the [...] economically meaningful extraction and use of decision-relevant findings from qualitatively diverse and differently structured information that is subject to rapid change and occurs in previously unknown quantities” (BITKOM 2012, p. 7). The explanation not only shows the technical background, but focuses on the entrepreneurial aspect that lies behind Big Data. It is about using the new tools for strategic and operational business. Big Data provides tools that support business-critical applications such as sales control or production monitoring and with which new business models can be developed on the basis of available data. Since not all data can be evaluated, it is important for controlling to deal with the essential information. Visualization tools have created new possibilities here.

Many companies use Big Data technologies primarily to accelerate established business processes such as reporting, customer analysis or behavior analysis. However, Big Data offer significantly more growth potential if management is willing to invest in innovative new business models and processes.

6.5.3 Cloud Computing

Clarification of terms

One of the newer terms in IT practice is cloud computing. Often, experts also associate cloud computing with the offerings of the large “hyperscalers” such as Amazon Web Services (AWS) or Google. Unfortunately, the different variants of cloud computing are often presented in a greatly simplified manner, so that one could say superficially that cloud computing is the use of IT resources via the Internet.

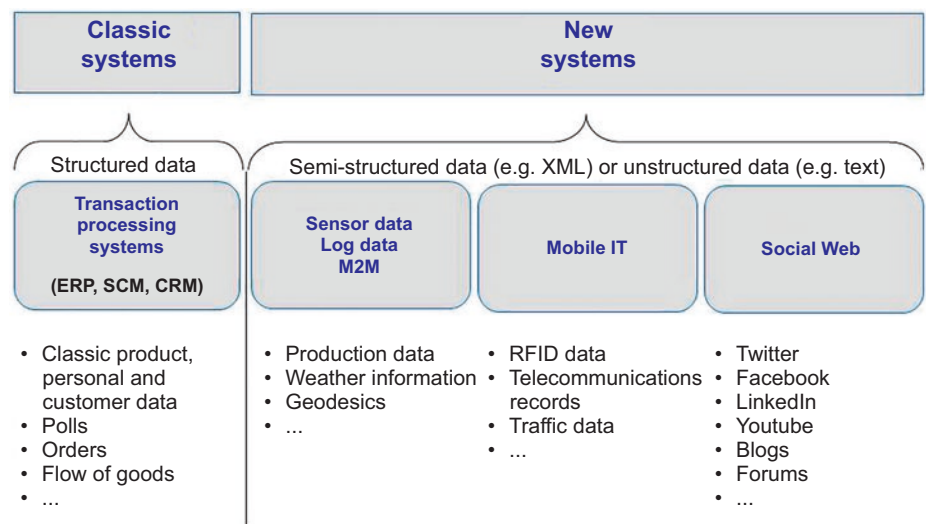


Fig. 6.17 Big Data Data Sources

At its core, cloud computing is about different “sourcing options” when using or procuring IT services (see Brassel and Gadatsch 2019 for details). Cloud computing stands for the use of virtualized hardware.

Fig. 6.17 shows five stages of possible options for providing IT services. In practice, there are also a large number of other variants. The following variants are to be distinguished.

- **On Premise:** The IT infrastructure is operated with **own** personnel on **own** resources (e.g. in a data center). The use is on dedicated physical hardware (e.g. servers). The user of the software is also the operator of the hardware.
- **Private Cloud:** The IT infrastructure is (also) operated with **own** personnel on **own** resources (e.g. in a data center). However, “virtualized” hardware (e.g. servers) is used. The user of the software is also the operator of the (virtualized) hardware.
- **Managed Private Cloud:** The IT infrastructure is operated with **foreign** personnel on own resources (e.g. in a data center). Only “virtualized” hardware (e.g. servers) is used. The user of the software is (only) the owner of the (virtualized) hardware, but no longer the operator. The Managed Private Cloud is therefore an externally operated Private Cloud
- **Outsourced Private Cloud:** The IT infrastructure is operated with **foreign** personnel on **foreign** resources (e.g. in a data center). Again, only “virtualized” hardware (e.g. servers) is used. The user of the software is now neither the owner nor the operator of the (virtualized) hardware. He is “only” a dedicated user of a virtualized environment.

- **Public Cloud:** The IT infrastructure is operated with **foreign** personnel on **foreign** resources (e.g. in a data center). Again, only “virtualized” hardware (e.g. servers) is used. The user of the software is still a “tenant” in the (virtualized) highly standardized environment.

Typical examples of Public Cloud offerings are “Amazon Web Services” or Microsoft’s Office services (Office 365). The following aspects can be added:

- **Private Clouds** are not public, they refer to the provision of cloud computing services for a defined group of users. Access is restricted to authorized persons and is usually via an intranet or VPN.
- **‘Managed Private Cloud’** is operated based on Service Level Agreements (SLAs for short, i.e. agreements on certain IT services) in the customer’s premises by a service provider.
- In the case of the **‘Outsourced Private Cloud’**, the service provider builds or takes over a cloud infrastructure which then remains physically with him.

Properties of Cloud Services

The Cloud-Computing can be assigned the following properties: On-demand access, pay-per-use and elasticity (cf. Biebl [2012](#), p. 24).

- **On-Demand Access:** The user can automatically “book and cancel” IT resources. The provision and termination of the services takes place without direct interaction in a very short time.
- **Pay-per-use:** The IT services are invoiced according to use. Fixed costs usually do not or only to a comparatively small extent. Common billing units are, for example, data transfer volume or data storage volume. The user can trace the calculation basis himself.
- **Elasticity:** The user can rely on seemingly unlimited resources. The provision of the services takes place in the shortest possible time. This is of interest in the event of short-term peaks in demand when large amounts of data have to be analyzed in a short time.

This distinguishes Cloud Computing from classical outsourcing models, which are statically shaped. Classical outsourcing models require lengthy contract negotiations, have longer terms and are only difficult to reverse for the company, which often leads to an undesirable dependence on the IT service provider.

Technical view

Cloud computing is usually considered on four hierarchical levels: “Human as a Service”, “Software as a Service”, “Platform as a Service” and “Infrastructure as a Service”.

The lowest level of the **“Infrastructure as a Service”** provides access to virtual hardware. Typical examples are Amazon’s services for providing virtual servers (“Elastic Compute Cloud”) or providing mass storage by Google (“Cloud Storage”). These services relieve customers from having to operate their own data centers with the necessary security infrastructure.

The layer above is primarily aimed at software developers. **“Platform as a Service”** provides developers with complete development environments. This includes, for example, Microsoft’s “Azure” offering, a platform for creating cloud applications.

The layer most familiar to the end user is **“Software as a Service”**. This includes applications that are primarily aimed at the end user (private or business). The number of possible examples is enormous. Typical applications are email services (web.de), search services (google), office solutions (Microsoft 365) or complete enterprise resource planning systems (SAP Business ByDesign).

The uppermost layer of the **“Human as a Service”** from the user’s point of view is a crowd-sourcing approach. It uses cloud solutions to transfer tasks to human resources. A typical example is Amazon’s “Mechanical Turk” service, which can distribute and monitor micro-tasks to a large number of “crowdworkers”. More recent examples are “transport journeys by private car”, “delivery of drinks and food by bicycle courier”.

Simplified organizational forms

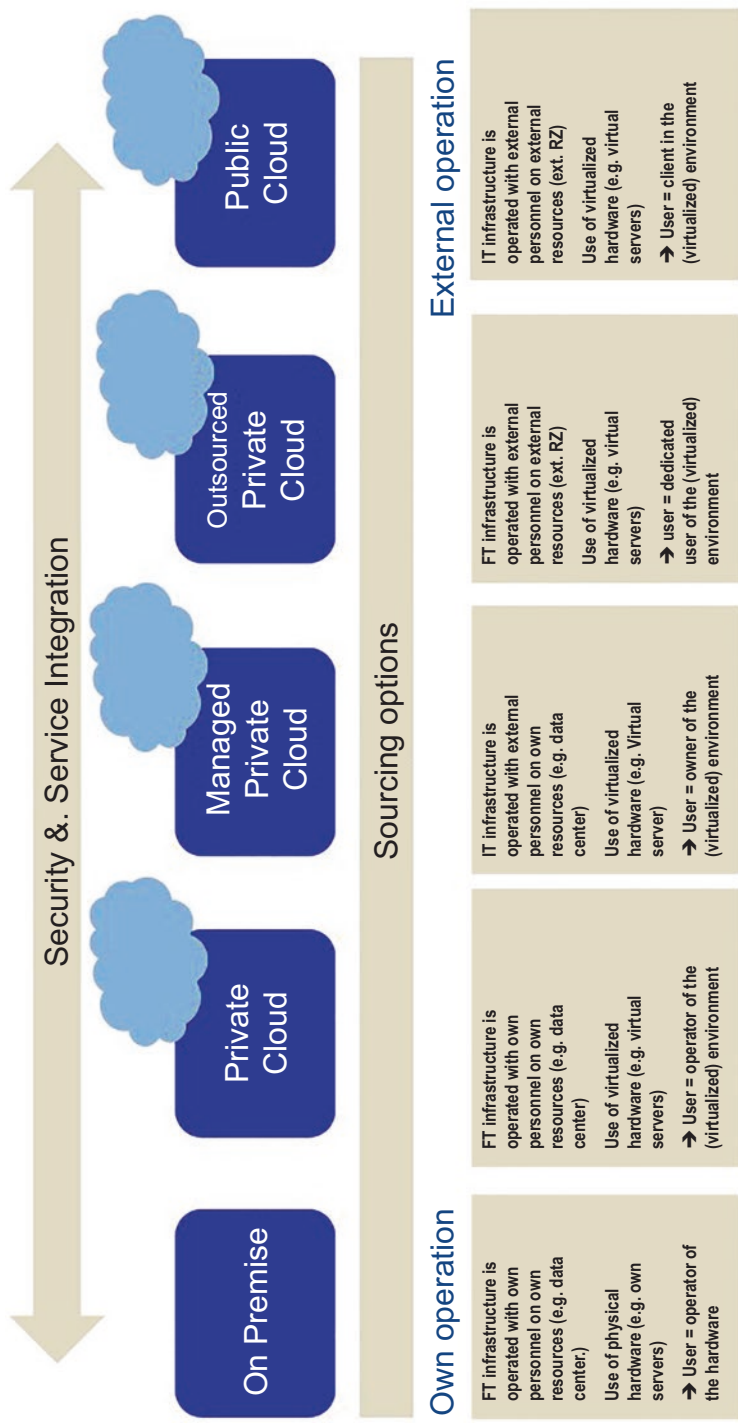
The organization of cloud services is usually greatly simplified with the categories “private cloud”, “community cloud”, “public cloud” and “hybrid cloud” (cf. Fig. 6.18).

In the “private cloud”, providers (e.g. internal IT department) and users (departments of the company) belong to the same user organization (in the example of Fig. 6.18 to the “user organization”). The main motivation for this concept is the security aspect: the data remains completely under the control of the user. However, a private cloud requires enormous investments in hardware, software and personnel.

The “community cloud” allows uniform services to be offered to user organizations with similar requirements in terms of security, compliance or functionality. The operation can be carried out by one or more user organizations, an external organization or any combination thereof. For highly regulated industries such as financial services or health-care, such a solution may be of interest, but also for the public sector.

The “public cloud” is the standard case for cloud computing. Providers and users belong to different user organizations. Access to the cloud is often via an internet-based portal. The use requires a contract between the parties, which is often concluded online in a uncomplicated way.

Under hybrid clouds, any mixed forms are understood. A typical application scenario is the provision of resources for peaks by a provider from the external public cloud. In business intelligence (BI), for example, content from internal (“private cloud”) and external sources (“public cloud”) is mixed and made accessible to external users at least partially (Seufert and Bernhardt 2010).



Private clouds are not public, they denote the provision of cloud computing services for a defined group of users. Access is restricted to authorized persons and is usually via an intranet or VPN. Managed Private Cloud' is operated on the customer's premises by a service provider on the basis of SLAs. In the case of the outsourced private cloud, the service provider builds or takes over a cloud infrastructure which then also remains physically on its premises.

Fig. 6.18 Cloud sourcing options. (Adapted from Münzl et al. 2009; and Brassel and Gadatsch 2019)

The increasing use of cloud services has far-reaching consequences for companies that are difficult to assess, because every employee can become active without the involvement of the IT department, if he has an internet connection and can provide the necessary funds for fee-based services. The proportion of companies that have outsourced at least some of their applications to the “cloud” has increased dramatically (see Chow et al. 2009).

This trend towards “shadow IT” has already led to changes. In the past, the CIO was primarily responsible for supporting users, but his role is changing to advisory, strategic and regulatory activities, which are referred to as IT governance (see Bremmer 2014).

Cloud portfolio for decision makers

Often, the company is faced with the task of finding a suitable solution for the respective application case. The portfolio shown in Fig. 6.19 can be used for this purpose. The relevant processes can be classified according to the criteria “requirements for service integration” and “requirements for data autonomy”. Requirements for service integration relate to aspects such as flexibility, access on demand, payment according to the principle of “pay as you use” and elasticity or scalability. Requirements for data autonomy relate to questions such as the possibility of access to one’s own IT personnel and, above all, control options of who does what with the data when. Accordingly, the five sourcing options from Fig. 6.17 can be classified (see Fig. 6.19). An important aspect of outsourcing or cloud computing is that activities move from the inside to the outside and that management of interfaces to the provider must be given increased attention. Otherwise, there is a risk of increased susceptibility to process chain disruptions (Kühl 2015, p. 116) (Fig. 6.20).

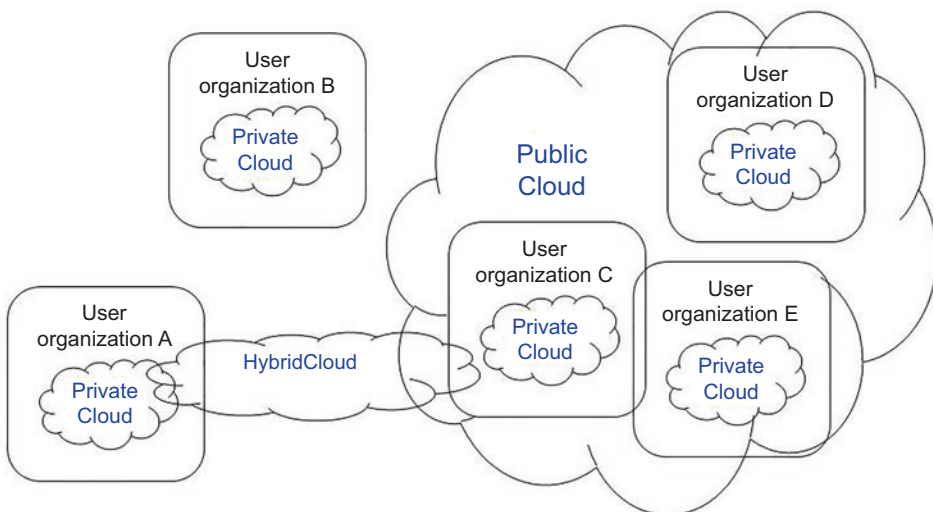


Fig. 6.19 Cloud organization. (Based on Baun et al. 2010, p. 26)

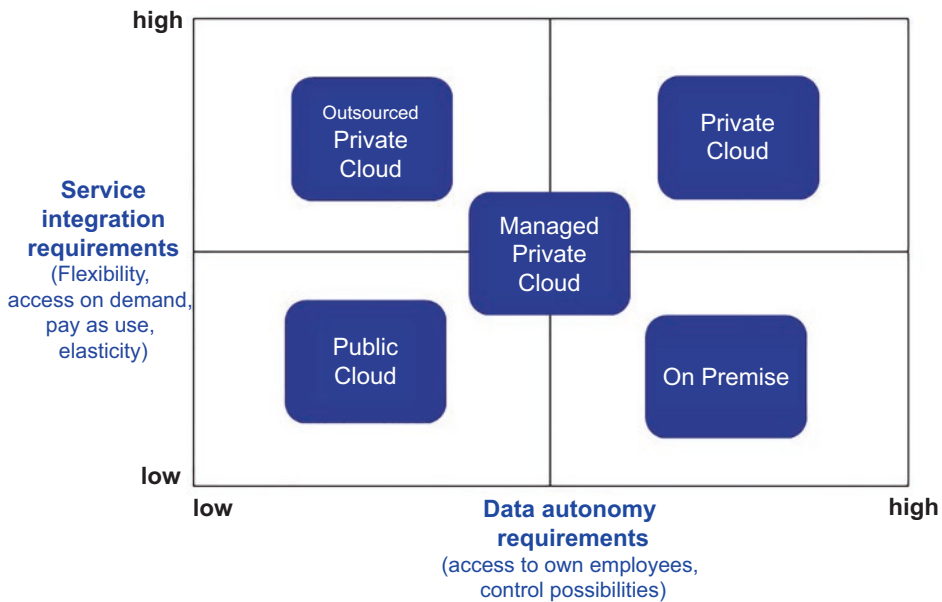


Fig. 6.20 Cloud portfolio for decision makers

6.5.4 Industry 4.0/Internet of Things

The term “Industry 4.0” is closely related to the “Internet of Things”. This refers to networked production resources (machines, devices, buildings, etc.), people and intelligent objects that know their status, use and history. All objects are merged into Cyber-Physical-Systems (CPS) in a Smart Factory, which execute production processes flexibly (cf. Working Group Industry 4.0 2013).

The trend topic “Industry 4.0” or “Internet of Things” not only causes comprehensive activities in industry, but also the German federal government is trying to strengthen the location Germany by flank measures. The Federal Ministry of Economics and Energy has commissioned studies on the exploitation of the potentials of the application of “Industry 4.0” in small and medium-sized enterprises by external specialists and is setting up Industry 4.0 competence centers to support small and medium-sized enterprises in the introduction and use of Industry 4.0. The task of the centers is to translate “Industry 4.0” into the language of small and medium-sized enterprises in order to make them more flexible. This should enable a change in the often family-oriented management of small and medium-sized enterprises.

Despite the current intensive discussion, the maturity level of Industry 4.0 is rather modest. The technologies discussed in the context of Industry 4.0, such as Augmented Reality, Machine-to-Machine Communication, Virtual Reality, Enterprise 3D-Printing, will reach the necessary maturity level for productive use in regular operation in 10 years (cf. Siepmann and Roth 2016, p. 251).

6.6 Review Questions and Exercises

6.6.1 Questions

1. Explain the difference between graphic programs and special modeling tools.
2. What are the central questions to be considered when choosing a modeling tool?
3. Describe the core functionality of a workflow management system.
4. Explain the term ERP system and list some areas of a company that are typically supported by ERP systems.
5. Explain why the introduction and use of ERP systems lead to a feedback loop (life cycle).
6. Why can ERP systems also be classified as process control systems?
7. Discuss the claim that when using individual software, the dependence on manufacturers is lower than when using standard software.
8. For what reasons do many companies increasingly use standard software?
9. Explain the need to use reference process models within the life cycle model for the introduction of standard software.
10. What speaks in the following cases for or against an individual development of software?
 - Replacement of a 30-year-old logistics software that no longer meets today's requirements.
 - Replacement of a 15-year-old payroll system.
 - Introduction of a modern email software for internal communication.
11. Which introduction strategy do you recommend in the following cases?
 - Introduction of a centrally used accounting software at a food discounter?
 - Introduction of a logistics system at an electronics retailer with branches and online shop?
 - Introduction of a campus management system at a university with one location?
12. How do new trends, such as digitization, affect process management?
13. Explain possible sourcing options for IT services?
14. What is the difference between a "Private Cloud" and a "Managed Private Cloud"?

6.6.2 Case Study

The case study looks at a plant engineering company with around 1400 employees. Around 75% of them work at the headquarters in Germany. The rest of the staff work worldwide in regional locations, sales offices and branches. The annual turnover is 640 million euros.

DESCRIPTION OF THE SITUATION

The central department of organization and IT is responsible for organization and IT planning, the data center and application development as well as the PC user service and reports to the commercial board of directors. The PC user service is provided by an external service provider. For the implementation of the currently largest IT project “Introduction of an accounting standard software”, a large software house with corresponding experience in such projects was commissioned. The company introduces a complex accounting standard software. The project budget is about 2 million euros without costs for new hardware. So far, a largely self-developed software has been used, which does not meet the requirements of the company. The old system has hardly been further developed in recent years for cost reasons. The goal of the project is the complete replacement of the old system and the most comprehensive use of the standard software. With the implementation of the introduction project, a software house was commissioned, as no specific expertise is available in the company. As part of the project, the company’s own employees should be trained so that they can independently take over the later support and further development of the standard software. The project was divided into five functionally oriented sub-projects and thus adapted to the organizational responsibilities in the company: accounting, personnel, logistics and production, sales as well as a cross-sectional sub-project technology. The project manager is an employee of the IT department who has a business education and many years of experience. He reports to the head of organization and IT, who also chairs the steering committee. The steering committee includes the heads of the organizational units accounting, personnel, etc. The competent commercial board of directors receives serious indications from his employees after only a few months about the project progress, which are briefly outlined in the following sections.

Status of work

The subject matter concepts of the sub-projects accounting and personnel have been largely created, since the responsible executives could agree on the largely use of the standard functions of the software. Due to the strong integration of the standard software modules, there are still several cross-departmental tasks relating to the sub-project logistics and production as well as the sub-project sales to be regulated. For example, the procurement process successively passes through the departments of purchasing, goods receipt, invoice verification, accounts payable and general ledger. Since the overall process must be coordinated, regulations must be made in several subject matter concepts. The subject matter concepts for the sub-projects logistics and production as well as sales are incomplete. Key business processes are still under discussion. The reason for this is that the current workflows in these areas of responsibility are very far from the reference processes of the standard software and no agreement has yet been reached on a process restructuring. The heads of the specialist departments insist in the discussion with the consultants of the software house on the transfer of historically grown workflows into the standard software and in particular on the retention of the previous organizational

responsibilities. The software consultants argue that the company's processes can be solved within the scope of the standard software with a greater willingness to business reengineering. However, they cannot always prevail in the discussion with the responsible employees of the specialist departments. The sub-project technology includes technical tasks in the narrower sense (construction and commissioning of the hardware, networking, etc.) as well as the creation of a authorization concept. This includes the organizational and subject-related regulation of responsibilities for processes (e.g. Who is allowed to carry out the creditor payment run?, Who is allowed to create and change supplier and customer master data?) And objects (access to individual cost centers, materials, personnel data, etc.) and their technical implementation in system tables. Due to the still incomplete description of the subject-specific concepts, not all authorizations have been able to be specified and implemented so far.

Project organization

The members of the project team are distributed across several locations. Project meetings take place weekly in different meeting rooms that can be rented individually. There is no central project office available. Short-term meetings with more than four people are often not possible due to the lack of suitable meeting rooms. Numerous employees from the specialist departments are not released from their regular activities. This has led to numerous schedule conflicts in the past, with the result that everyday business has taken precedence over project activities on several occasions. Individual consultants of the software company are active in other projects of other customers. In the sub-project logistics and production, complaints from employees of the specialist department about the unavailability of individual consultants are accumulating. Some sub-project managers of the specialist departments are not allowed to make binding decisions for the project, as their respective managers have reserved important decisions for themselves. This regularly leads to delays in project work when difficult questions must be answered, e.g. when business processes and organizational regulations have to be changed, since the software consultants have to convince several employees of the specialist department.

TASK

The commercial board would like to gain an independent impression of the situation of the project and has commissioned an independent external consultant to develop solutions to improve the situation.

SOLUTION PROPOSAL

A basic problem of the project is the disregard of the connection between business reengineering and the use of information technology. The introduction of standard software usually only leads to success if the company adapts its processes to the intended possibilities of the standard software. The insistence on traditional solutions increases the introduction costs and the later maintenance costs, for example when changing releases.

Modern business standard software usually requires a process organization that is obviously not present in this case. The Board should stop the project and take a restructuring phase in which, first a suitable reorganization oriented towards the reference processes of the selected standard software is thought through. If the standard software does not cover the business objectives in the core area of the company (production, logistics and sales), the selection decision may also have to be reconsidered. The functional cut of the project favors departmental thinking and departmental egoism. This also affects the chosen project organization, which is a mirror image of the organizational structure.

After the concept for the process organization (see above) has been created, the project organization should not be structured according to functional tasks, but according to the most comprehensive process chains (e.g. subprojects for order processing, spare parts business, etc.). The project members must be given the competence for decisions by the responsible managers (process responsible). For the duration of the project, the core team must receive a central project office with conference and work rooms. The consultants of the commissioned software house must be available throughout the project period. The project manager should report to the Board of Directors, as this is a company-critical project. The steering committee is to be newly appointed, depending on the future process organization.

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